# Quantum with Unity Pro
## Hardware
## Reference Manual

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When devices are used for applications with technical safety requirements, the relevant instructions must be followed.

Failure to use Schneider Electric software or approved software with our hardware products may result in injury, harm, or improper operating results.

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# Safety Information

## Important Information

### NOTICE

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

**⚠ DANGER**

**DANGER** indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

**⚠ WARNING**

**WARNING** indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

**⚠ CAUTION**

**CAUTION** indicates a hazardous situation which, if not avoided, **could result** in minor or moderate injury.

***NOTICE***

***NOTICE*** is used to address practices not related to physical injury.

## PLEASE NOTE

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

## BEFORE YOU BEGIN

Do not use this product on machinery lacking effective point-of-operation guarding. Lack of effective point-of-operation guarding on a machine can result in serious injury to the operator of that machine.

### ⚠ WARNING

**UNGUARDED EQUIPMENT**

- Do not use this software and related automation equipment on equipment which does not have point-of-operation protection.
- Do not reach into machinery during operation.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

This automation equipment and related software is used to control a variety of industrial processes. The type or model of automation equipment suitable for each application will vary depending on factors such as the control function required, degree of protection required, production methods, unusual conditions, government regulations, etc. In some applications, more than one processor may be required, as when backup redundancy is needed.

Only you, the user, machine builder or system integrator can be aware of all the conditions and factors present during setup, operation, and maintenance of the machine and, therefore, can determine the automation equipment and the related safeties and interlocks which can be properly used. When selecting automation and control equipment and related software for a particular application, you should refer to the applicable local and national standards and regulations. The National Safety Council's Accident Prevention Manual (nationally recognized in the United States of America) also provides much useful information.

In some applications, such as packaging machinery, additional operator protection such as point-of-operation guarding must be provided. This is necessary if the operator's hands and other parts of the body are free to enter the pinch points or other hazardous areas and serious injury can occur. Software products alone cannot protect an operator from injury. For this reason the software cannot be substituted for or take the place of point-of-operation protection.

Ensure that appropriate safeties and mechanical/electrical interlocks related to point-of-operation protection have been installed and are operational before placing the equipment into service. All interlocks and safeties related to point-of-operation protection must be coordinated with the related automation equipment and software programming.

**NOTE:** Coordination of safeties and mechanical/electrical interlocks for point-of-operation protection is outside the scope of the Function Block Library, System User Guide, or other implementation referenced in this documentation.

## START-UP AND TEST

Before using electrical control and automation equipment for regular operation after installation, the system should be given a start-up test by qualified personnel to verify correct operation of the equipment. It is important that arrangements for such a check be made and that enough time is allowed to perform complete and satisfactory testing.

**⚠ WARNING**

**EQUIPMENT OPERATION HAZARD**

- Verify that all installation and set up procedures have been completed.
- Before operational tests are performed, remove all blocks or other temporary holding means used for shipment from all component devices.
- Remove tools, meters, and debris from equipment.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

Follow all start-up tests recommended in the equipment documentation. Store all equipment documentation for future references.

**Software testing must be done in both simulated and real environments.**

Verify that the completed system is free from all short circuits and temporary grounds that are not installed according to local regulations (according to the National Electrical Code in the U.S.A, for instance). If high-potential voltage testing is necessary, follow recommendations in equipment documentation to prevent accidental equipment damage.

Before energizing equipment:

- Remove tools, meters, and debris from equipment.
- Close the equipment enclosure door.
- Remove all temporary grounds from incoming power lines.
- Perform all start-up tests recommended by the manufacturer.

## OPERATION AND ADJUSTMENTS

The following precautions are from the NEMA Standards Publication ICS 7.1-1995 (English version prevails):

- Regardless of the care exercised in the design and manufacture of equipment or in the selection and ratings of components, there are hazards that can be encountered if such equipment is improperly operated.
- It is sometimes possible to misadjust the equipment and thus produce unsatisfactory or unsafe operation. Always use the manufacturer's instructions as a guide for functional adjustments. Personnel who have access to these adjustments should be familiar with the equipment manufacturer's instructions and the machinery used with the electrical equipment.
- Only those operational adjustments actually required by the operator should be accessible to the operator. Access to other controls should be restricted to prevent unauthorized changes in operating characteristics.

# About the Book

## At a Glance

### Document Scope

This documentation is a reference for the hardware of the Quantum automation system with Unity Pro.

### Validity Note

This document is valid for Unity Pro 13.0 or later.

### Product Related Information

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

The application of this product requires expertise in the design and programming of control systems. Only persons with such expertise should be allowed to program, install, alter, and apply this product.

Follow all local and national safety codes and standards.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

# Part I
# Introduction

## Introduction

The following part provides an overview of the Quantum Automation System

## What Is in This Part?

This part contains the following chapters:


| Chapter | Chapter Name | Page |
|---|---|---|
| 1 | System | 19 |
| 2 | System Components | 23 |
| 3 | Network Support | 35 |
| 4 | Quantum Configurations | 59 |
| 5 | Module Configuration | 73 |
| 6 | Hardware Installation and Maintenance | 77 |

# Chapter 1
# System

## Purpose

This chapter provides an overview on the Quantum system.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| System Overview | 20 |
| Typical Quantum System Configuration | 21 |

# System Overview

## Overview

The Quantum system is a special-purpose computing system with digital processing capabilities. Quantum is designed for real time control in industrial and manufacturing applications in a modular, expandable architecture employing the following modules:

- Controller Modules (CPU)
- Power Supply Modules (CPS)
- I/O Modules (Dxx, Axx)
- Network Interface Modules (including Field Bus Modules)
- Intelligent / Special Purpose Modules
- Simulator (XSM) and Battery (XCP) Modules
- Racks (XBP) and Rack Expander (XBE)
- CableFast Cabling (CFx)

## Expandable Architecture

Based on the local rack, the Quantum I/O system can be expanded by Network Interface modules with the following architecture:

| Network | Network Interface Modules | Media |
|---|---|---|
| Remote I/O (RIO) | RIO Head, RIO Drop | RIO Coax cable |
| Distributed I/O (DIO) | NOM, DIO Drop | Twisted Pair |

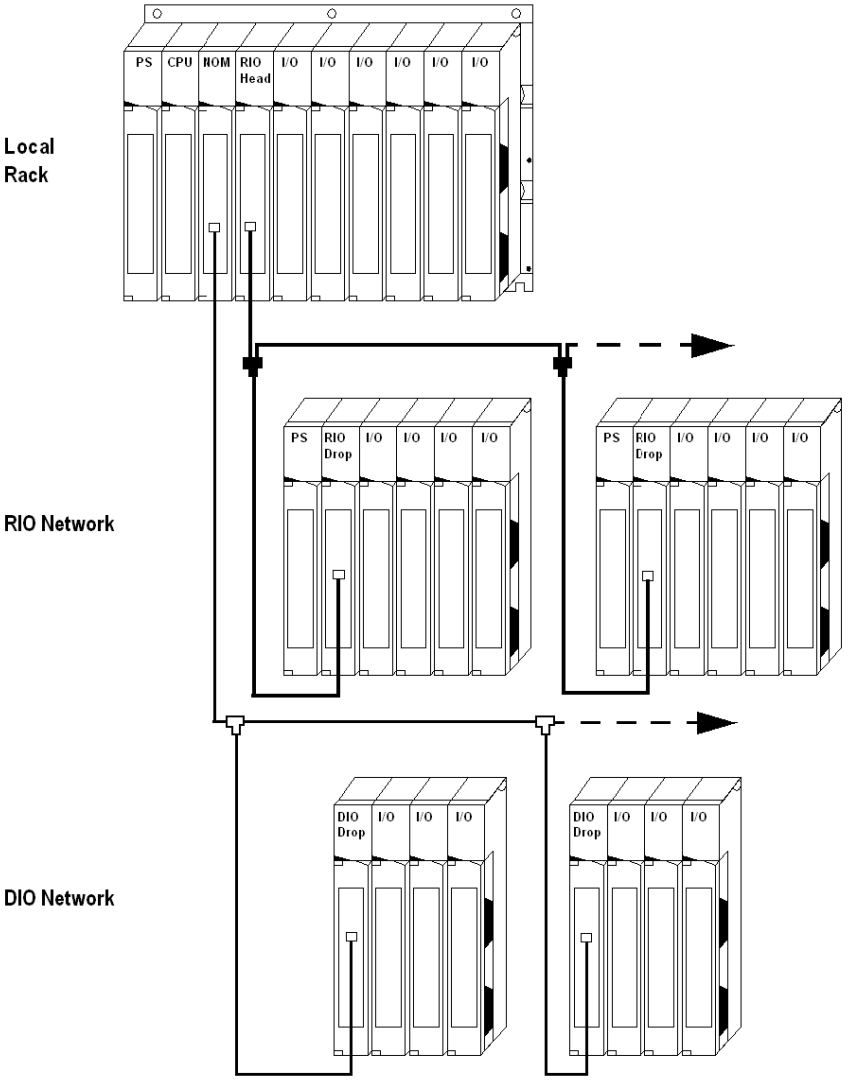
## Field Bus

By the means of field bus modules the Quantum I/O system supports the following field buses:

- AS-i

# Typical Quantum System Configuration

## Typical System Block Diagram



# Chapter 2
# System Components

## Purpose

This chapter provides an overview of the Quantum system components.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Controller Modules (CPUs) | 24 |
| Power Supply Modules (CPS) | 25 |
| I/O Modules | 26 |
| Network Interface Modules | 27 |
| Intelligent/Special Purpose I/O Modules | 28 |
| Simulator (XSM) and Battery (XCP) Module | 29 |
| Racks (XBP) and Rack Expander (XBE) | 30 |
| CableFast Cabling (CFx) for I/O Modules | 31 |
| Hot Standby System | 33 |

# Controller Modules (CPUs)

## Overview

The Quantum CPU serves as a bus master controlling the local, remote, and distributed I/O of the Quantum system.

The module is on the Quantum local I/O rack. It is a digitally operating electronic system, which uses a programmable memory for the internal storage of user instructions. These instructions are used to implement specific functions such as:

- Logic
- Process sequencing
- Timing
- Coupling
- Arithmetic

These instructions allow control through digital and analog outputs, for various types of machines and processes.

**NOTE:** For detailed Information see *Controller Modules (CPUs), page 91*

# Power Supply Modules (CPS)

## Overview

Quantum power supplies are used to supply system power to modules inserted into the rack, including:

- Quantum CPU modules
- Interface modules
- Quantum I/O modules

Depending upon the system configuration, the option exists of using the power supply in three different modes.

## Power Supply Modes Table

The following table shows the power supply modes.

| Power Supply Type | Usage |
|---|---|
| Standalone | For 3 A or 8 A configurations that do not require fault tolerant or redundant capabilities. |
| Standalone Summable | For configurations consuming more than the rated current of one supply, two summing power supplies can be installed in the same rack. |
| Standalone Redundant | For configurations requiring power for uninterrupted system operation. Two redundant power supplies are required for power redundancy. |

**⚠ WARNING**

**UNEXPECTED BEHAVIOR OF APPLICATION**

Use appropriate power supplies with the exceptions noted in the Power Supplies chapter.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

# I/O Modules

## Overview

Quantum I/O modules are electrical signal converters that convert signals to and from field devices to a signal level and format that can be processed by the CPU.

I/O modules are optically isolated to the bus. I/O modules are also software configurable.

## Field Devices

Typical field devices include:

- limit switches
- proximity switches
- temperature sensors
- solenoids
- valve actuators

## Further Information

**NOTE:** For detailed information see in the Quantum I/O Hardware Guide:

- Quantum I/O Analog IN Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*
- Quantum I/O Analog OUT Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*
- Quantum I/O Analog IN / OUT Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*
- Quantum I/O Discrete IN Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*
- Quantum I/O Discrete OUT Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*
- Quantum I/O Discrete IN / OUT Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*
- Quantum Intrinsically Safe Analog/Digital Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*
- Quantum Safety I/O Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*

# Network Interface Modules

## Overview

Different types of network interface modules are available and presented in the table below with their descriptions.

## Network Interface Modules Table

The following table shows the network interface modules.

| Type | | Description |
|---|---|---|
| RIO Part *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)* | | Single and dual channel Remote I/O interface modules (RIO heads and drops) connected via a coaxial cable network. |
| DIO Part *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)* | | Single and dual channel Distributed I/O interface modules connected via a twisted pair Modbus Plus cable network. |
| Modbus Plus Part *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)* | | • Single and dual channel Network Option Modules (NOM) connected via a twisted pair Modbus Plus cable network.<br>• Modbus Plus on fiber module connected via a fiber optic Modbus Plus cable network. |
| Ethernet Module *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)* | TCP/IP | Single channel TCP/IP Ethernet interface module connected via a twisted pair or fiber optic cable network. |
| | SY/MAX | SY/MAX Ethernet module connected via a twisted pair or fiber optic cable network. |
| Field Bus Part *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)* | Lonworks | Lonworks module connected via a twisted-pair network. |
| | Interbus | Interbus Interface module connected via a twisted pair network. |
| | Profibus | Profibus interface module connected via a Profibus port |
| | AS-i | The Quantum AS-i Master Module provides AS-i communications between the bus master module and the sensor/actuator slaves. |
| | Sercos | The SERCOS® Multi-Motion modules (MMS) are used to build a distributed automation solution, integrating motion applications with control applications |

# Intelligent/Special Purpose I/O Modules

## Overview

Quantum Intelligent/Special Purpose I/O modules operate with minimum intervention from the Quantum controller after initial downloading of module parameters or programs. The Quantum intelligent/special purpose I/O modules include the following:

- High Speed Counter modules (EHC)
- ASCII Interface module (ESI)
- High Speed Interrupt module (HLI)
- Time Stamp modules (ERT)
- Clock module (DCF)

# Simulator (XSM) and Battery (XCP) Module

## Overview

There are two types of simulator modules, as described below.

## Discrete and Analog Simulators Table

The following table shows discrete and analog simulators.

| Simulator | Points / Channels | Type | Description |
|---|---|---|---|
| Discrete Simulator | 16 Points IN | 140 XSM 002 00 | is used to generate up to 16 binary input signals to the 140 DAI 540 00 and the 140 DAI 740 00 AC input modules. |
| Analog Simulator | 2 channels IN<br>1 channel OUT | 140 XSM 010 00 | is used for simulating 4 ... 20 mA field current loops used with current input Quantum modules. |

## Battery Module (XCP)

The Quantum battery module provides RAM backup power for Quantum expert modules.

## Further Information

For detailed information see *Simulator (XSM) and Battery (XCP) Module, page 321*.

# Racks (XBP) and Rack Expander (XBE)

## Racks (XBP)

Quantum racks may be used in any locations of local, remote, or distributed I/O. There are six racks available in 2, 3, 4, 6, 10, and 16 slot versions.

## Rack Expander (XBE)

The 140 XBE 100 00 Rack Expander provides the Modicon Quantum with the capacity to expand Local and Remote I/O drops to a second rack. For improved I/O capacity and efficiency, the rack expander will save money by minimizing the number of Remote I/O drops. The Rack Expander also improves the overall performance of Remote I/O based systems by reducing the number of Remote I/O drops the Quantum CPU has to service. The Rack Expander effectively doubles the maximum number of Discrete I/O that can be serviced by a Quantum Remote I/O system.

## Further Information

**NOTE:** For detailed information see *Racks(XBP) and Rack Expander (XBE), page 305*

# CableFast Cabling (CFx) for I/O Modules

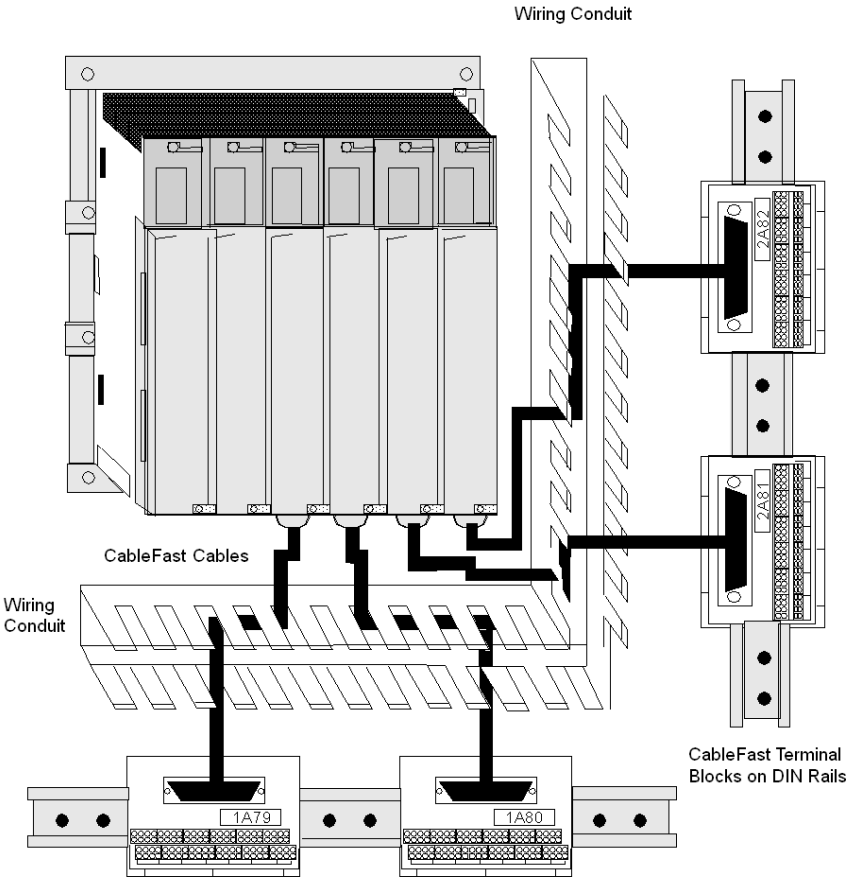
## Overview

The CableFast wiring system consists of pre-wired Quantum field wiring terminal strips for I/O modules.

## Features

It is available in various cable lengths, that are terminated with D-type connectors. The D connectors plug into DIN rail-mounted terminal blocks offered in straight through or special application versions. Cables and terminal blocks are ordered separately and the terminal blocks may be used with any cable length. Pigtail cable versions are also available.

## Quantum I/O Modules with CableFast Components




**NOTE:** For further details refer to *CableFast Cabling, page 395*.

# Hot Standby System

## Function

A Hot Standby (HSBY) system is based on two identically configured programmable logic controllers linked to each other and to the same remote I/O network. If one controller stops, the other assumes control of the I/O system.

## Primary and Standby Controller

The Quantum Hot Standby system is designed for use where downtime cannot be tolerated. The system delivers high availability through redundancy. Two racks are configured with identical hardware and software. One of the PLCs acts as the primary controller. It runs the application by scanning user logic and operating remote I/O. The other PLC acts as the standby controller. The primary controller updates the standby controller after each scan. The standby is ready to assume control within one scan if the primary stops. Primary and standby states are switchable. Either controller can be put into the primary state, but to do this, the other must be in the standby state. The remote I/O network is operated by the primary controller.

**NOTE:** A Quantum Hot Standby system supports only remote I/O. It does not support local I/O or distributed I/O (DIO).

For a detailed description of the Hot Standby (HSBY) system refer to the *Quantum HSBY User Manual*.

# Chapter 3
# Network Support

## Purpose

This chapter provides an overview of the Quantum Network Support.

## What Is in This Chapter?

This chapter contains the following sections:


| Section | Topic | Page |
|---|---|---|
| 3.1 | General Information | 36 |
| 3.2 | Remote I/O (RIO) and Distributed I/O (DIO) | 41 |
| 3.3 | Ethernet Networking Interfaces | 46 |
| 3.4 | Modbus/Modbus Plus Networking Interfaces | 49 |
| 3.5 | Fieldbus Networking Interfaces | 54 |

# Section 3.1
## General Information

### Purpose

This section provides general information on the Quantum Network Support.

### What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| Supported Networks Table | 37 |
| Quantum Network Interface Techniques | 39 |

# Supported Networks Table

## Supported Networks Table

The following table shows the Quantum supported networks.

| Service Description | Modbus | Modbus Plus | Remote I/O | Ethernet | | AS-i | Profibus | INTERBUS | SERCOS |
|---|---|---|---|---|---|---|---|---|---|
| | | | | TCP/IP | SY/MAX | | | | |
| Native to Quantum CPU | + | + | - | +(6) | - | - | - | - | - |
| Available on a Network Module | + | + | + | + | + | + | + | + | + |
| CPU Programming | +(1) | + | - | + | - | - | - | - | - |
| CPU Executive Firmware Loading Support | +(1) | +(1) | - | - | - | - | - | - | - |
| Module Firmware Loaded From CPU | + | + | + | + | + | - | + | -(5) | + |
| Report By Exception Communications | +(2) | + | - | + | + | - | - | - | - |
| Multi-node Broadcast Communications | - | +(1) | - | - | - | - | - | - | - |
| Synchronized I/O Scanning | - | - | + | - | - | - | - | - | + |
| NonSynchronized I/O Scanning | - | +(1) | - | - | - | - | - | + | - |
| Quantum I/O Drops | - | +(1) | + | - | - | - | - | - | - |
| Hot Standby Quantum I/O Drop Support | - | - | + | - | - | - | - | - | - |

**(1)** Refer to the Modbus Plus portion of the Quantum Specifier's guide section for details of available services on 140 NOM 2•• 00 Modbus Plus Network modules
**(2)** Service is only available on the native controller Modbus port when the XMIT block is used
**(3)** Available from third parties
**(4)** The SERCOS network standard is fiber optics.
**(5)** Module firmware loaded through serial port on module.
**(6)** Refers to HE-CPUs.

| Service Description | Modbus | Modbus Plus | Remote I/O | Ethernet | | AS-i | Profibus | INTERBUS | SERCOS |
|---|---|---|---|---|---|---|---|---|---|
| | | | | TCP/IP | SY/MAX | | | | |
| Hot Standby Data Communications Support | + | + | - | + | - | + | - | - | +(7) |
| Optional Dual Cabling | - | +(1) | + | - | - | - | - | - | - |
| Optional Fiber Optics | +(3) | +(1) | + | + | + | - | - | +(3) | +(4) |
| Momentum I/O Drops | - | +(1) | - | + | - | - | + | + | - |
| Variable Speed Drives | +(3) | +(1) | - | - | - | - | - | + | - |
| Servo Motion Control | - | +(1) | - | - | - | - | - | +(3) | + |
| HMI: Displays & Panels | + | +(1) | + | - | - | - | - | +(3) | - |
| HMI: Workstations | + | +(1) | - | + | + | - | - | - | - |

**(1)** Refer to the Modbus Plus portion of the Quantum Specifier's guide section for details of available services on 140 NOM 2•• 00 Modbus Plus Network modules
**(2)** Service is only available on the native controller Modbus port when the XMIT block is used
**(3)** Available from third parties
**(4)** The SERCOS network standard is fiber optics.
**(5)** Module firmware loaded through serial port on module.
**(6)** Refers to HE-CPUs.

# Quantum Network Interface Techniques

## Overview

Quantum communication and networking modules use a variety of different techniques to interface to the Quantum controller over the local rack.

## Direct CPU Driver

This technique allows the CPU to control high speed data transfers to and from the communication and networking modules, maximizing throughput and performance.

This technique is used extensively by the Remote I/O network and Hot Standby system to provide a highly deterministic synchronization of the CPU and I/O scans.

**NOTE:** Only one Remote I/O Head Interface is supported for each Quantum CPU.

## Option Module Interface

This technique allows the communication and networking modules to control data transfers to and from the CPU, maximizing the flexibility of the communications interface.

This technique is used extensively by the Modbus Plus and Ethernet peer-to-peer network modules. The number of option module interfaces supported by each CPU model is described in the following table.

## CPU Interface Support Table

The following table shows the summary of Quantum CPU option module interface support.

| Quantum Controller Model Number | Available Option Module Interfaces Supported Per CPU |
|---|---|
| 140 CPU 678 61 | 6 |
| 140 CPU 672 61 | 6 |
| 140 CPU 672 60 | 6 |
| 140 CPU 671 60S(1) | 6 |
| 140 CPU 671 60 | 6 |
| 140 CPU 670 60 | 3 |
| 140 CPU 658 60 | 6 |
| 140 CPU 652 60 | 6 |
| 140 CPU 651 60S(1) | 6 |
| 140 CPU 651 60 | 6 |
| 140 CPU 651 50 | 6 |
| (1) **NOTE:** Safety CPUs (140 CPU 651 60S and 140 CPU 671 60S) support only 140 NOE 771 11. | |

| Quantum Controller Model Number | Available Option Module Interfaces Supported Per CPU |
|---|---|
| 140 CPU 534 14A/B/U | 6 |
| 140 CPU 434 12A/U | 6 |
| 140 CPU 311 10 | 2 |
| (1) **NOTE:** Safety CPUs (140 CPU 651 60S and 140 CPU 671 60S) support only 140 NOE 771 11. | |

## Communications and Networking Table

The following table shows the Quantum communications and networking modules.

| Model Number | Description | Module Interface Technique | Rack Support | | | Bus Power mA |
|---|---|---|---|---|---|---|
| | | | Local | RIO | DIO | |
| 140 CRA 312 00 | Remote I/O Adapter | Direct CPU Driver | N | Y | N | 1,000 |
| 140 CRP 312 00 | Remote I/O Head | Direct CPU Driver | Y | Y | N | 1,000 |
| 140 CRP 931 00 | Remote I/O Head Interface, single cable | Direct CPU Driver | Y | N | N | 780 |
| 140 CRP 932 00 | Remote I/O Head Interface, dual cable | Direct CPU Driver | Y | N | N | 780 |
| 140 NOM 211 00 | Modbus Plus Options, single cable | Option Module | Y | N | N | 780 |
| 140 NOM 212 00 | Modbus Plus Option, dual cable | Option Module | Y | N | N | 780 |
| 140 NOM 252 00 | Modbus Plus Option, single channel fiber | Option Module | Y | N | N | 900 |
| 140 NOE 211 00 | Ethernet TCP/IP Twisted Pair | Option Module | Y | N | N | 1000 |
| 140 NOE 251 00 | Ethernet TCP/IP Fiber Optic | Option Module | Y | N | N | 1000 |
| 140 NOE 311 00 | One 10BASE-T Ethernet/ SY/MAX (RJ45) port. | Option Module | Y | N | N | 1000 |
| 140 NOE 351 00 | Two 10BASE-T Ethernet/ SY/MAX (RJ45) port. | Option Module | Y | N | N | 1000 |
| 140 NOE 771 •• | Ethernet TCP/IP Twisted Pair/Fiber Optic | Option Module | Y | N | N | 750 |
| 140 EIA 921 00 | AS-i Master | I/O Map (13/9) | Y | Y | Y | 250 |
| 140 MMS 425 00 | Multi-Axis Motion Controller w/SERCOS | Option Module | Y | N | N | 2500 |
| PTQ PDP MV1 | Profibus Option Module | Option Module | Y | N | N | 800 |

# Section 3.2
## Remote I/O (RIO) and Distributed I/O (DIO)

### Purpose

This section provides information on the Quantum (RIO) and (DIO).

### What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| Remote I/O (RIO) | 42 |
| Execution of Quantum Sections with Remote Inputs/Outputs | 43 |
| Distributed I/O (DIO) | 45 |

# Remote I/O (RIO)

## Overview

There are the following types of Quantum RIO Modules:

- RIO Heads:
  - 140 CRP 931 00 *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*
  - 140 CRP 932 00 *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*
- RIO Drops:
  - 140 CRA 931 00 *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*
  - 140 CRA 932 00 *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*

They use an S908-based networking I/O configuration. Communication is done via single or dual coaxial cabling up to 15,000 feet away. This configuration supports a mix of the following product lines:

- SY/MAX
- 800 Series
- Quantum I/O

When Quantum RIO is required, the Quantum controller may support up to 31 RIO drops. In an RIO configuration, an RIO head module is connected with coaxial cable to RIO drop modules at each remote drop.

**NOTE:** In a single-cable configuration, it is recommended to connect the cable in channel A on both Local I/O (CRP) and Remote I/O (CRA).

**NOTE:** In a dual-cable configuration, it is mandatory to connect channel A from Local I/O (CRP) with channel A from Remote I/O (CRA) and channel B from Local I/O (CRP) with channel B from Remote I/O (CRA).

# Execution of Quantum Sections with Remote Inputs/Outputs

## General

Quantum PLCs have a specific section management system. It applies to stations with remote inputs/outputs.

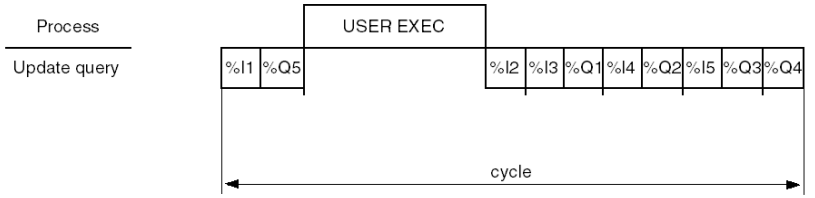
These stations are used with following RIO modules:

- 140 CRA 931 00
- 140 CRA 932 00

This system allows remote inputs/outputs to be updated on sections with optimum response times (without waiting for the entire task cycle before updating the inputs/outputs).

## Operation

The following diagram shows the IO phases when 5 drops are associated to client task sections.




**%Ii** inputs of drop No. i
**%Qi** outputs of drop No. i
**i** drop number

Description

| Phase | Description |
|---|---|
| 1 | Request to update:<br>● the inputs of the first drop (i=1)<br>● the outputs of the last drop (i=5) |
| 2 | Processing the program |
| 3 | ● Updating the inputs of the first drop (i=1)<br>● Request to update the inputs of the second drop (i=2) |
| 4 | Request to update:<br>● the inputs of the third drop (i=3)<br>● the outputs of the first drop (i=1) |
| 5 | Request to update:<br>● the inputs of the fourth drop (i=4)<br>● the outputs of the second drop (i=2) |

| Phase | Description |
|---|---|
| 6 | Request to update:<br>● the inputs of the last drop (i=5)<br>● the outputs of the third drop (i=3) |
| 7 | Request to update the outputs of the fourth drop (i=4) |

## Adjustment of the Drop Hold-Up Time Value

In order for the remote outputs to be correctly updated and avoid fallback values to be applied, the drop hold-up time must be set to at least twice the mast task cycle time. Therefore the default value, 300 ms, must be changed if the MAST period is set to the maximum value, 255 ms. The adjustment of the Drop Hold-Up time *(see Modicon Quantum, Hot Standby System, User Manual)* must be done on all configured drops.

# Distributed I/O (DIO)

## Overview

Quantum DIO is implemented over a Modbus Plus network. The CPU or NOM modules may be the network head via their Modbus Plus ports.

Quantum DIO Modbus Plus drop adaptors are specifically designed to link Quantum I/O modules to the head via twisted pair shielded cable (Modbus Plus). The DIO drop modules also provide the I/O with power (maximum 3 A) from a 24 VDC or a 115/230 VAC source. Each DIO network supports up to 63 distributed drops using repeaters.

**NOTE:** For detailed information see Experts and Communication Reference Manual, Part Quantum Distributed I/O (DIO) Drop Modules *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*.

# Section 3.3
## Ethernet Networking Interfaces

### Purpose

This section provides information on the Quantum Ethernet Networking Interfaces.

**NOTE:** For details see Experts and Communication Reference Manual, Part Quantum Ethernet Modules *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*.

### What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| TCP/IP Ethernet | 47 |
| SY/MAX Ethernet | 48 |

# TCP/IP Ethernet

## Overview

Quantum TCP/IP Ethernet modules make it possible for a Quantum controller to communicate with devices on an Ethernet network using TCP/IP – the de facto standard protocol. An Ethernet module may be inserted into an existing Quantum system and connected to existing Ethernet networks via fiber optic or twisted pair cabling.

# SY/MAX Ethernet

## Overview

Quantum-SY/MAX Ethernet modules are Quantum modules that can be placed in a Quantum rack, to connect Quantum controllers to SY/MAX devices and applications.

# Section 3.4
## Modbus/Modbus Plus Networking Interfaces

### Purpose

This section provides information on the Quantum Modbus/Modbus Plus Networking Interfaces.

**NOTE:** For detailed information see Experts and Communication Reference Manual, Part Modbus Plus Network Option Modules (NOM) *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*.

### What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| General Information | 50 |
| Features | 51 |
| Modbus and Modbus Plus Services | 52 |
| Modbus/ModBus Plus Requests Per MAST Task Scan | 53 |

# General Information

## Overview

Quantum NOM modules provide extended communication capabilities for the Quantum system within a Modbus and a Modbus Plus configuration.

## Modbus

Modbus, a master/slave protocol, is a de facto industry standard with support from over 500 industrial suppliers.

On-line programming or data acquisition applications are easily supported directly from the serial port of any computer.

Modbus can be used in either a simple point-to-point manner with a pair of devices, or in a network architecture with up to 247 slave devices.

## Modbus Plus

Modbus Plus combines high speed, peer-to-peer communication and easy installation to simplify applications and reduce installation costs.

It allows host computers, controllers and other data sources to communicate as peers throughout the network via low-cost twisted pair cable or optional fiber optic cable.

As a deterministic token-passing network, Modbus Plus communicates at one megabaud for fast access to process data. It’s strength is its ability to control real-time control devices like I/O and drives, without degraded performance due to loading or traffic.

Bridging between Modbus and Modbus Plus is done automatically on CPUs and Modbus Plus network modules.

The bridge mode redirects Modbus messages onto the Modbus Plus network for easy connectivity between Modbus and Modbus Plus devices.

A summary of the available services on Quantum Modbus and Modbus Plus ports is given in the following table.

## Modbus Plus on Fiber Module

Quantum Modbus Plus on Fiber modules provides connectivity to Modbus Plus nodes by fiber cable without fiber optic repeaters.

The use of a 490 NRP 254 Fiber Optic Repeater allows the creation of a pure fiber optic network or a mixed fiber optic/twisted-pair network.

# Features

## Overview

Each Quantum CPU includes both a Modbus and Modbus Plus communications port. The features offered by both these communication protocols are listed in the following table.

## Modbus and Modbus Plus Features Table

The following table shows the Modbus and Modbus Plus features.

| Features | Modbus | Modbus Plus |
|---|---|---|
| Technique | Slaves polled by a master | Peer-to-peer, token rotation |
| Speed | 19.2 kbaud typical | 1 Mbaud |
| Electrical | RS-232, various others | RS-485 |
| Distance without repeater | RS-232, 50 ft. (15 m) | 1,500 ft. (457 m) |
| Media | Various | Twisted pair, Fiber optics |
| Max nodes per network | 247 | 64 |
| Max network traffic | 300 registers/sec @ 9.6 kbaud | 20,000 registers/sec |
| Programming | Yes | Yes |
| Read/Write data | Yes | Yes |
| Global data | No | Yes |
| Peer Cop | No | Yes |

# Modbus and Modbus Plus Services

## Modbus and Modbus Plus Services Table

The following table shows the Quantum Modbus and Modbus Plus services.

| Type | Service Description | Native CPU Ports | | NOM 1-2 Ports | | NOM 3-6 Ports1 | |
|---|---|---|---|---|---|---|---|
| | | Modbus | Modbus Plus | Modbus | Modbus Plus | Modbus | Modbus Plus |
| Modbus Services | Default Modbus Port Parameters | Yes | - | Yes | - | Yes | - |
| | Configurable Modbus Port Parameters | Yes | - | Yes | - | Yes(5) | - |
| | Modbus to Modbus Plus Bridging | Yes(1) | - | Yes(2) | - | Yes(2) | - |
| | Local CPU Programming | Yes(3) | - | Yes(3) | - | No | - |
| | Remote CPU Programming over Modbus Plus | Yes(3) | - | Yes(3) | - | Yes(1) | - |
| | Modbus access to local CPU | Yes | - | Yes | - | No | - |
| | Modbus access to remote CPU over Modbus Plus | Yes | - | Yes | - | Yes | - |
| | Modbus Network Slave Support | Yes | - | No | - | No | - |
| | Modbus Master support with XMIT block | Yes | - | No | - | No | - |
| | Executive Firmware Loading Support | Yes | - | No | - | No | - |
| Modbus Plus services | MSTR read/write register messaging(4) | - | Yes | - | Yes | - | Yes |
| | MSTR read/write Global Data messaging | - | Yes | - | Yes | - | Yes |
| | MSTR get/clear local/remote statistics | - | Yes | - | Yes | - | Yes |
| | Config Extension Global Data Support | - | Yes | - | Yes | - | No |
| | Config Extension Peer Cop Support | - | Yes | - | Yes | - | No |
| | Distributed I/O Support | - | Yes | - | Yes | - | No |
| | CPU Programming | - | Yes(3) | - | Yes(3) | - | Yes(3) |
| | Executive Firmware Loading Support | - | Yes | - | No | - | No |

**(1)** The native CPU Modbus port can be disabled from bridge mode operation with the native Modbus Plus Port.
**(2)** Modbus ports on NOMs are in bridge mode with their associated Modbus Plus port.
**(3)** Only one programmer connection can be logged in at a time to any CPU, and only one program monitor can be attached at a time to any CPU.
**(4)** Up to 4 MSTR read/write register instructions can be serviced per CPU scan per Modbus Plus port.
**(5)** Modbus port parameters on NOMs 3-6 are defined by Modbus Port 3 when the comm parameter selector switch is in mem.

# Modbus/ModBus Plus Requests Per MAST Task Scan

## Maximum Request Served

The maximum number of requests served depends on limitations at various levels:

- Module port limitation: The number of requests served depends on the module port capacity to serve those requests.
- CPU architecture limitation: The number of requests served depends on the CPU capacity to send and treat those requests through the PAC architecture.
- CPU com server limitation: Parameter set to define the maximum number of requests to be served.

The maximum number of requests is determined by the most restrictive limitation.

## Modbus/Modbus Plus Server Mode Communication

The following table provides the maximum number of served requests per MAST task scan for a Modicon Quantum Modbus or Modbus Plus communication in server mode:

| Priority (Service Order) | Requests Entry Port | Maximum Number of Requests | | |
|---|---|---|---|---|
| | | Module Port Limitation | CPU Architecture Limitation | CPU Com Server Limitation |
| 1 | Requests via a CPU embedded Modbus or Modbus Plus port. | 4 | 20 | 4...24 Set with %SW90 *(see Unity Pro, System Bits and Words, Reference Manual)* |
| 2 | Requests via NOM module. | 4 | | |
| 2 | Requests via NOE or NOC module in `4 x 256` communication mode. | 4 (1) | | |
| 3 | Requests via NOE or NOC module that support *n* `x 1024` communication mode, with *n* depending on module (*n* `x 1024` bytes is selected *(see Unity Pro, Operating Modes)*).<br>**NOTE:** Available for NOE with SV≥4.60. | *n* = 4, 8, or 12 (1) | 12 (high-end CPU)<br>8 (legacy CPU) | |
| 4 | Requests via USB port. | 4 | No limitation | |
| 5 | Requests via embedded Ethernet ports. | 16 | No limitation | |

**(1)** Limitation applies to other requests than FC03, FC16, or FC23 which are directly served by the module. In addition to the maximum requests per module, up to 8 requests of type FC03, FC16, or FC23 can be served per scan per module.

Example for requests via embedded Ethernet port:
**Limitation 1:** Module port limitation to serve requests via embedded Ethernet port = 16
**Limitation 2:** CPU setting: `%SW90` = 4
**Result:** Maximum number of served requests = 4.
In this case, the most restrictive limitation comes from the CPU setting.

# Section 3.5
## Fieldbus Networking Interfaces

### Purpose

This section provides information on Quantum Fieldbus Networking Interfaces.

**NOTE:** For detailed information see Experts and Communication Reference Manual, Part Quantum Field Bus Modules *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*.

### What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| INTERBUS (NOA) | 55 |
| Profibus (CRP) | 56 |
| AS-i (EIA) | 57 |
| Sercos (MMS) | 58 |

# INTERBUS (NOA)

## Overview

Quantum INTERBUS (NOA) modules provide connectivity between a Quantum controller and the INTERBUS network.

The INTERBUS is a fieldbus network designed for I/O blocks and intelligent devices used in manufacturing. It offers a master/slave topology which permits deterministic I/O servicing over its network which can be up to 12.8 km (8 miles) in length..

# Profibus (CRP)

## Overview

The PTQ PDP MV1 Profibus DP Communication Module provides connectivity to the Profibus.

Profibus DP is a fieldbus designed for I/O communication used in manufacturing. It supports baudrates up to 12 MBaud.

# AS-i (EIA)

## Overview

The Quantum AS-i Master Module provides AS-i communications between the bus master module and the sensor/actuator slaves. One master module can control 31 slaves. Multiple master modules can be used in a single control system. These sensor/actuators can be in the local CPU, an RIO, or a DIO drop adapter.

# Sercos (MMS)

## Overview

The SERCOS® MMS motion modules are used to build a distributed automation solution, integrating motion applications with control applications, based on Quantum PLCs. The axis modules and Quantum CPUs communicate either through the Quantum rack or by using the Modbus Plus network. The data transfer is transparent, and does not need any additional application program.

## Fiber optic interface

The physical interface between the SERCOS module and the servo drive is done by the SERCOS network, using fiber optic cable. This optic link is entirely digital, and provides communication parameters for the tuning, diagnostics, and operation of both motion control modules and servo drives.

# Chapter 4
# Quantum Configurations

## Purpose

This chapter provides information on the Quantum Configurations.

## What Is in This Chapter?

This chapter contains the following sections:


| Section | Topic | Page |
|---|---|---|
| 4.1 | Quantum Local I/O, Remote I/O and Distributed I/O | 60 |
| 4.2 | Quantum Local I/O | 63 |
| 4.3 | Quantum Remote I/O (RIO) | 66 |
| 4.4 | Quantum Distributed I/O (DIO) | 69 |

# Section 4.1
# Quantum Local I/O, Remote I/O and Distributed I/O

## Purpose

This section provides information on the features and configuration of Quantum Local I/O, Remote I/O and Distributed I/O.

## What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| Features | 61 |
| Local, RIO and DIO Configuration | 62 |

# Features

## Local, Remote and Distributed I/O Features

The following table shows features of the Local, Remote and Distributed I/O configurations.

| Feature | Configuration | | |
|---|---|---|---|
| | Local I/O | Remote I/O | Distributed I/O |
| Maximum I/O Words | | | |
| Per drop | unlimited I/O | 64 IN / 64 OUT | 30 IN / 32 OUT |
| Per network | | 1,984 IN / 1,984 OUT | 500 IN / 500 OUT |
| Maximum drops per network | | 31 | 63 (with repeater) |
| Media | | Coax | Twisted Pair |
| Speed | | 1.5 MHz | 1 MHz |
| Maximum distance without repeaters | | 15,000 ft. (4,573 m) | 1,500 ft. (457 m) |
| Scan synched I/O servicing | | Yes | No |
| Momentum I/O support | | No | Yes |
| Hot Standby support | | Yes | No |
| Modbus Plus compatible | | No | Yes |

# Local, RIO and DIO Configuration

## Overview

Valid Quantum configurations can be equipped with a combination of:

- Quantum CPUs
- Power supplies
- I/O interfaces
- Expert modules
- I/O modules

## Local, RIO, and DIO Configuration Table

The following table provides valid Quantum configurations, including racks and modules.

| If Configuration Type Is | Rack Types (Typical) Are | Required Modules Are | Optional Modules Are | Modules Not Permitted Are |
|---|---|---|---|---|
| Local | 6, 10, 16 slots | Power Supply CPU | RIO Head, I/O, NO•* | RIO Drop, DIO Drop |
| RIO** | 6, 10, 16 slots | Power Supply RIO Drop | I/O | CPU, RIO Head, DIO Drop NO•* |
| DIO | 2, 3, 4, 6 slots | DIO Drop | Power Supply, I/O | CPU, RIO Head, NO•* |
| * NOM, NOA, and NOE<br>** Remote I/O is typically used for large (number of modules) drops 6, 10, or 16 slot racks. Distributed I/O is typically used for small drops using 2, 3, 4, or 6 slot racks. | | | | |

**NOTE:** Every Quantum module requires power from the rack (except power supply and DIO modules). For a valid configuration, add up the required rack current (in mA) for every module, and verify that this number is less than the available power in the selected power supply.

# Section 4.2
# Quantum Local I/O

## Purpose

This chapter provides an Overview on the Quantum Local I/O.

## What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| Configuration | 64 |
| Example | 65 |

# Configuration

## Overview

A local I/O configuration is contained in one to two racks and includes Quantum modules mounted in a standard Quantum rack. Quantum Local I/O can be as few as one I/O module (in a three slot rack).

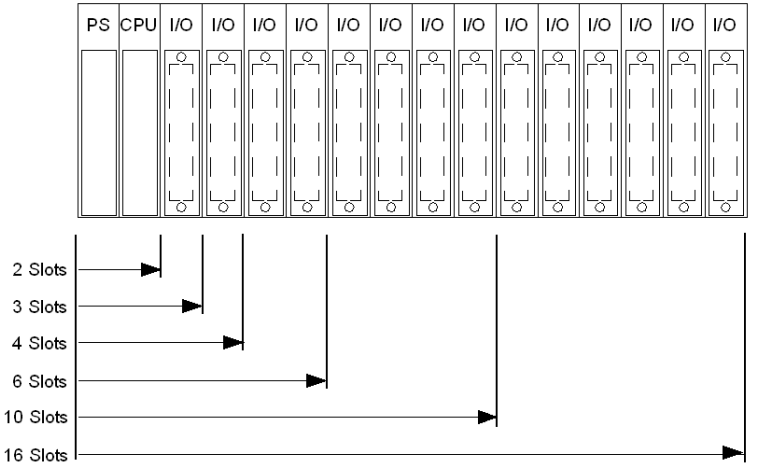
Or a maximum of 27 I/Os totalized as follows:

- 13 in the first rack along with CPU, power supply and expansion module (XBE)
- 14 in the expansion rack along with power supply and expansion module (XBE)

# Example

## I/O Configuration Figure

The following figure is an example of a typical local I/O configuration.



# Section 4.3
# Quantum Remote I/O (RIO)

## Purpose

This section provides an overview on the configuration of the Quantum Remote I/O (RIO).

**NOTE:** For detailed information see Experts and Communication Reference Manual, Part Quantum Remote I/O (RIO) Modules *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*.

## What Is in This Section?

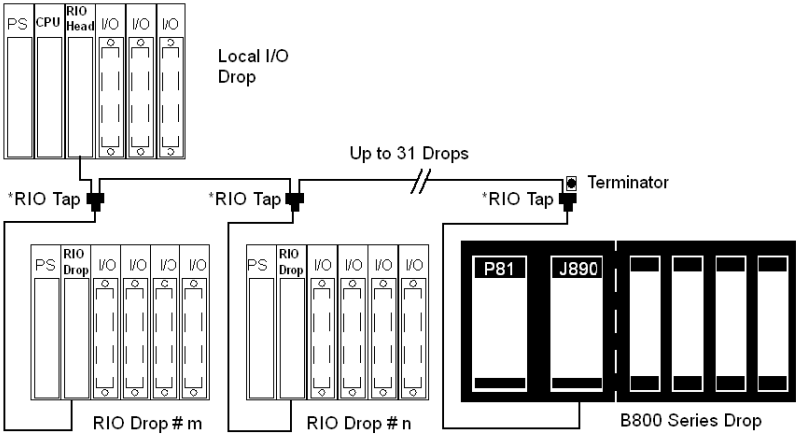
This section contains the following topics:


| Topic | Page |
|---|---|
| Single-cable Configuration | 67 |
| Dual-cable Configuration | 68 |

# Single-cable Configuration

## Single-cable RIO Configuration Figure

The following figure is an example of a single-cable Quantum RIO configuration.

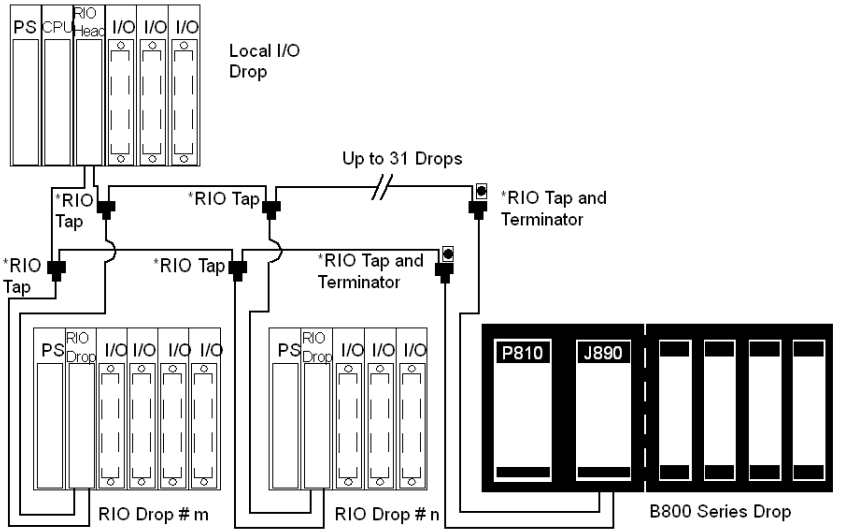



**NOTE:** In a single-cable configuration, it is recommended to connect the cable in channel A on both Local I/O (CRP) and Remote I/O (CRA).

# Dual-cable Configuration

## Dual-cable RIO Configuration Figure

The following figure is an example of a dual-cable Quantum RIO configuration.




**NOTE:** Dual cables provide systems with added protection against cable breaks or damaged connectors. With two cables connected between the host and each node, no single cable break will disrupt communications.

**NOTE:** In a dual-cable configuration, it is mandatory to connect channel A from Local I/O (CRP) with channel A from Remote I/O (CRA) and channel B from Local I/O (CRP) with channel B from Remote I/O (CRA).

# Section 4.4
# Quantum Distributed I/O (DIO)

## Purpose

This section provides an overview on the configuration of Quantum Distributed I/O (DIO).

**NOTE:** For detailed information see Experts and Communication Reference Manual, Part III Quantum Distributed I/O (DIO) Drop Modules *(see Quantum with Unity Pro, Experts and Communication, Reference Manual)*.

## What Is in This Section?

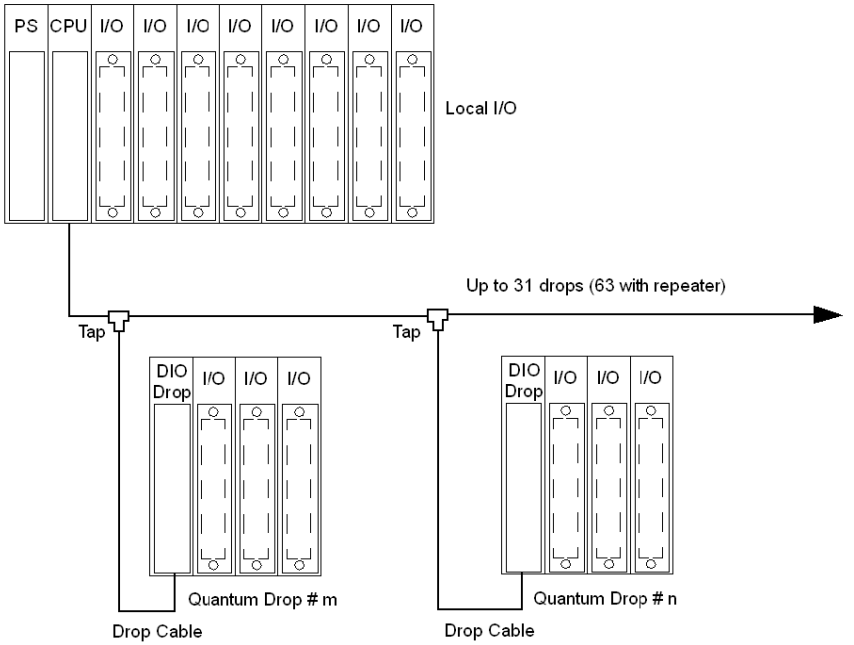
This section contains the following topics:


| Topic | Page |
|---|---|
| Single-cable Configuration | 70 |
| Dual-cable Configuration | 71 |

# Single-cable Configuration

## Single-cable DIO Configuration Figure

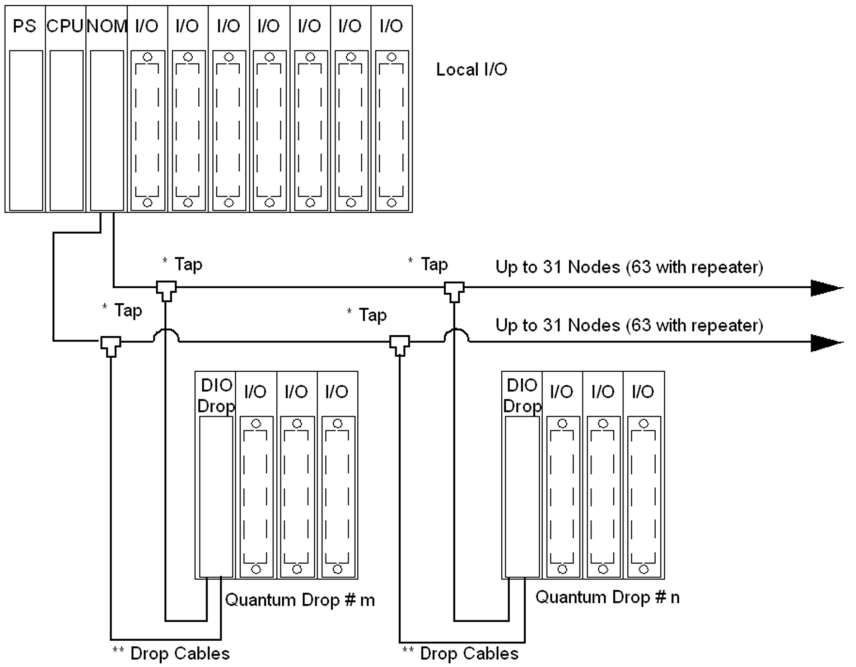
The following figure is an example of a single-cable Quantum DIO configuration.



# Dual-cable Configuration

## Dual-cable DIO Configuration Figure

The following figure is an example of a dual-cable Quantum DIO configuration.




**NOTE:** Dual cables provide systems with added protection against cable breaks or damaged connectors. With two cables connected between the host and each node, no single cable break will disrupt communications.

# Chapter 5
# Module Configuration

## Purpose

This chapter provides information on software configuration of the module.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Mapping a Local Quantum I/O Station | 74 |
| Open the Parameter Configuration | 76 |

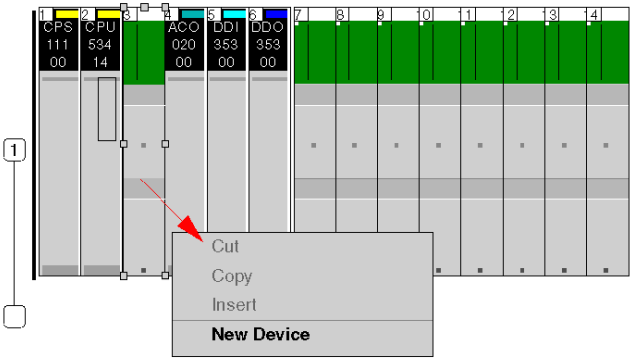
# Mapping a Local Quantum I/O Station

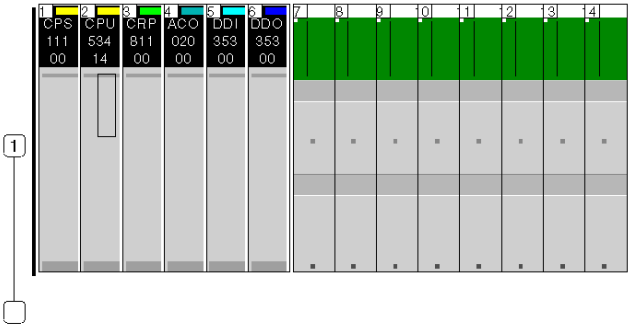
## Overview

Use the following dialog to map an existing local Quantum I/O station with a new module.

## Inserting a module (local)

This table shows the steps required for inserting a module in a local station.

| Step | Action |
|---|---|
| 1 | Call the Bus Editor |
| 2 | Mark a free slot in the local station (left mouse button) |
| 3 | Move the mouse pointer over the marked slot |
| 4 | Click on the right mouse button<br>**Result:** A shortcut menu is opened<br>  |
| 5 | Select **New Device**<br>**Result:** A dialog window opens that displays available modules |

| Step | Action |
|---|---|
| 6 | Select the desired module from the respective category in the Hardware catalog.<br>**Result:** The new module is inserted in the empty slot on the local station.<br>  |

# Open the Parameter Configuration

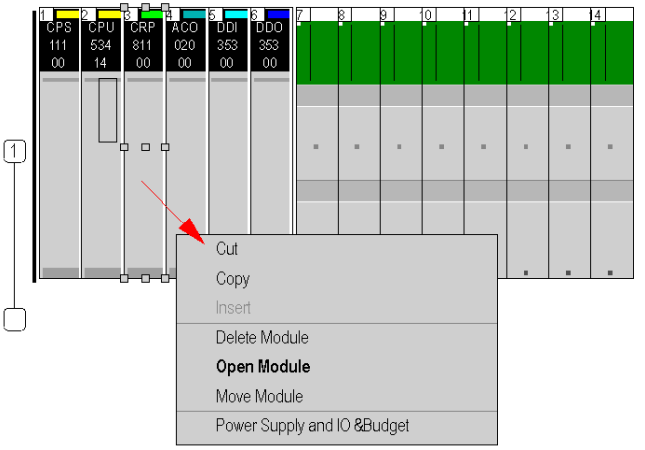
## Overview

The following dialog box enables you to call the parameter configuration for a module.

An explanation of the individual parameters can be found in the respective module chapters.

## Open the Parameter Configuration

This table shows the steps required to open the parameter configuration.

| Step | Action |
|---|---|
| 1 | Call the Bus Editor |
| 2 | Select the module |
| 3 | Click on the right mouse button<br>**Result:** A shortcut menu is opened<br>  |
| 4 | Select **Open Module**<br>**Result:** The module opens with the parameter configuration window |

# Chapter 6
# Hardware Installation and Maintenance

## Purpose

This chapter provides information on the Quantum hardware installation and maintenance.

## What Is in This Chapter?

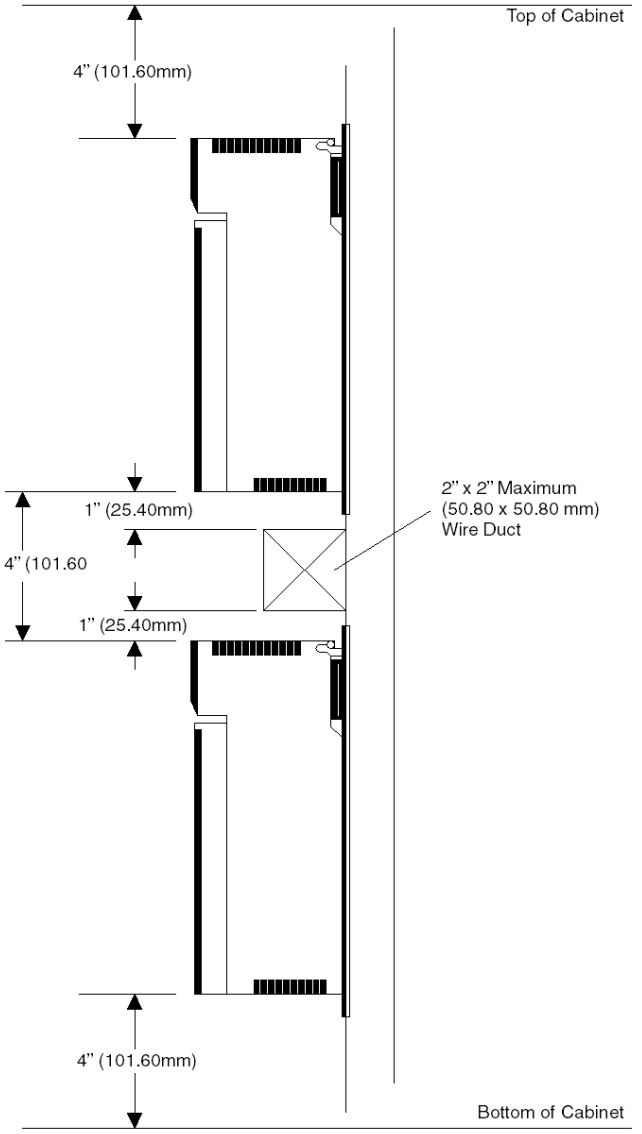
This chapter contains the following topics:


| Topic | Page |
|---|---|
| Space Requirements | 78 |
| Mounting Brackets | 80 |
| Mounting Quantum Modules | 83 |
| Mounting and Removing a Terminal Strip | 86 |
| Mounting Jumper Clips | 89 |
| Removing a Module Door | 90 |

# Space Requirements

## Spacing Requirements Figure

The following figure shows the Quantum system spacing requirements.



## Spacing Requirements Table

The following table gives a summary of the spacing requirements for a Quantum system.

| Minimum Spacing | Location |
|---|---|
| 4 in. (101.60 mm) | Between the top of the cabinet and the top of the modules in the upper rack. |
| 4 in. (101.60 mm) | Between the bottom of the cabinet and the bottom of the lower modules in the lower rack. |
| 4 in. (101.60 mm) | Between the upper and lower modules when the racks are mounted one above the other. |
| 1 in. (25.40 mm) | On either side between the cabinet walls and end modules. |
| **Note:** Wiring ducts up to 2 in. x 2 in. (50.80 mm x 50.80 mm) may be centered between back planes. If the duct extends further than 2 in. (50.80 mm) out from the mounting panel, there must be a 4 in. (101.60 mm) space between the modules and duct on the top and bottom. | |

**⚠ WARNING**

**UNEXPECTED EQUIPMENT OPERATION**

Install the racks lengthways and horizontally to facilitate ventilation and take the spacing requirements table into account.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

# Mounting Brackets

## Overview

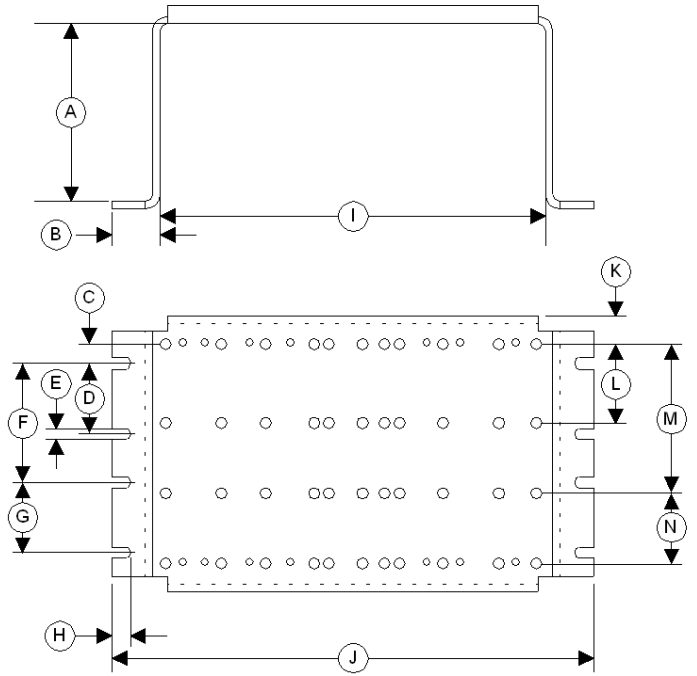
Mounting brackets are required when mounting racks in 19 inch NEMA cabinets. The mounting bracket supports the 2- through 10-position racks. The bracket mounts to rails using standard NEMA hardware.

Mounting brackets are offered in two sizes:

- 20 mm (0.79in) for back rail mounting
- 125 mm (4.92in) for front rail mounting

## 125 mm Mounting Bracket Figures

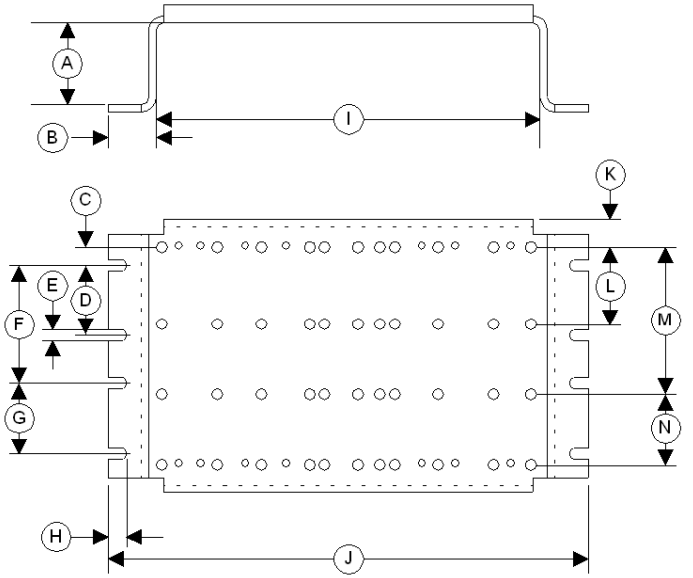
**NOTE:** Before installing a Quantum rack to a mounting bracket, ensure that the mounting holes of the bracket and rack are properly aligned.




Diameter of the mounting holes: 6.6 mm / 0.26 inches

**A** 125 mm (4.92 in)
**B** 22.83 mm (0.90 in)
**C** 17.5 mm (0.69 in)
**D** 88.9 mm (3.50 in)
**E** 7.1 mm (0.28 in)
**F** 146.1 mm (5.75 in)
**G** 88.9 mm (3.50 in)
**H** 14.7 mm (0.58 in)
**I** 436.6 mm (17.19 in)
**J** 482.25 mm (18.99 in)
**K** 20.2 mm (0.79 in)
**L** 94.5 mm (3.72 in)
**M** 175.5 mm (6.91 in)
**N** 94.5 mm (3.72 in)

## 20 mm Mounting Bracket Figure




Diameter of the mounting holes: 6.6 mm / 0.26 inches

**A** 20 mm (0.79 in)
**B** 22.83 mm (0.90 in)
**C** 17.5 mm (0.69 in)
**D** 88.9 mm (3.50 in)
**E** 7.1 mm (0.28 in)
**F** 146.1 mm (5.75 in)
**G** 88.9 mm (3.50 in)
**H** 14.7 mm (0.58 in)
**I** 436.6 mm (17.19 in)
**J** 482.25 mm (18.99 in)
**K** 20.2 mm (0.79 in)
**L** 94.5 mm (3.72 in)
**M** 175.5 mm (6.91 in)
**N** 94.5 mm (3.72 in)

# Mounting Quantum Modules

## Overview

Quantum modules can be inserted into any slot of any rack. Although Power supply modules should be installed in the first or last slots, to have a cooling effect. The modules can be removed under power (hot swapped) without damaging modules or the rack.

Refer to the following figures and procedure when mounting modules.

**NOTE:** To provide EMC level, the mounting area of the CPU has to provide metallic contact. Therefore remove any labels in the affected area and clean the surface using solvent. The Quantum PLC enclosures and terminal strips are made of polycarbonates. This material can be damaged by strong alkalis and various hydrocarbons, esters, halogens and ketones in combination with heat. Common products containing these include detergents, PVC products, petroleum products, pesticides, disinfectants, paint removers, and spray paints. Use care in selecting and using cleaning solvents..

**⚠ CAUTION**

**POSSIBILITY OF MODULE DAMAGE**

The inappropriate use of solvents, cutting oils, bug sprays and similar chemicals may cause the breakdown of module cases or terminal blocks.

**Failure to follow these instructions can result in injury or equipment damage.**

## Hot Swap Precautions

**⚠ WARNING**

**UNEXPECTED SYSTEM BEHAVIOR - CPU HOT SWAP CONSEQUENCES**

Do not Hot Swap Quantum CPU.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

Unplugging and re-plugging the Quantum CPU when the rack is under power is not recommended. You are strongly advised to shut the power down before attempting to remove the CPU for any reason.

**⚠ CAUTION**

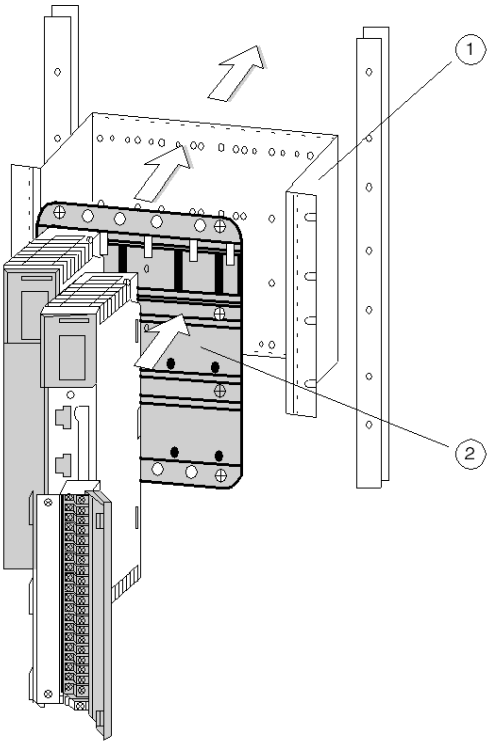
**UNINTENDED EQUIPMENT OPERATION**

Hot swapping an I/O module can generate an error code causing the module to stop functioning.

**Failure to follow these instructions can result in injury or equipment damage.**

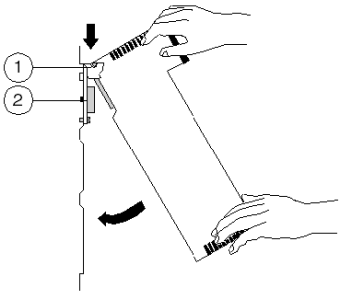
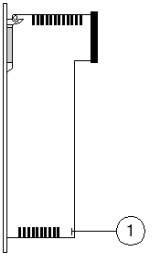
## Mounting Bracket and Rack

The following steps describe the mounting of bracket and rack

| Step | Action |
|---|---|
| 1 | If required for the application, select and install a 20 mm or 125 mm mounting bracket to the rack using standard hardware.<br>Front view:<br>  <br>**1** Mounting bracket<br>**2** Rack |
| 2 | Select and install the appropriate rack to the mounting bracket using standard hardware and remove the plastic rack connector dust covers. |

## Mounting a Module

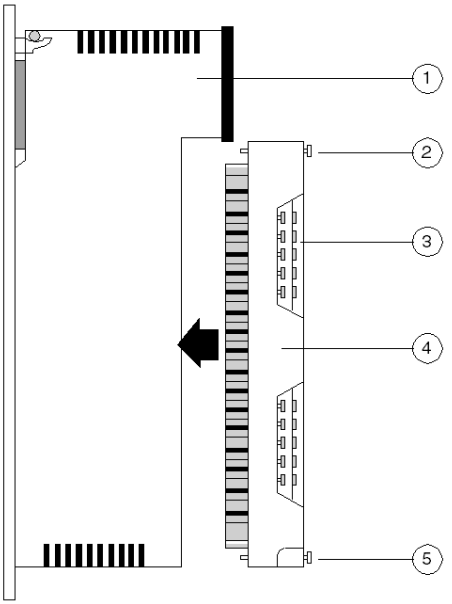
The following steps describe the mounting of a module

| Step | Illustration | Action |
|---|---|---|
| 1 | Side view:<br>  <br>**1** Module Hooks<br>**2** I/O Bus Connector | Mount the module at an angle on to the two hooks located near the top of the rack. |
| 2 | | Swing the module down to make an electrical connection with the rack I/O bus connector. |
| 3 | Side view:<br>  <br>**1** Mounting screw | Tighten the screw at the bottom of the module to fasten it to the rack.<br>**Note:** The maximum tightening torque for this screw is 2-4 in-lbs (0.23 - 0.45 Nm). |

# Mounting and Removing a Terminal Strip

## Mounting a Terminal Strip

The following steps describe the mounting of a Terminal Strip

| Step | Action |
|---|---|
| 1 | Install the appropriate Terminal Strip on the module.<br>Side view:<br>  <br>**1** Module<br>**2** Mounting screw (top)<br>**3** Terminal Strip I/O screws<br>**4** Terminal Strip<br>**5** Mounting screw (bottom) |
| 2 | Tighten with the mounting screws at the top and bottom of the terminal with a philips screwdriver.<br>**Note:** The tightening torque must be between 0.5 Nm and 0.8 Nm. |
| 3 | Make I/O connections with a philips screwdriver as shown in the individual Quantum module wiring diagrams.<br>**Note:** The tightening torque must be between 0.5 Nm and 0.8 Nm. |

## *NOTICE*

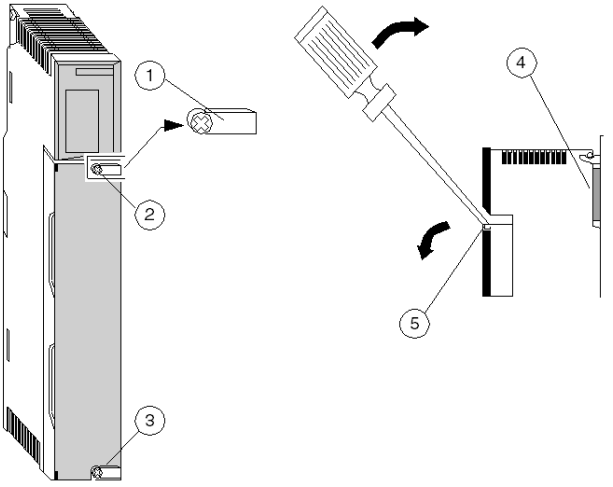
**DESTRUCTION OF ADAPTER**

- Before tightening the locknut to the torque 0.50...0.80 Nm, be sure to properly position the right-angle F adapter connector.
- During tightening, be sure to maintain the connector securely.
- Do not tighten the right-angle F adapter beyond the specified torque.

**Failure to follow these instructions can result in equipment damage.**

## Removing a Terminal Strip

The Quantum I/O terminal strips have been designed with a pry slot to assist in their removal. Follow the procedure below to remove the terminal strip.

| Step | Action |
|---|---|
| 1 | Loosen the Terminal Strip mounting screws located at the top and bottom of the Terminal Strip. |
| 2 | Locate the pry slot at the top of the Terminal Strip near the top mounting screw.<br>  <br>**1** Pry Slot (exploded view)<br>**2** Mounting screw (top)<br>**3** Mounting screw (bottom)<br>**4** Rack<br>**5** Pry Slot (side view) |
| 3 | Insert a flat edge screwdriver into the pry slot and lever the top of the Terminal Strip away from the module. |
| 4 | Once the top of the Terminal Strip has been loosened, the terminal strip may be removed by continuous outward pressure with the screwdriver. |

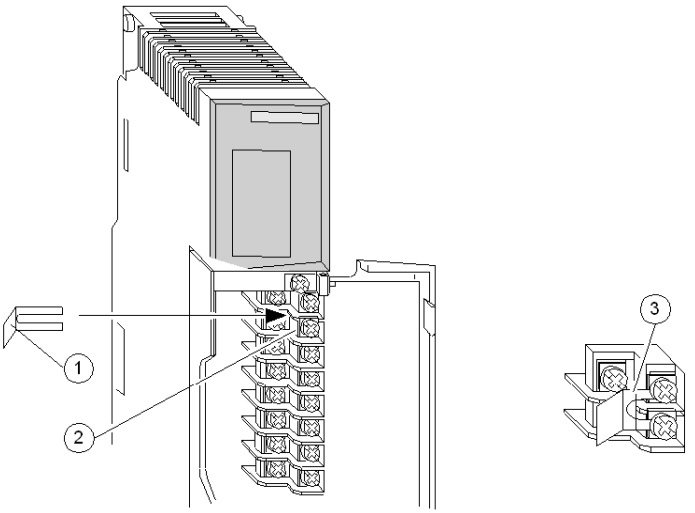
# Mounting Jumper Clips

## Overview

Terminal strip jumper clips are installed when contiguous I/O points need to be jumpered (for example, on the 140 AVO 020 00 analog out put module).

## Mounting Jumper Clips

The following steps describe the mounting of jumper clips.

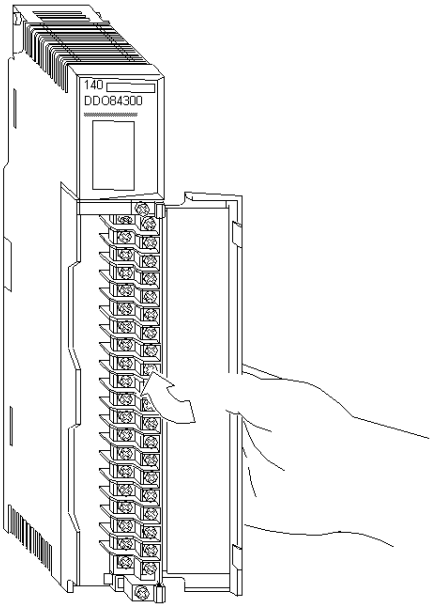
| Step | Action |
|---|---|
| 1 | Remove the power. |
| 2 | Remove the Terminal Strip from the module.<br>Front view:<br>  <br>**1** Jumper clip<br>**2** Terminal Strip I/O screw<br>**3** Inserted jumper clip (exploded view) |
| 3 | Loosen the terminal strip I/O screws of the points that you want jumpered. |
| 4 | Insert the jumper clip under the loosened screws (see exploded view). |
| 5 | Tighten the screws and reinstall the module. |

# Removing a Module Door

## Overview

The terminal strips have a removable door to facilitate access to the terminal strip. Remove the door before wiring a module.

## Removing a Module Door

| Step | Action |
|---|---|
| 1 | Open the module door. |
| 2 | Place your thumb near the middle of the door.<br>Front view:<br>  |
| 3 | With your thumb, apply pressure until the door bends and the door hinge pins pop out of the retaining holes at the top and bottom of the terminal strip. |
| 4 | After wiring the module, reinstall the door using the reverse of the above. |

# Part II
## Controller Modules (CPUs)

### Introduction

This part provides information on Quantum Controller Modules (CPU).

### What Is in This Part?

This part contains the following chapters:


| Chapter | Chapter Name | Page |
|---|---|---|
| 7 | General Information | 93 |
| 8 | Low End CPU | 99 |
| 9 | High End CPU | 135 |
| 10 | Memory Cards for High End CPU | 199 |

# Chapter 7
# General Information

## Purpose

This chapter provides general information on Quantum Controller (CPU) modules.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| CPU Overview | 94 |
| Machine Stop Codes | 95 |
| Battery Lifetimes for the Quantum CPUs | 97 |

# CPU Overview

## Characteristics

| | CPU | Max Internal Memory | Max Additional Memory (with PCMCIA) | Key Switch |
|---|---|---|---|---|
| Low End | 140 CPU 311 10 | 400 kBytes | NA | No |
| | 140 CPU 434 12• | 800 kBytes | NA | Yes |
| | 140 CPU 534 14• | 2.7 MBytes | NA | Yes |
| High End | 140 CPU 651 50 | 768 kBytes | 7168 kBytes | Yes |
| | 140 CPU 651 60 | 1024 kBytes | 7168 kBytes | Yes |
| | 140 CPU 651 60S(Safety) | 1024 kBytes | 7168 kBytes | Yes |
| | 140 CPU 652 60 | 3072 kBytes | 7168 kBytes | Yes |
| | 140 CPU 658 60 | 11264 kBytes | NA | Yes |
| | 140 CPU 670 60(1) | 512 kBytes | 7168 kBytes | Yes |
| | 140 CPU 671 60 | 1024 kBytes | 7168 kBytes | Yes |
| | 140 CPU 671 60S(Safety) | 1024 kBytes | 7168 kBytes | Yes |
| | 140 CPU 672 60 | 3072 kBytes | 7168 kBytes | Yes |
| | 140 CPU 672 61 | 3072 kBytes | 7168 kBytes | Yes |
| | 140 CPU 678 61 | 11264 kBytes | NA | Yes |

**(1)** This CPU is not sold worldwide. Please contact your local sales agency and/or support for more information.

# Machine Stop Codes

## Machine Stop Code Descriptions

| Stop Bit Code (hex) | Description |
|---|---|
| 0x7FFF | PLC unhealthy |
| 0x8000 | PLC stopped |
| 0x8400 | CPU has stopped because the firmware of the CRP module is not compatible with the application program in which the "Online Modification in RUN" check box has been selected |
| 0x4000 | Bad I/O traffic cop |
| 0x2000 | PLC in Dim awareness |
| 0x1000 | Bad port intervention |
| 0x0100 | Software error, PLC in HALT |
| 0x0400 | CRP firmware not compatible with CCOTF function |
| 0x0080 | Watchdog timer has expired |
| 0x0010 | RIO option error |
| 0x0001 | Bad configuration |

## Error Stopped Codes

- **PLC unhealthy:** This condition indicates that the CPU has not successfully completed one or more of its health diagnostics. Usually this means that the CPU needs to be replaced.
- **PLC stopped:** By itself, an 8000 hex is not an error but a CPU state. If, for example, a user issues a CPU stop command, the status register indicates 8000 hex. An error condition exists when 8000 is added with one or more of the previously defined errors (bits 0-14).
  **Example:**
  - 2 stop codes are added,
    the 8100 stop code suggests that a PLC has stopped because of a software error detection. The controller is in halt and the application is stopped.
  - 3 stop codes are added,
    the 8180 stop code suggests that a PLC has stopped because of a software error detection and that the watchdog timer has expired.
- **Bad I/O map:** This error occurs if the user declares more than one I/O drop in his configuration but does not have an RIO Head installed. This error may also occur if a drop configuration exceeds the maximum number of inputs/outputs allowable per drop.

- **PLC unconfigured:** Expect this condition if you are trying to log into the CPU for the first time. This error indicates that the CPU has not been configured. Write a configuration offline and transfer it to the CPU prior to logging into the CPU. If this error appears while you are seeking to communicate with a previously running CPU, this could indicate corrupted state memory in the CPU. You need to clear memory and attempt to reload the user logic program.
- **Bad modbus port intervention:** This error sometimes appears in conjunction with another error. The CPU is usually stopped when this error occurs. The error may also appear if you attempt to clear the system stop state. Try to clear user logic and reload.
- **Software error, PLC in HALT:** This error is usually caused by an incomplete or unsuccessful program load. Try to reload.
- **Watchdog timer has expired:** The error indicates that the CPU has taken too much time to complete its current scan. You may increase the Watchdog timer value. This error may indicate that the CPU is inoperable..
- **RIO option error:** The RIO option board (140 CRP 93• 00) is unhealthy. Replace the board. An error 8010 means that the CPU has stopped because RIO option board has to be replaced (unhealthy).
- **Bad configuration:** The most probable cause is that the memory has been modified through the MODBUS/MODBUS PLUS ports. If this error occurs during a program download, check your configuration data for values greater than the CPU's specified addressable range. This error can also appear if the CPU's memory is corrupt.

**NOTE:** The user can access the Stop codes and the sub Stop codes with the LCD keypad, available only with High-End processors, by selecting **System info** -> **Stop codes**.
If the sub Stop code is included between:

- 50 and 98, the machine stopped because of a Traffic Cop error..
- 101 and 137, the machine stopped because of an error of Peer cop.

You can see, in the words %SW124 and %SW125, information if it is a system error and the last default detected.

## HALT state causes

The causes of the HALT state are saved in the system words: %SW125, %SW126 and %SW127. When the PLC is in HALT mode, the application is no longer executed, the IO are no longer applied.

To quit the HALT state, the user must either initialize or reload the application into the PLC.

# Battery Lifetimes for the Quantum CPUs

## Purpose

The purpose of this document is to give detailed informations about the lifetime of batteries inside the Quantum CPUs. The estimation of these lifetimes are based on data from components manufacturers.

## Scope

The lifetime information is estimated for:

- Two ambient temperatures for the PLC location:
  - Typical lifetime value at 25°C,
  - Minimum liftime value at 60°C.
- Three different usage cases of the CPU: 92%, 66% and 33% of PLC power-up time. These values are for the following customer configurations:
  - 92%: PLC powered up all year long except during one month of maintenance,
  - 66%: PLC powered up all year long except during all weekends plus one month of maintenance,
  - 33%: PLC powered up all year long 12 hours a day, except during all weekends plus one month of maintenance.

## CPU Battery Lifetime

The table below presents the lifetime of CPU battery:

| Power-up time | 92% | | 66% | | 33% | |
|---|---|---|---|---|---|---|
| | Typical | Minimum | Typical | Minimum | Typical | Minimum |
| Lifetime in Years | 27.1 | 10.4 | 11.0 | 4.0 | 6.4 | 2.70 |

## Minimum Lifetimes of the CPU Battery, in a Powered Down PLC

In a powered down PLC, the minimum lifetimes of the main battery is **1.5 years** at 60 °C and **4.6 years** at 25 °C.

# Chapter 8
# Low End CPU

## Purpose

This chapter provides information on the Low End Controller modules.

## What Is in This Chapter?

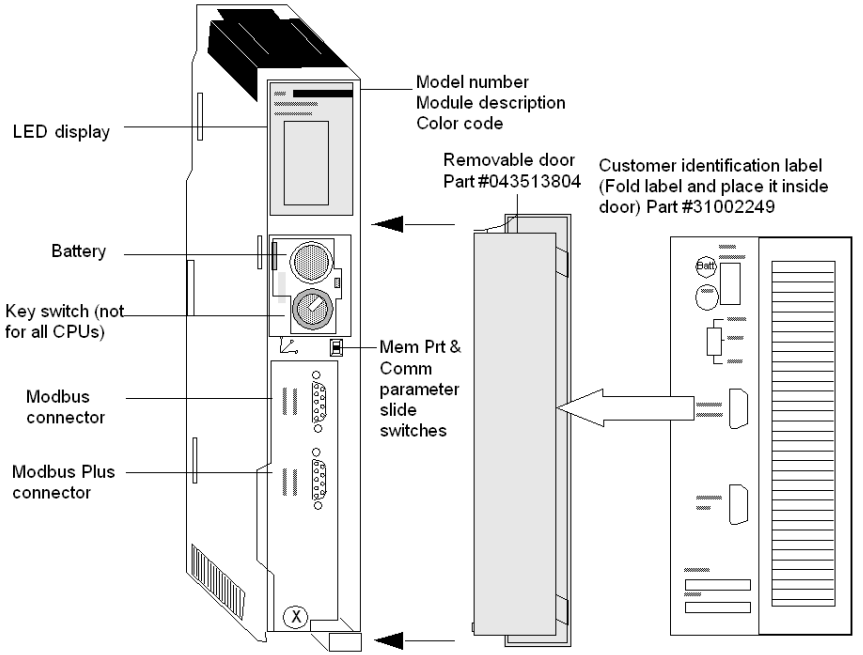
This chapter contains the following topics:


| Topic | Page |
|---|---|
| Presentation | 100 |
| Front Panel Switches | 101 |
| Rear Panel Switches | 104 |
| Key Switches | 105 |
| Modbus Connectors | 108 |
| Indicators | 110 |
| Processor Configuration Screen | 113 |
| 140 CPU 311 10 Specifications | 123 |
| 140 CPU 434 12A/U Specifications | 126 |
| 140 CPU 534 14A/U Specifications | 129 |
| 140 CPU 534 14B/U Specifications | 132 |

# Presentation

## Illustration

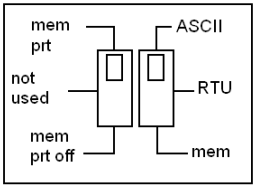
The following figure shows the Low End CPU module and its components.



# Front Panel Switches

## Front Panel Switches

Two three-position slide switches are located on the front of the CPU. The left switch is used for memory protection when in the top position and no memory protection in the middle and bottom positions. The switch on the right is used to select the Comm parameter settings for the Modbus (RS-232) ports.




**NOTE:** The memory switch selection is taken into account immediately.

**NOTE:** The modbus switch selection is taken into account after a Power off/on of the Quantum PLC

| ⚠ WARNING |
| --- |
| **UNINTENDED EQUIPMENT OPERATION**<br>Do not allow an automatic backup restore upon cycling power to the PLC. The backup restore function reloads in the PLC memory the version saved on the memory card, that can be different than the one that was running before the power off.<br>**Failure to follow these instructions can result in death, serious injury, or equipment damage.** |

## ASCII Comm Port Parameters Table

Setting the right slide switch to the top position assigns ASCII functionality to the port. The following ASCII Comm parameters are preset and cannot be changed:

| ASCII Comm Port Parameters | |
|---|---|
| Transmission Speed (Baud) | 2,400 |
| Parity | Even |
| Data Bits | 7 |
| Stop Bits | 1 |
| Device Address | Rear panel rotary switch setting |

**NOTE:** The CPU hardware defaults to bridge mode when the right slide switch is set to RTU mode. When networking controllers, a panel device connected to the CPU Modbus port can communicate with the controller to which it is connected, as well as log into any nodes on the Modbus Plus network.

## RTU Comm Port Parameters

Setting the right slide switch to the middle position assigns remote terminal unit (RTU) functionality to the port; the following comm parameters are set and cannot be changed.

| RTU Comm Port Parameters | |
|---|---|
| Transmission Speed (Baud) | 9,600 |
| Parity | Even |
| Data Bits | 8 |
| Stop Bits | 1 |
| Device Address | Rear panel rotary switch setting |

## Valid Comm Port Parameters Table

Setting the right slide switch to the bottom position gives you the ability to assign comm parameters to the port in software. The following parameters are valid.

| Item | Valid Comm Port Parameters |
|---|---|
| Mode | ASCII or RTU |
| Transmission Speed (Baud) | 19,200 |
| | 9,600 |
| | 7,200 |
| | 4,800 |
| | 3,600 |
| | 2,400 |
| | 2,000 |
| | 1,800 |
| | 1,200 |
| | 600 |
| | 300 |
| | 150 |
| | 134.5 |
| | 110 |
| | 75 |
| | 50 |
| Data Bits | ASCII: 7 Bit |
| | RTU: 8 Bit |
| Stop Bits | 1 / 2 |
| Parity | Enable/Disable Odd/Even |
| Device Address | 1 ... 247 |

# Rear Panel Switches

## Rear Panel Switches

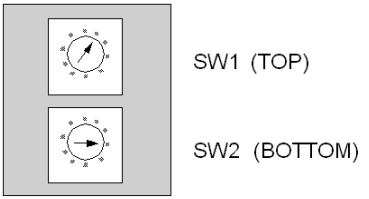
Two rotary switches (refer to the illustration and table below) are located on the rear panel of the CPU. They are used for setting the Modbus Plus node and Modbus port addresses.

**NOTE:** The highest address that may be set with these switches is 64.

SW1 (the top switch) sets the upper digit (tens) of the address; SW2 (the bottom switch) sets the lower digit (ones) of the address. The illustration below shows the correct setting for an example address of 13.

## SW1 and SW2 Switches Figure

The following figure shows the SW1 and SW2 settings.




## SW1 and SW2 Address Settings Table

The following table shows the SW1 and SW2 address settings.

| Node Address | SW1 | SW2 |
|---|---|---|
| 1 ... 9 | 0 | 1 ... 9 |
| 10 ... 19 | 1 | 0 ... 9 |
| 20 ... 29 | 2 | 0 ... 9 |
| 30 ... 39 | 3 | 0 ... 9 |
| 40 ... 49 | 4 | 0 ... 9 |
| 50 ... 59 | 5 | 0 ... 9 |
| 60 ... 64 | 6 | 0 ... 4 |

**NOTE:** If "0" or an address greater than 64 is selected, the Modbus + LED will be "on" steady, to indicate the selection of an invalid address.

# Key Switches

## Overview

The low end CPUs have 1,435 kb of Flash EPROM memory, which can be used to save the program and the variables' initial values. On power-up, if there is a program in Flash memory, you can choose one of three operating modes, using the PLC MEM switch on the CPU's front panel. The 140 CPU 434 12A and 140 CPU 534 12A are equipped with a key switch with the Start, Mem Prt, and Stop positions. The 140 CPU 311 10 features a slide switch with the Mem Prt On, Not Used, and Mem Prt Off options.

## Behavior on Power Up

The CPU's behavior on power up is determined by the key switch position. The behavior could be:

- Cold Restart
- Warm Restart

**⚠ WARNING**

**UNEXPECTED EQUIPMENT OPERATION**

Always perform a **Back Up Clear** after an OS upgrade of the Quantum (140 CPU 534 •, 140 CPU 434 14A, 140 CPU 311 10) CPUs.

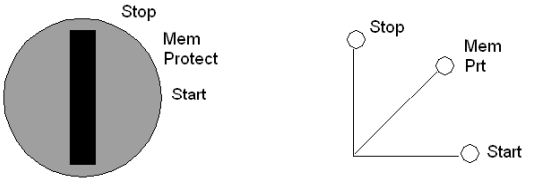
**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

**NOTE:** To ensure a warm restart of an application following an OS upgrade on Quantum CPUs, click **PLC → Project Backup... → Backup Clear**.

This section describes the three positions of the rotary key switch, and their respective meanings.

## Key Switch Figure

The key switch isolates memory from programming changes while the controller is in operation. The following figure shows the key switch.




**NOTE:** The key switch positions shown next to the left switch (above) are provided for reference only, and are marked on the module as indicated on the right.

**NOTE:** The 140 CPU 434 12A and 140 CPU 534 12A processors feature the key switch illustrated above, while the 140 CPU 311 10 has a slide switch.

## Key Switch Description Tables

The following table provides key/slider switch information for these three low end CPUs.

| CPU Type | Switch Position | Behaviour | Protected? | Accepts Stop or Start? | Key Switch Transition |
|---|---|---|---|---|---|
| **Quantum 140 CPU 311 10** | Mem Prt On | The application in Flash memory is not transferred to internal RAM; a warm restart of the application is triggered. | Y | N | From Mem Prt Off: does not modify last controller state and rejects programmer changes. |
| | Not used | Do not use this position, because it may lead to undefined operation | Y | N | n/a |
| | Mem Prt Off | The application in Flash memory is automatically transferred to internal RAM when the PLC is powered up. A cold restart of the application is triggered. | N | Y | From Mem Prt On : enables programmer changes and starts controller if stopped. |
| **Quantum 140 CPU 434 12A 140 CPU 534 14A** | Stop | The application in Flash memory is not transferred to internal RAM; a warm restart of the application is triggered. | Y | N | From Start or Mem Prt: stops controller, if running, and voids programmer changes. |
| | Mem Prt | The application in Flash memory is not transferred to internal RAM. A warm restart of the application is triggered. | Y | N | From Stop or Start: inhibits program changes, controller run status is unchanged. |
| | Start | The application in Flash memory is automatically transferred to internal RAM when the PLC is powered up. A cold restart of the application is triggered. | N | Y | From Stop: enables programmer changes, starts controller. From Mem Prt: accepts programmer changes, starts controller if stopped. |

The following table provides information on switch position for low end Quantum 140 CPUs with firmware < 3.2:

| Operation | Switch Position | | |
|---|---|---|---|
| | Stop | Mem Protect | Start |
| Programming mode | Allowed | Allowed | Allowed |
| Monitoring mode | Allowed | Allowed | Allowed |
| Upload application | Allowed | Allowed | Allowed |
| Download, online modification | Allowed | Forbidden | Allowed |
| Stop command from Unity Pro | – | Forbidden | Allowed |
| Run command from Unity Pro | Forbidden | Forbidden | Allowed |
| Init command from Unity Pro | Allowed | Allowed | Allowed |

The following table provides information on switch position for low end Quantum 140 CPUs with firmware ≥ 3.2 and Unity Pro V8:

| Operation | Switch Position | | |
|---|---|---|---|
| | Stop | Mem Protect | Start |
| Programming mode | Allowed(1) | Forbidden | Allowed(1) |
| Monitoring mode | Allowed(1) | Allowed(1) | Allowed(1) |
| Upload application | Allowed(1) | Forbidden | Allowed(1) |
| Download, online modification | Allowed(1) | Forbidden | Allowed(1) |
| Stop command from Unity Pro | – | Forbidden | Allowed(1) |
| Run command from Unity Pro | Forbidden | Forbidden | Allowed(1) |
| Init command from Unity Pro | Forbidden | Forbidden | Allowed(1) |
| (1): Password-protected if a password has been defined in Unity Pro application | | | |

# Modbus Connectors

## Modbus Connector Pinouts

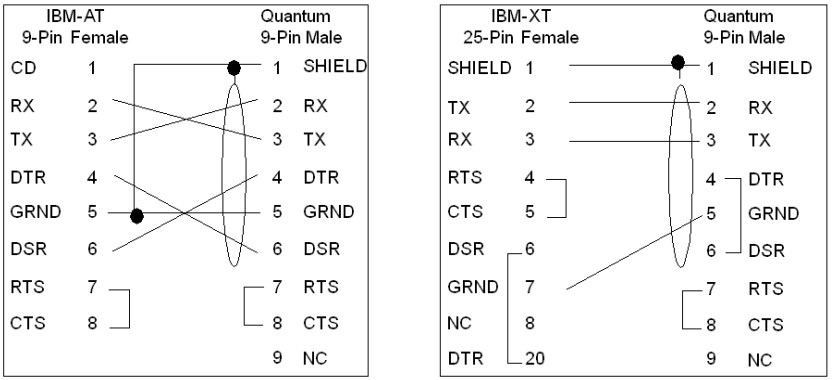
Quantum CPUs are equipped with a nine-pin RS-232C connector that supports Schneider Electric's proprietary Modbus communication protocol. The following is the Modbus port pinout connections for nine-pin and 25-pin connections.

Modbus Port 1 has full modem interfacing capabilities Modbus Port 2 RTS/CTS connections function properly for normal non-modem communications but do not support modems.

**NOTE:** Although the Modbus ports electrically support existing Modbus cables, it is recommended that a Modbus programming cable (Part # 990 NAA 263 20 or 990 NAA 263 50) be used. This cable has been designed to fit under the door of a Quantum CPU or NOM module.

## Modbus Ports Pinout Connections Figure

The following figure shows the Modbus port pinout connections for nine-pin and 25-pin connections.

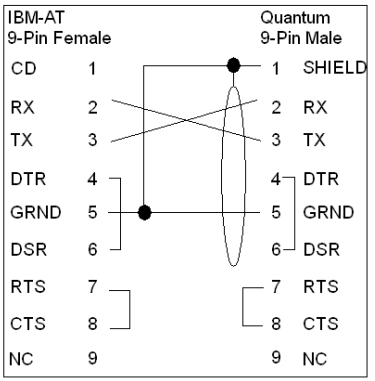



The following is the abbreviation key for the above figure.

| | |
|---|---|
| TX: Transmitted Data | DTR: Data Terminal Ready |
| RX: Received Data | CTS: Clear to Send |
| RTS: Request to Send | NC: No Connection |
| DSR: Data Set Ready | CD: Carrier Detect |

## Modbus Ports Pinout Connections for Portable Computers

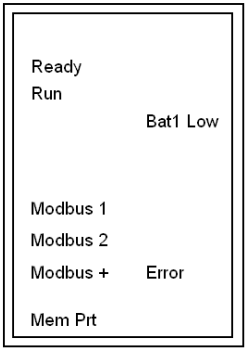
The following figure shows the Modbus port pinout connections for 9-pin portable computers.



# Indicators

## Illustration

The following figure shows the LED indicators.




## Description

The following table shows the LED description for the LE-CPU modules.

| LEDs | Color | Indication when On |
|---|---|---|
| Ready | Green | The CPU has passed powerup diagnostics. |
| Run | Green | The CPU has been started and is solving logic. (See the following table for *Run LED Error Codes, page 111*). |
| Modbus 1 | Green | Communications are active on the Modbus port 1. |
| Modbus 2 | Green | Communications are active on the Modbus port 2. |
| Modbus + | Green | Communications are active on the Modbus Plus port. |
| Mem Prt | Amber | Memory is write protected (the memory protect switch is on). |
| Bat 1 Low | Red | The battery needs replacing. |
| Error | Red | Indicates an communications error on the Modbus Plus port. |

## Run LED Error Codes

The following table shows the Run LED error codes for the LE-CPU modules.

| Number of Blinks | Code | Error |
|---|---|---|
| Continuous | 0000 | requested kernel mode |
| 2 | 80B | ram error during sizing |
| | 80C | run output active error |
| | 82E | MB command handler stack error |
| 3 | 769 | bus grant received |
| | 72A | not master asic on cpu |
| | 72B | master config write bad |
| | 72C | quantum bus DPM write error |
| | 72F | plc asic loopback test |
| | 730 | plc asic BAD_DATA |

| Number of Blinks | Code | Error |
|---|---|---|
| 4 | 604 | UPI timeout error |
| | 605 | bad UPI response opcode |
| | 606 | UPI bus diagnostic error |
| | 607 | modbus cmd-buffer overflow |
| | 608 | modbus cmd-length is zero |
| | 609 | modbus abort command error |
| | 614 | mbp bus interface error |
| | 615 | bad mbp response opcode |
| | 616 | timeout waiting for mbp |
| | 617 | mbp out of synchronization |
| | 618 | mbp invalid path |
| | 619 | page 0 not paragraph aligned |
| | 61E | bad external uart hardware |
| | 61F | bad external uart interrupt |
| | 620 | bad receive comm state |
| | 621 | bad transmit comm state |
| | 622 | bad comm state trn_asc |
| | 623 | bad comm state trn_rtu |
| | 624 | bad comm state rcv_rtu |
| | 625 | bad comm state rcv_asc |
| | 626 | bad modbus state tmr0_evt |
| | 627 | bad modbus state trn-int |
| | 628 | bad modbus state rcv-int |
| | 631 | bad interrupt |
| 5 | 503 | ram address test error |
| | 52D | P.O.S.T BAD MPU ERROR |
| 6 | 402 | ram data test error |
| 7 | 300 | EXEC not loaded |
| | 301 | EXEC Checksum |
| 8 | 8001 | Kernal prom checksum error |
| | 8002 | flash prog / erase error |
| | 8003 | unexpected executive return |

**NOTE:** Information in the Code column is visible only with the Flash download utility.

# Processor Configuration Screen

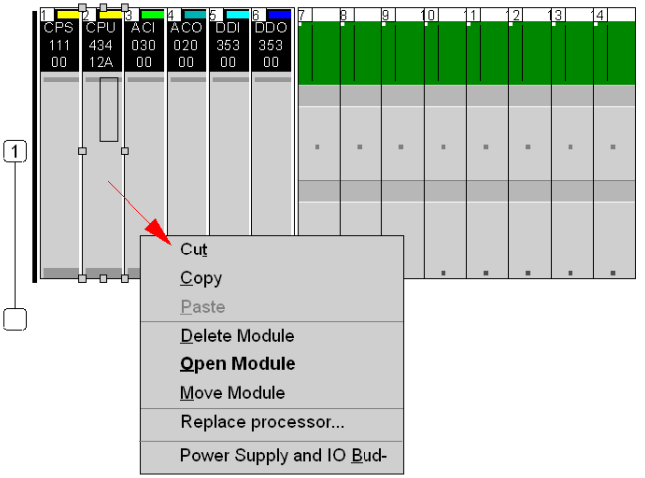
## Overview

There are the following possibilities to configure the Quantum LE-CPUs:

- Base configuration *(see page 113)*, including Modbus Ports
- Configuration of the communication type *(see page 122)*

## Base configuration

Steps to base configuration, including Modbus Ports

| Step | Action |
|---|---|
| 1 | Open the Bus Editor of Unity Pro |
| 2 | Select the CPU module |
| 3 | Click the right mouse button<br>**Result:** The context menu appears<br>  |

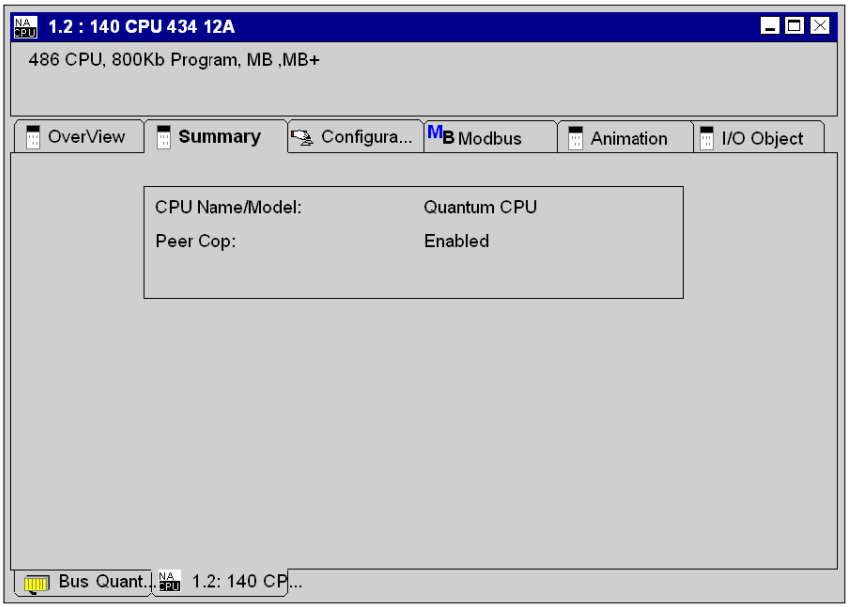
| Step | Action |
|---|---|
| 4 | Choose **Open Module**<br>**Result:** The module opens with the **Summary** tab |
| 5 | Choose one of the tabs:<br>● Overview *(see page 114)*<br>● Summary *(see page 114)*<br>● **Configuration** *(see page 115)*<br>● **Modbus Port** *(see page 118)*<br>● Animation *(see page 119)*<br>● I/O object *(see page 119)*<br>The tabs with configuration features are in bold type. |

## Overview

This screen contains the specification of the module, extracted from the Quantum Hardware Reference Guide.

## Summary

Summary Screen:

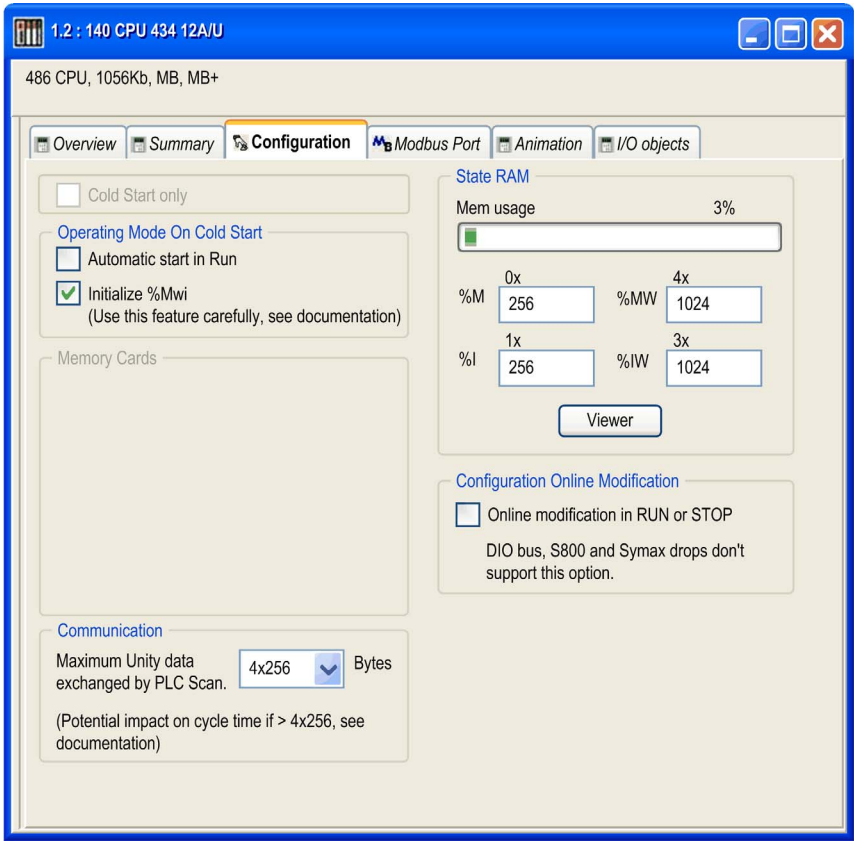


Description of the Summary Screen:

| Item | Detail | Option/Value | Description |
|---|---|---|---|
| CPU Name/Model: | Quantum CPU | | |
| Peer Cop: | Disabled | Enabled | Peer Cop=Enabled only possible in combination with NOM |
| | Time: | ..... | |

## Configuration

Configuration Screen:



Description of the Configuration Screen:

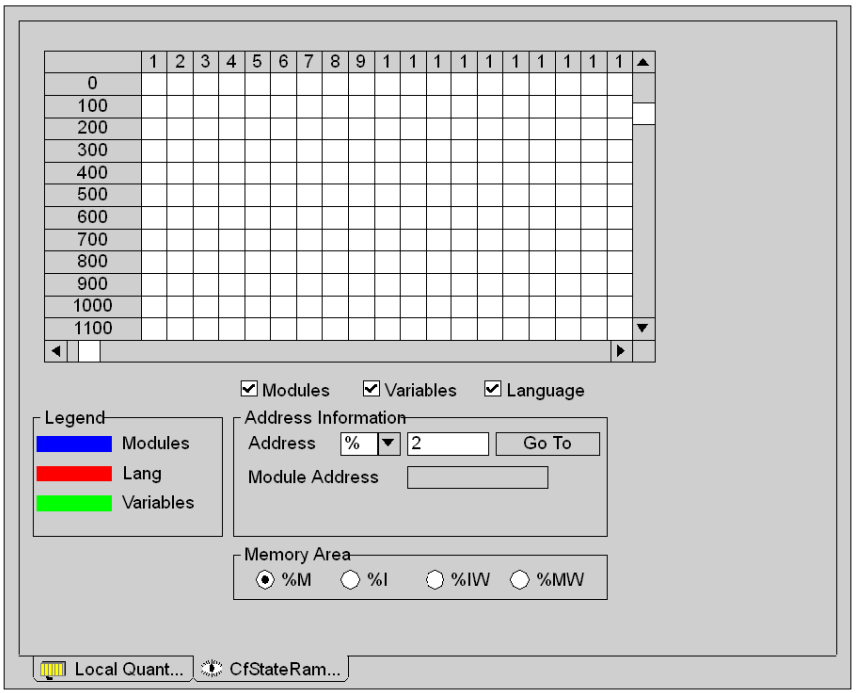
| Item | Option | Value | Description |
|---|---|---|---|
| Cold Start Only | | x | If you wish, enable the Cold Start Only feature. |
| Operating Mode On Cold Start | Automatic start in Run | x | Determines the operating condition during Cold Start |
| | Initialize %MWi | x | |
| Communication | By default, the bandwidth is 4x256 bytes, supported by the OS versions prior to V2.80 for the CPU and V4.60 for the NOE. | | The maximum data volume exchanged each cycle between the NOE and CPU modules. |
| | For Quantum processors:<br>● 140 CPU 311 10<br>● 140 CPU 534 14<br>● 140 CPU 434 12 | 4x256<br>4x1024 | |
| State RAM | Mem usage | (1) | A bar displays percent of memory used. |
| | %M-0x | (2) | Size of the different memory areas<br>**Note:** The values for %IW and %MW have to be divisible by 8. |
| | %MW-4x | (2) | |
| | %I-1x | (2) | |
| | %IW-3x | (2) | |
| | Viewer | N/A | Opens the State RAM Viewer tab, which displays the allocation of used memory. (See the illustration following.) |
| Configuration Online Modification | Online modification in RUN or STOP | x | This check box allows you to:<br>● Add or delete discrete or analog modules,<br>● Modify Parameters<br>**NOTE:** These modifications can be done in RUN. |
| **(1)** The value is expressed as a percentage and displayed on the scale.<br>**(2)** Enter the appropriate values. | | | |

For Quantum 140 CPU 3••, 140 CPU 4•• and 140 CPU 5•• the %MWi value is as follows:

| | | **Power ON: without cold start** | **Power ON: with cold start** |
|---|---|---|---|
| **Initialize %MWi** box | Unselected | %MWi keep their value | %MWi keep their value |
| | Selected | | %MWi take their initial value |

**NOTE:** %MWi resets with cold start %S0 or a programload. Cold start is typically after a program load, %S0 can be set with the user program to initiate a cold start.

Button to show the allocation of the used memory




**NOTE:** The state ram viewer can be directly accessible via the menu:

- **PLC → State Ram Viewer**

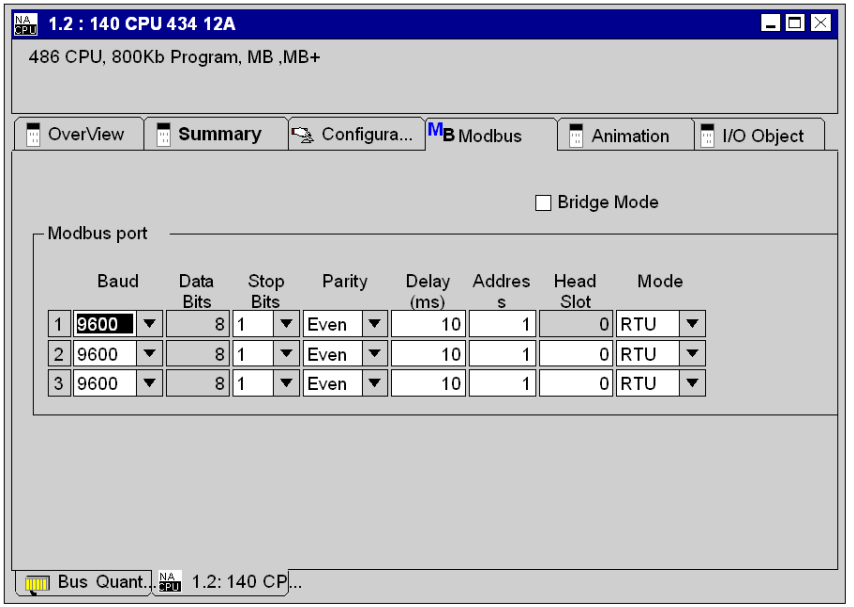
The content of the grid may be changed by setting the following two filters:

1. Memory user
   - Modules
   - Language
   - Variables

2. Memory area
   - %M
   - %I
   - %IW
   - %MW

## Modbus Port

Modbus Port Screen:




Description of the Modbus Port Screen:

| Item | Detail | Option/Value | Description |
|---|---|---|---|
| Modbus port | | | |

| Item | Detail | Option/Value | Description |
|---|---|---|---|
| Baud | 9600 | 50-19200 kBit/s | These data must be specified for every Link separately |
| Data Bits | 8 | 7 when ASCII mode is selected | |
| Stop Bits | 1 | 2 | |
| Parity | EVEN | ODD, NONE | |
| Delay (ms) | 10 ms | 10..1000 in 10 ms increments | |
| Address | 1 | 1..247 | |
| Head Slot | 0 | 1..16 | |
| Mode | RTU | ASCII | |

**Bridge Mode** must be checked if you are making a network link between Modbus and Modbus Plus.

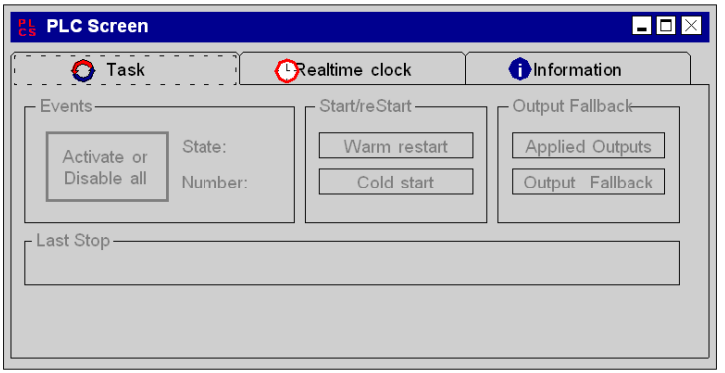
## Animation

Based on the animation window, there are the following windows, accessible by tabs:

- Task
- Realtime clock
- Information

**NOTE:** The windows are depicted in offline mode. Connected with a PLC their appearances change.

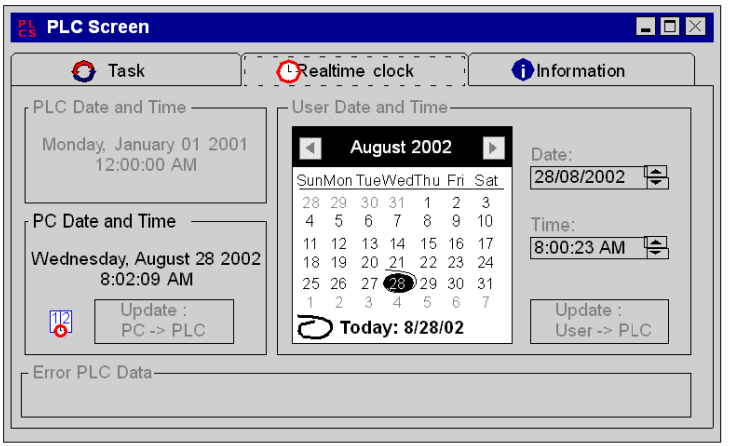
Animation Screen (Task):



Description of the Animation Screen (Task):

| Item | Detail | Option/Value | Description |
|---|---|---|---|
| Events | State:<br>Number: | ...<br>... | Online available<br>Status information of events |
| | Activate or Disable all | | Button to control the events |
| Start/reStart | Warm Start | | |
| | Cold Start | | |
| Output fallback | Applied Outputs | | Specifies the output behavior |
| | Output Fallback | | |
| Last Stop | | .../.../... | |

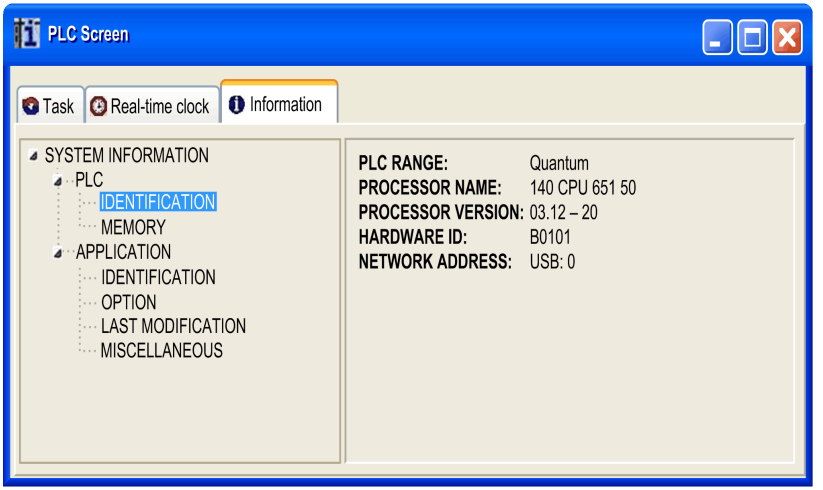
Animation Screen (Realtime clock):




Description of the Animation Screen (Realtime clock):

| Item | Detail | Description |
|---|---|---|
| PLC Date and Time | | Indication of the current PLC date and time |
| PC Date and Time | Update PC->PLC | To update the PLC with the PC system time |
| User Date and Time | Update User->PLC | To update the PLC with the time set by the user |

Animation Screen (Information):




Description of the Animation Screen (Information):

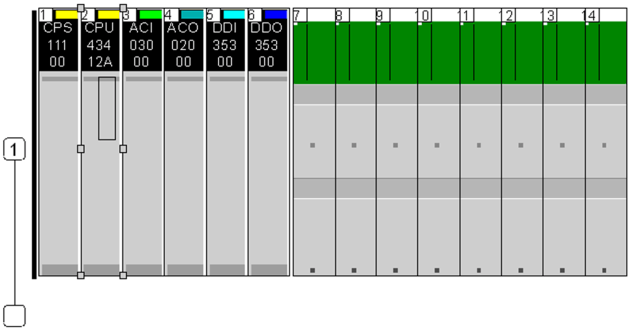
| Item | Detail | Option/Value | Description |
|---|---|---|---|
| System Information | PLC / Identification | PLC Range<br>Processor Name<br>Processor Version<br>Hardware ID<br>Network address | Only Online available |
| | PLC / Memory | RAM CPU | |
| | Application / Identification | Name<br>Creation Product<br>Date<br>Modification Product<br>Date<br>Version<br>Signature | |
| | Application / Option | Empty Terminal Support<br>Upload Information<br>Comments<br>Animation Table<br>Global Protection<br>Section Protection<br>Application Diagnostic<br>Forced Bits | |

## I/O Object

The I/O objects tab is used to associate variables with the module I/Os and to manage these different variables. This tab is described in I/O Management *(see Unity Pro, Operating Modes)*.

## Configuration of the communication type

Steps to configuration of the communication type

| Step | Action |
|---|---|
| 1 | Open the Bus Editor of Unity Pro |
| 2 | Move the mouse over the rectangle of the CPU module<br>**Result:** The mouse pointer changes to a hand<br>  |
| 3 | Double-click.<br>**Result:** A subdialog appears |
| 4 | Choose one of the following options:<br>● None<br>● DIO bus<br>● Peer Cop |

# 140 CPU 311 10 Specifications

## General Specifications

| | |
|---|---|
| Communication ports | 2 Modbus (RS-232)<br>1 Modbus Plus (RS-485) |
| Bus current required | 1250 mA |
| Max. number of NOM, NOC, NOE, PTQ PDP MV1 and MMS modules supported (any combination) | 2 |
| Key switch | No |

## Processor

| | |
|---|---|
| Model | Intel 486 |
| Math processor | No |
| Watchdog timer | 250 ms S/W adjustable |

## Memory

| | |
|---|---|
| Internal memory (max.) | 548 k |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 51712 |
| Registers | 9672 |

## Local I/O

| | |
|---|---|
| Max. I/O words | unlimited I/O |
| Max. I/O racks | 2 |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words per drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

## Ethernet Remote I/O

Not supported

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 IN and 500 OUT<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words per node | 30 in / 32 out |
| Max. number of DIO drops per network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Not supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V Lithium |
| Service life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity per year |
| Battery load current @ power-off | typical: 5 µA @ 25 °C (ambient temperature) |
| | maximum: 110 µA @ 60 °C |
| TOD clock | +/- 8.0 s/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive checksum<br>User logic check<br>Processor |
| Run time | RAM<br>RAM address<br>Executive checksum<br>User logic check |

# 140 CPU 434 12A/U Specifications

## General Specifications

- 140 CPU 434 12U
  This module is the version of the controller that is managed with Unity Pro programming software.
  **NOTE:** You can use only Unity Pro software to program the 140 CPU 434 12U controller.

  **NOTE:** The 140 CPU 434 12U controller is not compatible with Hot Standby topologies.
- **140 CPU 434 12A**
  This module is functionally identical to the non-A version. However, the following must be considered:
  - If you are using the module in a Hot Standby topology, then you must use either two non-"A" models or two A models.
  - The A version has a unique flash executive.
    **NOTE:** The "A" version and non-"A" flash executives are not interchangeable.
  - Schneider Electric software (Concept, ProWORX, and Modsoft) supports the "A" version. Any existing or new 140 CPU 434 12 program configuration will load into a 140 CPU 434 12A without any modifications.
    **NOTE:** You can flash the "A" version with the executive of the "U" version to become compatible with the Unity Pro software.

    **NOTE:** A 140 CPU 434 12A flashed with a "U" Unity executive is not compatible with Hot Standby topologies.

| | |
|---|---|
| Communication ports | 2 Modbus (RS-232) |
| | 1 Modbus Plus (RS-485) |
| Bus current required | 1250 mA |
| Max. number of NOM, NOC, NOE, PTQ PDP MV1 and MMS modules supported (any combination) | 6 |
| Key switch *(see page 105)* | Yes |

**NOTE:** This CPU can support up to 3 MODBUS networks.

## Processor

| | |
|---|---|
| Model | Intel 486 |
| Clock speed | 66 MHz |
| Math processor | Yes, on-board |
| Watchdog timer | 250 ms S/W adjustable |

## Memory

| | |
|---|---|
| Internal memory (max.) | 1056 k |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 |
| Registers | 64974 |

## Local I/O

| | |
|---|---|
| Max. I/O words | unlimited I/O |
| Max. I/O racks | 2 |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words per drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

## Ethernet Remote I/O

Not supported

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 in / 500 out<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words per node | 30 in / 32 out |
| Max. number od DIO drops/network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Not supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service Life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity per year |
| Battery load current @ power-off | typical: 7 µA @ 25 °C (ambient temperature) |
| | maximum: 210 µA @ 60 °C |
| TOD clock | +/- 8.0 s/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 534 14A/U Specifications

## General Specifications

This module is functionally identical to the non-A version. However, the following should be considered:

- If you are using the module in a Hot Standby topology, then you must use either two non-A models or two A/U models.
- The A/U model requires a new flash executive.
- The A/U and the non-A flash executives are not interchangeable.
- Schneider Electric software supports the A/U model. Any existing or new 140 CPU 534 14 program configuration will load into a 140 CPU 534 14A/U without any modifications.

| | |
|---|---|
| Communication ports | 2 Modbus (RS-232) |
| | 1 Modbus Plus (RS-485) |
| Bus current required | 1250 mA |
| Max. number of NOM, NOC, NOE, PTQ PDP MV1 and MMS modules supported (any combination) | 6 |
| Key switch *(see page 105)* | Yes |

**NOTE:** This CPU can support up to 3 modbus network

## Processor

| | |
|---|---|
| Model | Intel 586 DX |
| Clock speed | 133 MHz |
| On-board math processor | Yes, on-board |
| Watchdog timer | 250 ms S/W adjustable |

## Memory

| | |
|---|---|
| Internal memory (max.) | 2972 k |
| Max. HSBY Unlocated Data | 128 kbytes |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see the chapter Memory tab in the *Operating Modes Manual*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 |
| Registers | 64974 |

## Local I/O

| | |
|---|---|
| Max. I/O words | unlimited I/O |
| Max. I/O racks | 2 |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words per drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

## Ethernet Remote I/O

Not supported

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 in / 500 out<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words per node | 30 in / 32 out |
| Max. number of DIO drops per network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Not supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 210 µA @ 60 °C |
| TOD clock | +/- 8.0 s/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive checksum<br>User logic check<br>Processor |
| Run time | RAM<br>RAM address<br>Executive checksum<br>User logic check |

# 140 CPU 534 14B/U Specifications

## General Specifications

This module is functionally identical to the non-B version. However, the following should be considered:

- If you are using the module in a Hot Standby topology, then you must use either two non-B models or two B/U models.
- The B/U model requires a new flash executive.
- The B/U and the non-B flash executives are not interchangeable.
- Schneider Electric software supports the B/U model. Any existing or new 140 CPU 534 14 program configuration will load into a 140 CPU 534 14B/U without any modifications.

| | |
|---|---|
| Communication ports | 2 Modbus (RS-232) |
| | 1 Modbus Plus (RS-485) |
| Bus current required | 1250 mA |
| Max. number of NOM, NOC, NOE, PTQ PDP MV1 and MMS modules supported (any combination) | 6 |
| Key switch *(see page 105)* | Yes |

**NOTE:** This CPU can support up to 3 modbus network

## Processor

| | |
|---|---|
| Model | Intel 486 DX4 |
| Clock speed | 100 MHz |
| On-board math processor | Yes, on-board |
| Watchdog timer | 250 ms S/W adjustable |

## Memory

| | |
|---|---|
| Internal memory (maximum) | 2972 k |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 |
| Registers | 64974 |

## Local I/O

| | |
|---|---|
| Max. I/O words | unlimited I/O |
| Max. I/O racks | 2 |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words per drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

## Ethernet Remote I/O

Not supported

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 in / 500 out<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words per node | 30 in / 32 out |
| Max. number of DIO drops per network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Not supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 210 µA @ 60 °C |
| TOD clock | +/- 8.0 s/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive checksum<br>User logic check<br>Processor |
| Run time | RAM<br>RAM address<br>Executive checksum<br>User logic check |

# Chapter 9
# High End CPU

## Purpose

This chapter provides information on the High End Controller modules.

Quantum high-end CPUs (140 CPU 6•• •• *(see page 135)*) also support Ethernet backplanes (BME XBP •••• *(see Modicon M580, Hardware, Reference Manual)*) in the following network configurations:

- with a 140 CRP 312 00 remote I/O head module *(see Quantum EIO, Remote I/O Modules, Installation and Configuration Guide)* configured on the local rack *(see Quantum EIO, System Planning Guide)*
- with a BME CRA 312 •0 eX80 EIO adapter module *(see Modicon M580, RIO Modules, Installation and Configuration Guide)* configured on an Ethernet remote I/O drop *(see Quantum EIO, System Planning Guide)*

These Ethernet backplanes function in the same manner as when they are used in an M580 system. The eX80 EIO adapter modules also function in the same manner (including configuration, diagnostics, and performance) as when they are used in an M580 system.

### Quantum SIL3 Hot StandBy Offer

Read thoroughly the *Quantum Safety PLC - Safety Reference Manual* (part number 33003879) to build a safety PLC according to the safety certifications. Schneider Electric provides a range of products that are certified to be used in a IEC 61508 and SIL3 safety system.

This range includes:

- safety CPU modules (140 CPU 651 60S and 140 CPU 671 60S)
- safety I/O modules (140 SAI 940 00S, 140 SDI 953 00S, 140 SDO 953 00S)
- non-interfering modules
- Unity Pro XLS

**NOTE:** If remote racks or hot standby functionality is required, you can use the safety modules with existing Quantum remote I/O modules (140 CRP 932 00 and 140 CRA 932 00 only). If other I/O modules are present in the rack, they must either be certified non-interfering modules or they must be removed or exchanged for certified non-interfering modules that will provide similar capability.

## What Is in This Chapter?

This chapter contains the following topics:

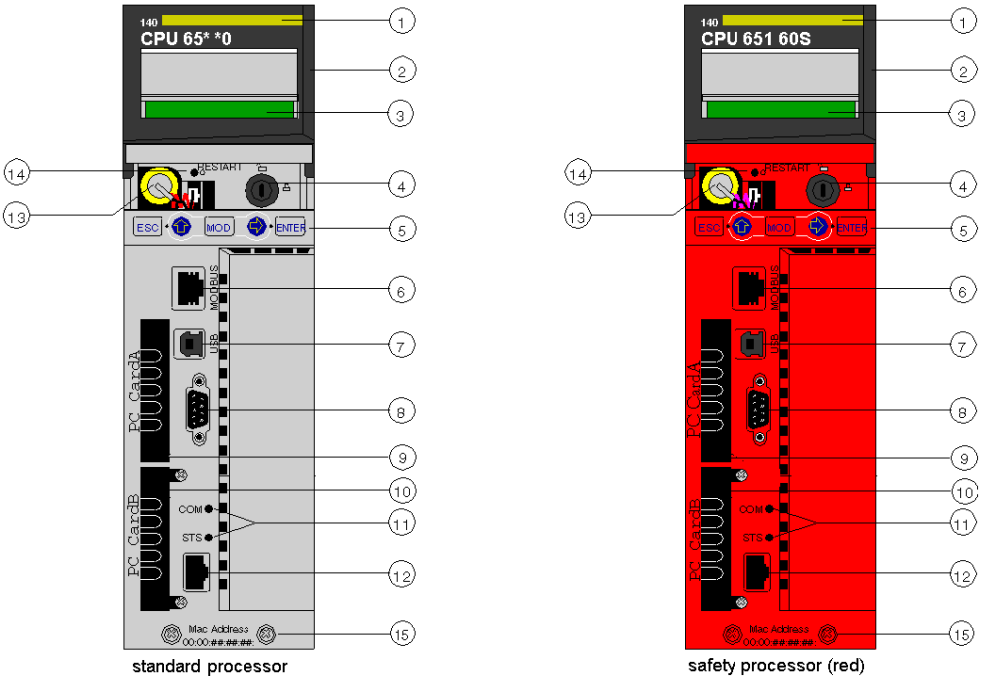

| Topic | Page |
|---|---|
| Physical Presentation and Mounting of Standard High End Modules | 137 |
| Standalone Safety CPU | 139 |
| Physical presentation and mounting of Hot StandBy High End modules | 141 |

| Topic | Page |
|---|---|
| Hot Standby Safety CPU Specifics | 142 |
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| Indicators | 149 |
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| Using the CPU LCD Display Screens | 153 |
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| 140 CPU 672 60 Specifications | 189 |
| 140 CPU 672 61 Specifications | 192 |
| 140 CPU 678 61 Specifications | 195 |

# Physical Presentation and Mounting of Standard High End Modules

## Illustration

The figure shows a standard High End module and its components.




standard processor

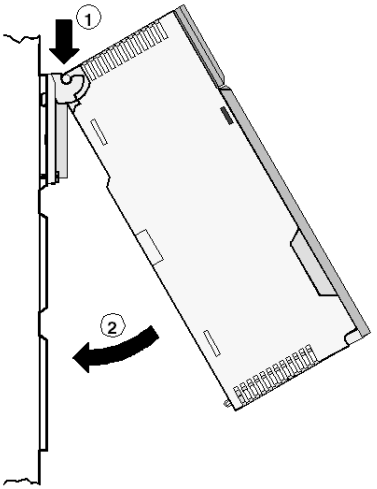
safety processor (red)

1 model number, module description, color code
2 lens cover (open)
3 LCD display (here covered by the lens cover)
4 key switch
5 keypad (with 2 red LED indicators)
6 modbus port (RS-232) (RS-485)
7 USB port
8 Modbus Plus port
9 PCMCIA slot A (depends on CPU reference)
10 PCMCIA slot B
11 LED indicators (yellow) for Ethernet communication
12 Ethernet port
13 battery (user installed)
14 reset button
15 2 screws

**NOTE:** Quantum High End processors are equipped with one PCMCIA slot (B) or two PCMCIA slots (A and B) in which to install Schneider PCMCIA cards (other cards are not accepted).

## Mounting

Mounting the module onto the central back plane:




**1** Hang the module.
**2** Screw the module to the back plane.

# Standalone Safety CPU

## Introduction

For use in standalone SIL3 solutions, the **140 CPU 651 60S** Quantum Safety CPU is certified.

The safety CPU includes a PCMCIA memory card, but its use and presence is not mandatory.

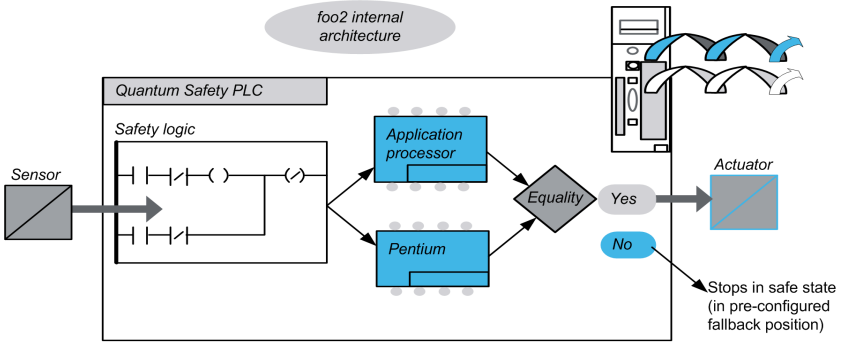
## Description of the Internal CPU Architecture

The Quantum Safety CPU contains 2 different processors, an Intel Pentium and an application processor. Each one executes the Safety logic in its own memory area and both compare the results of the execution at the end of each cycle.

Two CPUs are available:

- 140 CPU 651 60S (Standalone Safety CPU)
- 140 CPU 671 60S (Hot Standby Safety CPU)

The following figure shows the internal architecture of the Quantum Safety CPU:




## Benefits of the Double Code Generation and Execution

The 2 processors inside the Quantum Safety PLC allow double code generation and execution.

This diversity provides the following advantages in error detection:

- 2 executable codes are generated independently. The diversity of compilers allows the detection of systematic error in the code generation.
- The 2 generated codes are executed by 2 different processors. Thus, the CPU is able to detect both systematic errors in the code execution and random errors in the PLC.
- 2 independent memory areas are used for the 2 processors. Thus, the CPUs are able to detect random errors in the RAM and a full RAM test is not necessary at every scan.

## Description of the Watchdog

A hardware and a firmware watchdog check the PLC activity and the time needed to execute the user logic.

**NOTE:** You must configure the software watchdog (maximum PLC cycle time) to be consistent with the application execution time, the filtering of the I/O communication error, and the process Safety time (PST) targeted.

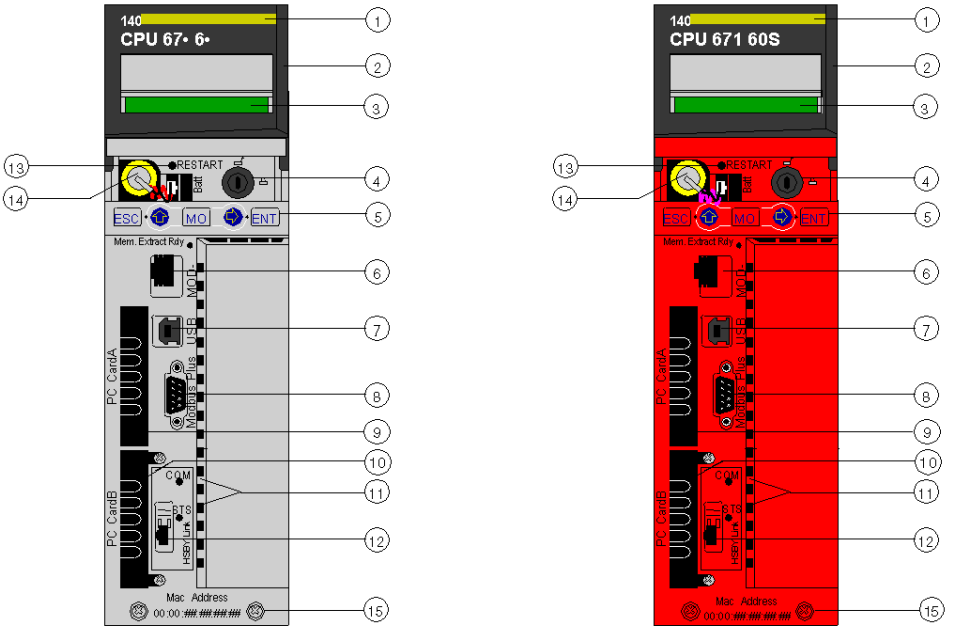
## Description of the Memory Check

Static memory areas, including the Flash memory, PCMCIA memory card (if any) and the RAM, are checked using the cyclic redundancy check (CRC) and the double code execution. Dynamic areas are protected by the double code execution and a periodic memory test. At cold start, these tests are re-initialized and fully performed before the CPU goes into Stop or Run mode.

# Physical presentation and mounting of Hot StandBy High End modules

## Illustration

The figure shows an Hot StandBy High End module and its components.




standard HSBY processor

safety HSBY processor (red)

1 model number, module description, color code
2 lens cover (open)
3 LCD Display (here covered by lens cover)
4 key switch
5 keypad (with 2 red LED indicators)
6 Modbus port (RS-232) (RS-485)
7 USB port
8 Modbus Plus port
9 PCMCIA slots A (depends on CPU reference)
10 PCMCIA slots B
11 LED indicators (yellow) for Ethernet communication
12 HSBY fiber optic communication port
13 reset button
14 battery (user installed)
15 2 screws

**NOTE:** Quantum High End processors are equipped with one PCMCIA slot (B) or two PCMCIA slots (A and B) in which to install Schneider PCMCIA cards (other cards are not accepted).

# Hot Standby Safety CPU Specifics

## Introduction

The 140 CPU 671 60S Quantum Safety CPU module is certified for use in Hot Standby SIL3 solutions compliant with the 61508 IEC standard. For more details about to the safety certifications, refer to the *Modicon Quantum Safety PLC Safety Reference Manual*.

In the Standalone Safety CPU, the Ethernet port is used to communicate with other devices using a normal Ethernet cable.

In the Hot Standby Safety CPU, the connection used to exchange data between the Primary CPU and the Standby CPU controller is a fiber optic link. Because the fiber optic link is not part of the Safety loop, the PFD and PFH values of the Hot Standby CPU are the same as those of the Standalone CPU.

Each Safety CPU can include a PCMCIA memory card, but its use and presence is not mandatory.

**NOTE:** This CPU cannot be used in a Quantum Ethernet I/O Hot Standby system.

## Description of a Safety Hot Standby Configuration

The Hot Standby configuration contains two identical local racks and at least one remote I/O drop because I/Os cannot be placed in the local rack of a Safety Hot Standby configuration.

Besides a power supply module (there must be at least one 140 CPS 124 20 or one 140 CPS 22 400), each local rack must contain:

- 140 CPU 671 60S module
- 140 CRP 932 00 module

Besides a power supply, I/O modules (including at least one 140 CPS 124 20 or one 140 CPS 22 400), the remote drop(s) must include a 140 CRA 932 00 module.

| ⚠ CAUTION |
| --- |
| **UNINTENDED EQUIPMENT OPERATION** |
| Use only high availability RIO modules with dual cabling in a Safety-Related System. |
| **Failure to follow these instructions can result in injury or equipment damage.** |

## Description of the Operating Modes

- **Safety Mode:** This is the default mode. It is a restricted mode in which modifications and maintenance activities are prohibited.
- **Maintenance Mode:** This is a temporary mode for modifying the project, debugging and maintaining the application program.

## State Compatibility with Safe and Maintenance Modes

A Quantum Hot Standby system has two states:

- **Redundant (1 CPU is Primary, 1 is Standby)**
  The Standby CPU controller mode follows the Primary CPU controller mode. For example, if you switch the Primary CPU controller from Safety to Maintenance mode, the Standby CPU controller switches from Safety to Maintenance mode at the start of the next cycle.
- **Non-redundant (at least 1 CPU Offline)**
  The two controllers are independent, one can be in Safety mode and the other one in Maintenance mode. For example, the Run Primary controller can be in Safety mode while the Stop Offline controller is in the Maintenance mode.

## Impact of the PLC Switchover on the Process Safety Time

If the Primary CPU detects an internal or external problem, it stops exchanging data with the Standby CPU and stops processing the I/O. As soon as the Standby CPU detects that there are no longer exchanges with the Primary CPU, it takes over the role of the Primary CPU, executing the user logic and processing the I/O. Therefore, the output modules must filter the lack of exchange with the Primary CPU to avoid glitches when a Switchover occurs. This is achieved by configuring the output module time-out. As a result, the PLC reaction time is greater than the time-out configured in the output module, thereby influencing the process Safety time.

**NOTE:** The behavior of the Hot Standby Safety CPU is equivalent to a Standalone Safety CPU.

In case of a detected error, the Safety PLC enters:

- Halt state when running in the Maintenance Mode
- Error state when running in the Safety Mode

## Availability of the Hot Standby Functions

In addition to the standard Hot Standby functions, you can use an EFB to program an automatic Switchover between Primary CPU and Standby CPU to verify the ability of the Standby CPU to take over from the Primary CPU. That means that the Standby CPU periodically becomes the Primary CPU and the Primary CPU becomes the Standby CPU.

It is recommended to avoid using the USB link during a Switchover.

The following table lists the available Hot Standby functions in Maintenance and Safety modes:

| Function | Maintenance Mode | Safety Mode |
|---|---|---|
| Hot Standby | yes | yes |
| Switchover | yes | yes |
| EFB Swap | no | yes |
| Keypad | yes | yes |
| Application mismatch | yes | no |
| OS Upgrade | yes, if Standby CPU is in Stop Offline | no |
| Application Transfer | yes | no |

**NOTE:** Applying the power simultaneously to Primary CPU and Standby CPU is allowed, but we recommend to do it sequentially.

# CPU Controls and Displays

## Lens Cover

The protective lens cover (2 on the CPU front panel *(see Modicon Quantum, Hot Standby System, User Manual)*) can be opened by sliding it upwards.

With the lens cover open you have access to the following items:

- key switch
- battery
- reset button

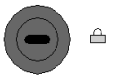
## Key Switch

The key switch (4) is a security feature and a memory protection switch. The key switch has two positions: locked and unlocked. The key switch is only read and deciphered by the PLC OS portion of the firmware and not by the OS loader portion.

The Quantum processors have a set of system menus that enable the operator to:

- perform PLC operations (i.e., start PLC, stop PLC)
- display module parameters (i.e., communications parameters)
- switch to the maintenance mode (in Safety processors)

The key positions are shown in the table below:

| Key Position | PLC Operation |
|---|---|
| unlocked:  | • System menu operations can be invoked and module parameters can be modified by the operator with the LCD and keypad.<br>• Memory protection is OFF.<br>• You can switch to Maintenance mode (in Safety processors). |
| locked:  | • No system menu operations can be invoked and module parameters are read-only.<br>• Memory protection is ON.<br>• Safe mode forced (in Safety processors). |

Switching the key switch position from locked to unlocked or vice versa turns on the LCD's backlight.

**NOTE:** For more explanations about Maintenance and Safe modes, refer to the *Quantum Safety PLC Safety Reference Manual*.

## Key Switch Description Tables

The following table provides information on switch position for high end Quantum 140 CPU 6•• ••• with firmware < 3.12:

| Operation | Switch Position | |
|---|---|---|
| | Unlocked | Locked |
| Programming mode | Allowed | Allowed |
| Monitoring mode | Allowed | Allowed |
| Upload application | Allowed | Allowed |
| Download, online modification | Allowed | Forbidden |
| Stop/Run/Init commands from Unity Pro | Allowed | Allowed |

The following table provides information on switch position for high end Quantum 140 CPU 6•• ••• with firmware ≥ 3.12 and Unity Pro < V8:

| Operation | Switch Position | |
|---|---|---|
| | Unlocked | Locked |
| Programming mode | Allowed | Forbidden |
| Monitoring mode | Allowed | Allowed |
| Upload application | Allowed | Forbidden |
| Download, online modification | Allowed | Forbidden |
| Stop/Run/Init commands from Unity Pro | Allowed | Forbidden |

The following table provides information on switch position for high end Quantum 140 CPU 6•• ••• with firmware ≥ 3.2 and Unity Pro ≥ V8:

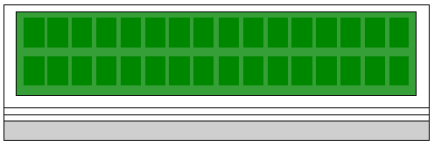
| Operation | Switch Position | |
|---|---|---|
| | Unlocked | Locked |
| Programming mode | Allowed(1) | Forbidden |
| Monitoring mode | Allowed(1) | Allowed(1) |
| Upload application | Allowed(1) | Forbidden |
| Download, online modification | Allowed(1) | Forbidden |
| Stop/Run/Init commands from Unity Pro | Allowed(1) | Forbidden |
| (1): Password-protected if a password has been defined in Unity Pro application | | |

## Reset Button

Pressing the reset button (12) forces a cold start of the PLC.

## LCD Display

The liquid crystal display (LCD - 3) has 2 lines with 16 characters each with changeable backlight state and contrast:



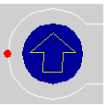
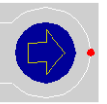
The backlight handling is entirely automated to save the life of the LCDs. The backlight turns on when one of the following occurs:

- a key is pressed
- the key switch state is changed
- an error message is displayed on the LCD

The backlight stays on for error messages as long as the error message is displayed otherwise, the backlight automatically turns off after five minutes.

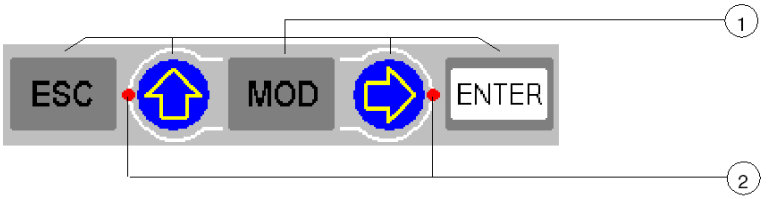
## Adjusting the Contrast

The contrast is adjustable from the keypad when the default screen is displayed:

| Step | Action | |
|---|---|---|
| 1 | Press the MOD key: |   |
| 2 | To adjust the contrast darker press: |  |
| 3 | To adjust the contrast lighter press: |  |
| 4 | To confirm the setting press: |   |

## Keypad

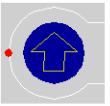
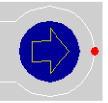
The keypad (5) has five keys that are mapped to hardware addresses. Each of the two arrow keys includes an LED:




**1** 5 keys
**2** 2 LEDs

## Using the Keys

Keypad functions:

| Key | Function | |
|---|---|---|
| ESC | To cancel an entry, suspend or stop an action in progress<br>To display the preceding screens successively (move up the menu tree) | |
| ENTER | To confirm a selection or an entry | |
| MOD | To set a field on the display into the modify mode | |
|  | LED: on | Key active:<br>● to scroll through menu options<br>● to scroll through modify mode field options |
| | LED: flashing | Key active: The modify mode field can be scrolled. |
| | LED: off | Key inactive: No menu options and no field options. |
|  | LED on | Key active:<br>● to move around in a screen, field to field<br>● to go to the sub-menu |
| | LED flashing | Key active: Used to move digit to digit in a modify mode field. |
| | LED off | Key inactive, there is no:<br>● sub-menu for menu option<br>● scrolling around a screen<br>● scrolling around a field |

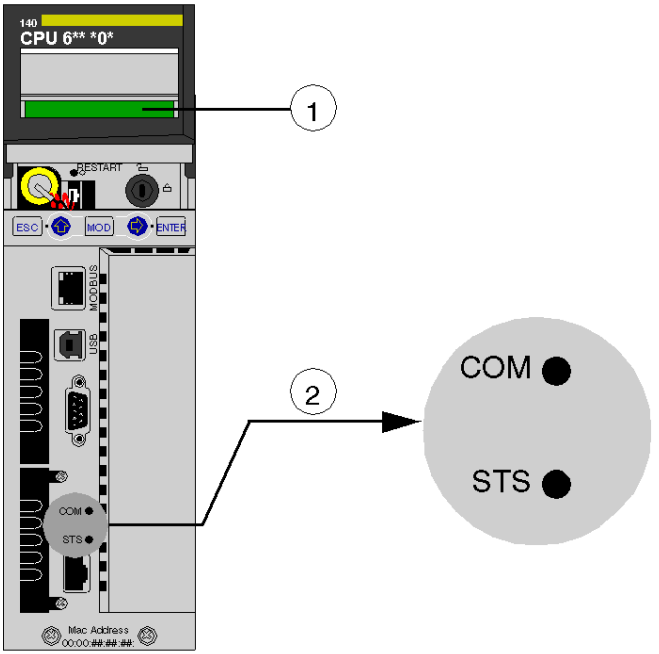
# Indicators

## Overview

The High End processors use two types of indicators:

1. LCD display: The Default Screen *(see page 155)* serves as a Controller status screen.
2. LED Indicators: The functionality of the LEDs is described in a table after the figure.

The following figure shows the two types of indicators.




High End processors

**1** LCD Display (lens cover closed)
**2** LED Indicators

## LED Description

The following table shows the description for the LED indicators of the different High End CPU modules.

<table>
<tr><th rowspan="2">LEDs</th><th colspan="3">Indication</th></tr>
<tr><th colspan="2">Standard Processors 140 CPU 65• •0 / 140 CPU 651 60S</th><th>Hot StandBy Processors 140 CPU 67• 6• / 140 CPU 671 60S</th></tr>
<tr><td>COM (yellow)</td><td colspan="2">Controlled by the Coprocessor hardware<br>● Indicates Ethernet activity</td><td>Controlled by the Coprocessor hardware<br>● Indicates Primary or Standby activity</td></tr>
<tr><td rowspan="10">STS (yellow)</td><td colspan="2">Controlled by the Coprocessor software</td><td rowspan="10">Controlled by the Coprocessor firmware<br>● Blinking: system is redundant and data are exchanged from the Primary to Standby controller<br>● ON: system not redundant / Copro booting from power-on to end of self-tests<br>● OFF: Copro auto tests were not successfull</td></tr>
<tr><td>ON</td><td>Normal</td></tr>
<tr><td>OFF</td><td>Copro auto tests unsuccessful. Possible hardware problem.</td></tr>
<tr><td colspan="2">Flashing:</td></tr>
<tr><td>1 Flash</td><td>Configuration in progress.<br>Temporary situation.</td></tr>
<tr><td>2 Flashes</td><td>Invalid MAC address</td></tr>
<tr><td>3 Flashes</td><td>Link not connected</td></tr>
<tr><td>4 Flashes</td><td>Duplicate IP Address. Module is set to its default IP address.</td></tr>
<tr><td>5 Flashes</td><td>Waiting for IP address from address server</td></tr>
<tr><td>6 Flashes</td><td>Invalid IP address. Module is set to its default IP address.</td></tr>
<tr><td>7 Flashes</td><td>Firmware incompatibility between PLC OS and Copro firmware</td></tr>
</table>

# Modbus port

## Overview

In front of the CPU the Modbus port is of type RJ45.

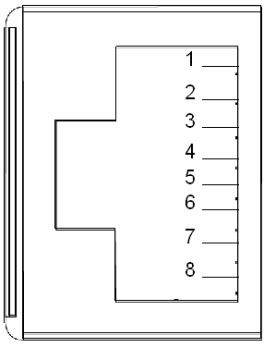
The network topology is determined by a protocol:

- for RS-232, it's a point to point topology,
- for RS-485, it's a bus topology with processor as master.

Connections on RJ45 are different, according to selected protocol. In configuration window of a 140 CPU 6•• •• Quantum PLC, the protocol is selected in tab **Port Modbus**.

## Pinouts

Illustration:




The table following presents the pinouts of RS-232 and RS485 protocols:

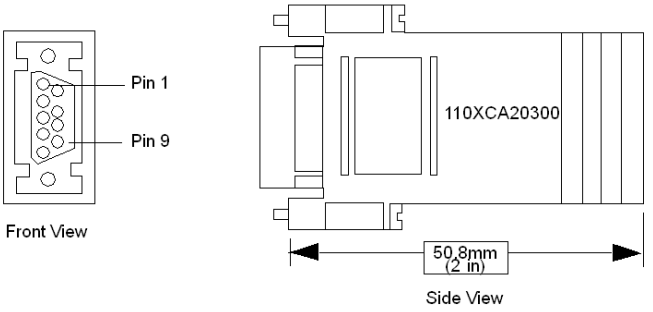
| Pin | RS-232 Signal | RS-485 Signal |
|---|---|---|
| 1 | DTR | D- |
| 2 | DSR | D+ |
| 3 | TxD | |
| 4 | RxD | Not used |
| 5 | GND | GND |
| 6 | RTS | |
| 7 | CTS | Not used |
| 8 | GND (optional) | GND (optional) |

**NOTE:** For RS-485 protocol pins 1 and 6 must be short circuited as well as pins 2 and 3.

## RS232/RJ45 adapter

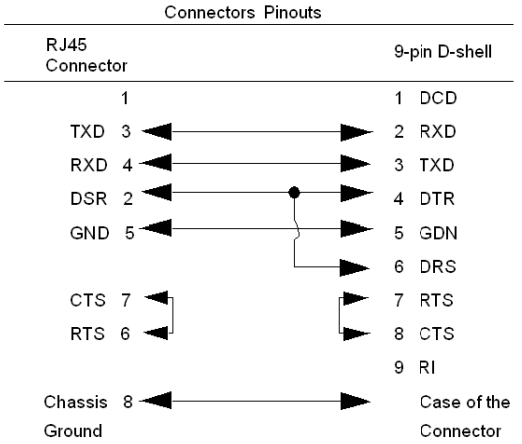
To connect PC-AT computers that have a 9-pin RS-232 port to the 8-pin RJ45 Modbus port on the 140 CPU 651 •0, 140 CPU 670 60, 140 CPU 671 60, 140 CPU 672 60 and 140 CPU 672 61, you must connect the 110 XCA 020 300 adapter (9-pin/RJ45) on the PC with the straight 110 XCA 28 202 cable (8-pin RJ45 to 8-pin RJ45).

The following figures show the 9-pin adapter front view (left) and side view (right).




## Connector Pinouts Figure

The following figure shows the 9-pin RJ45 connector schematic.



# Using the CPU LCD Display Screens

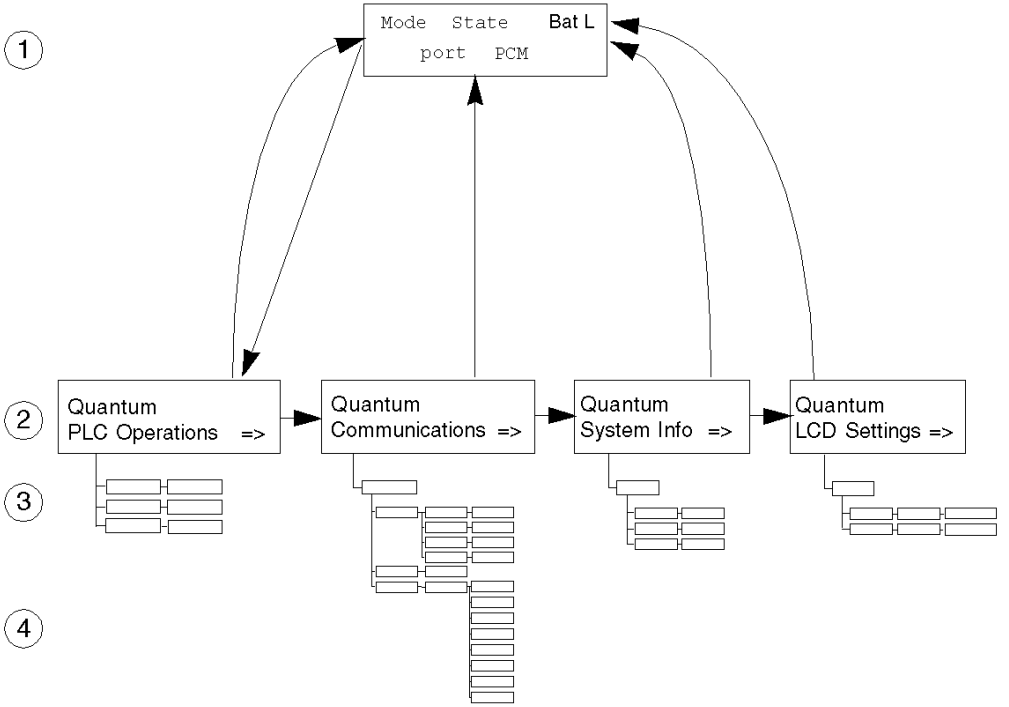
## Introduction

The controller LCD displays messages. These messages indicate the controller status. There are four levels of menus and submenus. Use the keypad *(see page 148)* on the front of the controller to access the menus.

For detailed information about the menus and submenus see:

- PLC Operations Menus and Submenus *(see page 156)*
- Using the Communications Menus and Submenus *(see page 159)*
- Using the LCD Settings Menus and Submenus *(see page 161)*
- Using the System Info Menus and Submenus *(see page 162)*

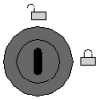
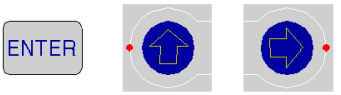
Structure: LCD display menus and submenus:




**1** default screen
**2** system menus
**3** sub menus
**4** sub screens

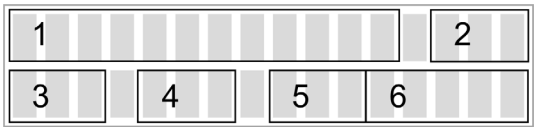
## Accessing the Screens

Use the keys on the keypad to access the system menus and submenus:

| Step | Action |
|---|---|
| 1 | To access the screens, confirm that the key switch is in the unlocked position.<br>  |
| 2 | To step down to a lower menu, operate one of the following keys:<br>  |
| 3 | To return to the previous menu, press:<br>  |

## 140 CPU 65• •• Default Screen

The default screen is read-only and contains the following fields:

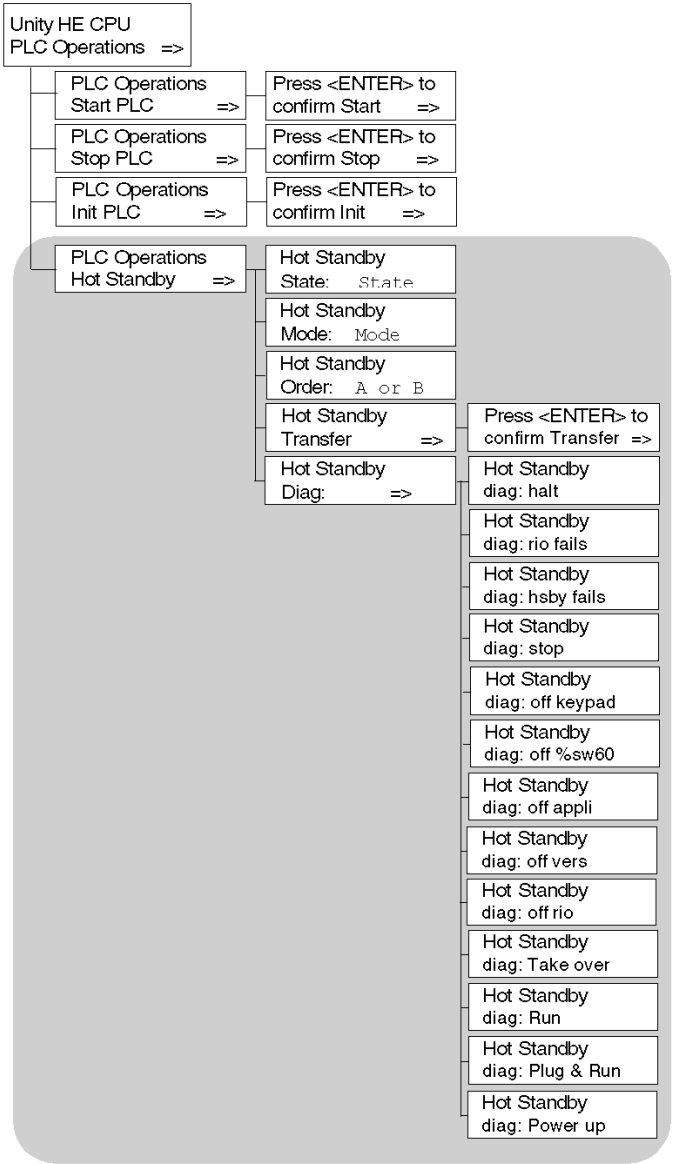



The default screen displays the following information:

| Field | | Display | Description |
|---|---|---|---|
| 0 | CPU mode | M | maintenance mode (only on safety processors) |
| | | S | safe mode (only on safety processors) |
| 2 | CPU state | RUN | Application program is running. |
| | | STOP | Application program is **NOT** running. |
| | | | STOP offline |
| | | No Conf | Processor has no application program. |
| | | Halt | detected state error (in maintenance mode for safety modules) |
| BatL | | | Indicates battery health:<br>● steady = Battery is low.<br>● no message = Battery is OK. |
| Port | USB | | Indicates that the port has activity. |
| | Modbus Plus | MB+ | Indicates Modbus Plus activity. |
| | | mb+ | no activity |
| | | Dup | duplicate MB+ address |
| | | ERR | detected Modbus communications error |
| | | INI | initial network search |
| | Modbus | 232 | serial port activity for RS-232 |
| | | 485 | serial port activity for RS-485 |
| | PCM | 1 | Displayed status indicates battery health of the PCMCIA card in slot 1:<br>● steady = Battery is OK.<br>● flashing = Battery is low (only for green PCMCIAs (version <04)) (1). |
| | | 2 | Displayed status indicates battery health of the PCMCIA card in slot 2:<br>● steady = Battery is OK.<br>● flashing = Battery is low (only for green PCMCIAs (version < 04)) (1). |
| | | **(1)** With blue PCMCIAs (version >= 04), when main battery is low, there is no flash. | |

## PLC Operations Menu

The structure of the PLC Operations menu and submenus are:



Submenu for PLC Operations: Start, Stop and Init:

| Start, Stop, Init Screens Display | Fields Available | Description |
|---|---|---|
| Start PLC | Press <ENTER> to confirm Start. | Pressing <ENTER> starts the controller. |
| Stop PLC | Press <ENTER> to confirm Stop. | Pressing <ENTER> stops the controller. |
| Init PLC | Press <ENTER> to confirm Init. | Pressing <ENTER> initializes the controller.<br>On safety processors, this command is only available in the maintenance mode. |

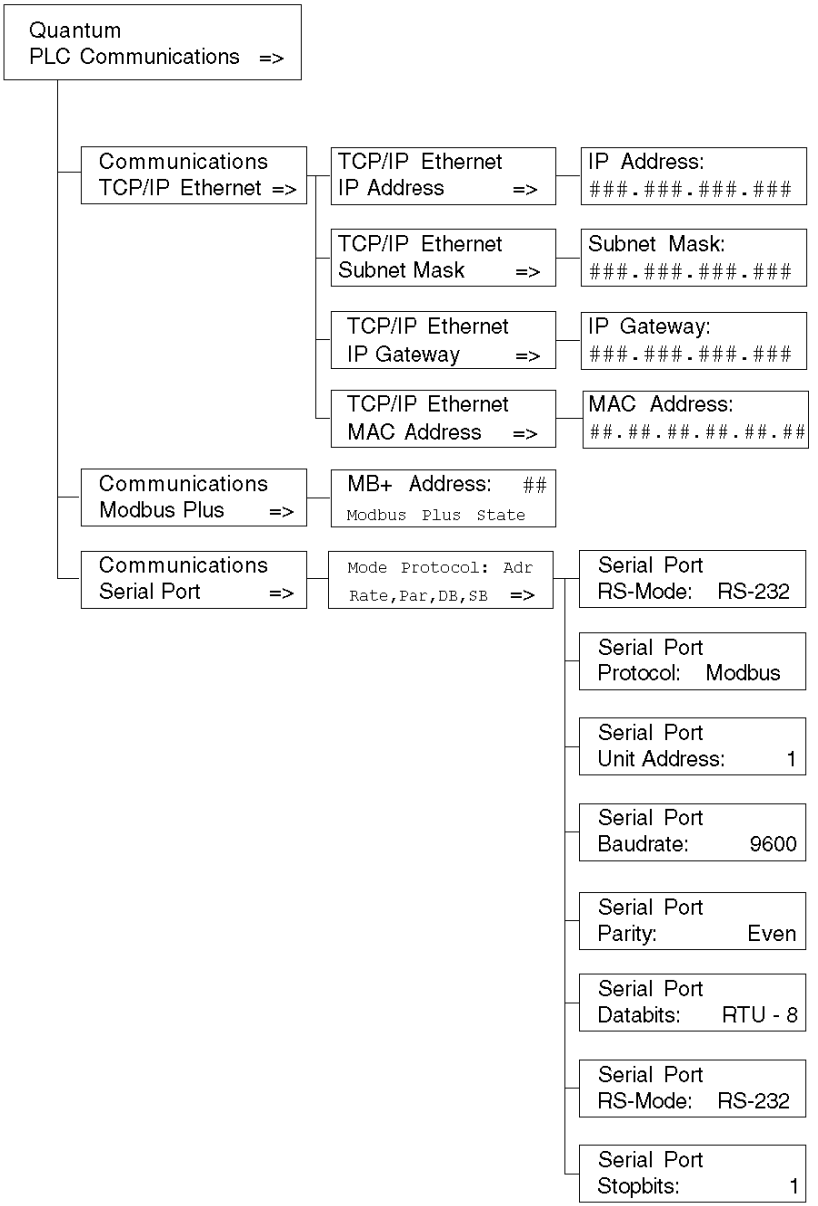
Submenu for PLC Operations Hot Standby CPU:

| Screen | Field | Option | | Description |
|---|---|---|---|---|
| Hot Standby State: | `State`<br>read-only | PRIMARY CPU | | Controller serves as primary unit. |
| | | STANDBY CPU | | Controller serves as standby unit. |
| | | Offline | | Controller is neither primary nor standby unit. |
| Hot Standby Mode: | `Mode`<br>(modifiable only if the key switch is in the unlocked position) | RUN | STS steady | Controller is active and is either serving as primary PLC or able to take over the primary PLC role if needed. |
| | | | STS flashing | Controller is transferring/updating. When the transfer is done, RUN stays on steady. |
| | | OFFLINE | STS steady | Controller is taken out of service without stopping it or disconnecting it from power.<br>If the controller is the primary PLC when the mode is changed to offline, control switches to the standby PLC.<br>If the standby PLC changes to offline, the primary PLC continues to operate without a backup. |
| | | | STS flashing | Controller is transferring/updating.When the transfer is done, OFFLINE stays on steady. |
| Hot Standby Order: | `A or B`<br>(modifiable only if the key switch is in the unlocked position) | FIRST | | Hot Standby power order<br>**NOTE:** To change the A/B order, confirm that the PLC is in STOP mode. |
| | | SECOND | | |
| Hot Standby Transfer: | -<br>(This menu option is enabled only when the key switch is in the unlocked position.) | | | Pressing the <ENTER> key confirms the transfer. The transfer initiates the request of a program update from the primary PLC. Pressing any other key cancels the transfer initiation and returns to the Hot Standby Transfer menu option screen.<br>**NOTE:** Transfer by keypad is based on %SW60.5 *(see Unity Pro, System Bits and Words, Reference Manual)* use: modifying %SW60.5 by application and requiring simultaneously a transfer by Keypad, could lead to some issues (no transfer or transfer retry). |

| Screen | Field | Option | Description |
|---|---|---|---|
| Hot Standby Diag: | The order of diagnostic screen varies with the operation. | | |
| | Halt | | user task in halt mode |
| | RIO fails | | detected error reported by RIO head |
| | HSBY fails | | detected error reported by optical link |
| | Stop | | stop command sent |
| | Off keypad | | offline command entered on keypad |
| | Off %SW60 | | offline command set in command register |
| | Off appli | | offline due to application mismatch |
| | Off vers | | offline due to PLC or co-processor OS mismatch |
| | Off RIO | | offline due to detectedRIO error |
| | Take over | | standby CPU switched to primary CPU mode |
| | Run | | run command sent |
| | Plug & Run | | sun-link operational and standby CPU started |
| | Power up | | no message: PLC just started |

## Communications Menu

Communications menu and submenus:



Submenu for TCP/IP Ethernet PLC Communications submenus:

| TCP/IP Ethernet Screen Displays | Fields Available | Options Available | Description |
|---|---|---|---|
| TCP/IP Ethernet IP Address[1,2] | ###.###.###.### | decimal numbers | Displays IP address. |
| TCP/IP Ethernet Subnet Mask[1,2] | ###.###.###.### | decimal numbers | Displays subnetwork mask address. |
| TCP/IP Ethernet IP Gateway[1,2] | ###.###.###.### | decimal numbers | Displays Ethernet IP gateway address. |
| TCP/IP Ethernet MAC Address | ##.##.##.##.##.## (read-only) | hexadecimal numbers | Displays MAC address. |

[1)]Parameters can be modified only if no applications have been downloaded (in NO CONF state).

[2)]When a new PLC application has been downloaded, the Ethernet address on the screen is updated only after accessing the highest level of the menu structure.

Modbus Plus PLC Communications submenus:

| Fields Available | Options Available | Description |
|---|---|---|
| ## (modifiable only if the key switch is in the unlocked position) | 1-64 | Enter a valid Modbus Plus address. |
| Modbus Plus State | Monitor Link | Modbus Plus state |
| | Normal Link | |
| | Sole Station | |
| | Duplicate Address | |
| | No Token | |

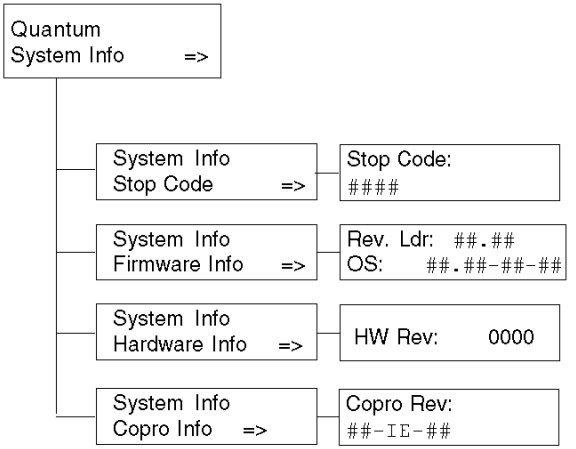
Serial PLC Communications submenus:

| Fields Available* | Options Available | Description |
|---|---|---|
| Mode | 232 | RS mode |
| | 485 | |
| Protocol | ASCII | protocols available |
| | RTU | |
| Adr | 1 - 247 | unit address |
| | for Modbus Switchover<br>Primary CPU 1-119<br>Standby CPU 129 - 247 | |

| Fields Available* | Options Available | Description |
|---|---|---|
| Rate | 50, 75, 110, 134.5, 150, 300, 600, 1200, 1800, 2400, 3600. 4800, 7200, 9600, 19200 bits/s | baud rate |
| Par | NONE | parity |
| | ODD | |
| | EVEN | |
| DB | 7,8 | data bits: If protocol is Modbus, then RTU-8 or ASCII-7. |
| SB | 1,2 | stop bits |
| *If the key switch is in the unlocked position, fields are modifiable. | | |

## System Info Menu

Structure of System Info menus and submenus:



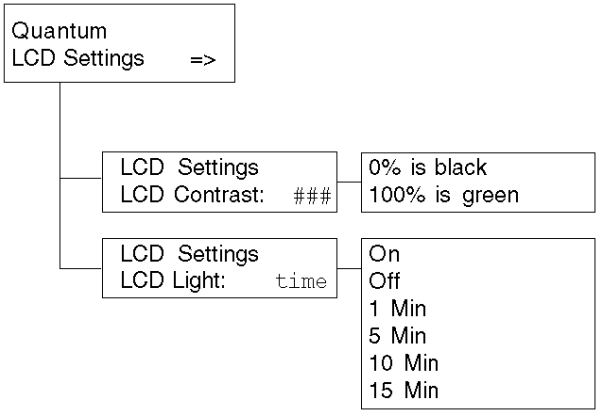

System Info, PLC Communications submenus:

| System Info Screen Displays | Fields Available* | Option Available | Description |
|---|---|---|---|
| Stop Code | #### | | machine stop code |
| | Description | | description of the machine stop code |
| Firmware Info | Rev.Ldr: ##.## | | OS revision |
| | OS: ##.##-##-## | | OSLoader revision |

| System Info Screen Displays | Fields Available* | Option Available | Description |
|---|---|---|---|
| Hardware Info | HW Rev: #### | | hardware revision |
| Copro Info | ##-IE-## | | co-processor revision |
| *Fields are read-only. | | | |

## LCD Settings Menu

LCD Settings menus and submenus:




LCD Contrast settings submenu:

| LCD Screen Contrast Screen Displays | Fields Available | Description |
|---|---|---|
| LCD Contrast: | #### | Use the arrow keys to adjust the setting:<br>• Up arrow increases percent (brighter).<br>• Right arrow decreases percent (darker). |

LCD Light setting submenus:

| Screen Displays | Fields Available | Description |
|---|---|---|
| LCD Light: | On | LCD remains on permanently or until changed. |
| | Off | LCD remains off permanently or until changed. |
| | 1 Min | LCD remains on for 1 minute. |
| | 5 Min | LCD remains on for 5 minutes. |
| | 10 Min | LCD remains on for 10 minutes. |
| | 15 Min | LCD remains on for 15 minutes. |

# How to change the battery of a 140 CPU 6xx xx CPU

## Procedure

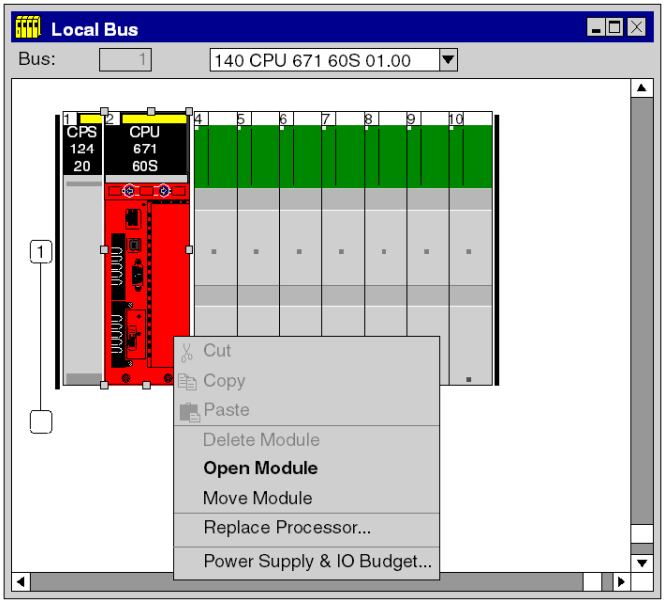
The table below summarizes the procedure for changing the battery of a 140 CPU 6xx xx CPU.

| Step | Action |
|---|---|
| 1 | While CPU is running, slide the front plastic cover of the LCD screen to open it. |
| 2 | Find the battery connector and pull the cable to remove it from its slot. |
| 3 | Remove the battery. |
| 4 | Insert a new battery. |
| 5 | Connect the battery connector in the same slot as the previous one. |
| 6 | Check that the **Batt Low** message on the LCD screen has disappear. |

# Processor Configuration Screen

## Accessing with Unity Pro

After starting Unity Pro, go to the Local Bus in the Structural View of the Project Browser.

| Step | Action |
|---|---|
| 1 | Double-click on Local Bus to open the Local Bus configuration editor. |
| 2 | Select the CPU module and right-click.<br>The context menu appears.<br>  |

| Step | Action |
|---|---|
| 3 | Select Open Module.<br>The editor appears. |
| 4 | Choose one of these tabs:<br>● Overview<br>● Summary<br>● Configuration *(see Unity Pro, Operating Modes)* Quantum / Configuration *(see Modicon Quantum, Hot Standby System, User Manual)* Quantum Hot Standby PLCs<br>● Modbus Port *(see Unity Pro, Operating Modes)* Quantum / Modbus Port *(see Modicon Quantum, Hot Standby System, User Manual)* Quantum Hot Standby PLCs / Modbus *(see Unity Pro, Operating Modes)* Quantum Safety PLCs<br>● Animation Port *(see Unity Pro, Operating Modes)* Quantum / Animation Port *(see Modicon Quantum, Hot Standby System, User Manual)* Quantum Hot Standby PLCs<br>● Hot Standby<br>● Faults<br>● I/O Objects |

**NOTE:** %MWi will reset with the cold start %S0 or a loading program. Cold start is typically after a program load, %S0 can be set with the user program to initiate a cold start

# 140 CPU 651 50 Specifications

## General Specifications

| Element | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet |
| Bus current required | 2160 mA |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00.**2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00.**2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Function | Description |
|---|---|
| Model | Pentium |
| Clock speed | 166 MHz |
| Coprocessor | Yes, built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal user memory (available) | 768 kByte without PCMCIA card<br>512 kByte with PCMCIA card |
| Additional memory (with PCMCIA card) | 7168 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Program Execution Time

| Kilo Instruction executed per millisecond (Kins/ms) | | Execution time per instruction (ms/Kins) | |
|---|---|---|---|
| 100% Boolean | 65% Boolean + 35% digital | 100% Boolean | 65% Boolean + 35% digital |
| 10.28 | 9.91 | 0.097 | 0.101 |

**NOTE:** When considering the execution time with the RAM or the PCMCIA card, the values are identical as the program execution takes place within the CACHE memory.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## Local I/O

| | |
|---|---|
| Max. I/O words | 1024 bits/module with no limit on total Local I/O words |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 651 50 is **not able** to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>● up to 31 Quantum drops (140 CRA 312 00)<br>● up to 16 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 651 50 is **not able** to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN is **not supported** by the 140 CPU 651 50.

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 in / 500 out<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words per Node | 30 in / 32 out |
| Max. number of DIO Drops per Network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service Life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 seconds/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 651 60 Specifications

## General Specifications

| Element | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet |
| Bus current required | 2760 mA |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00**.2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00**.2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Function | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal user memory | 1024 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Program Execution Time

| Kilo Instruction executed per millisecond (Kins/ms) | | Execution time per instruction (ms/Kins) | |
|---|---|---|---|
| 100% Boolean | 65% Boolean + 35% digital | 100% Boolean | 65% Boolean + 35% digital |
| 10.28 | 10.07 | 0.097 | 0.099 |

**NOTE:** When considering the execution time with the RAM or the PCMCIA card, the values are identical as the program execution takes place within the CACHE memory.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## Local I/O

| | |
|---|---|
| Max. I/O words | 1024 bits/module with no limit on total LIO words |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 651 60 is **not able** to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>● up to 31 Quantum drops (140 CRA 312 00)<br>● up to 16 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 651 60 is **not able** to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN is **not supported** by the 140 CPU 651 60.

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 in / 500 out<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words/node | 30 in / 32 out |
| Max. number of DIO drops/network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 seconds/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 651 60S Specifications

## General Specifications

| | |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet |
| Bus current required | 2760 mA |
| Max. number of NOE 771 11 modules supported | 6 |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| | |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal user memory | 1024 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## Local I/O

| | |
|---|---|
| Max. I/O words | 1024 bits/module with no limit on total LIO words |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

## Ethernet Remote I/O

Not supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 seconds/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up & Run Time | RAM<br>RAM address<br>Executive CRC<br>User Logic Check<br>Processors<br>Clock |

# 140 CPU 652 60 Specifications

## General Specifications

| Element | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet |
| Bus current required | 2760 mA |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00.**2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00.**2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Function | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal user memory | 3072 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## Local I/O

| | |
|---|---|
| Max. I/O words | 1024 bits/module with no limit on total LIO words |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 652 60 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>● up to 31 Quantum drops (140 CRA 312 00)<br>● up to 31 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 652 60 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN **is possible** with the 140 CPU 652 60.

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 in / 500 out<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words/node | 30 in / 32 out |
| Max. number of DIO drops/network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 seconds/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 658 60 Specifications

## General Specifications

| Element | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet |
| Bus current required | 2760 mA |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00.**2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00.**2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Function | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal memory | 11264 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## Local I/O

| | |
|---|---|
| Max. I/O words | 1024 bits/module with no limit on total LIO words |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 658 60 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>• up to 31 Quantum drops (140 CRA 312 00)<br>• up to 31 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 658 60 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN **is possible** with the 140 CPU 658 60.

## Distributed I/O

| | |
|---|---|
| Number of networks per system | 1 (3**) |
| Max. words per network | 500 in / 500 out<br>For every DIO drop, there is a minimum of two words input of overhead |
| Max. words/node | 30 in / 32 out |
| Max. number of DIO drops/network | 64 |
| ** Requires the use of the 140 NOM 21• 00 option module. | |

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Type | 3 V lithium |
| Service life | 1200 mAh |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 seconds/day @ 0...60 °C |

## Diagnostics

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 670 60 Specifications

## Module Specifications

| Component | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet (used as a Hot Standby port) |
| Transmission Mode | Multi mode |
| Bus current required | 2.5 A |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 3 including:<br>● up to 3 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00.**2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00.**2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Feature | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, Built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Memory

| | |
|---|---|
| Internal user memory | 512 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |
| Max. HSBY Unlocated Data | 254 kByte |
| Max. HSBY Located Data | 128 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Program Execution Time

| Kilo Instruction executed per millisecond (Kins/ms) | | Execution time per instruction (ms/Kins) | |
|---|---|---|---|
| 100% Boolean | 65% Boolean + 35% digital | 100% Boolean | 65% Boolean + 35% digital |
| 10.28 | 10.07 | 0.097 | 0.099 |

**NOTE:** When considering the execution time with the RAM or the PCMCIA card, the values are identical as the program execution takes place within the CACHE memory.

## Reference Capacity

| | |
|---|---|
| Discretes | 64 kByte max. |
| Registers | 64 kByte max. |

## S908 Remote I/O

Not supported

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | up to 6 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 670 60 is **not able** to handle the S908 remote I/O and Ethernet remote I/O architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN **is not supported** by the 140 CPU 670 60.

## Change Configuration On The Fly (CCOTF)

Not supported

## Battery and Clock

| | |
|---|---|
| Battery type | 3 V Lithium |
| Service life | 1.2 Ah |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 s/day @ 0...60 °C |

## Diagnostic

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 671 60 Specifications

## Module Specifications

| Element | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet (used as a Hot Standby port) |
| Transmission Mode | Multi mode |
| Bus current required | 2.5 A |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00.**2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00.**2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Function | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, Built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

**NOTE:** For this processor with OS (Executive) version 2.8 or higher, the Hot Standby Primary/Standby system synchronization (the sum of the MAST task duration and the watchdog value) must not exceed 2000 ms (2 seconds).

## Memory

| | |
|---|---|
| Internal user memory | 1024 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |
| Max. HSBY Unlocated Data | 512 kByte |
| Max. HSBY Located Data | 128 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Program Execution Time

| Kilo Instruction executed per millisecond (Kins/ms) | | Execution time per instruction (ms/Kins) | |
|---|---|---|---|
| 100% Boolean | 65% Boolean + 35% digital | 100% Boolean | 65% Boolean + 35% digital |
| 10.28 | 10.07 | 0.097 | 0.099 |

**NOTE:** When considering the execution time with the RAM or the PCMCIA card, the values are identical as when the program execution takes place within the CACHE memory.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 671 60 is **not able** to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>● up to 31 Quantum drops (140 CRA 312 00)<br>● up to 16 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 671 60 is **not able** to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN **is not supported** by the 140 CPU 671 60.

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Battery type | 3 V lithium |
| Service life | 1.2 Ah |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 s/day @ 0...60 °C |

## Diagnostic

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 671 60S Specifications

## Module Specifications

| Component | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet (used as a Hot Standby port) |
| Transmission Mode | Multi mode |
| Bus current required | 2.5 A |
| Maximum number of NOE 771 11 modules supported | 6 |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Feature | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, Built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal user memory | 1024 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |
| Max. HSBY Unlocated Data | 385 kByte |
| Max. HSBY Located Data | 128 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

## Ethernet Remote I/O

Not supported

## Battery and Clock

| | |
|---|---|
| Battery type | 3 V Lithium |
| Service life | 1.2 Ah |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 s/day @ 0...60 °C |

## Diagnostic

| | |
|---|---|
| Power-up | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM<br>RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 672 60 Specifications

## Module Specifications

| Component | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet (used as a Hot Standby port) |
| Transmission Mode | Multi mode |
| Bus current required | 2.5 A |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00.**2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00.**2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Feature | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, Built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal user memory | 3072 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |
| Max. HSBY Unlocated Data | 1536 kByte |
| Max. HSBY Located Data | 128 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 672 60 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>• up to 31 Quantum drops (140 CRA 312 00)<br>• up to 31 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 672 60 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Battery type | 3 V Lithium |
| Service life | 1.2 Ah |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 s/day @ 0...60 °C |

## Diagnostic

| | |
|---|---|
| Power-up | RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 672 61 Specifications

## Module Specifications

| Component | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet (used as a Hot Standby port) |
| Transmission Mode | Single mode |
| Bus current required | 2.5 A |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ one 140 NOC 781 00 module<br>– or –<br>❍ one 140 NOC 781 00**.2** module<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00**.2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Feature | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, Built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal user memory | 3072 kByte |
| Additional memory (with PCMCIA card) | 7168 kByte |
| Max. HSBY Unlocated Data | 1536 kByte |
| Max. HSBY Located Data | 128 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see Memory usage *(see Unity Pro, Operating Modes)*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 672 61 is able to handle the mix of S908 remote I/O and Ethernet remote I/o architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>• up to 31 Quantum drops (140 CRA 312 00)<br>• up to 31 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 672 61 is able to handle the mix of S908 remote I/O and Ethernet remote I/o architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN **is possible** with the 140 CPU 672 61.

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Battery type | 3 V Lithium |
| Service life | 1.2 Ah |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 s/day @ 0...60 °C |

## Diagnostic

| | |
|---|---|
| Power-up | RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM address<br>Executive Checksum<br>User Logic Check |

# 140 CPU 678 61 Specifications

## Module Specifications

| Component | Description |
|---|---|
| Communication ports | 1 Modbus (RS-232/RS-485)<br>1 Modbus Plus (RS-485)<br>1 USB<br>1 Ethernet (used as a Hot Standby port) |
| Transmission Mode | Single mode |
| Bus current required | 2.5 A |
| Max. number of communication modules supported at a time:<br>**NOTE:** This includes any module combination of: NOM, NOC, NOE, PTQ PDP MV1 and MMS | 6 including:<br>● up to 6 Quantum EIO distributed I/O modules 140 NOC 780 00<br>● up to 2 Quantum EIO control head modules in the following configurations only:<br>❍ two 140 NOC 781 00 modules<br>– or –<br>❍ two 140 NOC 781 00.**2** modules<br>– or –<br>❍ one 140 NOC 781 00 module and one 140 NOC 781 00.**2** module<br>**NOTE:** You can enable IP forwarding in only **one** of the control head modules. |
| Key switch | Yes |
| Key pad | Yes |

## Processor

| Feature | Description |
|---|---|
| Model | Pentium |
| Clock speed | 266 MHz |
| Coprocessor | Yes, Built-in Ethernet |
| Watchdog timer | 250 ms software adjustable |

## Memory

| | |
|---|---|
| Internal memory | 11264 kByte |
| Max HSBY Unlocated Data | 1536 kByte |
| Max HSBY Located Data | 128 kByte |

**NOTE:** Not all of the Internal memory is available for the user program due to memory overhead, such as user, system, configuration, diagnostic data, etc. For more information see the chapter Memory tab in the *Operating Modes Manual*.

## Reference Capacity

| | |
|---|---|
| Discretes | 65528 kByte max. |
| Registers | 64976 kByte max. |

## S908 Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 64 in / 64 out* |
| Max. number of remote drops | 31 |
| * This information can be a mix of discrete or register I/O. For each word of configured I/O, one of the I/O words must be subtracted from the total available. | |

**NOTE:** The 140 CPU 678 61 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

## Ethernet Remote I/O

| | |
|---|---|
| Max. I/O words/drop | 400 in + 400 out |
| Max. number of remote drops | 31 including:<br>● up to 31 Quantum drops (140 CRA 312 00)<br>● up to 31 X80 drops (BMX CRA 312 •0) |
| Max. number of I/O per Quantum drop | No limit |
| Max. number of I/O per X80 drop (BMX CRA 312 00) | up to 16 analog I/O |
| | up to 128 discrete I/O |
| Max. number of I/O per X80 drop (BMX CRA 312 10) | up to 184 analog I/O |
| | up to 1024 discrete I/O |

**NOTE:** The 140 CPU 678 61 is able to handle the mix of S908 remote I/O and Ethernet remote I/O architectures.

**NOTE:** Adding a complete Ethernet remote I/O drop online while PLC is in RUN **is possible** with the 140 CPU 678 61.

## Change Configuration On The Fly (CCOTF)

Supported

## Battery and Clock

| | |
|---|---|
| Battery type | 3 V lithium |
| Service life | 1.2 Ah |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery load current @ power-off | typical: 14 µA @ 25 °C (ambient temperature) |
| | maximum: 420 µA @ 60 °C |
| TOD clock | +/-8.0 s/day @ 0...60 °C |

## Diagnostic

| | |
|---|---|
| Power-up | RAM address<br>Executive Checksum<br>User Logic Check<br>Processor |
| Run Time | RAM address<br>Executive Checksum<br>User Logic Check |

# Chapter 10
# Memory Cards for High End CPU

## Purpose

This chapter provides information on the memory cards for High End Controller modules.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Memory Cards for High End CPUs | 200 |
| Installing/Extracting PCMCIA Extension Cards on Advanced Quantum Processors | 203 |
| Changing the Batteries of a PCMCIA Memory Card | 205 |
| Battery Lifetimes for the PCMCIA Memory Card | 209 |

# Memory Cards for High End CPUs

## Standard Memory Cards for PLCs

Standard memory cards for PLCs are classified in two groups:
- saved RAM memory extension cards
- Flash Eprom memory extension cards

## Saved RAM Memory Extension Cards

Saved RAM memory extension cards are typically used when generating and debugging an application program. The memory is saved by a removable battery integrated in the memory card.

## Flash Eprom Memory Extension Cards

Flash Eprom memory extension cards are typically used after the process of debugging the application program is completed. Such cards allow only the global transfer of the application; their main purpose is to avoid risks often associated with battery back-ups.

## References for Standard Memory Extension Cards

The following table indicates the compatibility of the cards with the various processors:

| Product Reference | Type/Capacity | |
|---|---|---|
| | Application | File |
| TSX MFP P 512K | Flash Eprom 512 kb | 0 |
| TSX MFP P 001M | Flash Eprom 1024 kb | 0 |
| TSX MFP P 002M | Flash Eprom 2048 kb | 0 |
| TSX MFP P 004M | Flash Eprom 4096 kb | 0 |

## Application + Files Type Memory Extension Cards

In addition to the conventional application storage area (program + constants), these memory cards also maintain a file area used by the program to archive and/or restore data.

Here are two sample applications:
- automatic storage of application data and remote consultation via modem
- storage of manufacturing formulas.

There are two types of memory cards:
- Saved RAM memory extension cards: application + files. The memory is saved by a removable battery built into the memory card.
- Flash Eprom memory extension cards: application + files. In this instance, the data storage area is maintained in saved RAM, which implies that this type of card must be equipped with a back-up battery.

## Card Reference Numbers

The following table provides the card reference numbers for the applications + file-type memory extension card, and the compatibility of these cards with the processors:

| Product reference | Technology | Capacity | |
|---|---|---|---|
| | | Application area | File area (RAM type) |
| TSX MRP C 768K (1) | RAM | 768 kb | |
| | | 192 to 768 kb | 0 to 576 kb |
| TSX MRP C 001M (1) | RAM | 1024 kb | |
| | | 192 to 1024 kb | 0 to 832 kb |
| TSX MRP C 001M7 (1) | RAM | 1792 kb | |
| | | 192 to 1792 kb | 0 to 1600 kb |
| TSX MRP C 002M (1) | RAM | 2048 kb | |
| | | 192 to 2048 kb | 0 to 1856 kb |
| TSX MRP C 003M (1) | RAM | 3072 kb | |
| | | 192 to 3072 kb | 0 to 2880 kb |
| TSX MRP C 007M (1) | RAM | 7168 kb | |
| | | 192 to 7168kb | 0 to 6976 kb |
| TSX MCP C 512K | Flash Eprom | 512 kb | 512 kb |
| TSX MCP C 001M | Flash Eprom | 1024 kb | 512 kb |
| TSX MCP C 002M | Flash Eprom | 2048 kb | 1024 kb |
| TSX MCP C 004M | Flash Eprom | 4096 kb | 2048 kb |
| (1) PCMCIA having their applications memory areas and floating capacity and unfrozen files | | | |

## File-type Memory Extension Cards without Application

These memory cards contain data. There is no application field (program + constants). These memory extension file storage cards are of the *saved RAM* type. The memory is saved by a removable battery built into the memory card.

## Card Reference Numbers

The following table provides reference numbers for memory extension file-type storage cards (without application), and indicates the compatibility of these cards with the various processors:

| Product reference | Technology | Capacity | |
|---|---|---|---|
| | | Application area | File area (RAM type) |
| TSX MRP F 004M | RAM | 4096 kb | |
| | | 0 | 4096 kb |
| TSX MRP F 008M | RAM | 8192 kb | |
| | | 0 | 8192 kb |

# Installing/Extracting PCMCIA Extension Cards on Advanced Quantum Processors

## Overview

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

Ensure that the protective cover is closed when the processor is running to maintain enclosure environmental ratings.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Memory Cards Located in Slot A (Top)

The extraction (or absence) of the cover, or of a data- or file-type memory card and case has no effect on PLC operations. In this case, the read/write functions of the memory card indicate an error if the application is in RUN mode.

The extraction of the application-type memory card and the card case causes the PLC to stop without saving the application context. In this case, the module outputs change to fallback mode. Inserting the case and the memory card containing the application causes a cold start.

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

Ensure that the right user application is contained in the memory card before inserting it into the PLC. If the application contained in the memory card includes the RUN AUTO option, then the PLC will automatically restart in RUN mode with this application as soon as the inserted memory card content is restored to the PLC.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Memory Cards Located in Slot B (Bottom)

The PCMCIA memory card and its case can be inserted in slot B of the processor when the PLC is turned on.

Slot B for data and file-type memory cards is not allowed to be used in a Quantum safety CPU, because this data storage is not available for safety projects.

**⚠ WARNING**

**LOSS OF ABILITY TO PERFORM SAFETY FUNCTIONS**

Do not use slot B for memory cards with Safety projects in Quantum Safety PLC.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Installing/Extracting PCMCIA Cards

The following tables provide step-by-step instructions for installing and extracting PCMCIA cards. Installing the memory card on the Quantum advanced CPU module requires a case.

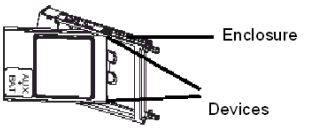
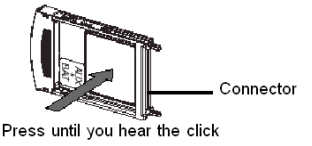
## Position of the PCMCIA Card in the Processor

The following table describes the possible slots for the different types of PCMCIA cards in the various PLC processors:

| PCMCIA card | Slot A | Slot B |
|---|---|---|
| Standard: TSX MRPP and MFPP | Yes | No |
| Application and files: TSX MRPC and MCPC | Yes | No |
| Data or Files: TSX MRPF | Yes | Yes |

## Installing the Card in the Case

The following steps are to be performed regardless of the PCMCIA card type:

| Step | Action | Illustration |
|---|---|---|
| 1 | Position the memory card in the case at an oblique angle using the two attachment tabs. |  |
| 2 | Fully insert the memory card in the case. Now the card is solidly attached to the case. |  |

## Installing the Card in the PLC

Before installing the card, note that if the program contained in the PCMCIA memory card includes the RUN AUTO option, the processor will automatically restart in RUN mode immediately after the card is inserted. Carry out the following steps to install the memory card into the processor:

| Step | Action |
|---|---|
| 1 | To remove the protective cover, unlock it and then remove it from the PLC. |
| 2 | Place the PCMCIA card/case assembly in the open slot. Fully insert the card/case assembly, then press lightly on the case to connect the card. |

# Changing the Batteries of a PCMCIA Memory Card

## General Points

Memory cards:

- TSX MRP P• standard RAM
- TSX MRP C• RAM for files and application and TSX MCP C• Flash EPROM
- TSX MRP F• data and file-type

have 2 backup batteries, TSX BAT M02 (main) and TSX BAT M03 (auxiliary), which need to be changed periodically.

Two methods are possible:

- one is preventive, based on a periodic change of batteries, without first checking their status,
- the other is predictive, based on the signal sent by a system but, but is only possible for certain memory cards.

## Preventive Method

This method is valid for memory cards versions and the PLCs that use those cards. Change both batteries according to the PV of the PCMCIA card, the PLC use, and the lifetime of batteries *(see page 209)*. It does not matter which battery you change first: the application is preserved by the memory card. For the operate mode for changing the batteries: see the service instructions provided with the memory cards.

**NOTE:**

- Batteries must not be removed from their positions simultaneously. One battery backs up the data and applications while the other is being replaced.
- Install the batteries as shown in the following diagrams, noting the correct polarity (+ and -)
- the memory card must not exceed 24 hours without its main battery in working order,
- to save the auxiliary batteries, you can replace them every 1.5 years. In this case, the maintenance procedure is a bit more complex, because for some memory cards you must remember to change the auxiliary battery only one time out of 3.
- the service lives shown above were calculated for the most unfavorable case: ambient temperature around the PLC at 60 °C, and the PLC switched on for 21% of the time in the year (which corresponds to an 8 hour rotation per day with 30 maintenance stoppage days per year).

## Predictive Method

This is maintenance based on using bits `%S67` and `%S75` and using the PCMCIA indicator on the Quantum interface. This method assumes that the auxiliary battery is changed preventively every 1.5 years. It is only possible:

- with Unity Pro ≥ 2.02,
- if the memory card is installed in the upper or lower PCMCIA slot on Quantum processors 140 CPU 6•• ••.

When system bit %S67 (card in the top slot) or %S75 (card in the bottom slot) changes to 1 or the PCMCIA indicator on the processor interface flashes, it means that the main battery is weak. You have 8 days to replace the battery, as shown in the following tables.

**NOTE:** Before turning on the PLC or removing the memory card, save the project in Unity Pro. If the PLC must be left off or if the memory card must be outside the PLC for more than 8 days, and you have exceeded the service life of the main battery, then back up the application in Unity Pro.

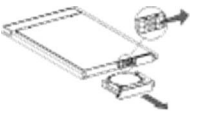
## Changing the Batteries

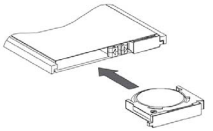
Carry out the following steps:

| Step | Action |
|---|---|
| 1 | Take the card out of its slot *(see Premium and Atrium using Unity Pro, Processors, racks and power supply modules, Implementation manual)*. |
| 2 | Separate the PCMCIA card *(see Premium and Atrium using Unity Pro, Processors, racks and power supply modules, Implementation manual)* from its clip (or caddy). |
| 3 | Hold the PCMCIA card so you can access the battery slot. This is at the end of the card without the connector. |
| 4 | Replacement of the TSX BAT M02 battery: see table 1.<br>Replacement of the TSX BAT M03 battery: see table 2. |
| 5 | Attach the PCMCIA card *(see Premium and Atrium using Unity Pro, Processors, racks and power supply modules, Implementation manual)* to its clip (or caddy). |
| 6 | Place the card back in the PLC. *(see Premium and Atrium using Unity Pro, Processors, racks and power supply modules, Implementation manual)* |

## Procedure for the TSX BAT M02 Battery:

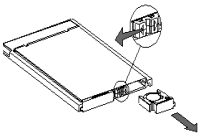
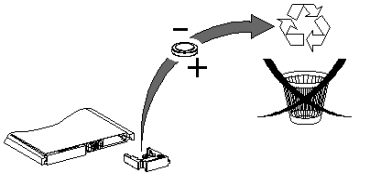
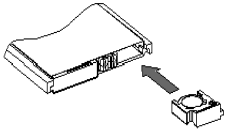
Table 1

| Step | Action | Illustration |
|---|---|---|
| 1 | Toggle the changeover lever toward the **TSX BAT M02** (MAIN) battery in order to remove drawer from the main battery. |  |
| 2 | Remove the used battery from its holder: |  |

| Step | Action | Illustration |
|---|---|---|
| 3 | Place the new battery in the holder, taking care to respect the polarity. |  |
| 4 | Insert the holder containing the battery in the card. |  |

## Procedure for the TSX BAT M03 Battery:

Table 2

| Step | Action | Illustration |
|---|---|---|
| 1 | Toggle the changeover lever toward the **TSX BAT M03** (AUX) battery in order to remove drawer from the battery. |  |
| 2 | Remove the used battery from its holder: |  |
| 3 | Place the new battery in the holder, taking care to respect the polarity. |  |
| 4 | Insert the holder containing the battery in the card. |  |

# Battery Lifetimes for the PCMCIA Memory Card

## Purpose

The purpose of this document is to give detailed information about the lifetime of batteries inside PCMCIA memory cards. The estimation of these lifetimes are based on data from component manufacturers.

## Scope

The lifetime information is estimated for:

- RAM PCMCIA memory cards,
- The three different cases of Product Version (PV): PV1/2/3, PV4/5 and PV6,
- Four ambient temperatures for the PLC location: 25°C / 40°C / 50°C / 60°C,
- Four different usage cases of the PCMCIAs: 100%, 92%, 66% and 33% of PLC power-up time. These values are for the following customer configurations:
  - 100%: PLC powered up all year long or during 51 weeks,
  - 92%: PLC powered up all year long except during one month of maintenance,
  - 66%: PLC powered up all year long except during all weekends plus one month of maintenance,
  - 33%: PLC powered up all year long 12 hours a day, except during all weekends plus one month of maintenance.
- A Min (minimum) and a Typical lifetime value:
  - The Min value comes from the most unfavorable characteristics given by the component manufacturers. The actual observed lifetime will be greater than this value.
  - The typical value comes from the typical characteristics of the component.

## Main Battery Lifetime of PV1/2/3 PCMCIA (in Years)

The table below presents the lifetime of main battery TSX BAT M01(PV1/2/3) for PCMCIA memory cards:

| PV1/2/3 | For a 25°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 7.10 | 7.10 | 6.71 | 5.58 | 5.77 | 3.36 | 4.82 | 2.20 |
| TSX MCP C 512K | 7.10 | 7.10 | 6.71 | 5.65 | 5.77 | 3.46 | 4.82 | 2.28 |
| TSX MCP C 002M | 7.10 | 7.10 | 6.29 | 3.82 | 4.66 | 1.57 | 3.45 | 0.88 |
| TSX MRP P128K | 7.10 | 7.10 | 6.71 | 5.58 | 5.77 | 3.36 | 4.82 | 2.20 |
| TSX MRP P224K | 7.10 | 7.10 | 6.71 | 5.65 | 5.77 | 3.46 | 4.82 | 2.28 |
| TSX MRP P384K | 7.10 | 7.10 | 6.71 | 4.99 | 5.77 | 2.60 | 4.82 | 1.59 |
| TSX MRP C448K | 7.10 | 7.10 | 6.29 | 4.65 | 4.66 | 2.24 | 3.45 | 1.33 |

| PV1/2/3 | For a 25°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MRP C768K | 7.10 | 7.10 | 6.29 | 4.65 | 4.66 | 2.24 | 3.45 | 1.33 |
| TSX MRP C001M | 7.10 | 7.10 | 5.91 | 3.95 | 3.91 | 1.66 | 2.68 | 0.94 |
| TSX MRP C01M7 | 7.10 | 7.10 | 5.58 | 3.43 | 3.36 | 1.32 | 2.20 | 0.72 |
| TSX MRP C002M | 7.10 | 7.10 | 5.91 | 3.34 | 3.91 | 1.26 | 2.68 | 0.69 |
| TSX MRP C003M | 7.10 | 7.10 | 5.58 | 2.60 | 3.36 | 0.87 | 2.20 | 0.47 |
| TSX MRP C007M | 7.10 | 7.10 | 4.56 | 1.59 | 2.16 | 0.46 | 1.27 | 0.24 |
| TSX MRP F004M | 7.10 | 7.10 | 5.58 | 2.60 | 3.36 | 0.87 | 2.20 | 0.47 |
| TSX MRP F008M | 7.10 | 7.10 | 4.56 | 1.59 | 2.16 | 0.46 | 1.27 | 0.24 |

| PV1/2/3 | For a 40°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 3.55 | 3.55 | 3.54 | 3.20 | 3.54 | 2.46 | 3.48 | 1.87 |
| TSX MCP C 512K | 3.55 | 3.55 | 3.54 | 3.22 | 3.54 | 2.51 | 3.48 | 1.93 |
| TSX MCP C 002M | 3.55 | 3.55 | 3.42 | 2.53 | 3.08 | 1.34 | 2.71 | 0.82 |
| TSX MRP P128K | 3.55 | 3.55 | 3.54 | 3.20 | 3.54 | 2.46 | 3.48 | 1.87 |
| TSX MRP P224K | 3.55 | 3.55 | 3.54 | 3.22 | 3.54 | 2.51 | 3.48 | 1.93 |
| TSX MRP P384K | 3.55 | 3.55 | 3.54 | 3.00 | 3.54 | 2.02 | 3.48 | 1.41 |
| TSX MRP C448K | 3.55 | 3.55 | 3.42 | 2.87 | 3.08 | 1.80 | 2.71 | 1.20 |
| TSX MRP C768K | 3.55 | 3.55 | 3.42 | 2.87 | 3.08 | 1.80 | 2.71 | 1.20 |
| TSX MRP C001M | 3.55 | 3.55 | 3.30 | 2.59 | 2.74 | 1.40 | 2.21 | 0.87 |
| TSX MRP C01M7 | 3.55 | 3.55 | 3.20 | 2.35 | 2.46 | 1.15 | 1.87 | 0.69 |
| TSX MRP C002M | 3.55 | 3.55 | 3.30 | 2.31 | 2.74 | 1.11 | 2.21 | 0.65 |
| TSX MRP C003M | 3.55 | 3.55 | 3.20 | 1.93 | 2.46 | 0.80 | 1.87 | 0.45 |
| TSX MRP C007M | 3.55 | 3.55 | 2.84 | 1.31 | 1.75 | 0.44 | 1.16 | 0.24 |
| TSX MRP F004M | 3.55 | 3.55 | 3.20 | 1.93 | 2.46 | 0.80 | 1.87 | 0.45 |
| TSX MRP F008M | 3.55 | 3.55 | 2.84 | 1.31 | 1.75 | 0.44 | 1.16 | 0.24 |

| PV1/2/3 | For a 50°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 2.35 | 2.35 | 2.42 | 2.25 | 2.69 | 2.02 | 3.10 | 1.75 |
| TSX MCP C 512K | 2.35 | 2.35 | 2.42 | 2.26 | 2.69 | 2.05 | 3.10 | 1.81 |
| TSX MCP C 002M | 2.35 | 2.35 | 2.36 | 1.90 | 2.42 | 1.20 | 2.47 | 0.80 |
| TSX MRP P128K | 2.35 | 2.35 | 2.42 | 2.25 | 2.69 | 2.02 | 3.10 | 1.75 |
| TSX MRP P224K | 2.35 | 2.35 | 2.42 | 2.26 | 2.69 | 2.05 | 3.10 | 1.81 |
| TSX MRP P384K | 2.35 | 2.35 | 2.42 | 2.15 | 2.69 | 1.71 | 3.10 | 1.34 |
| TSX MRP C448K | 2.35 | 2.35 | 2.36 | 2.09 | 2.42 | 1.55 | 2.47 | 1.15 |
| TSX MRP C768K | 2.35 | 2.35 | 2.36 | 2.09 | 2.42 | 1.55 | 2.47 | 1.15 |
| TSX MRP C001M | 2.35 | 2.35 | 2.31 | 1.93 | 2.20 | 1.25 | 2.05 | 0.85 |
| TSX MRP C01M7 | 2.35 | 2.35 | 2.25 | 1.80 | 2.02 | 1.04 | 1.75 | 0.67 |
| TSX MRP C002M | 2.35 | 2.35 | 2.31 | 1.77 | 2.20 | 1.01 | 2.05 | 0.64 |
| TSX MRP C003M | 2.35 | 2.35 | 2.25 | 1.54 | 2.02 | 0.75 | 1.75 | 0.44 |
| TSX MRP C007M | 2.35 | 2.35 | 2.07 | 1.12 | 1.51 | 0.42 | 1.11 | 0.23 |
| TSX MRP F004M | 2.35 | 2.35 | 2.25 | 1.54 | 2.02 | 0.75 | 1.75 | 0.44 |
| TSX MRP F008M | 2.35 | 2.35 | 2.07 | 1.12 | 1.51 | 0.42 | 1.11 | 0.23 |

| PV1/2/3 | For a 60°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 1.57 | 1.57 | 1.63 | 1.56 | 1.91 | 1.54 | 2.40 | 1.50 |
| TSX MCP C 512K | 1.57 | 1.57 | 1.63 | 1.56 | 1.91 | 1.56 | 2.40 | 1.54 |
| TSX MCP C 002M | 1.57 | 1.57 | 1.61 | 1.38 | 1.77 | 1.01 | 2.00 | 0.74 |
| TSX MRP P128K | 1.57 | 1.57 | 1.63 | 1.56 | 1.91 | 1.54 | 2.40 | 1.50 |
| TSX MRP P224K | 1.57 | 1.57 | 1.63 | 1.56 | 1.91 | 1.56 | 2.40 | 1.54 |
| TSX MRP P384K | 1.57 | 1.57 | 1.63 | 1.51 | 1.91 | 1.36 | 2.40 | 1.19 |
| TSX MRP C448K | 1.57 | 1.57 | 1.61 | 1.47 | 1.77 | 1.25 | 2.00 | 1.04 |
| TSX MRP C768K | 1.57 | 1.57 | 1.61 | 1.47 | 1.77 | 1.25 | 2.00 | 1.04 |
| TSX MRP C001M | 1.57 | 1.57 | 1.58 | 1.40 | 1.65 | 1.05 | 1.72 | 0.78 |
| TSX MRP C01M7 | 1.57 | 1.57 | 1.56 | 1.33 | 1.54 | 0.90 | 1.50 | 0.63 |
| TSX MRP C002M | 1.57 | 1.57 | 1.58 | 1.31 | 1.65 | 0.87 | 1.72 | 0.60 |

| PV1/2/3 | For a 60°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MRP C003M | 1.57 | 1.57 | 1.56 | 1.18 | 1.54 | 0.67 | 1.50 | 0.42 |
| TSX MRP C007M | 1.57 | 1.57 | 1.47 | 0.92 | 1.23 | 0.40 | 1.00 | 0.23 |
| TSX MRP F004M | 1.57 | 1.57 | 1.56 | 1.18 | 1.54 | 0.67 | 1.50 | 0.42 |
| TSX MRP F008M | 1.57 | 1.57 | 1.47 | 0.92 | 1.23 | 0.40 | 1.00 | 0.23 |

## Main Battery Lifetime of PV4/5 PCMCIA (in Years)

The table below presents the lifetime of main battery TSX BAT M02 (PV4/5) for PCMCIA memory cards:

| PV4/5 | For a 25°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 7.22 | 7.22 | 7.15 | 6.27 | 7.02 | 4.48 | 6.76 | 3.23 |
| TSX MCP C 512K | 7.22 | 7.22 | 7.15 | 6.33 | 7.02 | 4.59 | 6.76 | 3.35 |
| TSX MCP C 002M | 7.22 | 7.22 | 6.83 | 4.69 | 5.90 | 2.25 | 4.96 | 1.33 |
| TSX MRP P128K | 7.22 | 7.22 | 7.15 | 6.27 | 7.02 | 4.48 | 6.76 | 3.23 |
| TSX MRP P224K | 7.22 | 7.22 | 7.15 | 6.33 | 7.02 | 4.59 | 6.76 | 3.35 |
| TSX MRP P384K | 7.22 | 7.22 | 7.15 | 5.77 | 7.02 | 3.57 | 6.76 | 2.36 |
| TSX MRP C448K | 7.22 | 7.22 | 6.83 | 5.47 | 5.90 | 3.12 | 4.96 | 1.99 |
| TSX MRP C768K | 7.22 | 7.22 | 6.83 | 5.47 | 5.90 | 3.12 | 4.96 | 1.99 |
| TSX MRP C001M | 7.22 | 7.22 | 6.54 | 4.82 | 5.09 | 2.37 | 3.91 | 1.41 |
| TSX MRP C01M7 | 7.22 | 7.22 | 6.27 | 4.30 | 4.48 | 1.91 | 3.23 | 1.10 |
| TSX MRP C002M | 7.22 | 7.22 | 6.54 | 4.20 | 5.09 | 1.83 | 3.91 | 1.04 |
| TSX MRP C003M | 7.22 | 7.22 | 6.27 | 3.41 | 4.48 | 1.29 | 3.23 | 0.71 |
| TSX MRP C007M | 7.22 | 7.22 | 5.39 | 2.21 | 3.02 | 0.70 | 1.91 | 0.37 |
| TSX MRP F004M | 7.22 | 7.22 | 6.27 | 3.41 | 4.48 | 1.29 | 3.23 | 0.71 |
| TSX MRP F008M | 7.22 | 7.22 | 5.39 | 2.21 | 3.02 | 0.70 | 1.91 | 0.37 |

| PV4/5 | For a 40°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 4.63 | 4.63 | 4.72 | 4.32 | 5.09 | 3.61 | 5.59 | 2.94 |
| TSX MCP C 512K | 4.63 | 4.63 | 4.72 | 4.35 | 5.09 | 3.68 | 5.59 | 3.04 |
| TSX MCP C 002M | 4.63 | 4.63 | 4.58 | 3.51 | 4.48 | 2.00 | 4.30 | 1.28 |
| TSX MRP P128K | 4.63 | 4.63 | 4.72 | 4.32 | 5.09 | 3.61 | 5.59 | 2.94 |
| TSX MRP P224K | 4.63 | 4.63 | 4.72 | 4.35 | 5.09 | 3.68 | 5.59 | 3.04 |
| TSX MRP P384K | 4.63 | 4.63 | 4.72 | 4.08 | 5.09 | 2.99 | 5.59 | 2.20 |
| TSX MRP C448K | 4.63 | 4.63 | 4.58 | 3.93 | 4.48 | 2.68 | 4.30 | 1.87 |
| TSX MRP C768K | 4.63 | 4.63 | 4.58 | 3.93 | 4.48 | 2.68 | 4.30 | 1.87 |
| TSX MRP C001M | 4.63 | 4.63 | 4.45 | 3.58 | 4.00 | 2.10 | 3.49 | 1.35 |
| TSX MRP C01M7 | 4.63 | 4.63 | 4.32 | 3.29 | 3.61 | 1.73 | 2.94 | 1.06 |
| TSX MRP C002M | 4.63 | 4.63 | 4.45 | 3.23 | 4.00 | 1.66 | 3.49 | 1.01 |
| TSX MRP C003M | 4.63 | 4.63 | 4.32 | 2.74 | 3.61 | 1.21 | 2.94 | 0.69 |
| TSX MRP C007M | 4.63 | 4.63 | 3.89 | 1.91 | 2.60 | 0.67 | 1.80 | 0.36 |
| TSX MRP F004M | 4.63 | 4.63 | 4.32 | 2.74 | 3.61 | 1.21 | 2.94 | 0.69 |
| TSX MRP F008M | 4.63 | 4.63 | 3.89 | 1.91 | 2.60 | 0.67 | 1.80 | 0.36 |

| PV4/5 | For a 50°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 2.58 | 2.58 | 2.69 | 2.56 | 3.12 | 2.50 | 3.89 | 2.39 |
| TSX MCP C 512K | 2.58 | 2.58 | 2.69 | 2.56 | 3.12 | 2.53 | 3.89 | 2.45 |
| TSX MCP C 002M | 2.58 | 2.58 | 2.64 | 2.25 | 2.88 | 1.61 | 3.22 | 1.16 |
| TSX MRP P128K | 2.58 | 2.58 | 2.69 | 2.56 | 3.12 | 2.50 | 3.89 | 2.39 |
| TSX MRP P224K | 2.58 | 2.58 | 2.69 | 2.56 | 3.12 | 2.53 | 3.89 | 2.45 |
| TSX MRP P384K | 2.58 | 2.58 | 2.69 | 2.47 | 3.12 | 2.18 | 3.89 | 1.88 |
| TSX MRP C448K | 2.58 | 2.58 | 2.64 | 2.41 | 2.88 | 2.01 | 3.22 | 1.63 |
| TSX MRP C768K | 2.58 | 2.58 | 2.64 | 2.41 | 2.88 | 2.01 | 3.22 | 1.63 |
| TSX MRP C001M | 2.58 | 2.58 | 2.60 | 2.28 | 2.68 | 1.67 | 2.74 | 1.23 |
| TSX MRP C01M7 | 2.58 | 2.58 | 2.56 | 2.15 | 2.50 | 1.42 | 2.39 | 0.98 |
| TSX MRP C002M | 2.58 | 2.58 | 2.60 | 2.13 | 2.68 | 1.38 | 2.74 | 0.94 |

| PV4/5 | For a 50°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MRP C003M | 2.58 | 2.58 | 2.56 | 1.90 | 2.50 | 1.05 | 2.39 | 0.66 |
| TSX MRP C007M | 2.58 | 2.58 | 2.40 | 1.46 | 1.97 | 0.62 | 1.58 | 0.35 |
| TSX MRP F004M | 2.58 | 2.58 | 2.56 | 1.90 | 2.50 | 1.05 | 2.39 | 0.66 |
| TSX MRP F008M | 2.58 | 2.58 | 2.40 | 1.46 | 1.97 | 0.62 | 1.58 | 0.35 |

| PV4/5 | For a 60°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 1.75 | 1.75 | 1.84 | 1.78 | 2.21 | 1.88 | 2.95 | 2.00 |
| TSX MCP C 512K | 1.75 | 1.75 | 1.84 | 1.78 | 2.21 | 1.90 | 2.95 | 2.04 |
| TSX MCP C 002M | 1.75 | 1.75 | 1.82 | 1.62 | 2.09 | 1.33 | 2.55 | 1.06 |
| TSX MRP P128K | 1.75 | 1.75 | 1.84 | 1.78 | 2.21 | 1.88 | 2.95 | 2.00 |
| TSX MRP P224K | 1.75 | 1.75 | 1.84 | 1.78 | 2.21 | 1.90 | 2.95 | 2.04 |
| TSX MRP P384K | 1.75 | 1.75 | 1.84 | 1.73 | 2.21 | 1.70 | 2.95 | 1.63 |
| TSX MRP C448K | 1.75 | 1.75 | 1.82 | 1.71 | 2.09 | 1.59 | 2.55 | 1.44 |
| TSX MRP C768K | 1.75 | 1.75 | 1.82 | 1.71 | 2.09 | 1.59 | 2.55 | 1.44 |
| TSX MRP C001M | 1.75 | 1.75 | 1.80 | 1.64 | 1.98 | 1.37 | 2.24 | 1.11 |
| TSX MRP C01M7 | 1.75 | 1.75 | 1.78 | 1.57 | 1.88 | 1.20 | 2.00 | 0.91 |
| TSX MRP C002M | 1.75 | 1.75 | 1.80 | 1.56 | 1.98 | 1.17 | 2.24 | 0.87 |
| TSX MRP C003M | 1.75 | 1.75 | 1.78 | 1.44 | 1.88 | 0.92 | 2.00 | 0.62 |
| TSX MRP C007M | 1.75 | 1.75 | 1.70 | 1.17 | 1.56 | 0.57 | 1.40 | 0.34 |
| TSX MRP F004M | 1.75 | 1.75 | 1.78 | 1.44 | 1.88 | 0.92 | 2.00 | 0.62 |
| TSX MRP F008M | 1.75 | 1.75 | 1.70 | 1.17 | 1.56 | 0.57 | 1.40 | 0.34 |

## Main Battery Lifetime of PV6 PCMCIA (in Years)

The table below presents the lifetime of main battery TSX BAT M02 (PV6) for PCMCIA memory cards:

| PV6 | For a 25°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 7.2 | 7.2 | 7.2 | 6.3 | 7.0 | 4.5 | 6.8 | 3.2 |
| TSX MCP C 512K | 7.2 | 7.2 | 7.2 | 6.5 | 7.0 | 5.1 | 6.8 | 3.9 |
| TSX MCP C 002M | 7.2 | 7.2 | 6.8 | 5.8 | 5.9 | 3.6 | 5.0 | 2.4 |
| TSX MRP P128K | 7.2 | 7.2 | 7.2 | 6.3 | 7.0 | 4.5 | 6.8 | 3.2 |
| TSX MRP P224K | 7.2 | 7.2 | 7.2 | 6.5 | 7.0 | 5.1 | 6.8 | 3.9 |
| TSX MRP P384K | 7.2 | 7.2 | 7.2 | 6.5 | 7.0 | 5.1 | 6.8 | 3.9 |
| TSX MRP C448K | 7.2 | 7.2 | 6.8 | 5.8 | 5.9 | 3.6 | 5.0 | 2.4 |
| TSX MRP C768K | 7.2 | 7.2 | 6.8 | 5.8 | 5.9 | 3.6 | 5.0 | 2.4 |
| TSX MRP C001M | 7.2 | 7.2 | 6.5 | 5.2 | 5.1 | 2.8 | 3.9 | 1.7 |
| TSX MRP C01M7 | 7.2 | 7.2 | 6.3 | 4.7 | 4.5 | 2.3 | 3.2 | 1.4 |
| TSX MRP C002M | 7.2 | 7.2 | 6.5 | 5.2 | 5.1 | 2.8 | 3.9 | 1.7 |
| TSX MRP C003M | 7.2 | 7.2 | 6.3 | 4.7 | 4.5 | 2.3 | 3.2 | 1.4 |
| TSX MRP C007M | 7.2 | 7.2 | 5.4 | 3.5 | 3.0 | 1.3 | 1.9 | 0.7 |
| TSX MRP F004M | 7.2 | 7.2 | 6.3 | 4.7 | 4.5 | 2.3 | 3.2 | 1.4 |
| TSX MRP F008M | 7.2 | 7.2 | 5.4 | 3.5 | 3.0 | 1.3 | 1.9 | 0.7 |

| PV6 | For a 40°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 4.6 | 4.6 | 4.7 | 4.3 | 5.1 | 3.6 | 5.6 | 2.9 |
| TSX MCP C 512K | 4.6 | 4.6 | 4.7 | 4.4 | 5.1 | 4.0 | 5.6 | 3.5 |
| TSX MCP C 002M | 4.6 | 4.6 | 4.6 | 4.1 | 4.5 | 3.0 | 4.3 | 2.2 |
| TSX MRP P128K | 4.6 | 4.6 | 4.7 | 4.3 | 5.1 | 3.6 | 5.6 | 2.9 |
| TSX MRP P224K | 4.6 | 4.6 | 4.7 | 4.4 | 5.1 | 4.0 | 5.6 | 3.5 |
| TSX MRP P384K | 4.6 | 4.6 | 4.7 | 4.4 | 5.1 | 4.0 | 5.6 | 3.5 |
| TSX MRP C448K | 4.6 | 4.6 | 4.6 | 4.1 | 4.5 | 3.0 | 4.3 | 2.2 |
| TSX MRP C768K | 4.6 | 4.6 | 4.6 | 4.1 | 4.5 | 3.0 | 4.3 | 2.2 |

| PV6 | For a 40°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MRP C001M | 4.6 | 4.6 | 4.4 | 3.8 | 4.0 | 2.4 | 3.5 | 1.6 |
| TSX MRP C01M7 | 4.6 | 4.6 | 4.3 | 3.5 | 3.6 | 2.0 | 2.9 | 1.3 |
| TSX MRP C002M | 4.6 | 4.6 | 4.4 | 3.8 | 4.0 | 2.4 | 3.5 | 1.6 |
| TSX MRP C003M | 4.6 | 4.6 | 4.3 | 3.5 | 3.6 | 2.0 | 2.9 | 1.3 |
| TSX MRP C007M | 4.6 | 4.6 | 3.9 | 2.8 | 2.6 | 1.2 | 1.8 | 0.7 |
| TSX MRP F004M | 4.6 | 4.6 | 4.3 | 3.5 | 3.6 | 2.0 | 2.9 | 1.3 |
| TSX MRP F008M | 4.6 | 4.6 | 3.9 | 2.8 | 2.6 | 1.2 | 1.8 | 0.7 |

| PV6 | For a 50°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 2.6 | 2.6 | 2.7 | 2.6 | 3.1 | 2.5 | 3.9 | 2.4 |
| TSX MCP C 512K | 2.6 | 2.6 | 2.7 | 2.6 | 3.1 | 2.7 | 3.9 | 2.7 |
| TSX MCP C 002M | 2.6 | 2.6 | 2.6 | 2.5 | 2.9 | 2.2 | 3.2 | 1.9 |
| TSX MRP P128K | 2.6 | 2.6 | 2.7 | 2.6 | 3.1 | 2.5 | 3.9 | 2.4 |
| TSX MRP P224K | 2.6 | 2.6 | 2.7 | 2.6 | 3.1 | 2.7 | 3.9 | 2.7 |
| TSX MRP P384K | 2.6 | 2.6 | 2.7 | 2.6 | 3.1 | 2.7 | 3.9 | 2.7 |
| TSX MRP C448K | 2.6 | 2.6 | 2.6 | 2.5 | 2.9 | 2.2 | 3.2 | 1.9 |
| TSX MRP C768K | 2.6 | 2.6 | 2.6 | 2.5 | 2.9 | 2.2 | 3.2 | 1.9 |
| TSX MRP C001M | 2.6 | 2.6 | 2.6 | 2.4 | 2.7 | 1.9 | 2.7 | 1.5 |
| TSX MRP C01M7 | 2.6 | 2.6 | 2.6 | 2.3 | 2.5 | 1.6 | 2.4 | 1.2 |
| TSX MRP C002M | 2.6 | 2.6 | 2.6 | 2.4 | 2.7 | 1.9 | 2.7 | 1.5 |
| TSX MRP C003M | 2.6 | 2.6 | 2.6 | 2.3 | 2.5 | 1.6 | 2.4 | 1.2 |
| TSX MRP C007M | 2.6 | 2.6 | 2.4 | 1.9 | 2.0 | 1.1 | 1.6 | 0.7 |
| TSX MRP F004M | 2.6 | 2.6 | 2.6 | 2.3 | 2.5 | 1.6 | 2.4 | 1.2 |
| TSX MRP F008M | 2.6 | 2.6 | 2.4 | 1.9 | 2.0 | 1.1 | 1.6 | 0.7 |

| PV6 | For a 60°C PLC ambient temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 100% powered up | | 92% PU (30d maint. | | 66% PU (WE. 30d maint. | | 33% PU (12h.WE. 30d maint. | |
| | Typical | Min | Typical | Min | Typical | Min | Typical | Min |
| TSX MCP C 224K | 1.8 | 1.8 | 1.8 | 1.8 | 2.2 | 1.9 | 3.0 | 2.0 |
| TSX MCP C 512K | 1.8 | 1.8 | 1.8 | 1.8 | 2.2 | 2.0 | 3.0 | 2.2 |
| TSX MCP C 002M | 1.8 | 1.8 | 1.8 | 1.7 | 2.1 | 1.7 | 2.5 | 1.6 |
| TSX MRP P128K | 1.8 | 1.8 | 1.8 | 1.8 | 2.2 | 1.9 | 3.0 | 2.0 |
| TSX MRP P224K | 1.8 | 1.8 | 1.8 | 1.8 | 2.2 | 2.0 | 3.0 | 2.2 |
| TSX MRP P384K | 1.8 | 1.8 | 1.8 | 1.8 | 2.2 | 2.0 | 3.0 | 2.2 |
| TSX MRP C448K | 1.8 | 1.8 | 1.8 | 1.7 | 2.1 | 1.7 | 2.5 | 1.6 |
| TSX MRP C768K | 1.8 | 1.8 | 1.8 | 1.7 | 2.1 | 1.7 | 2.5 | 1.6 |
| TSX MRP C001M | 1.8 | 1.8 | 1.8 | 1.7 | 2.0 | 1.5 | 2.2 | 1.3 |
| TSX MRP C01M7 | 1.8 | 1.8 | 1.8 | 1.6 | 1.9 | 1.3 | 2.0 | 1.1 |
| TSX MRP C002M | 1.8 | 1.8 | 1.8 | 1.7 | 2.0 | 1.5 | 2.2 | 1.3 |
| TSX MRP C003M | 1.8 | 1.8 | 1.8 | 1.6 | 1.9 | 1.3 | 2.0 | 1.1 |
| TSX MRP C007M | 1.8 | 1.8 | 1.7 | 1.4 | 1.6 | 0.9 | 1.4 | 0.6 |
| TSX MRP F004M | 1.8 | 1.8 | 1.8 | 1.6 | 1.9 | 1.3 | 2.0 | 1.1 |
| TSX MRP F008M | 1.8 | 1.8 | 1.7 | 1.4 | 1.6 | 0.9 | 1.4 | 0.6 |

## Minimum Lifetime of the Main Battery, in a Powered Down PLC

In a powered down PLC, the minimum lifetime of the main battery is 6 months in PV6 PCMCIAs.

## Auxiliary Battery Lifetime

The auxiliary battery TSX BATM 03 is included in the PCMCIA product. Whatever the usage cases and ambient temperature, the lifetime of the auxiliary battery is:

- 5 years in PV1/2/3
- 1.7 years in PV4/5
- 5 years in PV6

# Part III
## Power Supply Modules (CPS)

### Introduction

This part provides information on the Quantum Power Supply modules.

### What Is in This Part?

This part contains the following chapters:


| Chapter | Chapter Name | Page |
|---|---|---|
| 11 | General Information | 221 |
| 12 | 140 CPS 111 00: 115/230 VAC Standalone 3 A Power Supply Module | 235 |
| 13 | 140 CPS 111 00 (PV 01 or greater): 115/230 VAC Standalone Power Supply Module | 239 |
| 14 | 140 CPS 114 00: 115/230 VAC Standalone 8 A Power Supply Module | 243 |
| 15 | 140 CPS 114 10: 115/230 VAC Standalone/Summable 8 A Power Supply Module | 247 |
| 16 | 140 CPS 114 20: 115/230 VAC Standalone/Summable 11 A Power Supply Module | 251 |
| 17 | 140 CPS 124 00: 115/230 VAC Standalone/Redundant 8 A Power Supply Module | 257 |
| 18 | 140 CPS 124 20: 115/230 VAC Standalone/Redundant 11 A Power Supply Module | 263 |
| 19 | 140 CPS 211 00: 24 VDC Standalone 3 A Power Supply Module | 269 |
| 20 | 140 CPS 214 00: 24 VDC Standalone/Summable 7–8 A Power Supply Module | 273 |
| 21 | 140 CPS 224 00: 24 VDC Standalone/Redundant 6–8 A Power Supply Module | 279 |
| 22 | 140 CPS 414 00: 48 VDC Standalone/Summable 7–8 A Power Supply Module | 285 |
| 23 | 140 CPS 424 00: 48 VDC Standalone/Redundant 6–8 A Power Supply Module | 291 |
| 24 | 140 CPS 511 00: 125 VDC Standalone 3 A Power Supply Module | 297 |
| 25 | 140 CPS 524 00: 125 VDC Standalone/Redundant 8 A Power Supply Module | 301 |

# Chapter 11
# General Information

## Purpose

This chapter provides general information of the Quantum Power Supplies.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Table of Power Supplies | 222 |
| System Design Considerations for Quantum Power Supplies | 223 |
| Compatibility | 228 |
| Presentation and Wiring | 230 |
| Indicators | 232 |
| Displaying the Power Supply Budget | 233 |

# Table of Power Supplies

## Overview

| Type | Part Numbers | Source Voltage | Bus Current |
|---|---|---|---|
| Standalone | 140 CPS 111 00 | 115/230 Vac | 3 A |
| | 140 CPS 114 00 | 115/230 Vac | 8 A |
| | 140 CPS 211 00 | 24 Vdc | 3 A |
| | 140 CPS 511 00 | 125 Vdc | 3 A |
| Standalone/Summable | 140 CPS 114 10 | 115/230 Vac | 8 A |
| | 140 CPS 114 20 | 115/230 Vac | 11 A |
| | 140 CPS 214 00 | 24 Vdc | 7–8 A |
| | 140 CPS 414 00 | 48/72 Vdc | 7–8 A |
| Standalone/Redundant | 140 CPS 124 00 | 115/230 Vac | 8 A |
| | 140 CPS 124 20 | 115/230 Vac | 11 A |
| | 140 CPS 224 00 | 24 Vdc | 6–8 A |
| | 140 CPS 424 00 | 48/72 Vdc | 6–8 A |
| | 140 CPS 524 00 | 125 Vdc | 8 A |

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

Use appropriate power supplies with the exceptions noted in this chapter.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

# System Design Considerations for Quantum Power Supplies

## Overview

There exists some important design differences between various models of Quantum power supplies that require careful consideration by the system designer in order to achieve maximum system performance. The principal difference lies in the generation within the power supply of important rack signals related to the health of the power supply and the status of the input power.

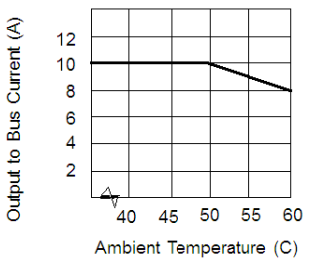
Quantum power supplies include on-board early power detection logic which is used to signal other modules on the rack that input power is lost. This signal is called POK (power OK) and is active high (i.e., when the signal is high, power is OK).

There is both an internal (to the power supply) and an external (as seen by the rack and other modules) version of the POK signal. The internal POK signal is represented by the Pwr ok LED (light emitting diode) on the front panel of the power supplies.

The system POK signal is generated so that there is sufficient time between the negative going edge of system POK (power outage) and the actual interruption of power to the rack. This first notification of power lost is necessary for the Quantum executive to perform an orderly system shutdown.

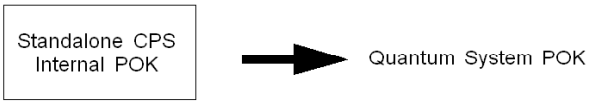
## Standalone Power Supplies

Four models of standalone power supplies are offered:

| Power Supply | Range | Bus Current |
|---|---|---|
| 140 CPS 111 00 | 115/230 Vac | 3 A(60°C) |
| 140 CPS 114 00 | 115/230 Vac | Operating curve:<br>  <br> |
| 140 CPS 211 00 | 24 Vdc | 3 A(60°C) |
| 140 CPS 511 00 | 125 Vdc | 3 A(60°C) |

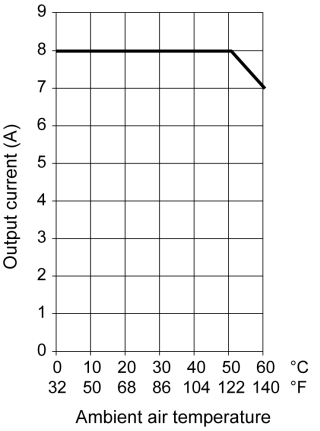
| ⚠ CAUTION |
| --- |
| **UNINTENDED EQUIPMENT OPERATION**<br>When a rack is equipped with a standalone power supply do not install any other power supply.<br>When the power supply must be redundant, install redundant power supply units.<br>**Failure to follow these instructions can result in injury or equipment damage.** |

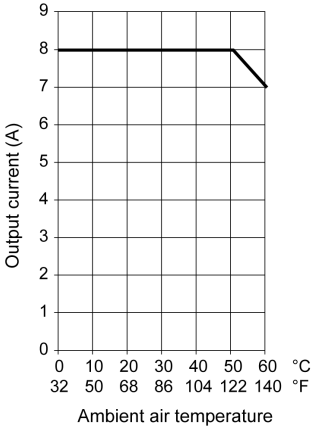
The following figure shows the direction of the internal POK to the Quantum system POK.




## Summable Power Supplies

Four summable power supply models are offered:

| Power Supply | Range | Bus Current |
| --- | --- | --- |
| 140 CPS 114 10 | 115/230 Vac | 8 A (60°C) |
| 140 CPS 114 20 | 115/230 Vac | 11 A (60°C) |
| 140 CPS 214 00 | 24 Vdc | Operating curve:<br>  <br> |

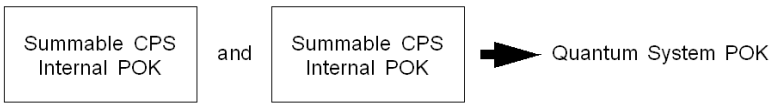
| | | |
|---|---|---|
| 140 CPS 414 00 | 48/72 Vdc | Operating curve:<br>  <br> |

A summable power supply may be used as a standalone power supply without reservation in any Quantum system.

For systems configured with a mix of CPS, NOM, expert, and I/O modules, whose total current consumption from the rack exceeds the current provided by one summable supply, two summable supplies may be used in a single rack. In such a system, the total current available on the rack is the sum of the capability of both supplies (i.e., 2 x 8 A = 16 A). The summable supplies are designed so that they split the current supplied to the load almost equally, which also has the added benefit of increasing total system MTBF, and to distribute the thermal load across the rack. Summable supplies should be installed in the end module positions in the Quantum rack to maximize the system thermal performance.

The Quantum system POK signal in systems powered by two summable power supplies is only true (power is OK) when both internal POK signals (in the 140 CPS •14 •0) are true. Quantum summable power supplies are not hot-swappable.

The following figure shows the direction of the Quantum POK summable power supplies.

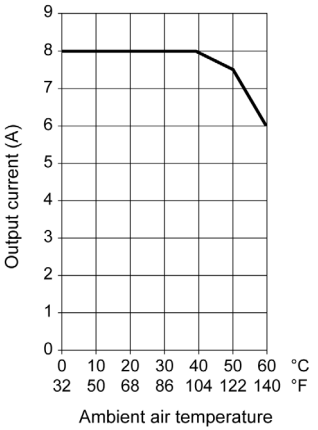
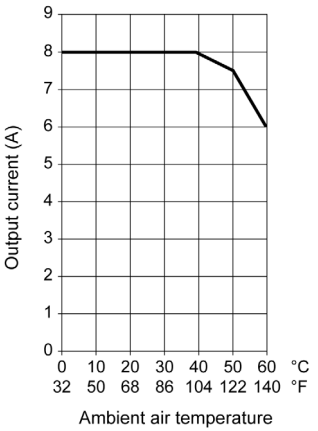



The proper method for starting systems powered by summable power supplies is to insert both supplies in the rack in an unpowered state, and then apply power to each supply. There is no requirement to turn on each supply simultaneously. The system designer must realize that the operation of the summable supply described above is independent of total rack load, i.e., even if the total load on the rack is less than 8 A, if there are two summable supplies installed in the rack, the system POK is generated as shown in this section.

For the special case of a single summable supply used as a standalone, the system POK generation reverts to the standalone case as shown previously in this section.

## Redundant Power Supplies

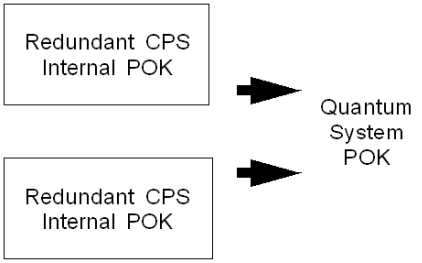
Five redundant power supply models are offered:

| Power Supply | Range | Bus Current |
|---|---|---|
| 140 CPS 124 00 | 115/230 Vac | 8 A (60°C) |
| 140 CPS 124 20 | 115/230 Vac | 11 A (60°C) |
| 140 CPS 224 00 | 24 Vdc | Operating curve:<br>  |
| 140 CPS 424 00 | 48/72 Vdc | Operating curve:<br>  |
| 140 CPS 524 00 | 125 Vdc | 8 A (60°C) |

Similar to the summable supplies, the Quantum redundant power supplies also contain circuitry which forces the installed power supplies to share output current almost equally. An important difference between the summable and the redundant supply lies in the system POK generation circuitry.

The Quantum system POK signal in systems powered by redundant power supplies is true (power is OK) if either or both internal POK is true.

The following figure shows the Quantum system POK redundant power supply direction:




**NOTE:** Redundant power supply module health may be monitored in an I/O module health status word.

Another important difference from the summable system is the total available system rack loading. If there are N redundant power supplies installed in a rack, the total rack load must not exceed the capability of N-1 supplies. (Nmax=3)

For example, if two 8 A power supplies are installed in the rack (N=2), the maximum rack load available for redundant operation is the current sourced by N-1 ( = 1) supplies, which is 8 A. If three 8 A redundant power supplies are installed (N = 3), the maximum rack load available for redundant operation is the current sourced by N-1 ( = 2) supplies, which is 16 A.

If these constraints are observed, then in a system of two or three redundant supplies, one supply (it doesn't matter which one) is hot-swappable. This is possible because there is excess capacity in the N-1 remaining supplies to source the rack current while the Nth supply is being swapped.

An obvious extension to this argument is that a single redundant power supply may be used as a standalone supply (but the cheapest solution is obtained by using a summable or standalone supply for this application).

# Compatibility

## Table of Combination

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

Use appropriate power supplies with the exceptions noted in the following tables.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

Quantum standalone power supplies cannot be combined together. List of Quantum standalone power supplies:

- 140CPS11100
- 140CPS21100
- 140CPS51100

Quantum summable power supplies combination:

| Combination with 1 power supply: | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 |
|---|---|---|---|---|
| 140CPS11420 | Yes | Yes | No | No |
| 140CPS11410 | Yes | Yes | No | No |
| 140CPS21400 | No | No | Yes | Yes |
| 140CPS41400 | No | No | Yes | Yes |

Quantum redundant power supplies combination:

| Combination with 1 or 2 power supplies: | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
|---|---|---|---|---|---|
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

## Compatibility Issues

Power supplies:

- Do not use a standalone power supply in combination with any other supply in the same backplane.
- With the exception of standalone models, power supplies with the same model number are compatible when installed in the same backplane.

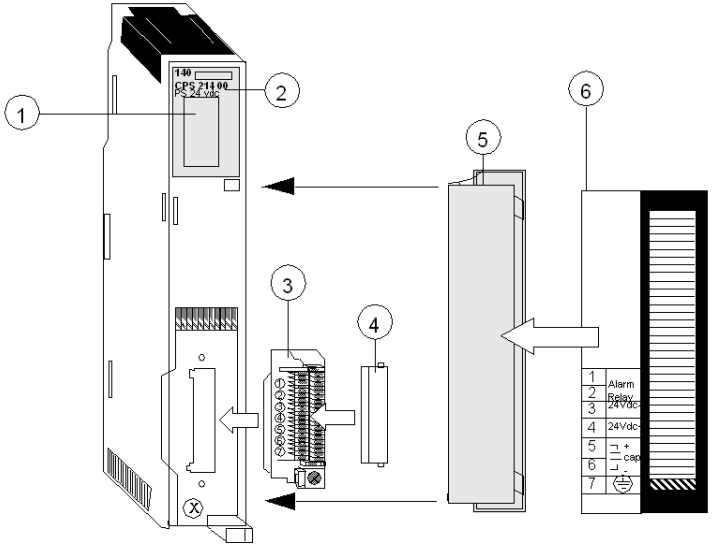
DIO:

- While it is possible to use a standalone or a summable power supply with a DIO drop (as long as the DIO input is left unpowered), it is not possible to use a redundant supply with the DIO drop.
- The added power supply does not need to be of the same type as the DIO adapter. AC powered supplies may be used with DC type adapters and vice-versa.
- DIO module current load with an added power supply is typically 200 mA.

# Presentation and Wiring

## Illustration

The following figure shows the power supply module.




1 LED Area
2 Model Number, Module Description, Color Code
3 Field Wiring Connector
4 Field Wiring Connector Cover
5 Removable door
6 Customer Identification Label, (Fold label and place it inside door)

**NOTE:** When field wiring the power supply module, the maximum wire size that should be used is 1 - 14 AWG or 2 - 16 AWG; the minimum is 20 AWG.

**NOTE:** The tightening torque must be between 0.5 Nm and 0.8 Nm.

# NOTICE

**DESTRUCTION OF ADAPTER**

- Before tightening the locknut to the torque 0.50...0.80 Nm, be sure to properly position the right-angle F adapter connector.
- During tightening, be sure to maintain the connector securely.
- Do not tighten the right-angle F adapter beyond the specified torque.

**Failure to follow these instructions can result in equipment damage.**

| | | | | | |
|---|---|---|---|---|---|
| |  |  |  |  |  |
| $mm^2$ | 0.14…1.5 | 0.16…0.75 | 0.14…2.5 | 0.14…4 | 0.16…1.5 |
| AWG | 20…16 | 20…18 | 20…14 | 20…12 | 20…16 |

| | | | |
|---|---|---|---|
|  Ø 3,5 |  C  | Nm | 0.6 |
| | | pound-inch | 5.4 |

# Indicators

## Illustration

The following figure shows the Power Supply LED indicator.




## Description

The following table shows the Power Supply LED description.

| LEDs | Color | Indication when On |
|---|---|---|
| Pwr ok | Green | Power is supplied to the bus. |

# Displaying the Power Supply Budget

## Overview

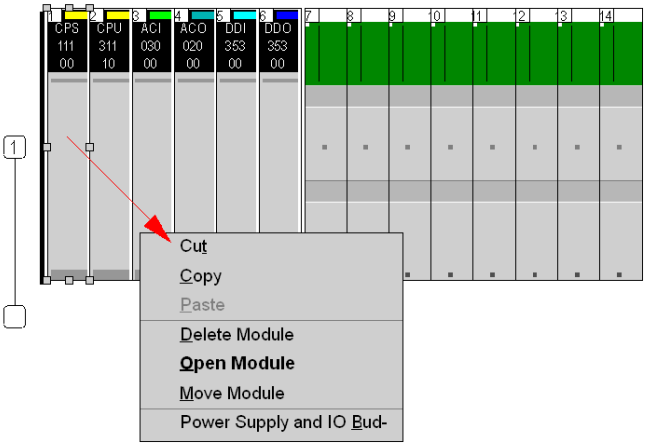
If one or more power supply modules is configured, the Unity Pro software displays the available load and the consumption of all configured modules. Power supply availability is displayed as a bar graph. Each time you add a module, the green bar increases. An overload condition turns the display red. The first time you overshoot the mark, a message is displayed. Nevertheless, you can analyze and generate.

If there is no power supply, the software displays the power consumption requirement in red.

In the case of redundant power supplies, where the full load is applied to each power supply module, the software displays the power supply budget as if there is only one power supply. For summable power supplies, the load is distributed between the power supplies.

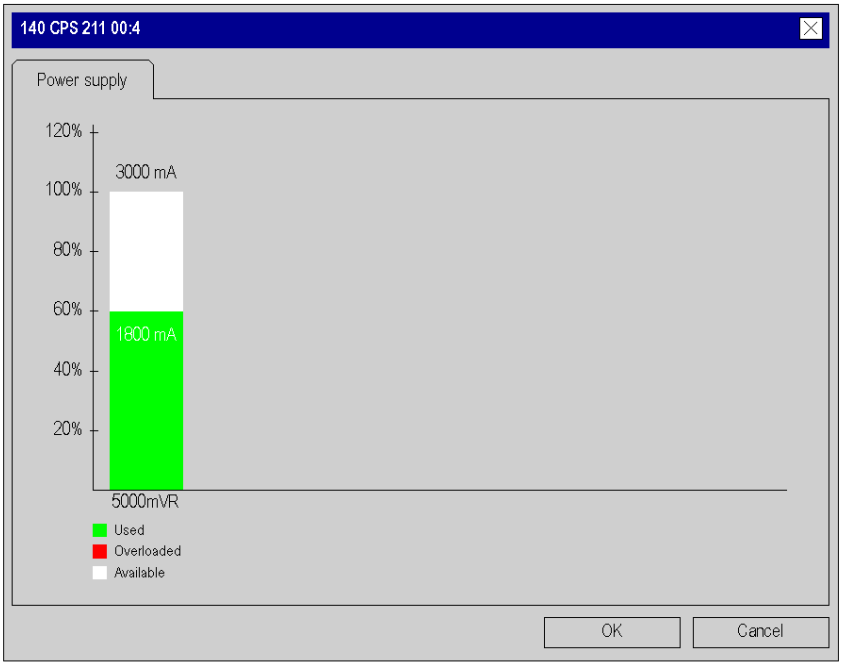
## Procedure

To access the Power Supply and I/O Budget screen:

| Step | Action |
|---|---|
| 1 | Open the Bus Editor of Unity Pro. |
| 2 | Select the power supply module. |
| 3 | Open the context menu with a click on the right mouse button.<br>**Result:**<br>  |
| 4 | Choose **Power Supply and IO Budget**. |

## Presentation

Here is a sample screen:



# Chapter 12
# 140 CPS 111 00: 115/230 VAC Standalone 3 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 111 00 Power Supply module.

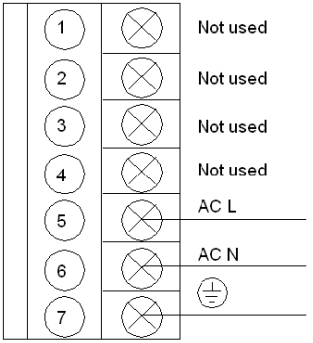
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 111 00 Wiring Diagram | 236 |
| 140 CPS 111 00 Specifications | 237 |

# 140 CPS 111 00 Wiring Diagram

## Wiring Diagram




**NOTE:** For closed system installations, you must use connector 140 XTS 005 00.

**⚠ DANGER**

**ELECTRIC SHOCK**

Ensure that the installation complies with wiring, grounding and electromagnetic standards defined in:

- national and local electrical code requirements,
- Grounding and Electromagnetic Compatibility of PLC Systems *(see Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual)* user manual.

**Failure to follow these instructions will result in death or serious injury.**

# 140 CPS 111 00 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| Internal power dissipation (electric losses) | 6.5 W |
| Network power consumption | Electric losses + (5.1 x Iout)[1] |
| Power factor | 0.5 |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 100 ... 276 VAC |
| Input frequency | 47 ... 63 Hz |
| Input voltage total harmonic distortion | Less than 10% of the fundamental rms value |
| Input current | 0.4 A @ 115 V AC<br>0.2 A @ 230 V AC |
| Inrush current | 10 A @ 115 V AC<br>20 A @ 230 V AC |
| VA rating | 50 VA |
| Fusing (external) | 1.5 A slow-blow fuse |
| Input power interruption | 1/2 cycle @ full load and minimum rated line voltage / frequency. No less than 1 s between interruptions. |

## Output to Bus

| | |
|---|---|
| Voltage | 5.1 VDC |
| Current | 3 A max. |
| | 0.3 A min. |
| Protection | Overcurrent, overvoltage |

# Chapter 13
# 140 CPS 111 00 (PV 01 or greater): 115/230 VAC Standalone Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 111 00 (PV 01 or greater) Power Supply module.

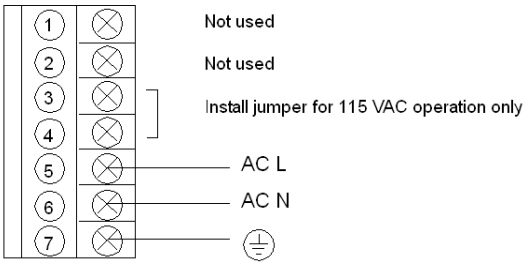
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 111 00 (PV 01 or greater) Wiring Diagram | 240 |
| 140 CPS 111 00 (PV 01 or greater) Specifications | 241 |

# 140 CPS 111 00 (PV 01 or greater) Wiring Diagram

## Wiring Diagram




**NOTE:** For closed system installations, you must use 140 XTS 005 00 connector.

**⚠ DANGER**

**ELECTRIC SHOCK**

Ensure that the installation complies with wiring, grounding and electromagnetic standards defined in:

- national and local electrical code requirements,
- Grounding and Electromagnetic Compatibility of PLC Systems *(see Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual)* user manual.

**Failure to follow these instructions will result in death or serious injury.**

# 140 CPS 111 00 (PV 01 or greater) Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| Internal power dissipation (electric losses) | 6.5 W |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| Power factor | 0.5 |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 100 ... 276 VAC |
| Input frequency | 47 ... 63 Hz |
| Input voltage total harmonic distortion | Less than 10% of the fundamental ms value |
| Input current | 0.4 A @ 115 VAC; 0.2 A @ 230 VAC |
| Inrush current | 10 A @ 115 VAC; 20 A @ 230 VAC |
| VA rating | 50 VA |
| Fusing (external) | 1.5 A slow-blow fuse |
| Input power interruption | 1/2 cycle @ full load and minimum rated line voltage / frequency. No less than 1 s between interruptions. |

## Output to Bus

| | |
|---|---|
| Voltage | 5.1 V dc |
| Current | 3 A max. |
| | 0.3 A min. |
| Protection | Overcurrent, overvoltage |

# Chapter 14
# 140 CPS 114 00: 115/230 VAC Standalone 8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 114 00 Power Supply module.

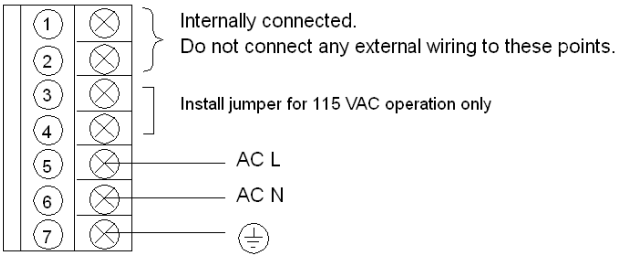
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 114 00 Wiring Diagram | 244 |
| 140 CPS 114 00 Specifications | 245 |

# 140 CPS 114 00 Wiring Diagram

## Wiring Diagram




**⚠ DANGER**

**ELECTRIC SHOCK**

Ensure that the installation complies with wiring, grounding and electromagnetic standards defined in:

- national and local electrical code requirements,
- Grounding and Electromagnetic Compatibility of PLC Systems *(see Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual)* user manual.

**Failure to follow these instructions will result in death or serious injury.**

# 140 CPS 114 00 Specifications

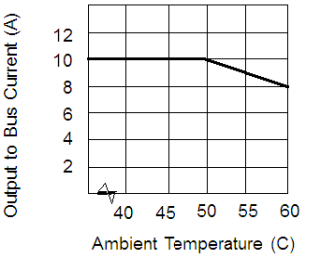
## General Specifications

| | |
|---|---|
| Operating Mode | Standalone |
| Internal power dissipation (in Watts) | 6.0 + 1.5 x $I_{OUT}$ (where $I_{OUT}$ is in A) |

## Input

| | |
|---|---|
| Input voltage | 93 ... 138 VAC |
| | 170 ... 276 VAC |
| Input frequency | 47 ... 63 Hz |
| Input voltage total harmonic distortion | Less than 10% of the fundamental rms value |
| Input current | 1.1 A @ 115 VAC |
| | 0.6 A @ 230 VAC |
| Inrush current | 38 A @ 115 VAC |
| | 19 A @ 230 VAC |
| VA rating | 130 VA |
| Fusing (external) | 2.0 A slow-blow |
| Input power interruption | 1/2 cycle @ full load and minimum rated line voltage / frequency. No less than 1 s between interruptions. |

## Output to Bus

| | |
|---|---|
| Voltage | 5.1 VDC |
| Current | 8 A @ 60 °C (see the operating curve)<br>min. current not required<br>Operating Curve:<br>  |
| Protection | Over-current, over-voltage |

# Chapter 15
# 140 CPS 114 10: 115/230 VAC Standalone/Summable 8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 114 10 Power Supply module.

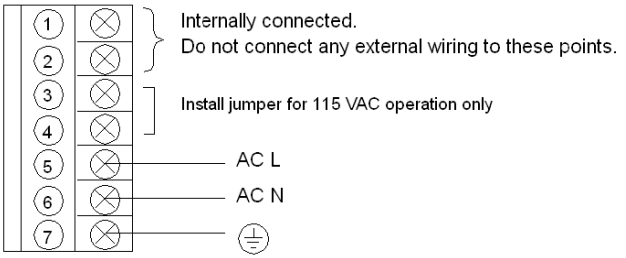
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 114 10 Wiring Diagram | 248 |
| 140 CPS 114 10 Specifications | 249 |

# 140 CPS 114 10 Wiring Diagram

## Wiring Diagram




**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 114 10 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Summable |
| Internal power dissipation (in W) | 6.0 + 1.5 x $I_{OUT}$ (where $I_{OUT}$ is in A) |

## Input

| | |
|---|---|
| Input voltage | 93 ... 138 VAC |
| | 170 ... 276 VAC |
| Input frequency | 47 ... 63 Hz |
| Input voltage total harmonic distortion | Less than 10% of the fundamental rms value |
| Input current | 1.1 A @ 115 VAC<br>0.6 A @ 230 VAC |
| Inrush Current | 38 A @ 115 VAC.<br>19 A @ 230 VAC |
| VA Rating | 130 VA |
| Fusing (external) | 2.0 A slow-blow recommended |
| Input Power Interruption | 1/2 cycle @ full load and minimum rated line voltage / frequency. No less than 1 second between interruptions. |

## Output to Bus

Output to Bus

| | |
|---|---|
| Voltage | 5.1 VDC |
| Current | 8 A @ 60 °C<br>min. Current not required |
| Protection | Over Current, Over Voltage |

# Chapter 16
# 140 CPS 114 20: 115/230 VAC Standalone/Summable 11 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 114 20 Power Supply module.

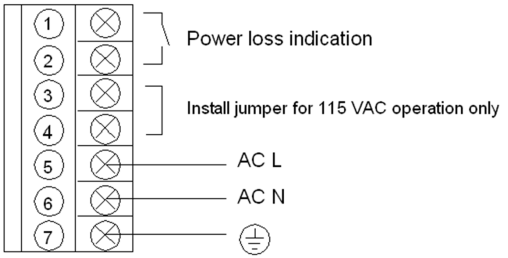
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 114 20 Wiring Diagram | 252 |
| 140 CPS 114 20 Specifications | 254 |

# 140 CPS 114 20 Wiring Diagram

## Wiring Diagram




**NOTE:** A normally closed relay contact rated at 220 VAC, 6A / 30 VDC, 5A is available on terminals 1 and 2 on the power terminal strip. This contact set may be used to signal input power OFF. The relay de-energizes when input power drops below:

- 70 VAC, when input voltage is set to 115VAC.
- 140 VAC, when input voltage is set to 230VAC.

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 114 20 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Summable |
| Internal power dissipation (electric losses) | 11 W |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| Power factor | 0.5 |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

140 CPS 114 20 power supplies are designed to not require the external EMI filter, ferrite bead and Olflex cable.

## Input

| | |
|---|---|
| Input voltage | 93 ... 138 VAC |
| | 170 ... 264 VAC |
| Input frequency | 47 ... 63 Hz |
| Input voltage total harmonic distortion | Less than 10% of the fundamental rms value |
| Input current | 1.2 A @ 115 VAC |
| | 0.7 A @ 230 VAC |
| Inrush current | ≤ 20 A @ 115 VAC |
| | ≤ 25 A @ 230 VAC |
| VA rating | 160 VA @ 11 A |
| Fusing (external) | 2.0 A slow-blow |
| Input power interruption | 1/2 cycle @ full load and minimum rated line voltage / frequency. No less than 1 s between interruptions. |

## Output to Bus

| | | | |
|---|---|---|---|
| Voltage | | | 5.1 VDC |
| Max. current | Standalone | | 11 A @ 60 °C |
| | Summable | a) Two 140 CPS 114 20(total load capacity) | 20 A @ 60 °C |
| | | b) One 140 CPS 114 20 and one 140 CPS 114 10 | 16 A @ 60 °C |
| Min. current | | | not required |
| Protection | | | Over-current, Over-voltage |

# Chapter 17
# 140 CPS 124 00: 115/230 VAC Standalone/Redundant 8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 124 00 Power Supply module.

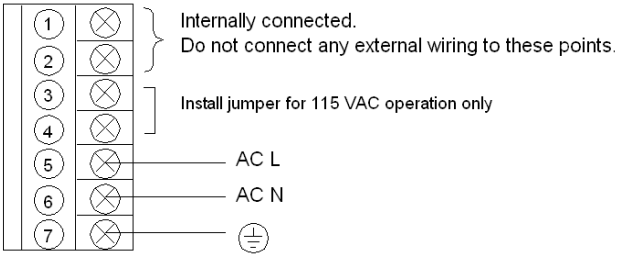
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 124 00 Wiring Diagram | 258 |
| 140 CPS 124 00 Specifications | 260 |

# 140 CPS 124 00 Wiring Diagram

## Wiring Diagram Figure




**NOTE:** When field wiring the power supply module, the maximum wire size that should be used is 1 - 14 AWG (2 $mm^2$) or 2 - 16 AWG (1.5 $mm^2$).

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 124 00 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Redundant |
| Internal power dissipation (electric losses) | 9 W |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| Power factor | 0.5 |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 93 ... 138 VAC |
| | 170 ... 276 VAC |
| Input frequency | 47 ... 63 Hz |
| Input voltage total harmonic distortion | Less than 10% of the fundamental rms value |
| Input current | 1.1 A @ 115 VAC |
| | 0.6 A @ 230 VAC |
| Inrush current | 38 A @ 115 VAC |
| | 19 A @ 230 VAC |
| VA rating | 130 VA |
| Fusing (external) | 2.0 A slow-blow |
| Input power interruption | 1/2 cycle @ full load and minimum rated line voltage / frequency. No less than 1 s between interruptions. |

## Output to Bus

| | | | |
|---|---|---|---|
| Voltage | | | 5.1 VDC |
| Max. Current | Standalone | | 8 A @ 60°C |
| | Redudant | a) Two 140 CPS 124 00 | 8 A @ 60°C |
| | | b) Three 140 CPS 124 00 | 16 A @ 60°C |
| | | c) One 140 CPS 124 00 and one 140 CPS 524 00 | 8 A @ 60°C |
| | | d) One 140 CPS 124 00 and two 140 CPS 524 00 | 16 A @ 60°C |
| | | e) Two 140 CPS 124 00 and one 140 CPS 524 00 | 16 A @ 60°C |
| Min Current | | | Not Required |
| Protection | | | Over-current, over-voltage |

# Chapter 18
# 140 CPS 124 20: 115/230 VAC Standalone/Redundant 11 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 124 20 Power Supply module.

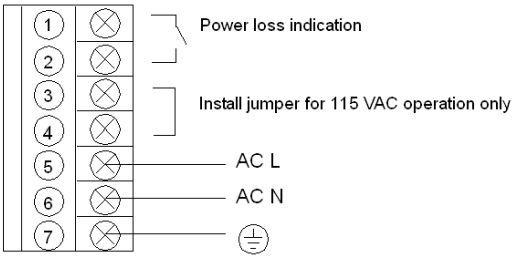
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 124 20 Wiring Diagram | 264 |
| 140 CPS 124 20 Specifications | 266 |

# 140 CPS 124 20 Wiring Diagram

## Wiring Diagram




**NOTE:** A normally closed relay contact rated at 220 VAC, 6 A / 30 VDC, 5 A is available on terminals 1 and 2 on the power terminal strip. This contact set may be used to signal input power OFF. The relay de-energizes when input power drops below:

- 70 VAC, when input voltage is set to 115 VAC.
- 140 VAC, when input voltage is set to 230 VAC.

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 124 20 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Redundant |
| Internal power dissipation (electric losses) | 11 W |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| Power factor | 0.5 |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

140 CPS 124 20 power supplies are designed to not require the external EMI filter, ferrite bead and Olflex cable.

## Input

| | |
|---|---|
| Input voltage | 93 ... 138 VAC |
| | 170 ... 263 VAC |
| Input frequency | 47 ... 63 Hz |
| Input voltage total harmonic distortion | Less than 10% of the fundamental rms value |
| Input current | 1.2 A @ 115 VAC |
| | 0.7 A @ 230 VAC |
| Inrush current | ≤20 A @ 115 VAC |
| | ≤25 A @ 230 VAC |
| VA Rating | 160 VA @ 11 A |
| Fusing (external) | 2.0 A slow-blow |
| Input power interruption | 1/2 cycle @ full load and minimum rated line voltage / frequency. No less than 1 s between interruptions. |

## Output to Bus

| | | | |
|---|---|---|---|
| Voltage | | | 5.1 VDC |
| Max. Current | Standalone | | 11 A @ 60°C |
| | Redundant | a) Two 140 CPS 124 20 | 10 A @ 60°C |
| | | b) Three 140 CPS 124 20 | 20 A @ 60°C |
| | | c) One 140 CPS 124 20 and one 140 CPS x24 00 | 6 A @ 60°C |
| | | | 9 A @ 40°C |
| | | d) One 140 CPS 124 20 and two 140 CPS x24 00 | 12 A @ 60°C |
| | | | 18 A @ 40°C |
| | | e) Two 140 CPS 124 20 and one 140 CPS x24 00 | 16 A @ 60°C |
| | | | 19 A @ 40°C |
| 140 CPS x24 00 means 140 CPS 224 00 or 140 CPS 424 00. | | | |
| Min. current | | | Not required |
| Protection | | | Over-current, over-voltage |

# Chapter 19
# 140 CPS 211 00: 24 VDC Standalone 3 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 211 00 Power Supply module.

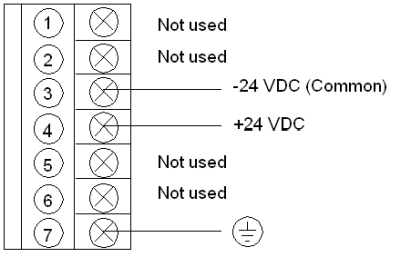
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 211 00 Wiring Diagram | 270 |
| 140 CPS 211 00 Specifications | 271 |

# 140 CPS 211 00 Wiring Diagram

## Wiring Diagram




**⚠ DANGER**

**ELECTRIC SHOCK**

Ensure that the installation complies with wiring, grounding and electromagnetic standards defined in:

- national and local electrical code requirements,
- Grounding and Electromagnetic Compatibility of PLC Systems *(see Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual)* user manual.

**Failure to follow these instructions will result in death or serious injury.**

# 140 CPS 211 00 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| Internal power dissipation (electric losses) | 2 + (1.8 x $I_{out}$) |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 20 ... 30 VDC |
| Input current | 1.6 A |
| Inrush current | 30 A |
| Fusing (external) | 2.5 A slow-blow |

## Output to Bus

| | |
|---|---|
| Voltage | 5.1 VDC |
| Max. current | 3 A |
| Min. current | 0.3 A |
| Protection | Over-current, over-voltage |

# Chapter 20
# 140 CPS 214 00: 24 VDC Standalone/Summable 7–8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 214 00 Power Supply module.

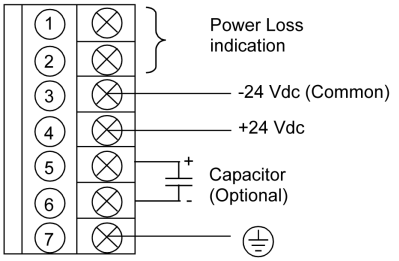
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 214 00 Wiring Diagram | 274 |
| 140 CPS 214 00 Specifications | 276 |

# 140 CPS 214 00 Wiring Diagram

## Wiring Diagram Figure




**NOTE:** 1. A normally closed relay contact rated at 220 Vac, 6 A / 30 Vdc, 5 A is available on terminals 1 and 2 on the power terminal strip. This contact set may be used to signal input power OFF. The relay de-energizes when input power drops below 18 Vdc.
2. Tolerance to input interruptions may be increased by adding a ≥ 50 Vdc electrolytic capacitor between terminals 5 and 6 on the power terminal strip. Refer to the hold-up capacitor timing chart for capacitor values *Input, page 276*.
3. When field wiring the power supply module, the maximum wire size that should be used is 1 - 14 AWG (2 $mm^2$) or 2 - 16 AWG (1.5 $mm^2$).

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

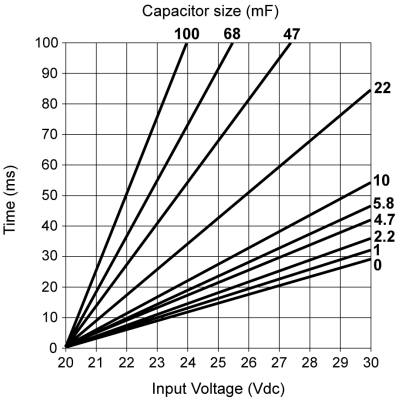
| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 214 00 Specifications

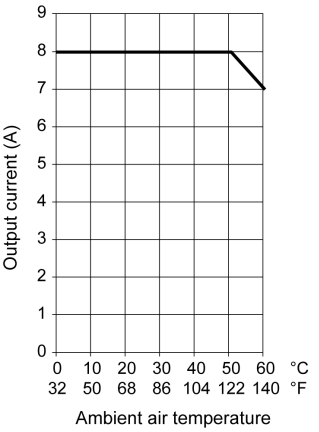
## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Summable |
| Internal power dissipation (electric losses) | 5 + (1.25 x $I_{out}$) |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 20 ... 30 Vdc |
| Input current | 3.8 A |
| Inrush current | 25 A @ 24 Vdc |
| Input ripple | 2.4 Vdc peak-to-peak, 94 ... 189 Hz |
| Input power interruption | 1 ms max. @ 24 Vdc (without capacitor)<br>Hold-up capacitor timing chart (with a capacitor):<br>  |
| Fusing (external) | 5.0 A slow-blow |

## Output to Bus

| | |
|---|---|
| Voltage | 5.1 Vdc |
| Max. current | Operating curve:<br>  <br> |
| Min. current | not required |
| Protection | Over-current, over-voltage |

# Chapter 21
# 140 CPS 224 00: 24 VDC Standalone/Redundant 6–8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 224 00 Power Supply module.

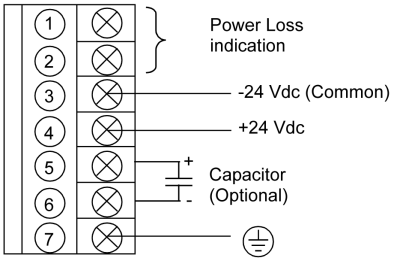
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 224 00 Wiring Diagram | 280 |
| 140 CPS 224 00 Specifications | 282 |

# 140 CPS 224 00 Wiring Diagram

## Wiring Diagram Figure




**NOTE:** 1. A normally closed relay contact rated at 220 Vac, 6 A / 30 Vdc, 5 A is available on terminals 1 and 2 on the power terminal strip. This contact set may be used to signal input power OFF. The relay de-energizes when input power drops below 18 Vdc.
2. Tolerance to input interruptions may be increased by adding a ≥ 50 Vdc electrolytic capacitor between terminals 5 and 6 on the power terminal strip. Refer to the hold-up capacitor timing chart for capacitor values *Input, page 282*.
3. When field wiring the power supply module, the maximum wire size that should be used is 1 - 14 AWG (2 $mm^2$) or 2 - 16 AWG (1.5 $mm^2$).

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

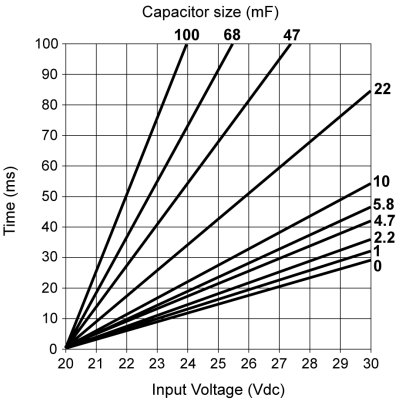
| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 224 00 Specifications

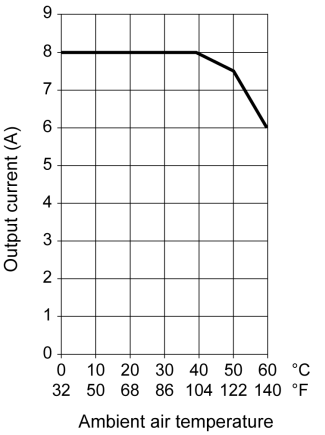
## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Redundant |
| Internal power dissipation (in W) | 7 + (1.5 x $I_{out}$[1]) |
| Network power consumption | Electric losses + (5.1 x $I_{out}$) |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 20 ... 30 Vdc |
| Input current | 3.8 A |
| Inrush current | 25 A @ 24 Vdc |
| Input ripple | 2.4 Vdc peak-to-peak, 94 ... 189 Hz |
| Input Power Interruption | 1 ms max. @ 24 Vdc (without capacitor)<br>Hold-up capacitor timing chart (with a capacitor):<br>  |
| Fusing (external) | 5.0 A slow-blow |

## Output to Bus

<table>
<tr><td>Voltage</td><td colspan="2"></td><td>5.1 Vdc</td></tr>
<tr><td rowspan="11">Max. Current</td><td>Standalone</td><td>Operating curve:<br> 
</td><td>8 A @ 40 °C (104 °F)<br>6 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="10">Redundant</td><td rowspan="2">a) Two 140 CPS x24 00</td><td>8 A @ 40 °C (104 °F)</td></tr>
<tr><td>6 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">b) Three 140 CPS x24 00</td><td>16 A @ 40 °C (104 °F)</td></tr>
<tr><td>12 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">c) One 140 CPS 224 00 and one 140 CPS x24 00</td><td>8 A @ 40 °C (104 °F)</td></tr>
<tr><td>6 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">d) One 140 CPS 224 00 and two 140 CPS x24 00</td><td>16 A @ 40 °C (104 °F)</td></tr>
<tr><td>12 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">e) Two 140 CPS 224 00 and one 140 CPS x24 00</td><td>16 A @ 40 °C (104 °F)</td></tr>
<tr><td>12 A @ 60 °C (140 °F)</td></tr>
<tr><td colspan="3">140 CPS x24 00 means 140 CPS 224 00 or 140 CPS 424 00.</td></tr>
<tr><td>Protection</td><td colspan="2"></td><td>Over-current, over-voltage</td></tr>
<tr><td>Surge withland</td><td colspan="2"></td><td>2.3 x max. rated input voltage for 1.3 ms</td></tr>
</table>

# Chapter 22
# 140 CPS 414 00: 48 VDC Standalone/Summable 7–8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 414 00 Power Supply module.

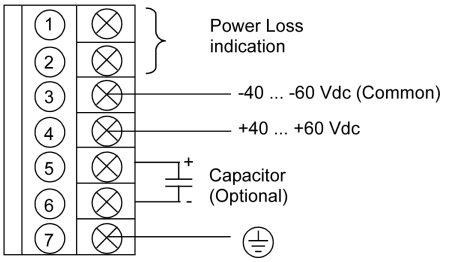
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 414 00 Wiring Diagram | 286 |
| 140 CPS 414 00 Specifications | 288 |

# 140 CPS 414 00 Wiring Diagram

## Wiring Diagram Figure




**NOTE:** 1. A normally closed relay contact rated at 220 Vac, 6 A / 30 Vdc, 5 A is available on terminals 1 and 2 on the power terminal strip. This contact set may be used to signal input power OFF or a power supply loss. The relay de-energizes when input power drops below 35 Vdc.
2. Tolerance to input interruptions may be increased by adding a ≥ 85 Vdc electrolytic capacitor between terminals 5 and 6 on the power terminal strip. Refer to the hold-up capacitor timing chart for capacitor values in the specification *Input, page 288*.
3. When field wiring the power supply module, the maximum wire size that should be used is 1 - 14 AWG (2 $mm^2$) or 2 - 6 AWG (1.5 $mm^2$).

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

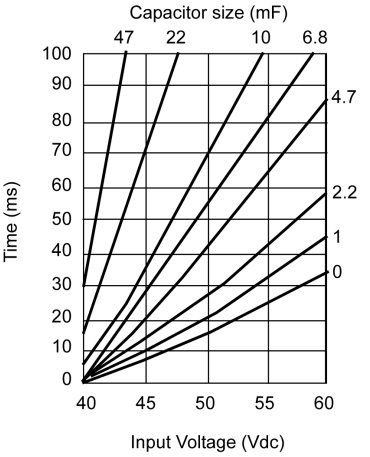
| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 414 00 Specifications

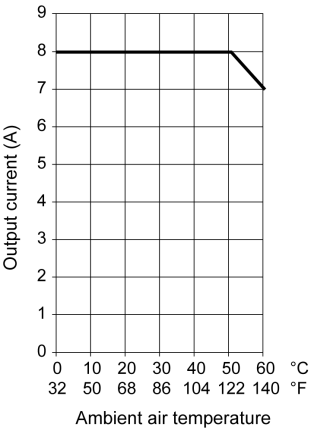
## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Summable |
| Internal power dissipation (in W) | 6 + (1.0 x $I_{out}$) |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 40 ... 60 Vdc |
| Input current | 1.2 A @ 48 Vdc |
| Inrush current | 25 A @ 48 Vdc |
| Input power interruption | 13 ms max. @ 48 Vdc (without capacitor)<br>Hold-up capacitor timing chart (with a capacitor):<br>  <br> |
| Fusing (external) | 2.5 A slow-blow |

## Output to Bus

| | |
|---|---|
| Voltage | 5.1 Vdc |
| Max. current | Operating curve:<br>  <br> |
| Protection | Over-current, over-voltage |

# Chapter 23
# 140 CPS 424 00: 48 VDC Standalone/Redundant 6–8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 424 00 Power Supply module.

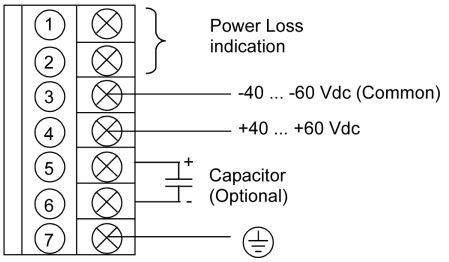
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 424 00 Wiring Diagram | 292 |
| 140 CPS 424 00 Specifications | 294 |

# 140 CPS 424 00 Wiring Diagram

## Wiring Diagram Figure




**NOTE:** 1. A normally closed relay contact rated at 220 Vac, 6 A / 30 Vdc, 5 A is available on terminals 1 and 2 of the power terminal strip. This contact set may be used to signal input power OFF or a power supply loss.
2. Tolerance to input interruptions may be increased by adding a ≥ 85 Vdc electrolytic capacitor between terminals 5 and 6 on the power terminal strip. Refer to the hold-up capacitor timing chart for capacitor values in the specification *Input, page 294*
3. When field wiring the power supply module, the maximum wire size that should be used is 1 - 14 AWG (2 $mm^2$) or 2 - 16 AWG (1.5 $mm^2$).

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

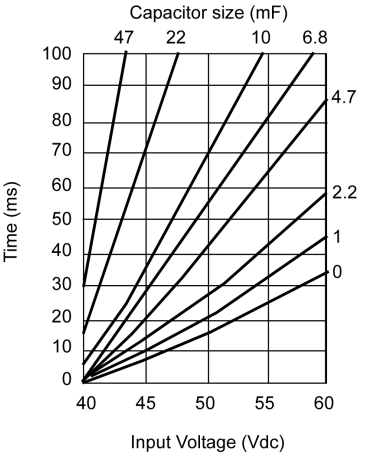
| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 424 00 Specifications

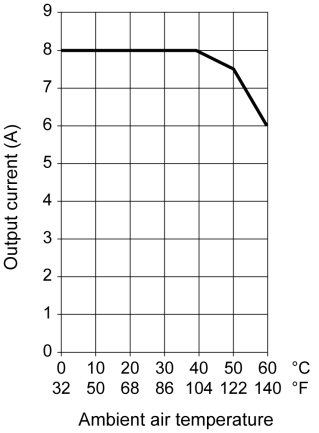
## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Redundant |
| Internal power dissipation (in W) | 7 + (1.5 x $I_{out}$)[1] |
| Network power consumption | Electric losses + (5.1 x $I_{out}$) |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 40 ... 60 Vdc |
| Input current | 1.3 A @ 48 Vdc |
| Inrush current | 25 A @ 48 Vdc |
| Input power interruption | 13 ms max. @ 48 Vdc (without capacitor)<br>Hold-up capacitor timing chart (with a capacitor):<br>  <br> |
| Fusing (external) | 2.5 A slow-blow |

## Output to Bus

<table>
<tr><td>Voltage</td><td colspan="2"></td><td>5.1 Vdc</td></tr>
<tr><td rowspan="11">Max. Current</td><td>Standalone</td><td>Operating curve:<br> 
</td><td>8 A @ 40 °C (104 °F)<br>6 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="10">Redundant</td><td rowspan="2">a) Two 140 CPS x24 00</td><td>8 A @ 40 °C (104 °F)</td></tr>
<tr><td>6 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">b) Three 140 CPS x24 00</td><td>16 A @ 40 °C (104 °F)</td></tr>
<tr><td>12 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">c) One 140 CPS 224 00 and one 140 CPS x24 00</td><td>8 A @ 40 °C (104 °F)</td></tr>
<tr><td>6 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">d) One 140 CPS 224 00 and two 140 CPS x24 00</td><td>16 A @ 40 °C (104 °F)</td></tr>
<tr><td>12 A @ 60 °C (140 °F)</td></tr>
<tr><td rowspan="2">e) Two 140 CPS 224 00 and one 140 CPS x24 00</td><td>16 A @ 40 °C (104 °F)</td></tr>
<tr><td>12 A @ 60 °C (140 °F)</td></tr>
<tr><td colspan="3">140 CPS x24 00 means 140 CPS 224 00 or 140 CPS 424 00.</td></tr>
<tr><td>Protection</td><td colspan="2"></td><td>Over-current, over-voltage</td></tr>
</table>

# Chapter 24
# 140 CPS 511 00: 125 VDC Standalone 3 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 511 00 Power Supply module.

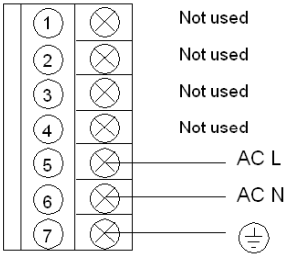
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 511 00 Wiring Diagram | 298 |
| 140 CPS 511 00 Specifications | 299 |

# 140 CPS 511 00 Wiring Diagram

## Wiring Diagram




**⚠ DANGER**

**ELECTRIC SHOCK**

Ensure that the installation complies with wiring, grounding and electromagnetic standards defined in:

- national and local electrical code requirements,
- Grounding and Electromagnetic Compatibility of PLC Systems *(see Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual)* user manual.

**Failure to follow these instructions will result in death or serious injury.**

# 140 CPS 511 00 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| Internal power dissipation (in W) | $2 + (1.8 \times I_{out})^1$ |
| Network power consumption | Electric losses + $(5.1 \times I_{out})$ |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 100 ... 150 VDC (including ripple) |
| Input current | 0.4 A |
| Inrush current | 10 A |
| Input power interruption | 1 ms max. |
| Fusing (external) | 1.5 A slow-blow |

## Output to Bus

| | |
|---|---|
| Voltage | 5.1 VDC |
| Max. current | 3 A |
| Protection | Over-current, over-voltage |

# Chapter 25
# 140 CPS 524 00: 125 VDC Standalone/Redundant 8 A Power Supply Module

## Purpose

This chapter provides information on the 140 CPS 524 00 Power Supply module.

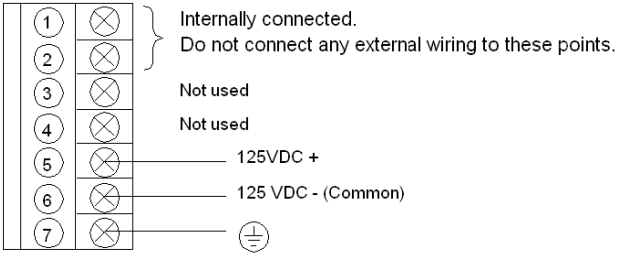
## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| 140 CPS 524 00 Wiring Diagram | 302 |
| 140 CPS 524 00 Specifications | 303 |

# 140 CPS 524 00 Wiring Diagram

## Wiring Diagram Figure




**NOTE:** When field wiring the power supply module, the maximum wire size that should be used is 1 - 14 AWG (2 $mm^2$) or 2 - 16 AWG (1.5 $mm^2$).

**⚠ WARNING**

**UNINTENDED EQUIPMENT OPERATION**

- Use only compatible combinations of power supplies in the Quantum backplane as indicated in the following compatibility matrix.
- Remove line power from this module prior to insertion or removal.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## Compatibility Matrix

For proper operation, refer to this table:

| Compatibility matrix | | | | | |
|---|---|---|---|---|---|
| Summable models | 140CPS11420 | 140CPS11410 | 140CPS21400 | 140CPS41400 | – |
| 140CPS11420 | Yes | Yes | No | No | – |
| 140CPS11410 | Yes | Yes | No | No | – |
| 140CPS21400 | No | No | Yes | Yes | – |
| 140CPS41400 | No | No | Yes | Yes | – |
| Redundant models | 140CPS12420 | 140CPS12400 | 140CPS22400 | 140CPS42400 | 140CPS52400 |
| 140CPS12420 | Yes | No | Yes | Yes | No |
| 140CPS12400 | No | Yes | No | No | Yes |
| 140CPS22400 | Yes | No | Yes | Yes | No |
| 140CPS42400 | Yes | No | Yes | Yes | No |
| 140CPS52400 | No | Yes | No | No | Yes |

# 140 CPS 524 00 Specifications

## General Specifications

| | |
|---|---|
| Operating mode | Standalone |
| | Redundant |
| Internal power dissipation (in W) | 13 W |
| Network power consumption | Electric losses + (5.1 x $I_{out}$)[1] |
| [1]: $I_{out}$: sum of currents consumption by all modules in rack (in A) | |

## Input

| | |
|---|---|
| Input voltage | 100 ... 150 VDC (including ripple) |
| Input current | 0.5 A @ 125 VDC |
| Inrush current | 28 A @ 125 VDC |
| Input power interruption | 1 ms max. |
| Fusing (external) | 2 A slow-blow |

## Output to Bus

| | | | |
|---|---|---|---|
| Voltage | | | 5.1 VDC |
| Max. Current | Standalone | | 8 A @ 60°C |
| | Redudant | a) Two 140 CPS 524 00 | 8 A @ 60°C |
| | | b) Three 140 CPS 524 00 | 16 A @ 60°C |
| | | c) One 140 CPS 124 00 and one 140 CPS 524 00 | 8 A @ 60°C |
| | | d) One 140 CPS 124 00 and two 140 CPS 524 00 | 16 A @ 60°C |
| | | e) Two 140 CPS 124 00 and one 140 CPS 524 00 | 16 A @ 60°C |
| Min Current | | | Not Required |
| Protection | | | Over-current, over-voltage |

# Part IV
# Racks(XBP) and Rack Expander (XBE)

## Introduction

This part provides information on the Quantum racks and rack expander.

## What Is in This Part?

This part contains the following chapters:


| Chapter | Chapter Name | Page |
|---|---|---|
| 26 | Selecting Racks (XBP) | 307 |
| 27 | 140 XBE 100 00: Rack Expander | 315 |

# Chapter 26
# Selecting Racks (XBP)

## Purpose

This chapter provides information on selecting Quantum Racks.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| General Information | 308 |
| 140 XBP 002 00: Two Position Rack | 309 |
| 140 XBP 003 00: Three Position Rack | 310 |
| 140 XBP 004 00: Four Position Rack | 311 |
| 140 XBP 006 00: Six Position Rack | 312 |
| 140 XBP 010 00: Ten Position Rack | 313 |
| 140 XBP 016 00: Sixteen Position Rack | 314 |

# General Information

## Overview

Racks are designed to mechanically fasten and electrically connect modules used in drops. The rack contains a passive circuit board which permits modules to communicate with each other and to identify their slot numbers without further switch settings.

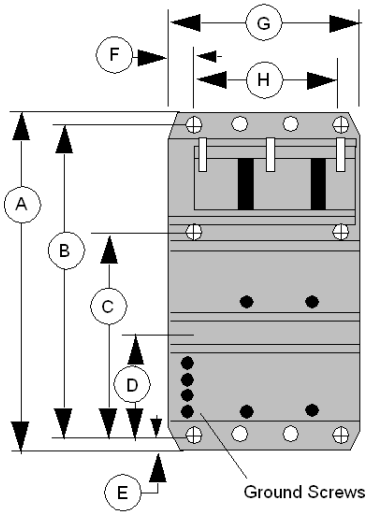
**NOTE:** To meet vibration/shock specifications, the rack must be mounted using all specified mounting holes. The rack is mounted using standard hardware (described below).

The recommended length for the mounting screws should be within the following range: 0.24 in. (6 mm) - 0.52 in. (13 mm)

The head height of the screws should not exceed 0.14 in. (3.5 mm).

# 140 XBP 002 00: Two Position Rack

## Two Position Rack




⊕ =Mounting Hole
Diameter: 8 mm (0.31 in)

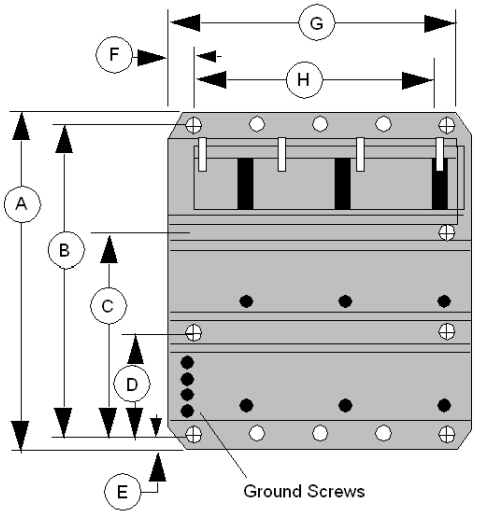
○ =Optional locations for Modbus Plus communication cable grounding.
Diameter: 8 mm (0.31 in)

● =Threaded mounting holes for half and full height modules.
Diameter: 4 mm (0.16 in)

**A** 290 mm (11.42 in)
**B** 270 mm (10.63 in)
**C** 175.5 mm (6.91 in)
**D** 94.5 mm (3.72 in)
**E** 10 mm (0.39 in)
**F** 15 mm (0.59 in)
**G** 102.61 mm (4.04 in)
**H** 72.44 mm (2.85 in)

# 140 XBP 003 00: Three Position Rack

## Three Position Rack




⊕ =Mounting Hole
Diameter: 8 mm (0.31 in)

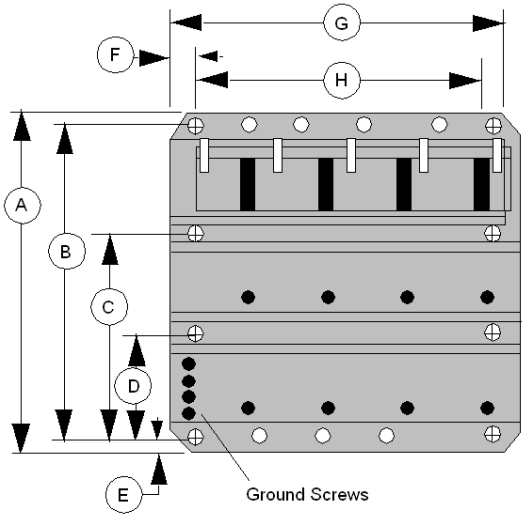
○ =Optional locations for Modbus Plus communication cable grounding.
Diameter: 8 mm (0.31 in)

● =Threaded mounting holes for half and full height modules.
Diameter: 4 mm (0.16 in)

**A** 290 mm (11.42 in)
**B** 270 mm (10.63 in)
**C** 175.5 mm (6.91 in)
**D** 94.5 mm (3.72 in)
**E** 10 mm (0.39 in)
**F** 15 mm (0.59 in)
**G** 143.13 mm (5.64 in)
**H** 113.08 mm (4.45 in)

# 140 XBP 004 00: Four Position Rack

## Four Position Rack




⊕ =Mounting Hole
Diameter: 8 mm (0.31 in)

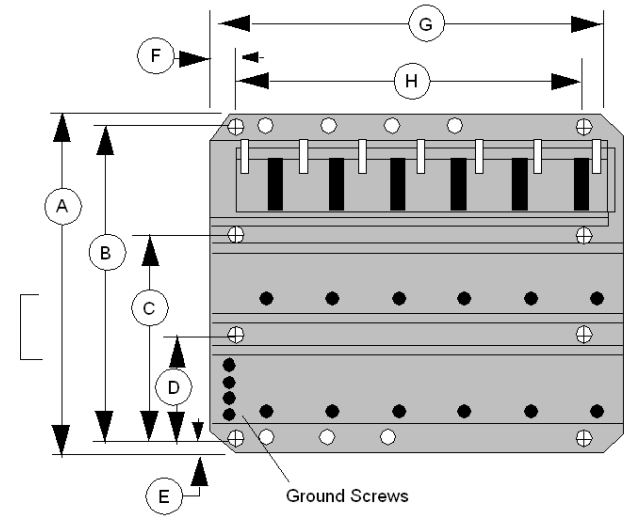
○ =Optional locations for Modbus Plus communication cable grounding.
Diameter: 8 mm (0.31 in)

● =Threaded mounting holes for half and full height modules.
Diameter: 4 mm (0.16 in)

**A** 290 mm (11.42 in)
**B** 270 mm (10.63 in)
**C** 175.5 mm (6.91 in)
**D** 94.5 mm (3.72 in)
**E** 10 mm (0.39 in)
**F** 15 mm (0.59 in)
**G** 183.69 mm (7.23 in)
**H** 153.72 mm (6.05 in)

# 140 XBP 006 00: Six Position Rack

## Six Position Rack




⊕ =Mounting Hole
Diameter: 8 mm (0.31 in)

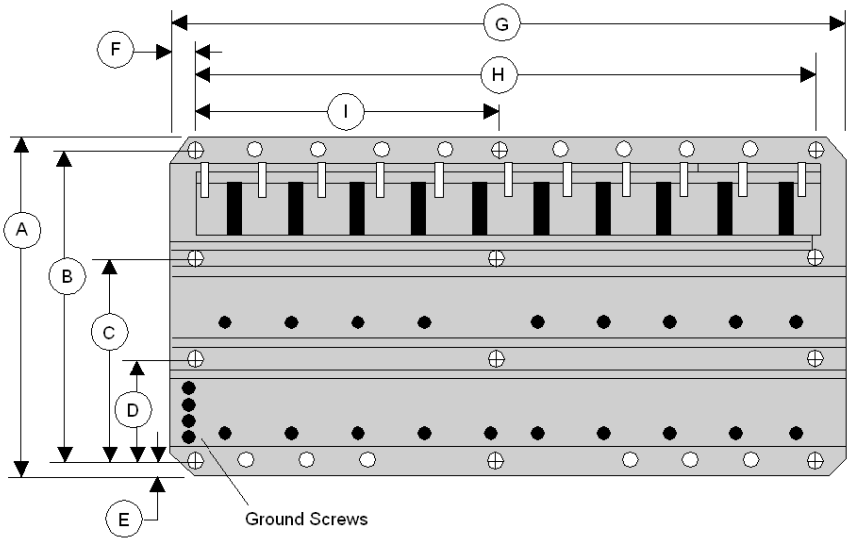
○ =Optional locations for Modbus Plus communication cable grounding.
Diameter: 8 mm (0.31 in)

● =Threaded mounting holes for half and full height modules.
Diameter: 4 mm (0.16 in)

**A** 290 mm (11.42 in)
**B** 270 mm (10.63 in)
**C** 175.5 mm (6.91 in)
**D** 94.5 mm (3.72 in)
**E** 10 mm (0.39 in)
**F** 15 mm (0.59 in)
**G** 265.1 mm (10.44 in)
**H** 235 mm (9.25 in)

# 140 XBP 010 00: Ten Position Rack

## Ten Position Rack




⊕ =Mounting Hole
Diameter: 8 mm (0.31 in)

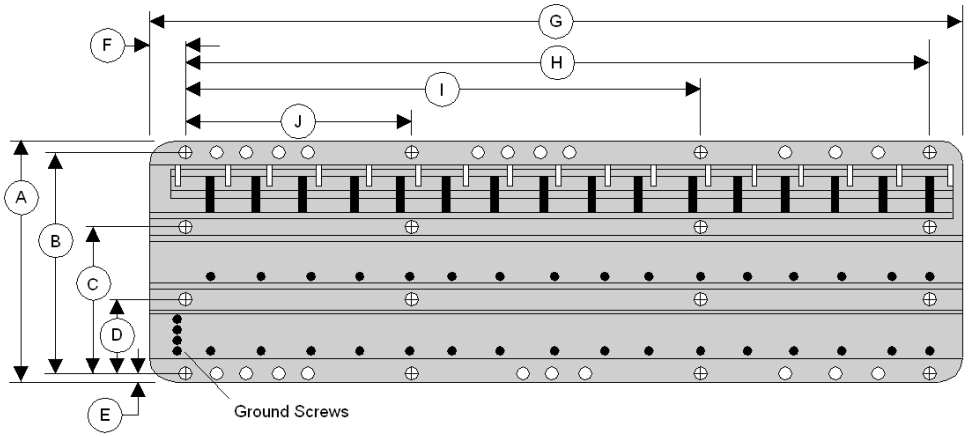
○ =Optional locations for Modbus Plus communication cable grounding.
Diameter: 8 mm (0.31 in)

● =Threaded mounting holes for half and full height modules.
Diameter: 4 mm (0.16 in)

**A** 290 mm (11.42 in)
**B** 270 mm (10.63 in)
**C** 175.5 mm (6.91 in)
**D** 94.5 mm (3.72 in)
**E** 10 mm (0.39 in)
**F** 15 mm (0.59 in)
**G** 427.66 mm (16.84 in)
**H** 397.56 mm (15.65 in)
**I** 198.78 mm (7.82 in)

# 140 XBP 016 00: Sixteen Position Rack

## Sixteen Position Rack




⊕ =Mounting Hole
Diameter: 8 mm (0.31 in)

○ =Optional locations for Modbus Plus communication cable grounding.
Diameter: 8 mm (0.31 in)

● =Threaded mounting holes for half and full height modules.
Diameter: 4 mm (0.16 in)

**A** 290 mm (11.42 in)
**B** 270 mm (10.63 in)
**C** 175.5 mm (6.91 in)
**D** 94.5 mm (3.72 in)
**E** 10 mm (0.39 in)
**F** 15 mm (0.59 in)
**G** 670.74 mm (26.42 in)
**H** 641.4 mm (25.25 in)
**I** 427.6 mm (16.83 in)
**J** 213.8 mm (8.42 in)

# Chapter 27
# 140 XBE 100 00: Rack Expander

## Purpose

This chapter provides information on the 140 XBE 100 00 module.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| General Information | 316 |
| Presentation | 318 |
| Operation Guidelines | 319 |
| 140 XBE 100 00 Specifications | 320 |

# General Information

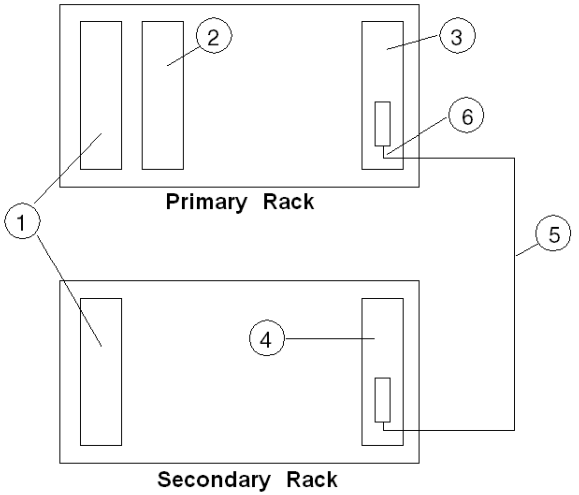
## Function

The Rack Expander acts as a repeater for data signals from a primary Quantum rack. Since power is not provided to the secondary rack over the expander cable, power supplies are required in both racks.

The Rack Expander does not have any LEDs to display its status. The active LEDs of the modules in the secondary rack and their associated diagnostic status bits indicate proper operation, including the Rack Expander.

## Basic Configuration

The rack containing the CPU or RIO drop adapter is called the Primary rack, and the adjacent rack is designated as the Secondary rack. Each rack requires its own power supply.




**1** Power Supplies
**2** CPU or RIO Adapter
**3** First Rack Expander (140 XBE 100 00)
**4** Second Rack Expander (140 XBE 100 00)
**5** Rack Expander Cable (140 XCA 717 0•)
**6** Cable end marked as "Primary"

**NOTE:** Install the cable when the racks are powered OFF.

| ⚠ CAUTION |
| --- |
| **UNINTENDED EQUIPMENT OPERATION**<br>● The communication cable is polarized. The cable end marked "Primary" must be connected to the rack that includes the CPU or RIO adapter.<br>● The communications cable must be connected to the Rack Expander module before it is hot swapped into a powered rack.<br>● Do not hot swap Rack Expanders modules while modules in secondary rack are operating.<br>**Failure to follow these instructions can result in injury or equipment damage.** |

## Features

- Enhances system performance requiring fewer remote I/O drops for the CPU to service.
- Enables a Quantum CPU to handle up to 27,648 discrete I/O.
- Makes it easy to plan for future I/O expansion by leaving an I/O slot empty for module insertion.
- Cost effective for local I/O by eliminating the need for a remote I/O head and drop.
- Cost effective by eliminating the need for an additional remote I/O drop.

## Part Numbers

The following table provides necessary part numbers.

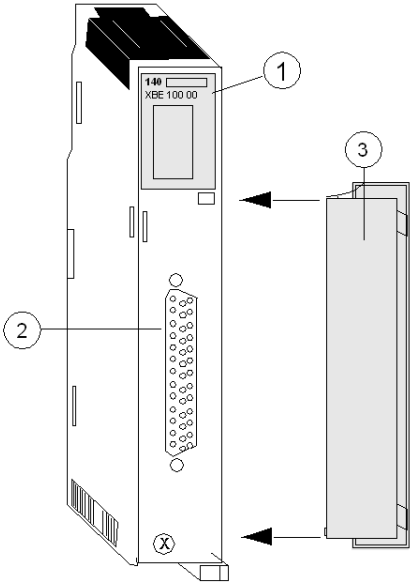
| | |
| --- | --- |
| Rack Expander | 140 XBE 100 00 |
| Rack Expander (conformally coated) | 140 XBE 100 00C |
| Expander Cable, 3' (1 m) | 140 XCA 717 03 |
| Expander Cable, 6' (2 m) | 140 XCA 717 06 |
| Expander Cable, 9' (3 m) | 140 XCA 717 09 |

# Presentation

## Function

With the 140 XBE 100 00 Rack Expander you can add a second rack to a local or remote drop. A custom communications cable, 3.0 m (9.84 ft) maximum, provides the data communication transfer.

## Rack Expander




**1** Module number, module description and color code
**2** Connector
**3** Removable door

## Operation Guidelines

### Rack Expander Guidelines

- The same 140 XBE 100 00 Rack Expander modules are used for the primary and secondary racks. The end of the Rack Expander cable marked "Primary" always connects to the Rack Expander module in the primary rack.
- The system can use any Quantum type power supply. Each rack can have a different type of power supply.
- Loss of power in the secondary rack will not shut down the entire drop. Only modules located in the "Secondary" rack will lose power.
- Rack expander modules can be located in any slot in the rack and do not have to be placed in corresponding slots in the primary and secondary racks.
- I/O modules that have downloadable executive firmware, such as the ESI module, are allowed in the secondary rack—except when downloading their execs. Executive firmware cannot be downloaded to modules in the secondary rack.
- It may be necessary to update the CPU or RIO drop executive firmware. See the Executive Firmware section in Specifications.
- The Rack Expander will not be recognized by the programming panel software. It will look like an unfilled slot in the PLC I/O map.
- The Rack Expander will allow configuration or I/O mapping of additional modules in the local drop containing a CPU or RIO drop adapter—up to the drop word limit or physical slot address limitation.
- Option modules, such as NOMs, NOEs and CHSs must reside in the primary rack.
- Any Interrupt module can be located in the secondary rack, but the interrupt mode is not supported.

If the primary rack power supply goes **OFF**, and the secondary rack power supply remains **ON**, the behavior of the outputs on the secondary rack is not guaranteed. Outputs could either go to the *Off* state or adopt their *Timeout* state as defined by the user.

If the power supply of the secondary rack goes **OFF**, and the primary rack power supply remains **ON**, the **I/O Health bit** of the modules located in this rack could indicate a healthy module.

**⚠ CAUTION**

**UNINTENDED EQUIPMENT OPERATION - UNKNOWN OUTPUT STATE**

- The Timeout State must be configured as USER DEFINED with a VALUE of 0 for all modules in the secondary rack regardless of the configuration.
- The same primary power source must be used for the primary and secondary racks.

**Failure to follow these instructions can result in injury or equipment damage.**

# 140 XBE 100 00 Specifications

## General Specifications

| | |
|---|---|
| Number of connected racks | 2 |
| Maximum distance | 3 m |
| rack requirements | For rack sizes (3-, 4-, 6-, 10- and 16-slot) |
| Slots used | 1 |
| Number of rack expander modules allowed | 1/rack |
| LEDs | None |
| Required addressing | The rack expander looks like an unfilled slot in the PLC I/O map. |
| Power consumption | 2.5 W |
| Bus current required | 500 mA |
| Connector | 3-pin D-type |

## Compatibility

| | |
|---|---|
| Primary rack | No restrictions |
| Secondary rack | Quantum I/O modules can be used in the secondary rack, unless otherwise noted in the I/O documentation. |
| Executive firmware | 140 CPU •13 0• - Version 2.2 |
| | 140 CPU •34 1• - Version 1.03 |
| | 140 CPU •34 1•A - Any version |
| | 140 CPU 424 02 - Version 2.15 |
| | 140 CPU 6•1 •0 - Any version |
| | 140 CRA 93• 0• - Version 1.2 |

## Max. Words/Drop

| | |
|---|---|
| Local I/O | 64 in / 64 out<br>**Note:**For Unity CPU (for example 140 CPU 65150) there is no limitation of I/O words on local rack. |
| Remote I/O | 64 in / 64 out |

# Part V
# Simulator (XSM) and Battery (XCP) Module

## Introduction

This part provides information on Quantum Simulator (XSM) and Battery (XCP) Module.

## What Is in This Part?

This part contains the following chapters:


| Chapter | Chapter Name | Page |
|---|---|---|
| 28 | Simulator Modules (XSM) | 323 |
| 29 | 140 XCP 900 00: Battery Module | 333 |

# Chapter 28
# Simulator Modules (XSM)

## Purpose

This chapter provides information on Quantum Simulator Modules (XSM).

## What Is in This Chapter?

This chapter contains the following sections:


| Section | Topic | Page |
|---|---|---|
| 28.1 | 140 XSM 010 00: Two Channels In / One Channel Out Analog Simulator | 324 |
| 28.2 | 140 XSM 002 00: 16 Point Discrete Input Simulator | 330 |

# Section 28.1
# 140 XSM 010 00: Two Channels In / One Channel Out Analog Simulator

## Purpose

This section provides information on the 140 XSM 010 00 analog simulator.

## What Is in This Section?

This section contains the following topics:


| Topic | Page |
|---|---|
| Presentation | 325 |
| 140 XSM 010 00 Wiring Diagram | 327 |
| 140 XSM 010 00 Specifications | 329 |

# Presentation

## Function

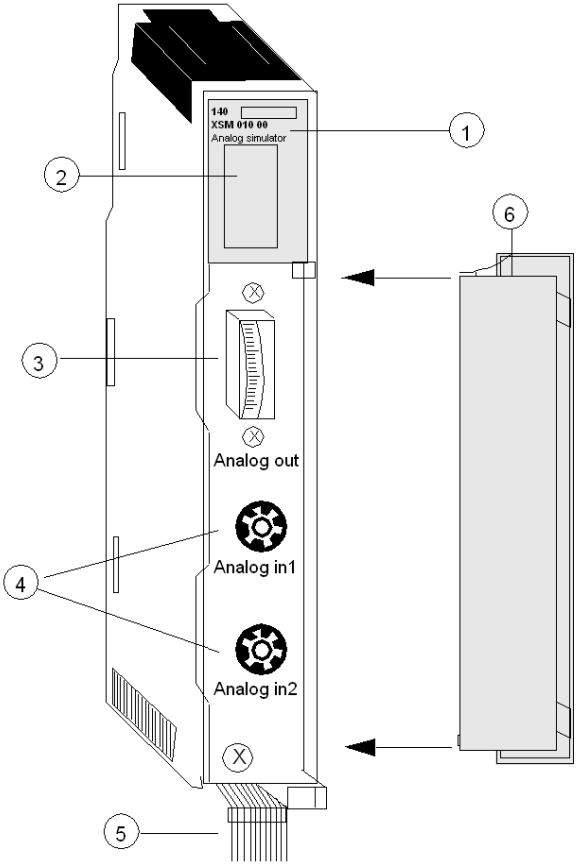
The 140 XSM 010 00 module simulates 4 ... 20 mA field current loops. It provides two adjustable 4 ... 20 mA analog signals and one fixed 24 VDC output. The simulator also measures and displays voltages from 0 ... 5 VDC.

The simulator module includes the following:

- An internal 24 VDC power supply
- A 0 ... 5 VDC meter
- Two 10-turn potentiometers

**NOTE:** The 140 XSM 010 00 is not a field-operational module. It should be used only for testing, simulating and calibrating Quantum current input modules.

## Illustration



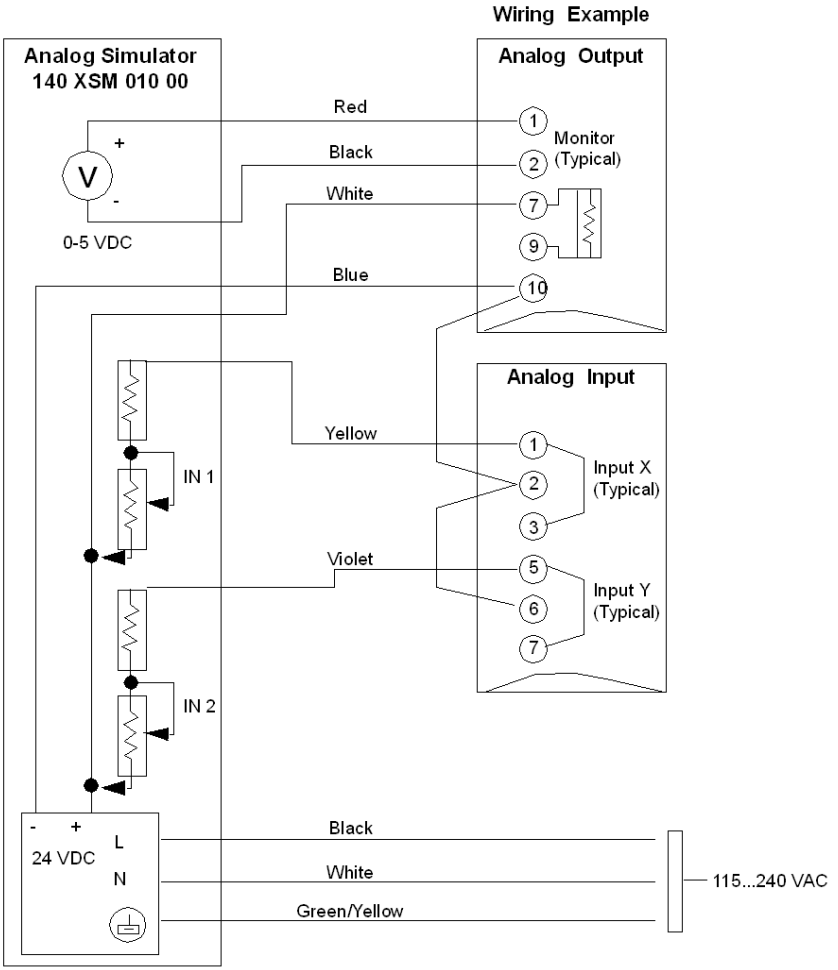

1 Model number, module description, color code
2 LED display
3 Voltmeter
4 Potentiometers
5 Color-coded leads
6 Removable door

**NOTE:** The 140 XSM 010 00 can be placed in any slot in the Quantum.

# 140 XSM 010 00 Wiring Diagram

## Sample Wiring Diagram

The following diagram shows a typical connection between the simulator, a 140 ACI 030 00 input module and a 140 ACO 020 00 output module. The simulator provides a variable 4 ... 20 mA input to the input module. The input can then be read by a Quantum CPU, and, if required, sent to an output module. For the output module to operate properly, the main current loop must be active, with 24 VDC supplied between terminals 9 and 10 with a 249 Ω voltage drop resistor.




**140 XSM 010 00 Generic Wiring Diagram for the 140 A•I 030 00 Input Modules, 140 A•O 020 00 Output Modules, and the 140 AMM 090 00 Input/Output Module**

**NOTE:** For a more detailed description of these modules, refer to the Discrete and Analog I/O Reference Manual Quantum I/O Analog IN Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*, Quantum I/O Analog OUT Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* and Quantum I/O Analog IN / OUT Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*.

# 140 XSM 010 00 Specifications

## Specifications

| | | |
|---|---|---|
| Voltage | Operating | 100 ... 240 VAC, 50/60 Hz |
| | Continuous output | 24 VDC, 400 mA max |
| Operating current | | 300 mA @ 120 VAC |
| Voltmeter range | | 0 ... 5 VDC |
| 10-turn potentiometer output | variable current | 4 ... 20 mA |
| | variable voltage | 1 ... 5 VDC |
| Internal fusing | | None |
| Bus current required | | None |

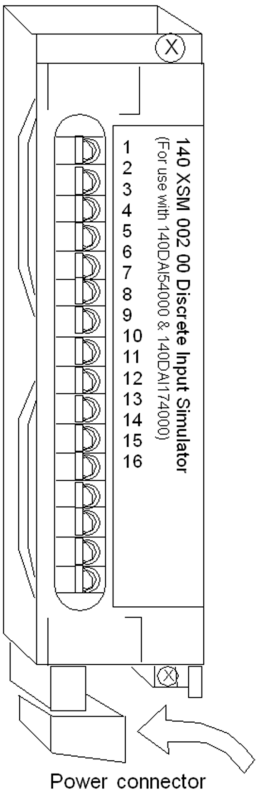
# Section 28.2
## 140 XSM 002 00: 16 Point Discrete Input Simulator

## Presentation

### Function

The 140 XSM 002 00 simulator consists of 16 toggle switches that can generate up to 16 binary input signals to the 140 DAI 540 00 and the 140 DAI 740 00 AC input modules.

### Illustration




| Switch position | Function |
|---|---|
| Right | Momentary signal |
| Middle | Off |
| Left | Constant signal |




Power connector label

**NOTE:** Voltage source range is 24 ... 240 VAC.

## ⚠ DANGER

**HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH**

When using this simulator module with the 140 DAI 540 00 or 140 DAI 740 00 input modules, do not to come in contact with the supplied 115 or 230 VAC located at the bottom of the simulator module.

**Failure to follow these instructions will result in death or serious injury.**

# Chapter 29
# 140 XCP 900 00: Battery Module

## Purpose

This chapter provides information on Quantum 140 XCP 900 00 Battery module.

## What Is in This Chapter?

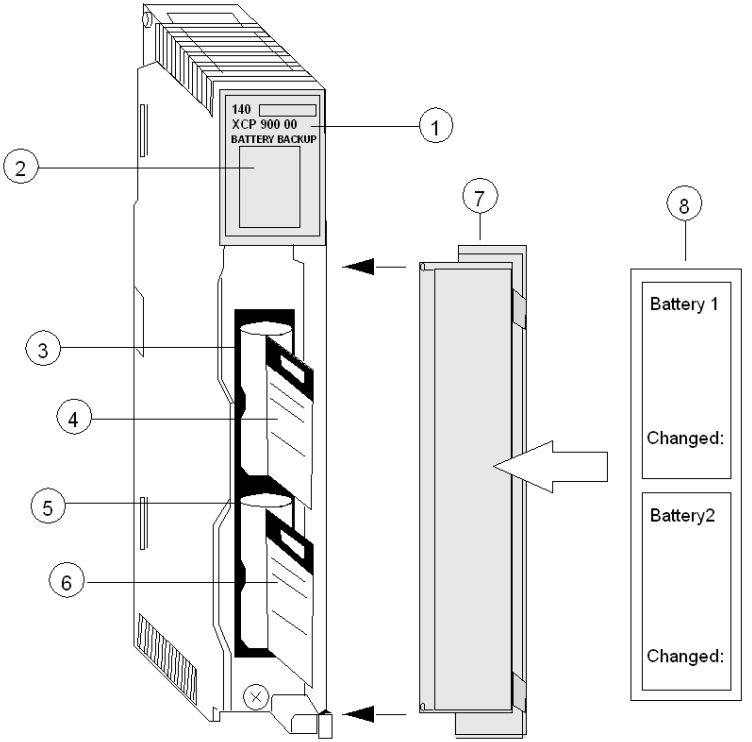
This chapter contains the following topics:


| Topic | Page |
|---|---|
| Presentation | 334 |
| 140 XCP 900 00 Indicators | 336 |
| Installation and Maintenance | 337 |
| 140 XCP 900 00 Specifications | 339 |

# Presentation

## Illustration

The following figure shows the battery module components.




**1** Model Number, Module Description, Color Code
**2** LED Display
**3** Battery Slot 1 (with battery installed)
**4** Battery Access Tape (for removal/replacement)
**5** Battery Slot 2 (with battery installed)
**6** Battery Access Tape (for removal/replacement)
**7** Removable Door
**8** Customer Identification Label (Fold label and place it inside door)

## Battery Backup

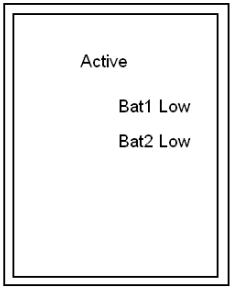
The 140 XCP 900 00 provides RAM backup power for expert modules. One non-rechargeable 3.6 V lithium battery is provided and is accessible from the front of the module in Battery Slot 1 (the upper slot) for easy removal when it is necessary to change it.

**NOTE:** Extended backup is provided when a second battery is installed in Battery Slot 2 (the lower slot).

# 140 XCP 900 00 Indicators

## Illustration

The battery module has three LED indicators.




## Description

| LEDs | Color | Indication when on |
|---|---|---|
| Active | Green | Bus communication is present. |
| Bat1 Low | Red | Battery 1 voltage is low. |
| Bat2 Low | Red | Battery 2 voltage is low. |

**NOTE:** The Bat1 Low and Bat2 Low LEDs turn on when a battery is:

- not installed
- installed backwards
- or in need of replacement

# Installation and Maintenance

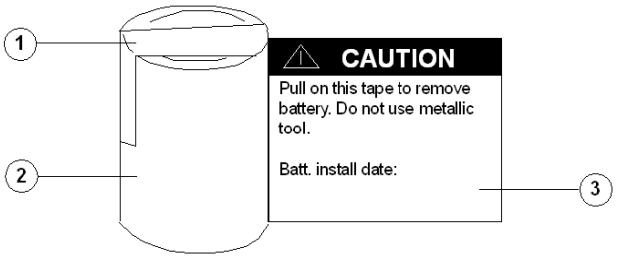
## Battery Installation and Replacement Considerations

The following procedure describes the installation of a battery.

| Step | Action |
|---|---|
| 1 | Remove the insulating strip from the plus (+) pole of the battery before inserting it into the module. This strip is used to insulate the battery when on the shelf. **Note:** The battery installed in the module, when shipped, has the insulating strip in place. Remove this strip and re-install the battery before operation. |
| 2 | When single battery backup is required, install the battery in Battery Slot 1. The circuitry is designed so Battery 1 supplies the current until it is used up. Battery 2 (when installed) then assumes the load requirement without interruption. Battery status is indicated via LEDs and status bytes. |
| 3 | When the controller is in operation, the batteries can be replaced at any time. **Note:** When the controller is powered OFF, battery replacement can be done without RAM loss only when a second functioning battery is installed. |

## Battery

This diagram shows the battery




**1** Insulating Strip
**2** Battery
**3** Access strip

## Installing/Removing a Battery

The following procedure describes how to install or remove a battery.

| Step | Action |
|---|---|
| 1 | Remove the insulating strip from the new battery. |
| 2 | If necessary, remove the old battery. Detach it from its housing (on the front of the module), by pulling the battery access strip until the battery pops out. |
| 3 | Replace it with the new battery using the reverse of the procedure in step 2. |

**⚠ CAUTION**

**EQUIPMENT DAMAGE**

- Use insulated objects to remove or replace a battery in this module.
- Maintain proper polarity when connecting and inserting new batteries into the 140 XCP 900 00.

**Failure to follow these instructions can result in injury or equipment damage.**

# 140 XCP 900 00 Specifications

## Specifications

| | |
|---|---|
| Battery type | C, 3 V lithium |
| Maximum load current | 100 mA |
| Service life | 5.5 Ah |
| Shelf life | 10 years with 0.5% loss of capacity/year |
| Battery part number | 990 XCP 990 00 |

## Service life

To determine the service life of a specific configuration, the current consumption of all available expert modules must be summed, refer to the following table. Place the value as determined in the equation below. The result will be the expected service life in days for a given system configuration.

$$ServiceLife = \frac{5.5Ah}{24h \times totalcurrent\,(inA)} = \frac{1}{4 \times I}$$

Where I is the total battery current load of all modules in the rack (in A).

The following table shows the modules and their associated current consumption:

| Module | Current Consumption (A) |
|---|---|
| ERT 854 00 | 70 |
| ESI 062 00 | 150 |
| ESI 062 10 | 150 |
| MMB 102 00 | 200 |
| MMB 104 00 | 200 |
| MMD 102 00 | 200 |
| MMD 104 00 | 200 |

## Calculating Service life example

Considering that the XCP 900 00 battery box provides battery backup for the RAM ESI 062 10 and two MMB 102 00 modules.

The service life can be calculated as follow:

$$\frac{5.5Ah}{24h \times (150 \times 10^{-6} + 2 \times 200 \times 10^{-6})A} = 417 days$$

**NOTE:** If a module is operated from two batteries, the service life is also doubled.

# Appendices



## Introduction

This appendices provide additional information on the Quantum Automation System.

## What Is in This Appendix?

The appendix contains the following chapters:


| Chapter | Chapter Name | Page |
|---|---|---|
| A | Miscellaneous Components | 343 |
| B | Power and Grounding Guidelines | 357 |
| C | Field Wiring Terminal Strip / Module Keying Assignment | 387 |
| D | CableFast Cabling | 395 |
| E | ConneXium Ethernet Cabling System | 469 |
| F | Agency Approvals and Conformal Coating | 475 |
| G | System Specifications | 487 |

# Appendix A
# Miscellaneous Components

## Introduction

This appendix provides information on the various miscellaneous components of the Quantum system, with illustrations and descriptions.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Racks and Brackets Tables | 344 |
| Cables | 345 |
| Spare Parts Table | 346 |
| Illustrated Miscellaneous Components | 347 |

# Racks and Brackets Tables

## Racks

| Part Number | Module Slots | Weight |
|---|---|---|
| 140 XBP 002 00 | 2 | 0.23 kg (0.5 lb) |
| 140 XBP 003 00 | 3 | 0.34 kg (0.75 lb) |
| 140 XBP 004 00 | 4 | 0.45 kg (1.0 lb) |
| 140 XBP 006 00 | 6 | 0.64 kg (1.4 lb) |
| 140 XBP 010 00 | 10 | 1.0 kg (2.2 lb) |
| 140 XBP 016 00 | 16 | 1.58 kg (3.5 lb) |

## Mounting Brackets

| Part Number | Description |
|---|---|
| 140 XCP 401 00 | 125 mm (4.92in) Bracket |
| 140 XCP 402 00 | 20 mm (0.79in) Bracket |

# Cables

## Cabling

| Description | Part Number | Length |
|---|---|---|
| RS-232 programming cable | 990 NAA 263 20 | 2.7 m (12 ft) |
| | 990 NAA 263 50 | 15.5 m (50 ft) |
| Modbus Plus trunk cable | 490 NAA 271 01 | 30 m (100 ft) |
| | 490 NAA 271 02 | 152 m (500 ft) |
| | 490 NAA 271 03 | 304 m (1000 ft) |
| | 490 NAA 271 04 | 456 m (1500 ft) |
| | 490 NAA 271 06 | 1520 m (5000 ft) |
| Modbus Plus drop cable | 990 NAD 211 10 | 2.4 m (8 ft) |
| | 990 NAD 211 30 | 6 m (20 ft) |
| Modbus Plus drop cable (left side) | 990 NAD 218 10 | 2.4 m (8 ft) |
| | 990 NAD 218 30 | 6 m (20 ft) |
| Modbus Plus drop cable (right side) | 990 NAD 219 10 | 2.4 m (8 ft) |
| | 990 NAD 219 30 | 6 m (20 ft) |
| Modbus Plus plastic DIO tap | 990 NAD 230 00 | |
| Modbus Plus ruggedized DIO tap | 990 NAD 230 10 | |
| Prefabricated RG-6 S908 RIO drop cable | AS-MBII-003 | 14 m (50 ft) |
| | AS-MBII-004 | 43 m (140 ft) |

# Spare Parts Table

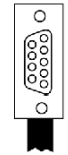
## Miscellaneous Spare Parts Table

The following table shows the miscellaneous spare parts for the Quantum modules.

| Spare Part Number | Description |
|---|---|
| 140 CPS 111 00 | AC Power Supply Door Label |
| 140 CPS 1•4 •0 | AC Power Supply Door Label |
| 140 CPS 211 00 | DC Power Supply Door Label |
| 140 CPS 2•4 00 | DC Power Supply Door Label |
| 140 XTS 005 00 | IP20 7 Pos Connector |
| 140 XTS 001 00 | IP20 40 Pos Connector |

# Illustrated Miscellaneous Components

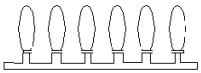
## 99 0 NAD 218•0 Connector Orientation



990NAD218X0

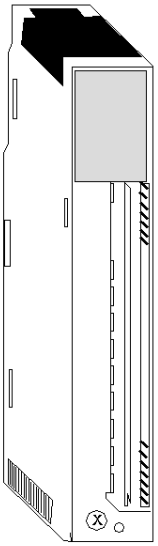
## 140 XCP 200 00 Coding Kit

The coding kit contains six white sets of plastic keys and 12 yellow sets.



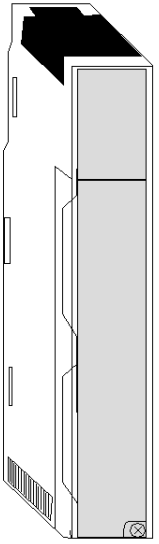
## 140 XCP 500 00 Empty Module

The empty module below is without a terminal strip.



## 140 XCP 510 00 Empty Module

The empty module below is without the terminal strip and with a door cover.



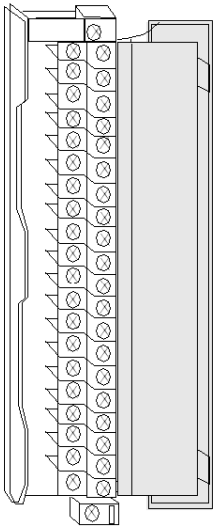
## 140 XCP 600 00 Terminal Strip Jumper Kit

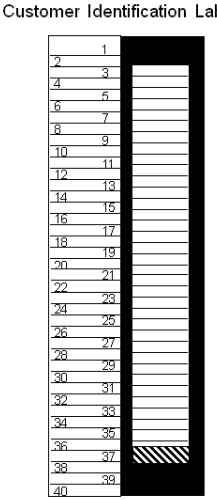
The terminal strip jumper kit contains 12 jumpers.



## 140 XTS 001 00 40-pin Field Wiring Terminal Strip

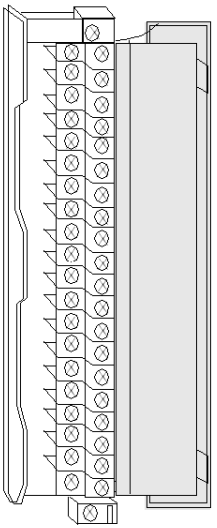
The 40-pin field wiring terminal strip has a screw-down bolted cover.

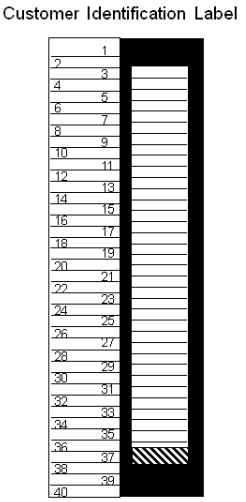




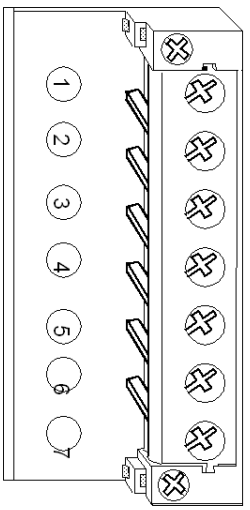
## 140 XTS 002 00 40-pin Field Wiring Terminal Strip

The 140 XTS 002 00 differs from the 140 XTS 001 00 in that it does not have the clear plastic cover over the screw-down bolts.

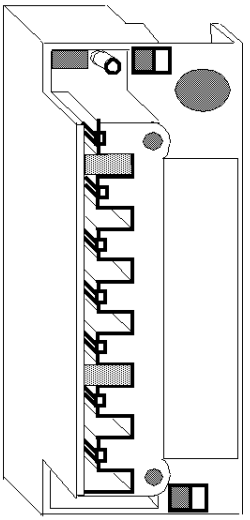




## 140 XTS 005 00 Seven-pin Field Wiring Terminal Strip




Front view

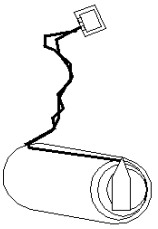


Rear view

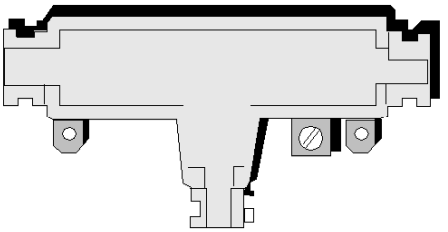
## 990 XCP 900 00 Battery



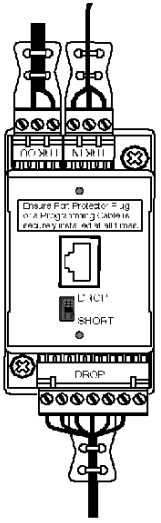
## 990 XCP 980 00 CPU Battery



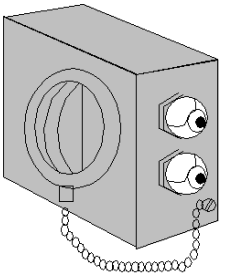
## 990 NAD 230 00 Modbus Plus Tap



## 990 NAD 230 20/21 Modbus Plus Super Tap



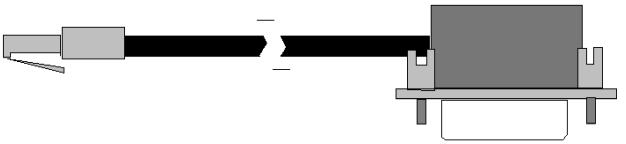
## 990 NAD 230 10 Ruggedized Modbus Plus Tap



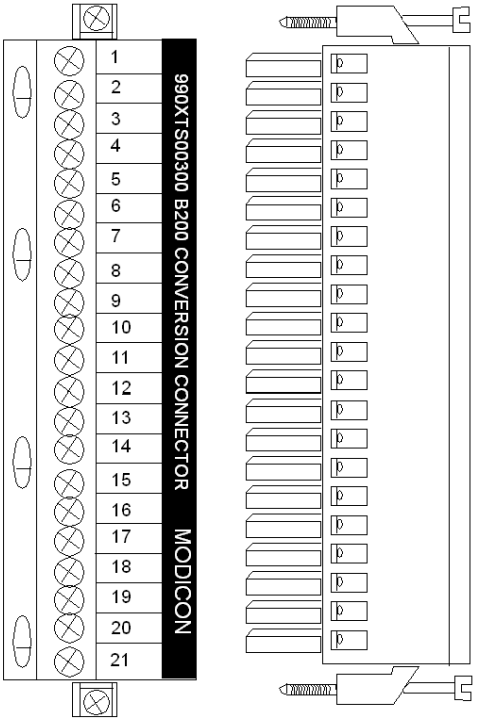
## 990 NAD 230 11 Terminator



## 990 NAA 215 10 Programming Cable

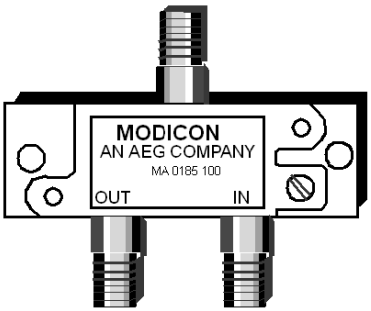


## 990 XTS 003 00 200 Series I/O Conversion Connector

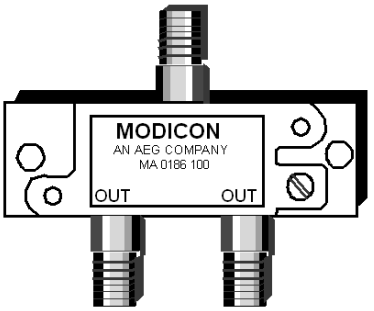



Front View Side View

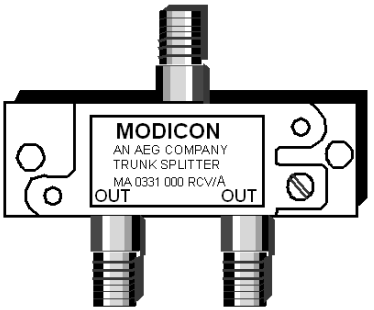
## MA-0185-100 Remote I/O Tap



## MA-0186-100 Remote I/O Splitter

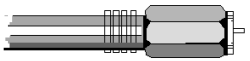



## MA-0331-100 Remote I/O Trunk Splitter



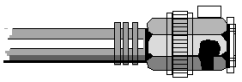

## MA-0329-001/MA-0329-002 Remote I/O F Connectors

The MA-0329-001 F connector supports quad shield RG 6 cable, and the MA-0329-002 F connector supports non-quad shield RG6 cable.



## 043509446/52-0487-000 Remote I/O BNC Connectors

The 043509446 BNC connector supports quad shield RG 6 cable, and the 52-0487-000 BNC connector supports non-quad shield RG 6 cable.



# Appendix B
# Power and Grounding Guidelines

## Introduction

This appendix provides information of power and grounding considerations for AC and DC powered systems, grounding and closed system installation.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| General Information | 358 |
| Batteries as DC power supplies | 360 |
| AC Power and Grounding Considerations | 361 |
| DC Power and Grounding Considerations | 365 |
| Closed System Installation | 371 |
| Modbus Plus Termination and Grounding | 373 |
| Fiber Repeaters | 376 |
| Grounding of RIO Networks | 378 |
| Analog Grounding Consideration | 381 |

# General Information

## Overview

**⚠ DANGER**

**ELECTRIC SHOCK**

The user is responsible for compliance with national and local electrical code requirements with respect to grounding of all equipment.Read, understand and implement the wiring and grounding requirements in this section.

**Failure to follow these instructions will result in death or serious injury.**

The required power and grounding configurations for AC powered and DC powered systems are shown in the following illustrations. Also shown are power and grounding configurations of AC and DC systems required for CE* compliance.

**NOTE:** Each rack shown has its own ground connection; that is, a separate wire returning to the main grounding point, rather than "daisy chaining" the grounds between power supplies or mounting plates.
The main grounding point is the local common connection of the panel ground, equipment ground, and earth grounding electrode.

## CE Compliance

The CE mark indicates compliance with the European Directive on Electromagnetic Compatibility (EMC) (89/336/EEC) and the Low Voltage Directive (73/23/EEC).

**NOTE:** In order to maintain CE compliance, the Quantum system must be installed in accordance to these instructions.

## Chassis Grounding

A chassis ground wire is required for each rack. The wire is connected between one of four ground screws (located on the rack) and the main ground point of the power system. This wire should be green (or green with a yellow stripe) and the AWG rating must be (at a minimum) sized to meet the fuse rating of the supply circuit.

## Power Supply Grounding

On each power supply connector there is a ground connection. The preferred connection is between the power supply connector ground terminal and one of the rack ground screws. This wire should be green (or green with a yellow stripe) and at a minimum the same AWG rating as the power connections to the supply.

In racks with multiple power supplies, each supply should have a ground connection between its input connector and the rack ground screws.

**NOTE:** It is recommended that the power supply, feeding the I/O modules, is grounded at the main ground point.

## Other Equipment Grounding

Other equipment in the installation should not share the grounding conductor of the system. Each piece of equipment should have its own grounding conductor returning to the main grounding point from which the equipment power originates.

## Systems with Multiple Power Feeds

In systems with multiple power feeds, the grounding should proceed in the same manner as single feed systems. However, a zero volt potential difference must be maintained between the equipment grounding conductors of the separate systems to prevent current flow on communication cables.

# Batteries as DC power supplies

## Overview

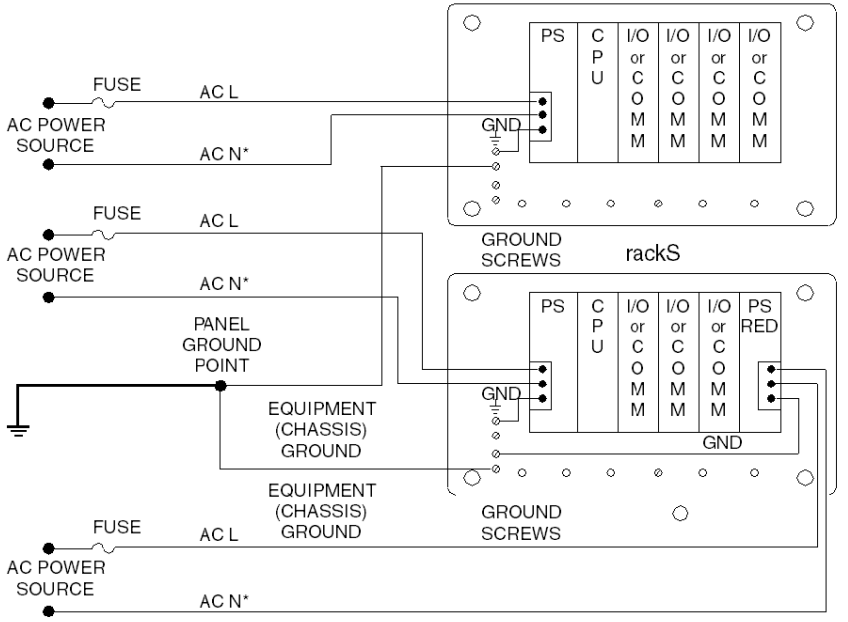
Power Supplies usually provide the adequate isolation from high **and** low frequency RF noise because of filtered outputs. Batteries provide only good filtering abilities against low frequency noise.

To protect battery powered networks, additional RFI filters are required such as:

- CURTIS F2800 RFI filters
- TRI-MAG, Inc. FL Series Filters or equivalent

# AC Power and Grounding Considerations

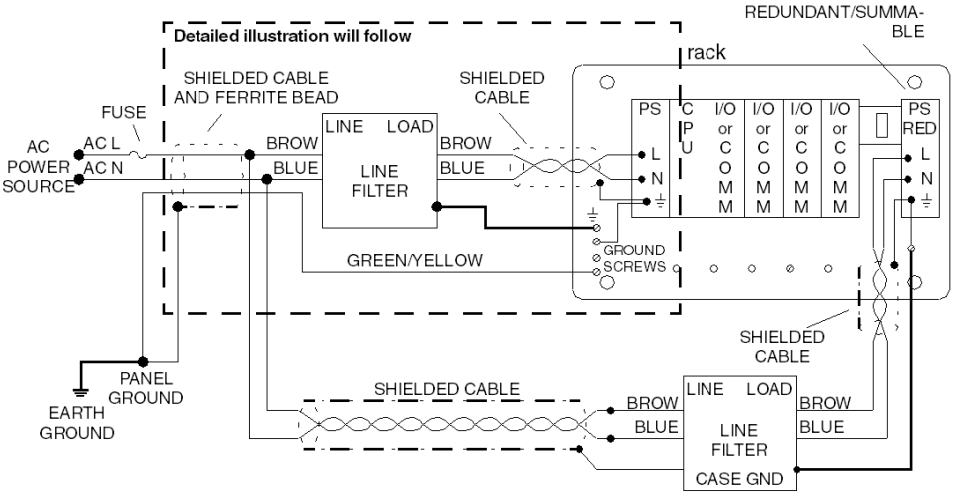
## AC Powered Systems




**NOTE:** *AC N should be earth grounded. If it is not earth grounded, it must be fused (refer to local codes).

## AC System with CE Compliance

**NOTE:** 140 CPS 111 00, 140 CPS 114 20 and 140 CPS 124 20 power supplies are designed to not require the external EMI filter, ferrite bead and Olflex cable.

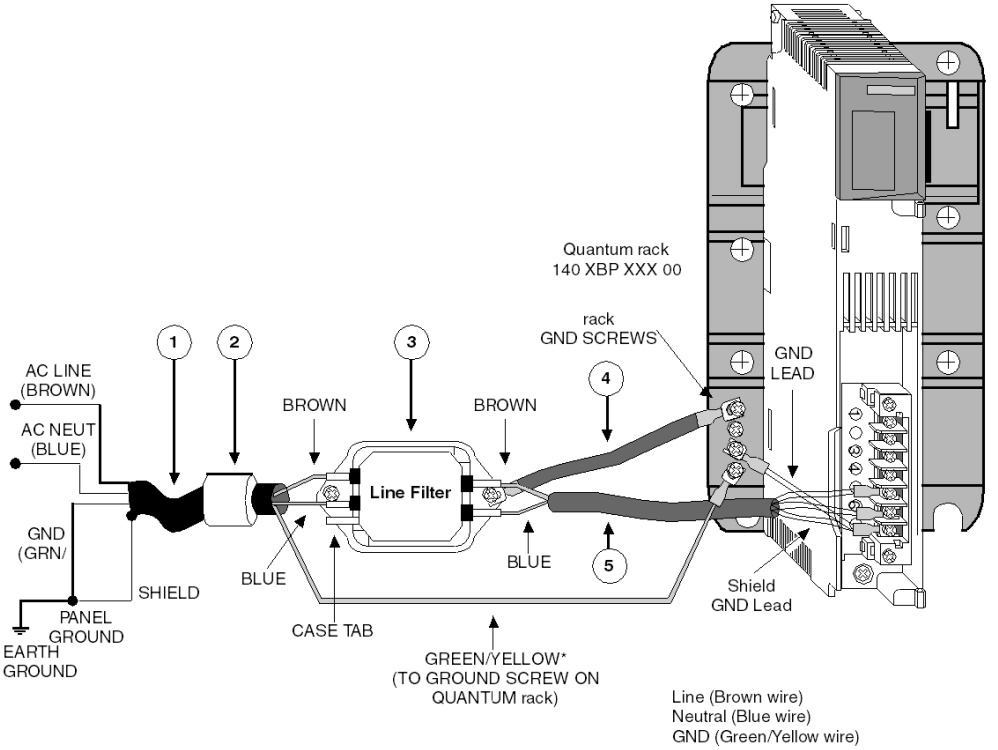



**NOTE:** To maintain CE compliance with the European Directive on EMC (89/336/EEC), the AC power supplies must be installed per the European Directive on EMC (89/336/EEC).

**NOTE:** 140 XTS 001 00 and 140 XTS 005 00 connector models must be used in systems that must meet closed system requirements as defined in EN 61131-2 (without relying upon an external enclosure).
External Line Filters must be protected by a separate enclosure which meets the requirements of IEC 529, Class IP20.

## Detail

The following figure shows the details of an AC powered system with CE compliance.




**NOTE:** Only one ground wire per rack is required. In redundant and summable systems, this lead is not connected for the additional line filter/power supply.

**NOTE:** For detailed wiring diagrams, refer to the part Power Supply Modules Power Supply Modules *(see page 219)*

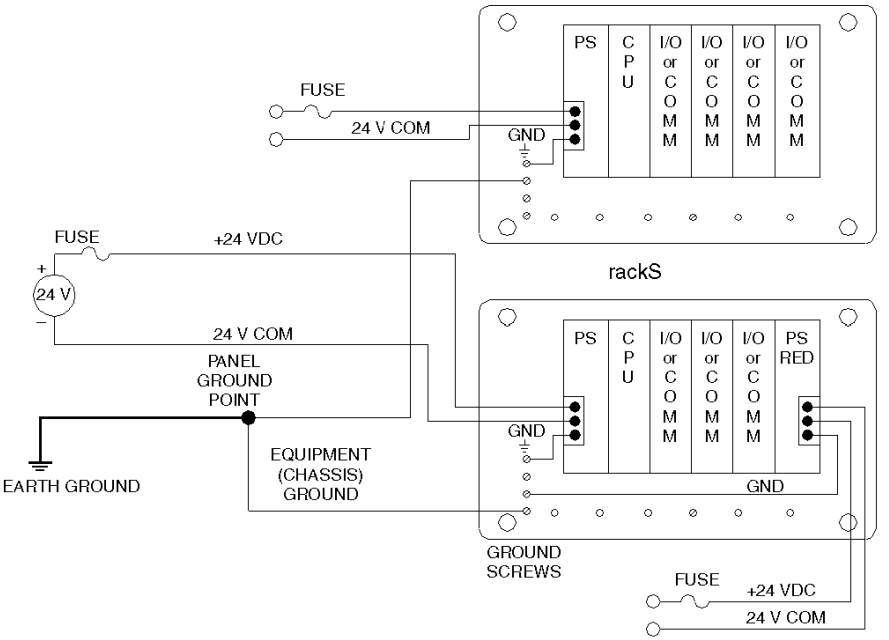
## Part List

| Callout | Vendor or equivalent | Part Number | Description | Instruction |
|---|---|---|---|---|
| 1 | Oflex-Series 100 cy | 35005 | Line Cord | Terminate the shield at panel ground; the filter end of the shield is not terminated. |
| 2 | Stewart<br>Fairite | 28 B 0686-200<br>2643665702 | Ferrite Bead | Install next to the filter and fasten with tie wraps at both ends of the ferrite bead. |
| 3 | Schaffner | FN670-3/06 | Line Filter (fast on terminals)<br>Dimensions:<br>Length:85 mm (3.4 in)<br>Width: 55 mm (2.2 in)<br>Height: 40 mm (1.6 in)<br>Mounting Holes: 5.3 mm (0.2 in) diameter<br>75 mm (3 in) centerline mounted.<br>Fast on terminals:<br>6.4 mm (0.25 in) | Install next to the power supply. |
| 4 | NA | NA | Ground Braid<br>Flat braid 134 mm (0.5 in) with a maximum length of 100 mm (4 in) | NA |
| 5 | Oflex Series 100cy | 35005 | Shield Cable<br>Maximum length:<br>215 mm (8.5 in) | Third lead (green/yellow) is not used; terminate the shield at the power supply ground terminal. |

# DC Power and Grounding Considerations

## 24 VDC Powered System Figure

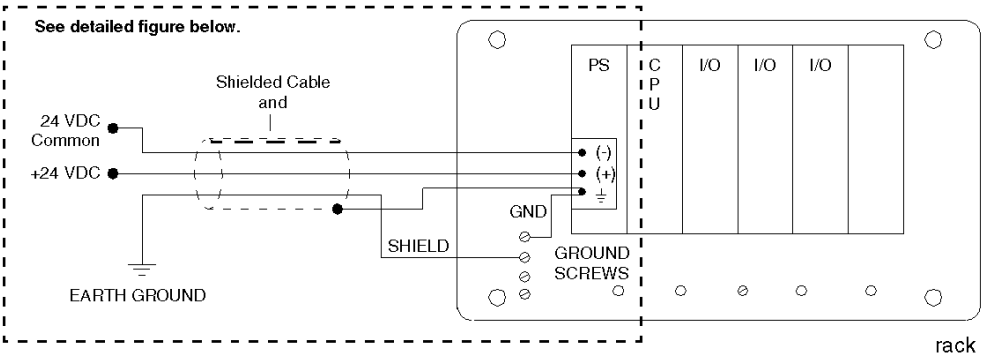
The following figure shows a 24 VDC powered system.




**NOTE:** It is recommended to earth ground the 24 VDC power supply.

## 24 VDC Powered System for CE Compliance

The following figure shows a 3 A, 24 VDC powered system for CE compliance.




## ⚠ CAUTION

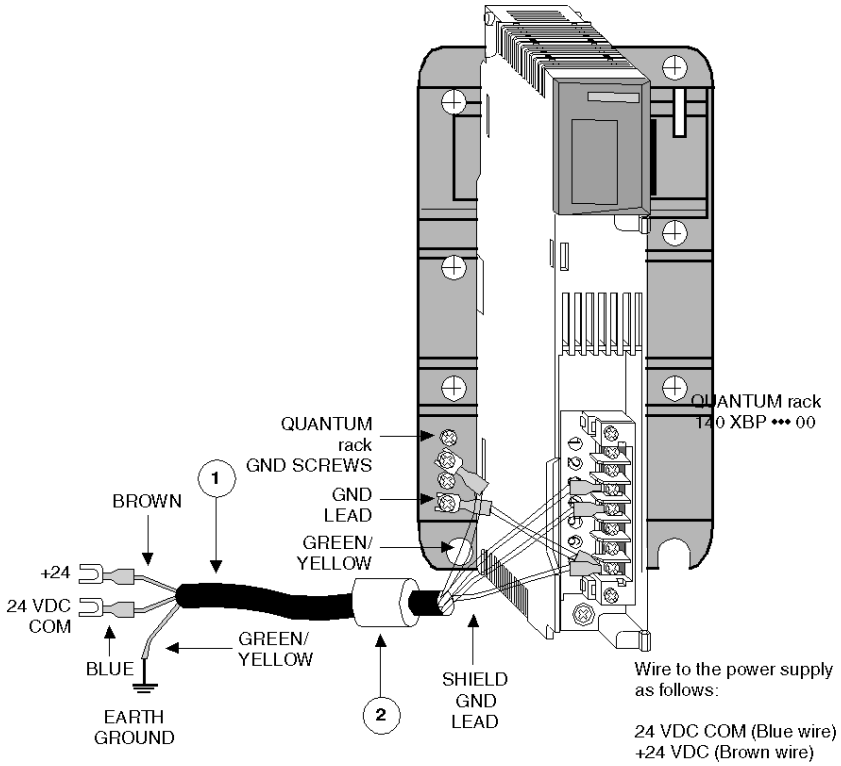
**CE MARK NONCOMPLIANCE**

The 140 CPS 211 00, the 140 CRA 211 20, and the 140 CRA 212 20 must be installed per the European Directive on EMC (89/336/EEC) and the Low Voltage Directive (73/23/EEC).

**Failure to follow these instructions can result in injury or equipment damage.**

## 24 VDC Detailed Figure

The following figure shows the detailed installation of a 3 A, 24 VDC powered system for CE compliance.



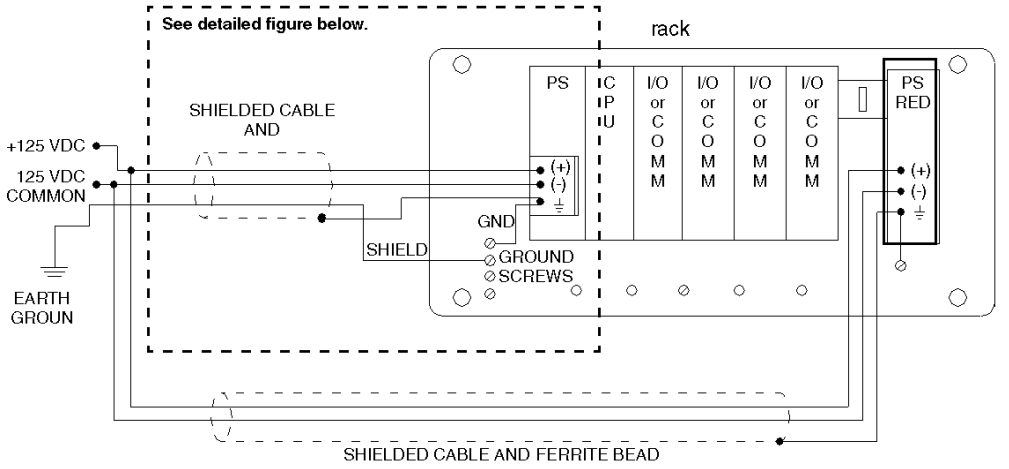

**NOTE:** For detailed wiring diagrams, refer to the part Power Supply Modules Power Supply Modules *(see page 219)*

Parts List.

| Callout | Vendor (or equivalent) | Part Number | Description | Instruction |
|---|---|---|---|---|
| 1 | Offlex Series 100cy | 35005 | Line Cord | Terminate the shield at the power supply ground terminal |
| 2 | Sreward Fairite | 28 BO686-200 2643665702 | Ferrite Bead | Install next to the filter and fasten with tie wraps at Both ends of the ferrite bead. |

## 125 VDC Powered System Figure

The following figure shows a 125 VDC powered system for CE compliance.




**⚠ CAUTION**

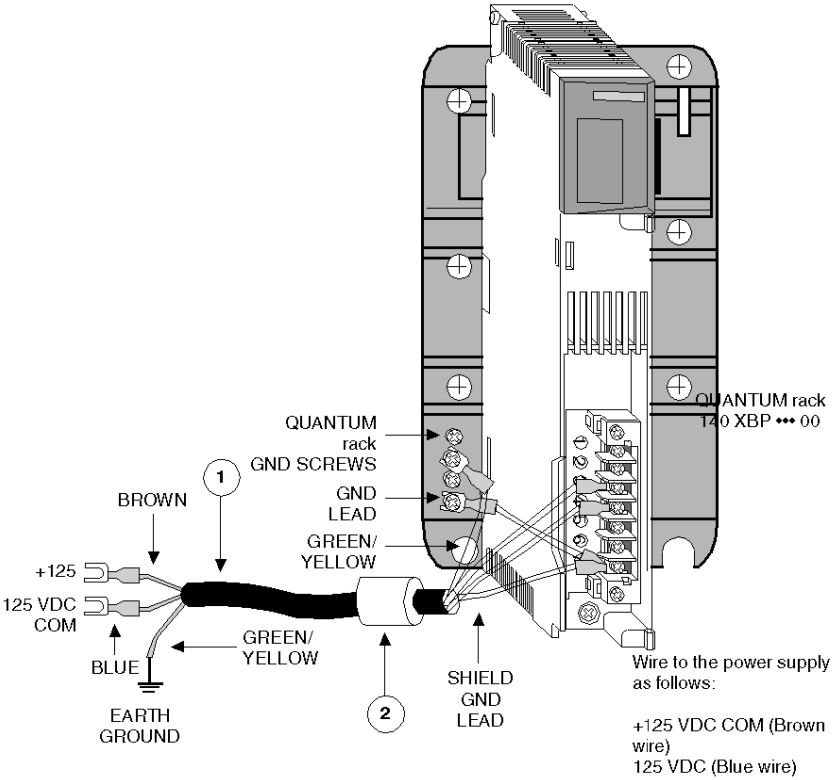
**CE MARK NONCOMPLIANCE**

The 140 CPS 511 00 & the 140 CPS 524 00 must be installed per the European Directive on EMC (89/336/EEC) and the Low Voltage Directive (73/23/EEC).

**Failure to follow these instructions can result in injury or equipment damage.**

## 125 VDC Detailed Figure

The following figure shows the detailed installation for the 125 VDC powered system for CE compliance.




**NOTE:** For detailed wiring diagrams of power supply modules, refer to the part Part Power Supplies *(see page 219)*

Parts List.

| Callout | Vendor (or equivalent) | Part Number | Description | Instruction |
|---|---|---|---|---|
| 1 | Offlex Series 100cy | 35005 | Line Cord | Terminate the shield at the power supply ground terminal |
| 2 | Sreward Fairite | 28 BO686-200 2643665702 | Ferrite Bead | Install next to the filter and secure with tie wraps at Both ends of the ferrite bead. |

## ⚠ CAUTION

**EUROPEAN COMPLIANCE**

The 140 CPS 511 00 & the 140 CPS 524 00 must be installed per the European Directive on EMC (89/336/EEC) and the Low Voltage Directive (73/23/EEC).

**Failure to follow these instructions can result in injury or equipment damage.**

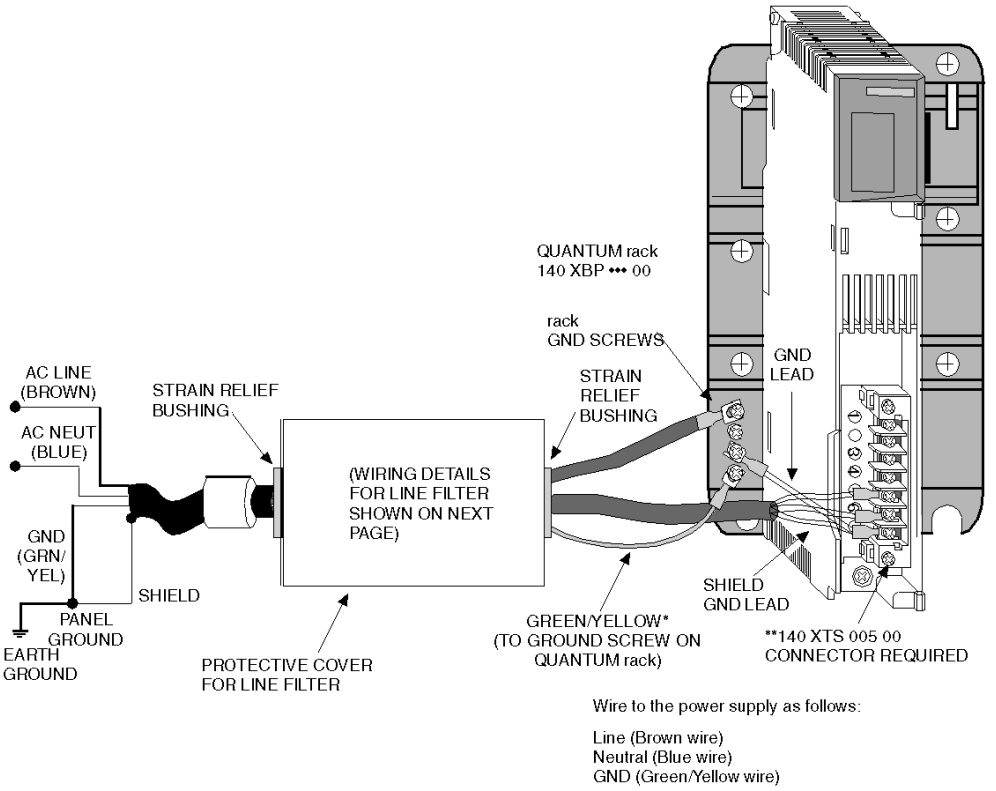
# Closed System Installation

## Overview

For installations that must meet closed system requirements as defined in EN 61131-2 (without relying upon an external enclosure) in which an external line filter is used, it must be protected by a separate enclosure that meets the requirements of IEC 529, Class IP20.

## AC/DC Installation

The following figure shows an AC/DC powered systems that meets CE closed system compliance.




* Only one ground wire per rack is required. In redundant and summable systems, this lead is not connected for the additional line filter/power supply

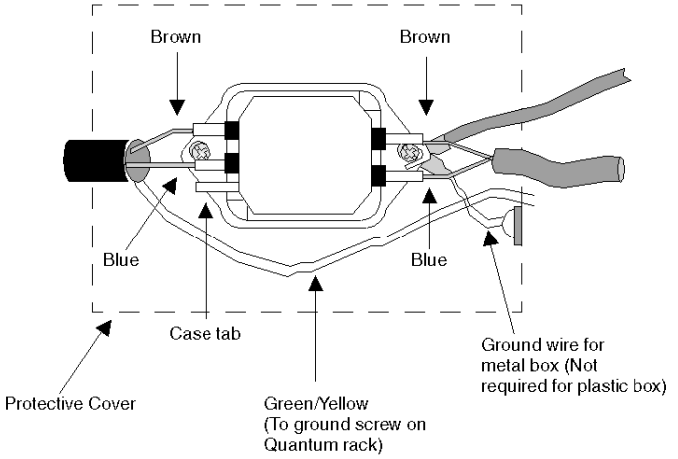
** Connectors 140 XTS 005 00 (for power supplies) and 140 XTS 001 00 (for I/O modules) must be ordered separately

**NOTE:** For detailed wiring diagrams, refer to the part Power Supply Modules Power Supply Modules *(see page 219)*

## Protective Cover

The protective cover must completely enclose the line filter. Approximate dimensions for the cover are 12.5 x 7.5 cm. Wire entry/exit shall be through strain relief bushings.

## Line Filter Connections

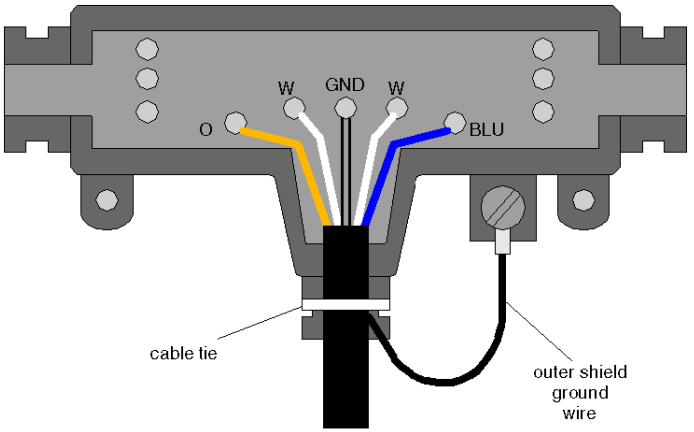


# Modbus Plus Termination and Grounding

## How Taps Have to Be Terminated

A tap is required at each site on the trunk cable to provide connections for the trunk cable and drop cable. Each tap contains an internal terminating resistor that can be connected by two jumpers. Two jumper wires are included in the tap package, but are not installed. At the taps at the two ends of a cable section, you must connect both of the jumpers to provide the proper terminating impedance for the network. Taps at inline sites must have both jumpers removed. The impedance is maintained regardless of whether a node device is connected to the drop cable. Any connector can be disconnected from its device without affecting the network impedance.

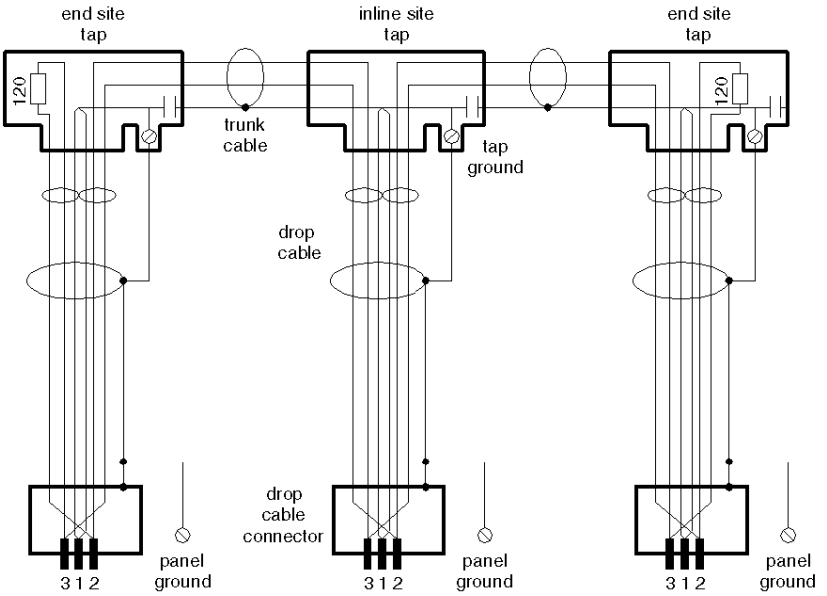
The diagram shows a Modbus Plus Network connection with terminating resistors and grounding.



## Grounding at the Tap

Each tap has a grounding screw for connection to the site panel ground. Schneider Electric drop cables have a grounding lug in the cable package. This must betightly crimped or soldered on the cable and connected to the grounding screw on the tap.

The diagram shows a drop cable, connected and grounded with a tap.

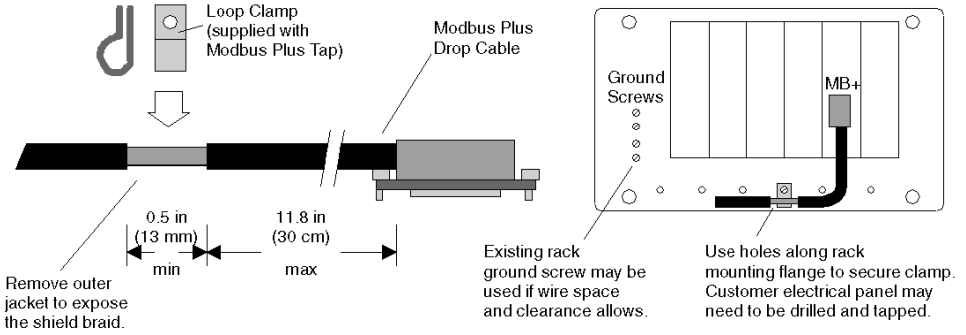



The node device end of the drop cable has a lug which must be connected to the node device's panel ground. The network cable must be grounded through this connection at each node site, even when the node device is not present. The ground point must not be left open. No other grounding method can be used.

## Grounding at the Device Panel

Modbus Plus network drop cables require a ground connection to the rack. The connection is made by means of a metal loop clamp that grounds the cable shield to the ground point.

The following figure shows the Modbus Plus grounding at the device panel.




**NOTE:** To maintain CE compliance with the European Directive on EMC (89/336/EEC), the Modbus Plus drop cables must be installed in accordance with these instructions.

## Preparing the Cable for Grounding

This table shows the steps to prepare the cable for grounding

| Step | Action |
|---|---|
| 1 | Determine the distance from the cable´s end connector to the intended ground point on your rack or panel |
| 2 | Stripping of the cable´s outer jacket<br>**Note:** Keep in mind, that the maximum allowable distance from the ground point to the cable´s end connector is 11.8 in (30 cm) |
| 3 | Remove 0.5 -1 in (13-25 mm) of the cable´s outer jacket to expose the shield braid as shown in the figure above. ) |
| 4 | If the panel has a suitable ground point for mounting the cable clamp, install the clamp at that point |

# Fiber Repeaters

## Grounding a Modbus Plus Fiber Repeater

| Step | Action |
|---|---|
| 1 | Connect the Repeater to the site ground<br>**Result:** The Repeater obtains it´s ground through the chassis ground screw or DC (-) wire. |
| 2 | Use a continuity tester to verify, that the repeater is grounded to the site ground. |

## Connecting AC Power to the Repeater

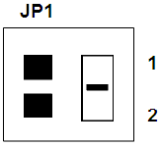
| Step | Action |
|---|---|
| 1 | Remove the power at it´s source. |
| 2 | If necessary install a different plug on the cable for the power source at your site.<br>**Note:** The AC power cable supplied with the repeater is keyed for North American 110-120 VAC outlets. |
| 3 | Remove the AC power cable from the repeater |
| 4 | Set the power selector plug to the 110-120 VAC or 220-240 VAC position for the power source at your site. To do this:<br>**1.** Remove the power selector plug by prying under it´s tab using a small screw driver.<br>**2.** Set the plug to the proper voltage position as shown on the plug body.<br>**3.** Reinsert the plug. |
| 5 | Insert the AC power cable in the rear panel connector. |
| 6 | Insert the AC power cable into the power source. |

## Connecting DC Power to the Repeater

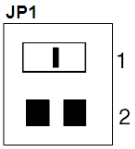
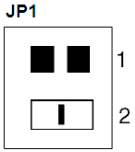
| Step | Action |
|---|---|
| 1 | Remove the power at it´s source. |
| 2 | Connect the source to the DC power terminals, observing the proper polarity. |

## RIO Shield-to-chassis Switch

RIO cable shield must be set to specify the NRP relationship to chassis ground. Jumper switch is shipped in neutral position as indicated on the picture below:




It have to be placed :

| Switch position | Function |
|---|---|
| 1 | NRP acts as a drop on CRP side (RIO cable shield is isolated from chassis ground by a capacitor) -i.e if low frequency is a problem.<br>  |
| 2 | NRP acts as a head on CRA side (RIO cable shield is connected directly to chassis ground) - i.e. the same ground as the main RIO head.<br>  |

# Grounding of RIO Networks

## Overview

The Remote I/O communication is based on single point grounding, that is located at the head. Coaxial cable and taps have no additional connection to the ground to help eliminate low frequency ground loops.

## Missing grounding

A cable system must be grounded at all times to ensure proper operation of the nodes on the network. The cable system is grounded by the RIO head processor. But if the cable is removed, the ground connection doesn't work anymore.

## Ground Blocks

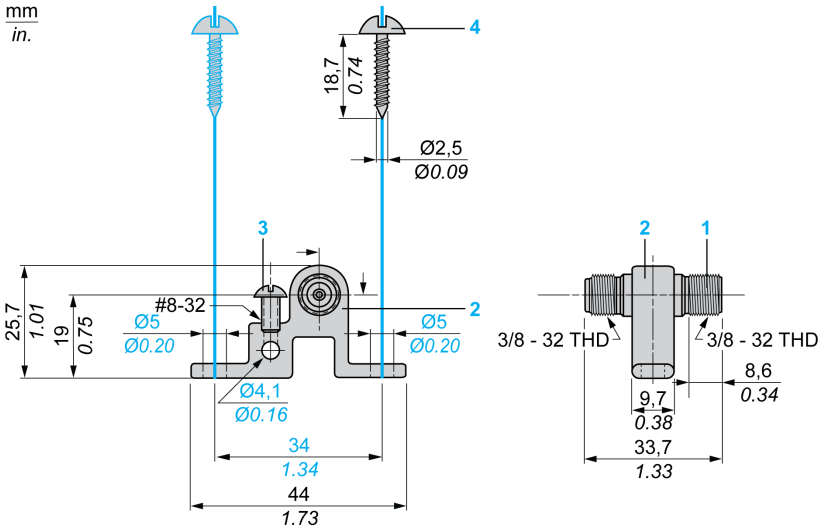
Ground blocks provide grounding, even if the cable is removed.

Additional properties are as follows:

- Low insertion loss
  Only if five or more are used, they have to be considered in the trunk attenuation with 0.2 dB each. The impedance is 75 Ω and the return loss >40 dB.
- Wide application frequency

## Ground Block structure

The ground block 60-0545-000 consists of two female in-line F connectors and a separate screw hole binding for attaching a ground wire. The grounding block has two mounting holes, allowing it to be mounted to a flat surface.




**1** Female in-line F connector
**2** Grounding block
**3** Locking screw (for ground wire)
**4** Mounting screw

**NOTE:** Local building codes may require the cable shield tied to ground, whenever the cable system exits and/or enters a new building (NEC Article 820-33).

## Surge protection

Surge protection is available for coaxial network trunks that span between buildings and are exposed to lightning. The recommended product has internal gas discharge surge protectors that absorb very high currents induced into the cable system by near-lightning strikes. The device indicated has insertion loss of less than 0.3 dB at the network operating frequency. The unused drop ports must be terminated with a 52-0402-000 Port Terminator. If desired, shrink tubing may be used to seal the F connections.

The device should be accessible for maintenance, and be protected from the elements if installed outside. The threaded stud should be connected to building ground.

The recommended product is Relcom Inc. p/n CBT-22300G. Contact information is:

Relcom Inc.

2221 Yew Street Forest Grove, Oregon 97116, USA

Tel: (800) 382-3765

www.relcominc.com

# Analog Grounding Consideration

## Overview

For the Analog Input Modules *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*, the earthing must be done by the analog wires. Analog wires must be grounded directly when entering the cabinet. You may use an analog cable grounding rail. This section describes this approach.

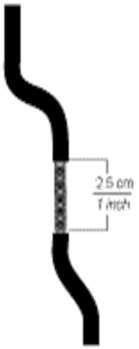
## Principle

High frequency interference can only be discharged via large surfaces and short cable lengths.

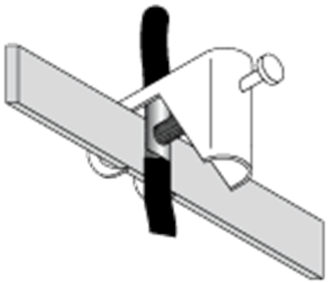
## Guidelines

Follow these wiring guidelines:

- Use shielded, twisted-pair cabling.
- Expose 2.5 cm (1 inch) as shown:.




- Make sure the wire is properly grounded (connection between the grounding bar and the clamps).

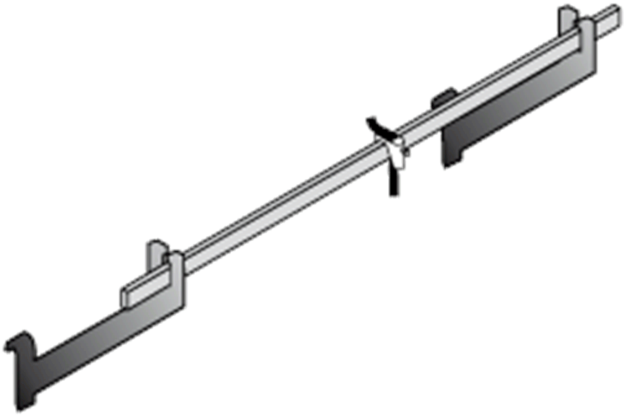


**NOTE:** It is strongly recommended to use the STB XSP 3000 grounding kit and, either the STB XSP 3010 or the STB XSP 3020 clamp kits.

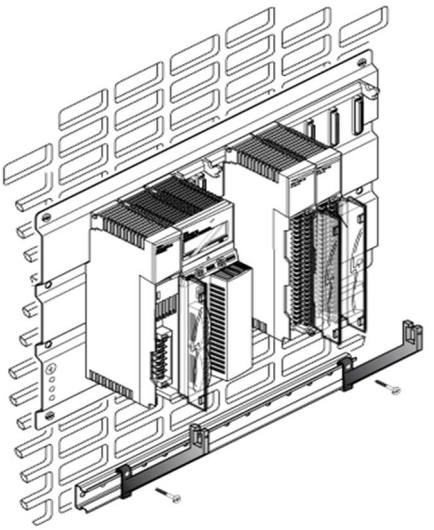
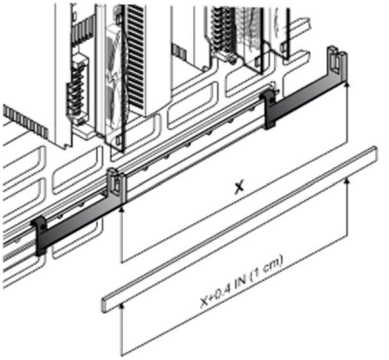
## Assembly of the STB XSP 3000 Kit

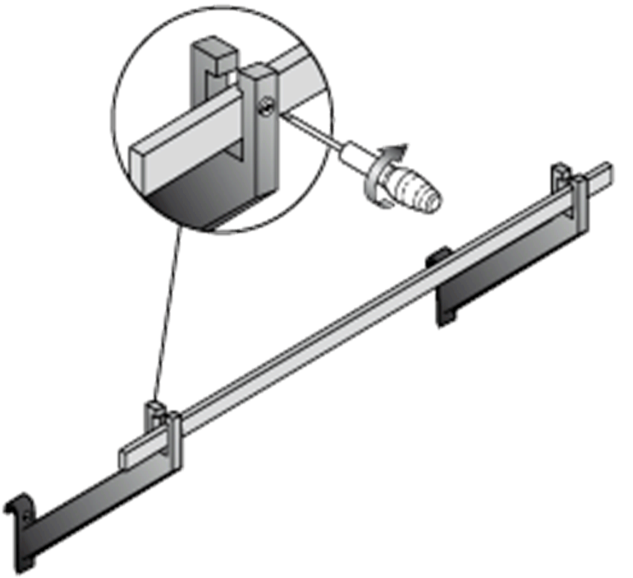
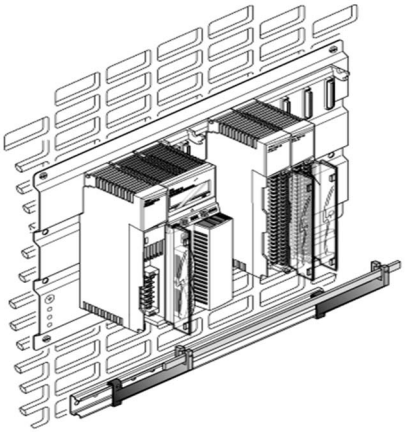
The followed kit is used to have a high quality of the analog signal.

The following table describes the STB XSP 3000 grounding kit.

| Elements | Description |
|---|---|
| Side Supports and grounding bar | The following illustration describes the assembly of the side supports and the grounding bar.<br>  |

The following table describes the step to assemble the STB XSP 3000 grounding kit:

| Step | Description |
|---|---|
| 1 | Push the side supports against the walls and tighten the screws.<br>  |
| 2 | Choose the length of the grounding bar.<br>  |
| 3 | Insert the functional grounding terminal block and tighten the screw<br>  |

| Step | Description |
|---|---|
| 4 | Fix the grounding bar at the side supports.<br>  |
| 5 | The following figure describes the final assembly of the STB XSP 3000.<br>  |

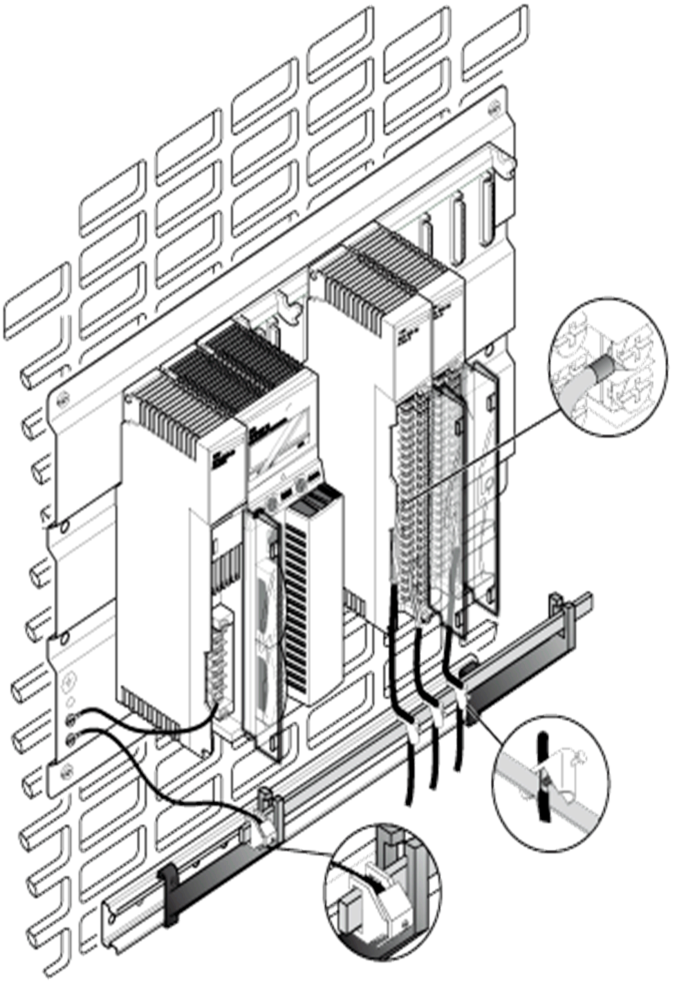
## STB XSP 3010 Kit and STB XSP 3020

The following table describes the different cable sections (in AWG and mm2):

| Reference | AWG | mm2 |
|---|---|---|
| STB XSP 3010 | 16 to 9 | 1.5 to 6.5 |
| STB XSP 3020 | 10 to 7 | 5 to 11 |

## Final Assembly

The following figure shows the final assembly:



# Appendix C
# Field Wiring Terminal Strip / Module Keying Assignment

## Purpose

This appendix provides information on Field Wiring Terminal Strip / Module Keying Assignment of I/O modules

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| General Information | 388 |
| Illustration | 389 |
| Primary Keying | 391 |
| Secondary Keying | 393 |

# General Information

## Location

Field wiring terminal strips and module housings are slotted on the left and right sides of the PCB card slot to accept keying pins (see *I/O Module Figure, page 389*).

## Purpose

To prevent plugging the terminal strip into the wrong module, once wiring connections have been made. Keying is implemented at the discretion of the user.

**NOTE:** Schneider Electric recommends that module key coding be part of the system installation procedure.

## Primary Keying

Primary keying is provided on the right side of the module, marked A through F (top and bottom positions are coded the same). Primary keying provides module class coding. Primary codes have been pre-defined, according to the following tables.

## Secondary Keying

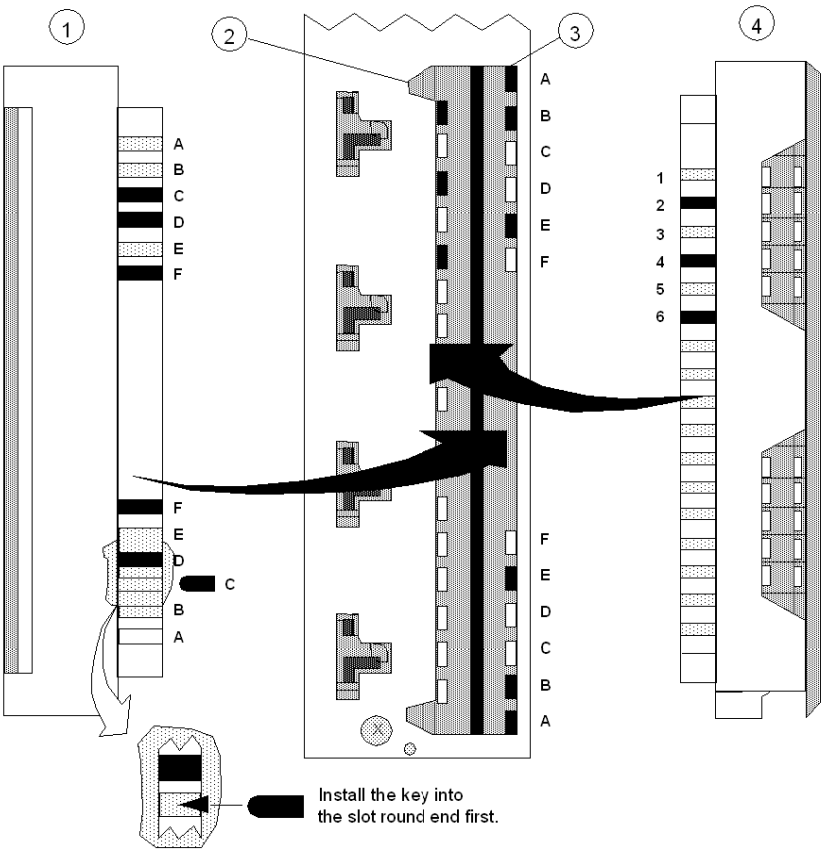
Secondary keying is provided on the left side of the module, marked 1 through 6. Secondary keying codes are user-definable and may be used to identify module personality within module classes, or other unique site requirements.

The user may also use personality keying to differentiate between like module types (i.e., 140 DAO 840 00 and 140 DAO 842 10 both have the same primary keying pin combinations), using the white keys for each code.

# Illustration

## I/O Module Figure

The following figure shows the I/O module and the terminal strip.




1 Right side of Terminal Strip
2 Secondary Key Slots
3 Primary Key Slots
4 Left side of Terminal Strip

**NOTE:** The primary / secondary keys shown (in black) in this example reflect the recommended coding of a 24 VDC module in slot 6 to its field wiring terminal strip.

To support keying, I/O modules accepting terminal strips come with 12 customer-installable primary keys (six yellow keys each for the module and terminal strip) and six secondary keys (three white keys each for the module and terminal strip). In the following table, check the Primary Module and Terminal Strip Coding columns for key locations.

To implement the user-optional secondary keying code (designed to help prevent the mismatching of terminal strips to I/O modules of identical type), 17 slot positions have been provided in modules and terminal strips to support a variety of coding schemes.

In addition (by using the secondary keying code), the user may key the field wiring terminal to the position where the module is installed in a rack, using the white keys for each code. To determine a unique module code and terminal strip code, refer to the *Primary Keying Table, page 391*.

# Primary Keying

## Primary Keying Table

The following table shows the primary module and terminal strip keying for the I/O modules.

| Module Class | Module Part Number | Module Coding | Terminal Strip Coding |
|---|---|---|---|
| 5 VDC | 140 DDI 153 10 | ABC | DEF |
| | 140 DDO 153 10 | | |
| 9 ... 12 VDC | Unassigned | ABD | CEF |
| 24 VDC | 140 DDI 353 00 | ABE | CDF |
| | 140 DDI 353 10 | | |
| | 140 DDM 390 00 | | |
| | 140 DDO 353 00 | | |
| | 140 DDO 353 10 | | |
| | 140 HLI 340 00 | | |
| | 140 SDI 953 00S | | |
| | 140 SDO 953 00S | | |
| 10 ... 60 VDC | 140 DDI 841 00 | ABF | CDE |
| | 140 DDI 853 00 | | |
| | 140 DDO 843 00 | | |
| 125 VDC | 140 DDI 673 00 | ACD | BEF |
| | 140 DDM 690 00 | | |
| | 140 DDO 885 00 | | |
| 24 VAC | 140 DAI 340 00 | ACE | BDF |
| | 140 DAI 353 00 | | |
| 48 VAC | 140 DAI 440 00 | ACF | BDE |
| | 140 DAI 453 00 | | |
| | 140 DAO 842 20 | | |
| 115 VAC | 140 DAI 540 00 | ADE | BCF |
| | 140 DAI 543 00 | | |
| | 140 DAI 533 00 | | |
| | 140 DAM 590 00 | | |
| | 140 DAO 840 10 | | |

| Module Class | Module Part Number | Module Coding | Terminal Strip Coding |
|---|---|---|---|
| 230 VAC | 140 DAI 740 00 | ADF | BCE |
| | 140 DAO 840 00 | | |
| | 140 DAO 842 10 | | |
| | 140 DRA 840 00 | | |
| Relay | 140 DRC 830 00 | AEF | BCD |
| Analog I/O | 140 ACI 030 00 | BCD | AEF |
| | 140 SAI 940 00S | | |
| | 140 AVI 030 00 | | |
| | 140 ACO 020 00 | BCE | ADF |
| | 140 AVO 020 00 | BCF | ADE |
| TC In / RTD / PT100 In | 140 ARI 030 10 | BDE | ACF |
| | 140 ATI 030 00 | | |
| Analog In/Out | 140 AMM 090 00 | BDF | ACE |
| Intelligent/ Special Purpose | 140 EHC 105 00 | BEF | ACD |
| | 140 EHC 202 00 | | |
| | Unassigned | CDE | ABF |
| | Unassigned | CDF | ABE |
| | Unassigned | CEF | ABD |
| | Unassigned | DEF | ABC |

# Secondary Keying

## Secondary Keying Table

The following table shows the secondary keying and rack positions.

| rack Position | Module Coding | Terminal Strip Coding |
|---|---|---|
| 1 | 123 | 456 |
| 2 | 124 | 356 |
| 3 | 125 | 346 |
| 4 | 126 | 345 |
| 5 | 134 | 256 |
| 6 | 135 | 246 |
| 7 | 136 | 245 |
| 8 | 145 | 236 |
| 9 | 146 | 235 |
| 10 | 156 | 234 |
| 11 | 234 | 156 |
| 12 | 235 | 146 |
| 13 | 236 | 145 |
| 14 | 245 | 136 |
| 15 | 246 | 135 |
| 16 | 256 | 134 |

# Appendix D
# CableFast Cabling

## Introduction

This appendix provides information to the CableFast cabling system.

## What Is in This Chapter?

This chapter contains the following topics:

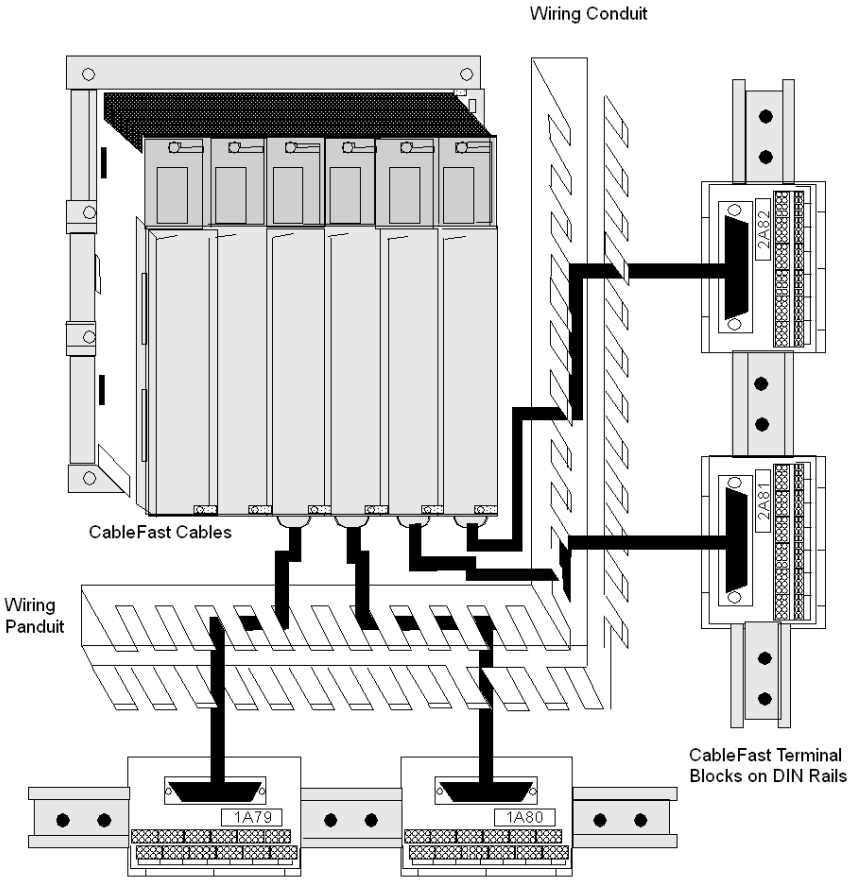

| Topic | Page |
|---|---|
| General Information | 396 |
| 140 CFA 040 00 Quantum CableFast Cabling Block | 402 |
| 140 CFB 032 00 Quantum CableFast Cabling Block | 405 |
| 140 CFC 032 00 Quantum CableFast Cabling Block | 409 |
| 140 CFD 032 00 Quantum CableFast Cabling Block | 417 |
| 140 CFE 032 00 Quantum CableFast Cabling Block | 421 |
| 140 CFG 016 00 Quantum CableFast Cabling Block | 423 |
| 140 CFH 008 00 Quantum CableFast Cabling Block | 431 |
| 140CFI00800 Quantum CableFast Cabling Block | 439 |
| 140CFJ00400 Quantum CableFast Cabling Block | 446 |
| 140CFK00400 Quantum CableFast Cabling Block | 454 |
| CableFast Cables | 462 |
| CableFast Accessories | 467 |

# General Information

## Overview

The CableFast wiring system consists of pre-wired Quantum field wiring terminal strips in various cable lengths, terminated with D-type connectors. The D connectors plug into DIN rail-mounted terminal blocks offered in straight-through or special application versions. Cables and terminal blocks are ordered separately. Terminal blocks may be used with any cable length. Pigtail cable versions are also available.

## Quantum I/O Modules with CableFast Components




**NOTE:** Make sure that the wiring conduits are large enough to support 3.65 m (12 ft) cables.

## CableFast System Specifications

| | | |
|---|---|---|
| Power ratings | 150 VAC/VDC @ 0.5 A/point | |
| | 150 VAC/VDC @ 2.0 A/point * | |
| | * Requires the 140 CFG 016 00 terminal block and the 140 XTS 012 •• cable | |
| Dielectric withstanding voltage | 1060 VAC and 1500 VDC | |
| Creepage and clearance | per IEC 1131, UL 508, CSA 22.2 #142-1987 | |
| Terminal block wire size/terminal | One wire | 2.5 $mm^2$ (12 AWG) |
| | Two wires) | 1.0 $mm^2$ (6 AWG) and above (See below for the maximum number of wires allowed per terminal. |
| | **Note:** It is recommended that no more than two wires be used at one time. | |
| | Wire Size | Number of Wires |
| | #24 | 4 |
| | #22 | 4 |
| | #18 | 3 |
| | #16 | 2 |
| | #14 | 1 |
| | #12 | 1 |
| Terminal screw | size | M3 |
| | screwdriver head size | 3.3 mm (0.13 in) flat head min. |
| | screw type | Captive |
| | screw finish | Tin plate (197 micro in min.) |
| | torque | 0.8 Nm (7.2 lb-in) |
| System flammability rating | 94 V-2 | |
| Temperature | Operating | 0 ... 60 °C (32 ... 140 °F) |
| | Storage | -40 ... +65 °C (-40 ... +149 °F) |
| Humidity | 0 ... 95% RH noncondensing | |
| Altitude | 6,666 ft. (2000 m) full operation | |
| Shock | +/-15 g peak, 11 ms, half sine wave | |
| Vibration | 10 ... 57 Hz @ 0.075 mm displacement | |
| | 57 ... 150 Hz @ 1 g, total 10 sweeps | |
| Mounting configuration | DIN rail mount, NS35/7.5 and NS32 | |

## Terminal Block Selection Guide

| | 140 CFA 040 00 | 140 CFB 032 00 | 140 CFC 032 00 | 140 CFD 004 00 | 140 CFE 032 00 | 140 CFG 016 00 | 140 CFH 008 00 | 140 CFI 008 00 | 140 CFJ 004 00 | 140 CFK 004 00 |
|---|---|---|---|---|---|---|---|---|---|---|
| 140 ACI 030 00 | X | | | | | | X | X | | |
| 140 ACO 020 00 | X | | | | | | | | X | X |
| 140 ACI 040 00 | X | | | | | | | | | |
| 140 ACO 130 00 | X | | | | | | | | | |
| 140 ARI 030 10 | X | | | | | | | | | |
| 140 ATI 030 00 | X (see note 3) | | | | | | | | | |
| 140 AMM 090 00 | X | | | | | | | | | |
| 14 AVI 030 00 | X | | | | | | X | X | | |
| 140 AVO 020 00 | X | | | | | | | | | X |
| 140 DAI 340 00 | X | | | | | X | | | | |
| 140 DAI 353 00 | X | X | X | X | | | | | | |
| 140 DAI 440 00 | X | | | | | X | | | | |
| 140 DAI 453 00 | X | X | X | X | | | | | | |
| 140 DAI 540 00 | X | | | | | X | | | | |
| 140 DAI 543 00 | X | | | | | | | | | |
| 140 DAI 553 00 | X | X | X | X | | | | | | |
| 140 DAI 740 00 | CableFast not allowed | | | | | | | | | |
| 140 DAI 753 00 | | | | | | | | | | |
| 140 DAM 590 00 | X see note 1 | | | | | | | | | |
| 140 DAO 840 00 | – | | | | | X (see note 2 | | | | |
| 140 DAO 840 10 | | | | | | X (see note 2 | | | | |
| 140 DAO 842 10 | – | | | | | X (see note 2 | | | | |
| 140 DAO 842 20 | – | | | | | X (see note 2 | | | | |
| 140 DAO 853 00 | X see note 1 | | | | | | | | | |
| 140 DDI 153 10 | X | | X | | | | | | | |
| 140 DDI 364 00 | not compatible with cable fast. See in the related chapter of the I/O reference guide for recommended cables | | | | | | | | | |

| | 140 CFA 040 00 | 140 CFB 032 00 | 140 CFC 032 00 | 140 CFD 004 00 | 140 CFE 032 00 | 140 CFG 016 00 | 140 CFH 008 00 | 140 CFI 008 00 | 140 CFJ 004 00 | 140 CFK 004 00 |
|---|---|---|---|---|---|---|---|---|---|---|
| 140 DDI 353 00 | X | X | X | | | | | | | |
| 140 DDI 353 10 | X | | | | | | | | | |
| 140 DDI 673 00 | X | | | | | | | | | |
| 140 DDI 841 00 | X | | | | | | | | | |
| 140 DDI 853 00 | X | X | X | X | | | | | | |
| 140 DDM 390 00 | X | | | | | | | | | |
| 140 DDM 690 00 | X see note 1 | | | | | | | | | |
| 140 DDO 153 10 | X | | | | | | | | | |
| 140 DDO 353 00 | X | | X | | X | | | | | |
| 140 DDO 353 01 | X | | X | | X | | | | | |
| 140 DDO 353 10 | X | | | | | | | | | |
| 140 DDO 364 00 | not compatible with cable fast. See in the related chapter of the I/O reference guide for recommended cables | | | | | | | | | |
| 140 DDO 843 00 | | | | | | X see note 2 | | | | |
| 140 DDO 885 00 | X see note 1 | | | | | | | | | |
| 140 DRA 840 00 | X see note 1 | | | | | | | | | |
| 140 DRC 830 00 | X see note 1 | | | | | | | | | |
| 140 DSI 353 00 | X | | | | | | | | | |
| 140 DVO 853 00 | X | | | | | | | | | |
| 140 SAI 940 00S | not compatible with cable fast. | | | | | | | | | |
| 140 SDI 953 00S | X | | | | | | | | | |
| 140 SDO 953 00S | X | | | | | | | | | |
| X = Valid Selections. | | | | | | | | | | |

**NOTE:** These are the maximum load current capacities of the 140 CFA 040 00 and 140 CFG 016 00:
1. When using the 140 CFA 040 00 terminal block, the indicated module outputs are limited to 0.5 A/point @ 150 VAC maximum and 0.5 A/point @ 150 VDC maximum.
2. When using the 140 CFG 016 00 terminal block and either the 140 XTS 012 •• or 140 XTS 102 •• high power cable, the indicated module output ratings are 2 A per point, 150 VAC maximum and 2 A per point, 150 VDC maximum.

3. The 140 CFA 040 00 block does not incorporate an isometric barrier and is not recommended for use with 140 ATI 030 00 TC module. Without such a barrier, temperature readings may vary up to 2 degrees from one end of the block to the other. If the application can tolerate this temperature difference, the block (and module) remote CJC may be used.

## CableFast Terminal Blocks

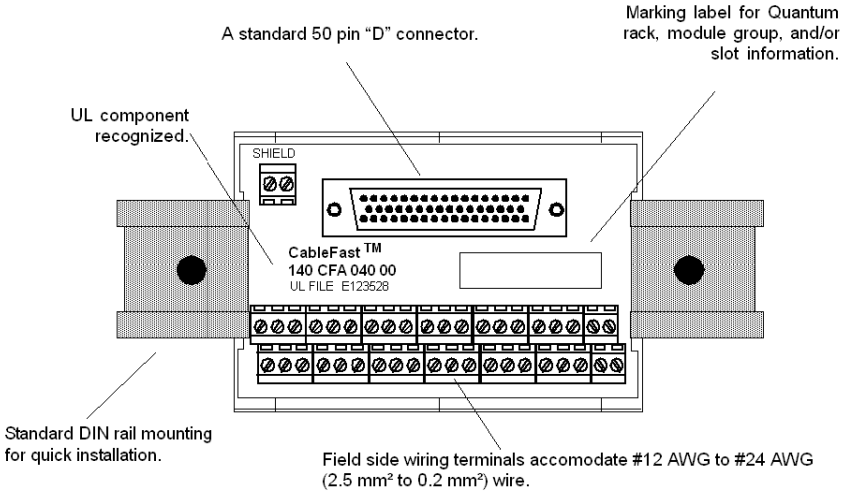
The following includes descriptions for the terminal blocks discussed in this Appendix.

This table includes descriptions for the following terminal blocks.

| Block Number | Block Description |
|---|---|
| 140 CFA 040 00 | The A block is a straight through point to point connection on the terminal block. Wiring of this block is identical to wiring the Quantum I/O connector (140 XTS 002 00). |
| 140 CFB 032 00 | The B block is used for individually fused 2-wire digital inputs. This terminal block is designed to help prevent a single point error from affecting the remaining inputs. It is not recommended for 1-wire inputs powered from the field. |
| 140 CFC 032 00 | The C block provides connection for 32 group fused input or output points. The block may be used for 1- or 2-wire inputs or outputs, and features a fuse per group, four groups total. Users select input or output mode via four switches located on the module. (The default is input mode.) |
| 140 CFD 032 00 | The D block is used for sensors requiring either 2- or 3-wire electrical interface. A fuse per group is supplied to accommodate the I/O module (4) groups. |
| 140 CFE 032 00 | The E block provides connection for 32 individually fused 24 VDC outputs. 1- and 2-wire interfacing may be selected. Field power must be supplied to the four groups. |
| 140 CFG 016 00 | The G block is a high power output block used on both AC and DC circuits requiring up to 2 A. Individual fusing is provided and may be used in both 1- and 2-wire installations. It is also used for isolated AC modules. |
| 140 CFH 008 00 | The H block is used for analog inputs, with individual fusing provided per channel. This interface provides plus, minus, shield, and power supply interface for both field and loop power configurations. |
| 140 CFI 008 00 | The I block is used for analog inputs. This interface provides plus, minus, shield, and power supply interface for both field and loop power configurations. |
| 140 CFJ 004 00 | The J block is used for analog outputs, with individual fusing provided per channel. This interface provides plus, minus, shield, and power supply interface for both field and loop power configurations. |
| 140 CFK 004 00 | The K block is used for analog outputs. This interface provides plus, minus, shield, and power supply interface for both field and loop power configurations. |

## CableFast Terminal Block Features

CableFast terminal blocks have the following features.




## CableFast Terminal Block Stacking Convention

The following figure and table show the stacking convention used by CableFast terminal blocks.

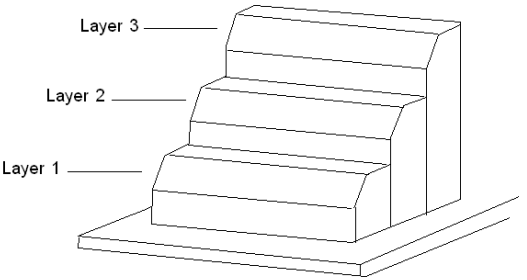



Table of Stacking Convention

| | | | |
|---|---|---|---|
| Signal | | | Layer 3 |
| Positive | Signal | Signal | Layer 2 |
| Negative | Positive | Negative | Layer 1 |

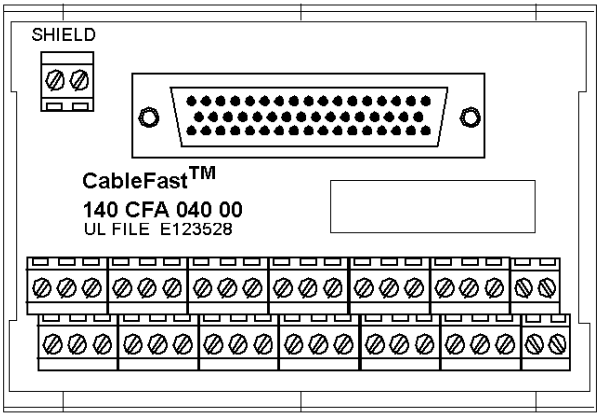
# 140 CFA 040 00 Quantum CableFast Cabling Block

## Overview

The A block is a straight-through point-to-point connection on the terminal block. Wiring this block (and other CableFast cabling blocks) is identical to wiring the Quantum I/O connector (140 XTS 002 00).

## 140 CFA 040 00 Terminal Block

The following terminal block is unique to the 140 CFA 040 00 module.



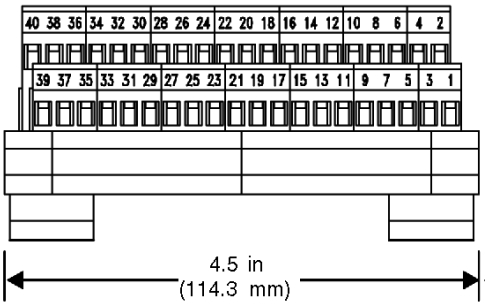

## Application Notes

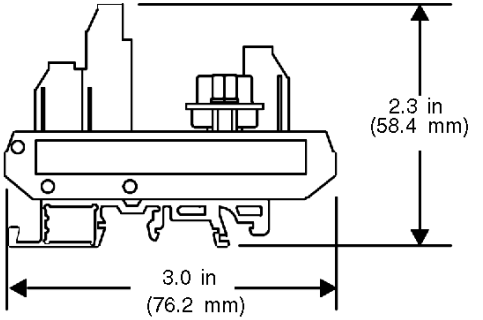
The following are the application notes for the 140 CFA 040 00 terminal block.

1. **Configuration** – two columns
2. **Compatibility** – this terminal block provides straight through (point-to-point) connection.

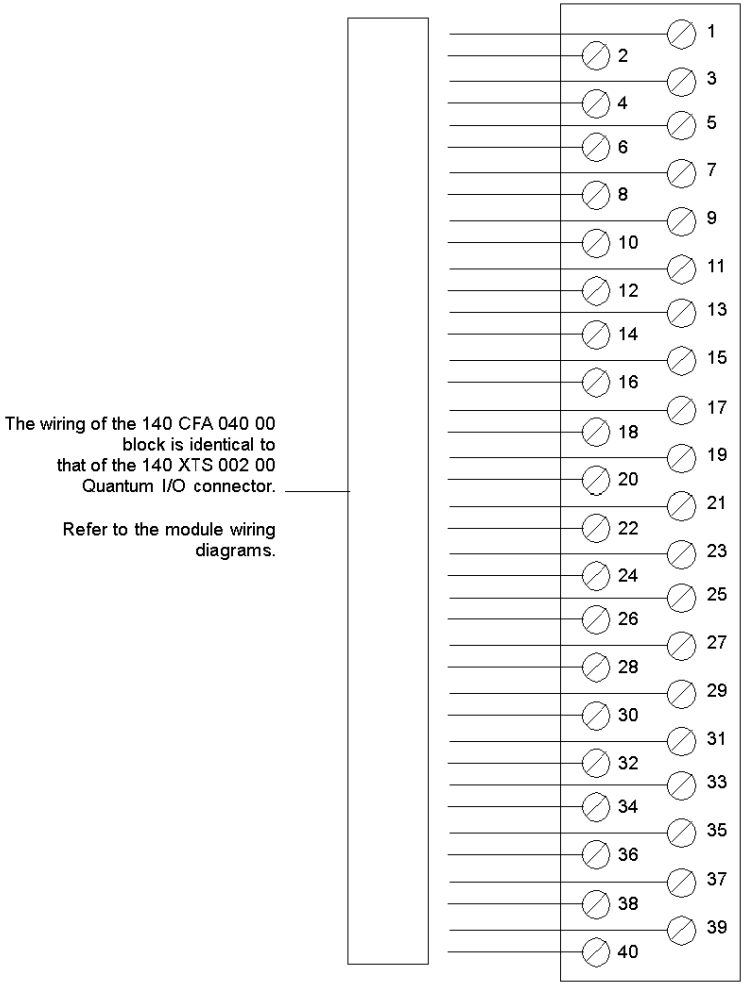
**NOTE:** You may use this terminal block with Quantum I/O modules except the 140 ATI 030 00 thermocouple module.

## 140 CFA 040 00 Dimensions



## 140 CFA 040 00 Wiring Diagram



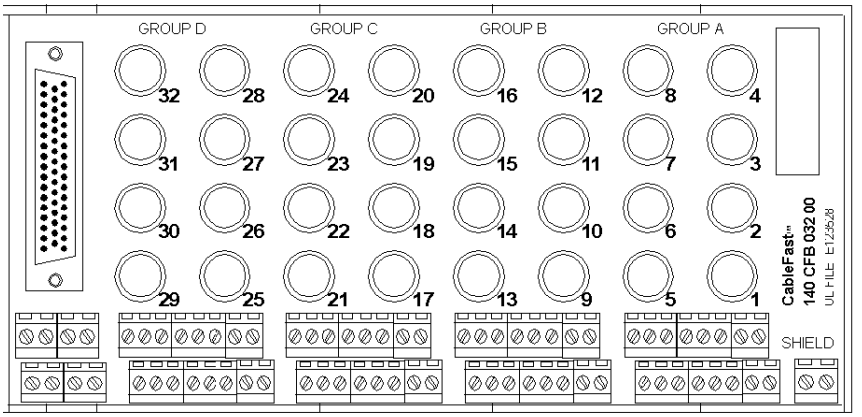
# 140 CFB 032 00 Quantum CableFast Cabling Block

## Overview

The B block is used for individually fused 2-wire digital inputs. This CableFast terminal block is designed to help prevent a single point error from affecting the remaining inputs. It is not recommended for sourced 1-wire inputs (powered from the field).

## Terminal Block

The following figure shows the terminal block for the 140 CFB 032 00 module.



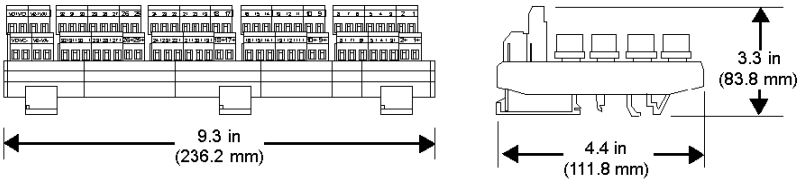

## Application Notes

The following are the application notes for the 140 CFB 0320 0 terminal block.

1. **Configuration** – arranged in four groups of eight I/O points. Two terminals per point help prevent disruption of service due to a single point error.
2. **Compatibility** – this terminal block provides individual 32 point 0.8 A fusing for the following input modules:140 DAI 353 00, 140 DAI 453 00, 140 DAI 553 00, 140 DDO 153 10, 140 DDI 353 00, and 140 DDI 853 00.

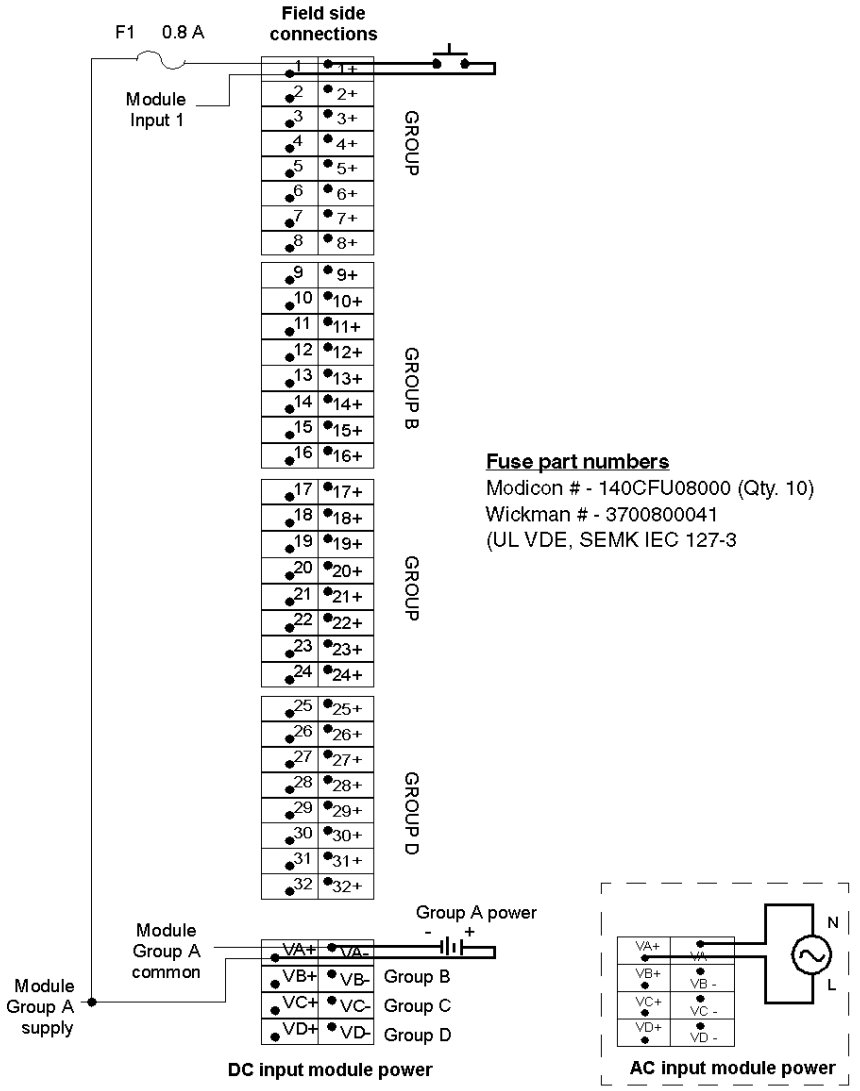
## Dimensions

The following figures show the dimensions for the 140CFB03200 terminal block.



## Wiring of Input Modules

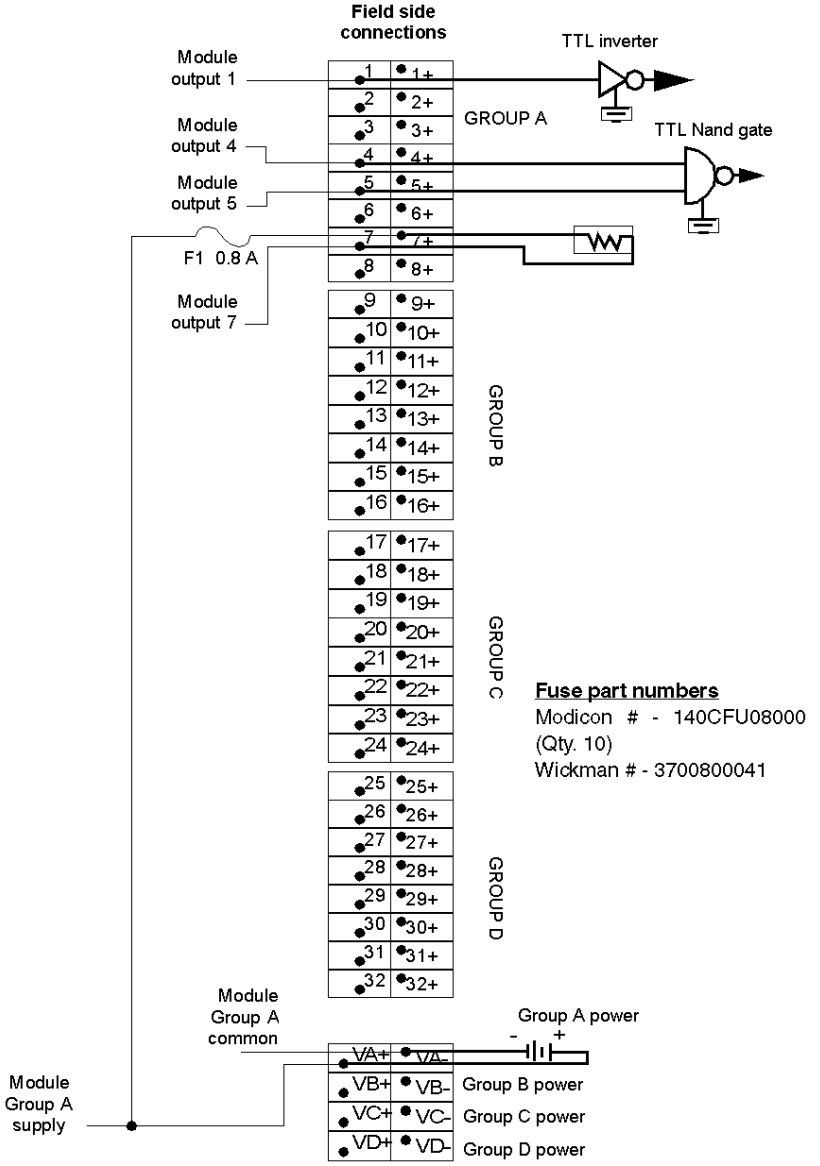
The following figure shows the 140 CFB 032 00 wiring for the following input modules: 140 DAI 353 00, 140 DAI 453 00, 140 DAI 553 00, 140 DDI 353 00, and 140 DDI 853 00.




**NOTE:** The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10) can be used to jumper the power between groups.

## Wiring of Output Module

The following figure shows the 140 CFB 032 00 wiring for the 140 DDO 153 10 output module.




**NOTE:** The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10) can be used to jumper the power between groups.

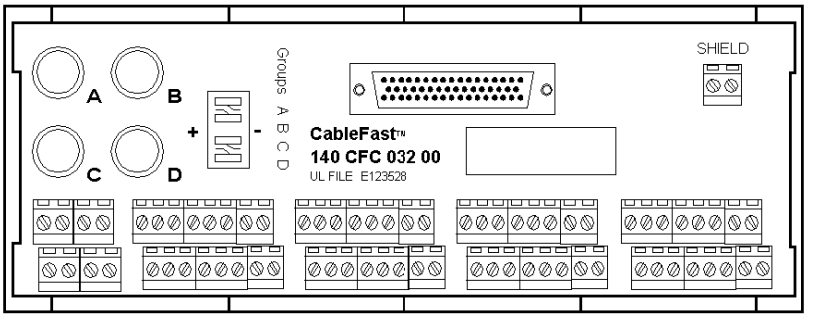
# 140 CFC 032 00 Quantum CableFast Cabling Block

## Overview

The C block provides connections for 32 group fused input or output points. You may use this CableFast terminal block for 1- or 2-wire inputs or outputs, and features a fuse per group – up to a total of four groups. Users select input or output mode via four switches located on the module. (The default is input mode.)

## Terminal Block

The following figure shows the terminal block for the 140 CFC 032 00 module.



## Application Notes

The following are the application notes for the 140 CFC 032 00 module.

1. **Configuration** – arranged in four groups of eight I/O points (two terminals per point). This block may be used for one- and two-wire inputs or outputs. The input and output mode is selected via four switches located on the block.
2. **Compatibility** – this terminal block provides 0.8 A group fusing for the following discrete modules:

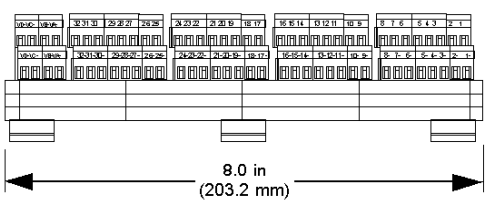
The following table shows the modules provided with 0.8 A group fusing.

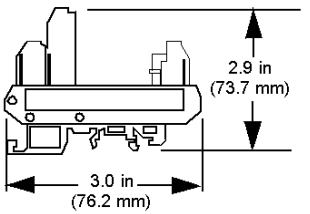
| Module | Mode | Switch setting | Fuse rating |
|---|---|---|---|
| 140 DAI 353 00 | Input | + | 0.8 A |
| 140 DAI 453 00 | Input | + | 0.8 A |
| 140 DAI 553 00 | Input | + | 0.8 A |
| 140 DDI 153 10 | Input | - | 0.8 A |
| 140 DDI 353 00 | Input | + | 0.8 A |
| 140 DDI 853 00 | Input | + | 0.8 A |
| 140 DDO 153 10 | Output | + | 4 A |
| 140 DDO 353 00 | Output | - | 4 A |

**NOTE:** Select input or output mode with the four switches located on the terminal.

## Dimensions

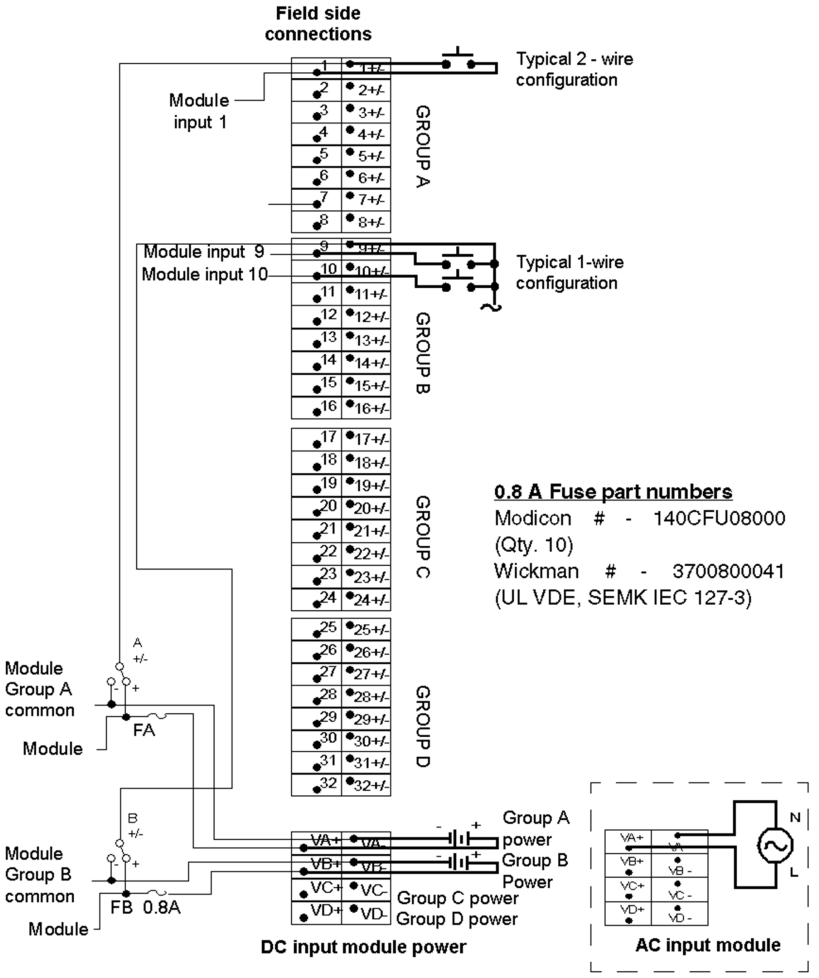
The following figures show the dimensions for the 140 CFC 032 00 terminal block block. All four switches must be set to the same position.



## Wiring for Input Modules

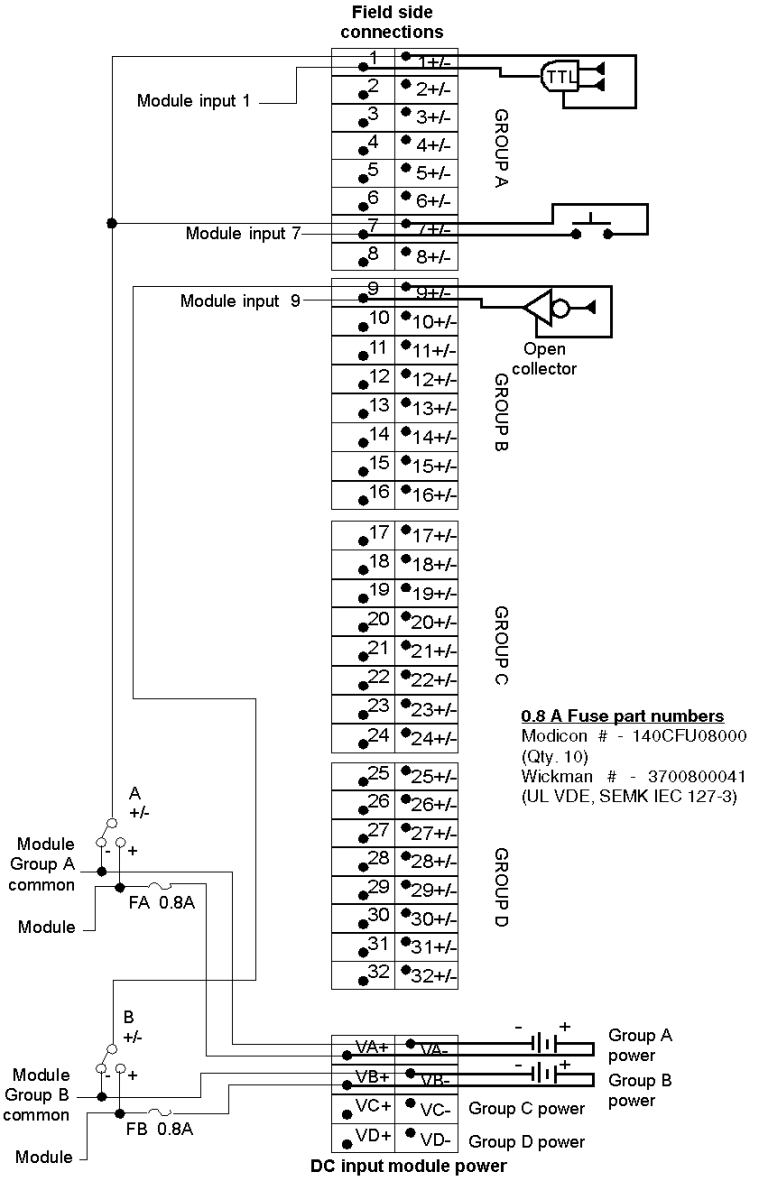
The following shows the 140 CFC 032 00 wiring for the following input modules: 140 DAI 353 00, 140 DAI 453 00, 140 DAI 553 00, 140 DDI 353 00, and 140 DDI 853 00.




**NOTE:** The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.

## Wiring for 140 DDI 153 10 Input Module

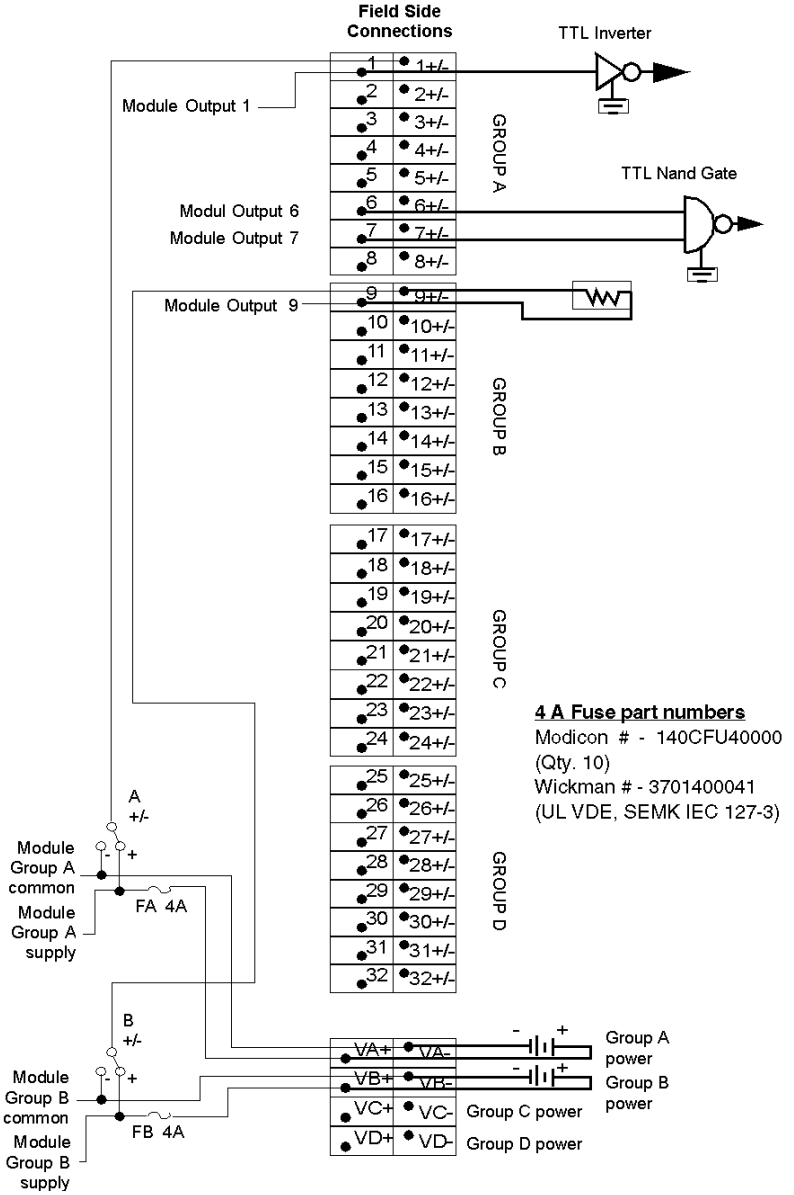
The following figure shows the 140 CFC 032 00 wiring for the 140 DDI 153 10 input module.




**NOTE:** The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.

## Wiring for 140 DDO 153 10 Output Module

The following figure shows the 140 CFC 032 00 wiring for the 140 DDO 153 10 output module.

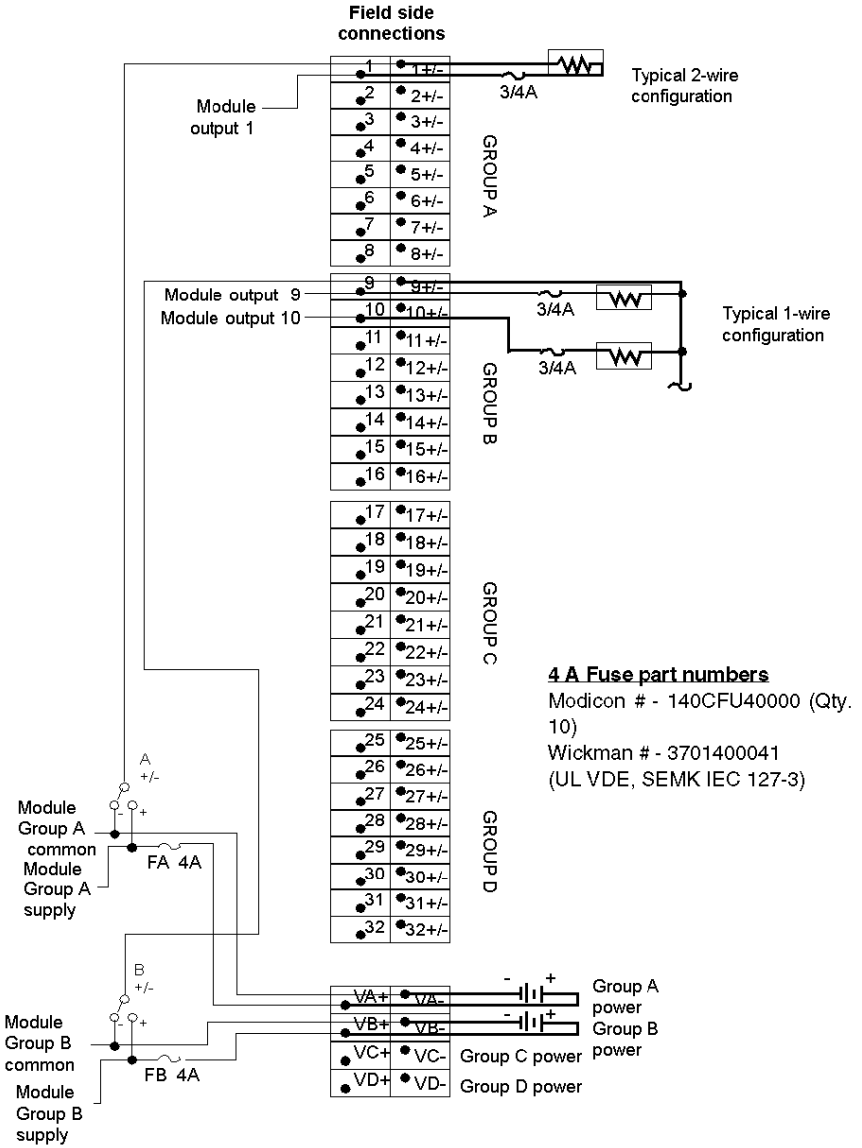


NOTE:

1. The 140 CFC 032 00 is shipped with Modicon the 140 CFU 080 00 (0.8 A) fuse installed. Check that the Modicon 140CFU40000 (4 A) fuse is installed when the 140 CFC 032 00 and the 140 DDO 153 00 are wired together.
2. The terminal block commoning strip, Modicon 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.

## Wiring for 140 DDO 353 0X Output Module

The following figure shows the 140 CFC 032 00 wiring for the 140 DDO 353 00 and 140 DDO 353 01 output modules.



NOTE:

1. The 140 CFC 032 00 is shipped with the Modicon 140 CFU 080 00 (0.8 A) fuse installed. Make sure the Modicon 140 CFU 400 00 (4 A) fuse is installed when the 140 CFC 032 00 and the 140 DDO 353 00 are wired together.
2. The terminal block commoning strip, Modicon 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.

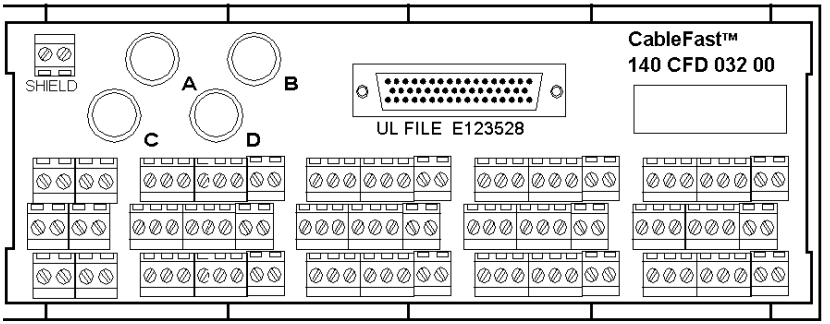
# 140 CFD 032 00 Quantum CableFast Cabling Block

## Overview

The CableFast cabling D block is used for sensors requiring either 2- or 3-wire electrical interfaces. A fuse per group is supplied to accommodate the I/O module (4) groups.

## Terminal Block

The following figure shows the 140 CFD 032 00 terminal block.



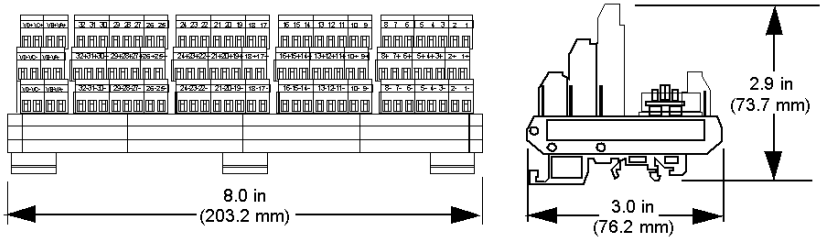

## Application Notes

The following are the application notes for the 140 CFD 032 00 module.

1. **Configuration** – arranged in four groups of eight I/O points. Each input is allocated three terminals.
2. **Compatibility** – this terminal block provides 0.8 A group fusing connection points for 3- and 2-wire proximity switches and is used with the following modules: 140 DAI 353 00, 140 DAI 453 00, 140 DAI 553 00, 140 DDI 353 00, and 140 DDI 853 00.

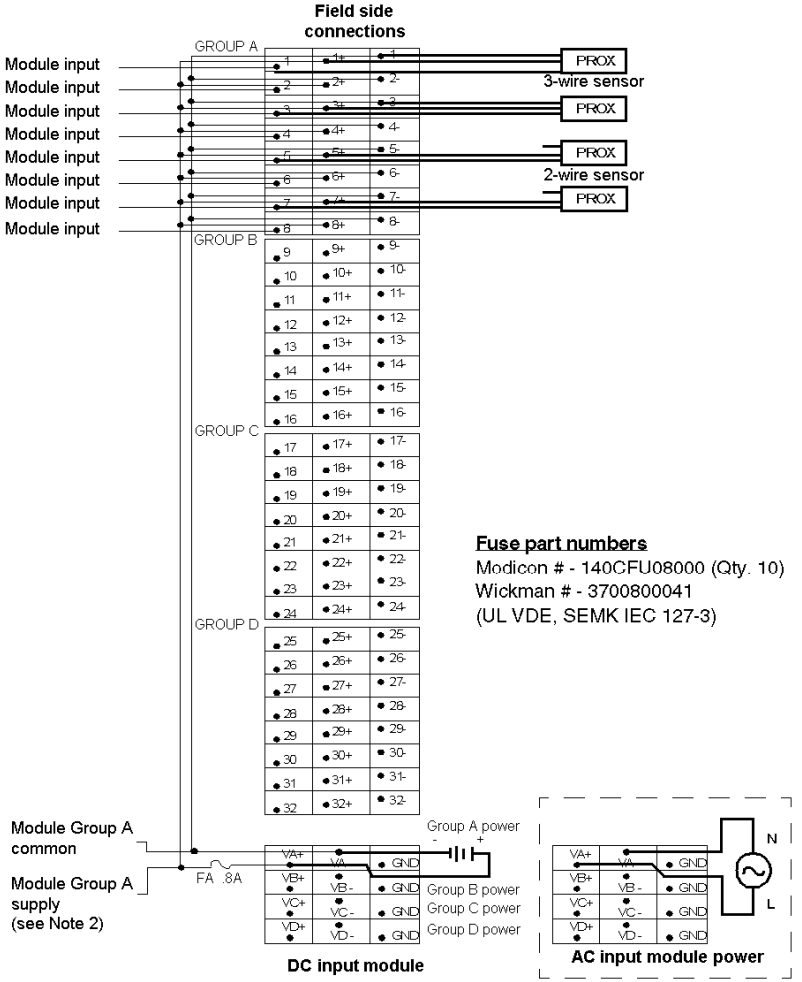
## Dimensions

The following shows the dimensions for the 140 CFD 032 00 module.



## Wiring

The following figure shows the wiring for the 140 CFD 032 00 module.



NOTE:

1. The GND (ground) terminal points are not connected.
2. The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.

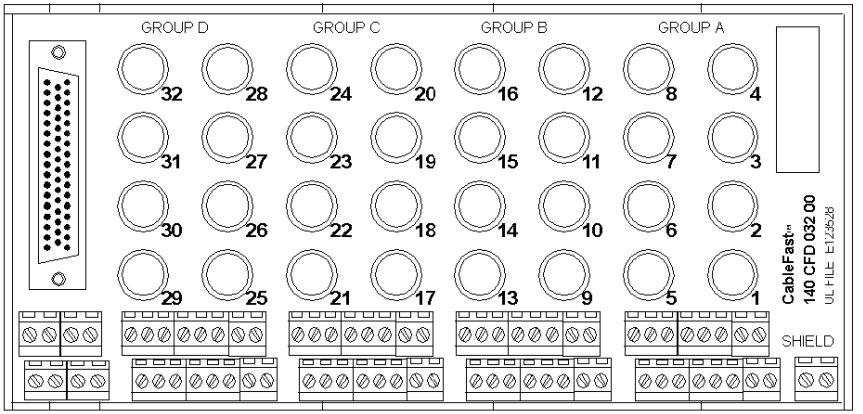
# 140 CFE 032 00 Quantum CableFast Cabling Block

## Overview

The CableFast E cabling block provides connections for 32 individually fused 24 VDC outputs. 1- and 2-wire interfacing may be selected. Field power must be supplied to the four groups.

## Terminal Block

The following figure shows the 140 CFE 032 00 terminal block.



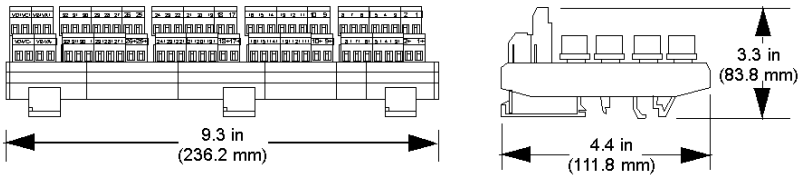

## Application Notes

The following are the application notes for the 140 CFE 032 00 module.

1. **Configuration** – arranged in four groups of eight I/O points. Two terminals per point help prevent disruption of service due to a single point error.
2. **Compatibility** – this terminal block provides individual 32-point 0.8 A fusing for the 140 DDO 353 00 and the 140 DDO 353 01 modules.

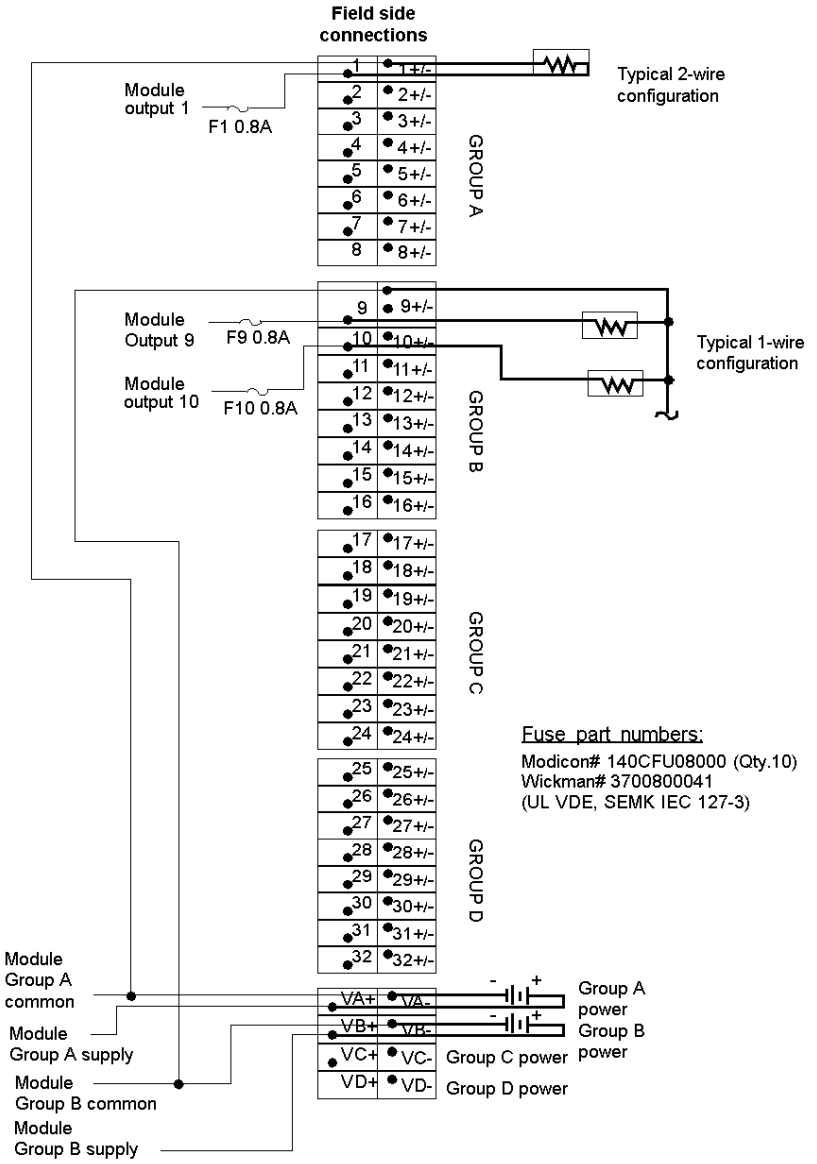
## Dimensions

The following shows the dimensions for the 140 CFE 032 00 module.



## Wiring Diagram

The following figure shows the wiring for the 140 CFE 032 00 module.




**NOTE:** The terminal block commoning strip, Modicon # 140 CFX 001 10 (QTY. 10), can be used to jumper the power between groups.

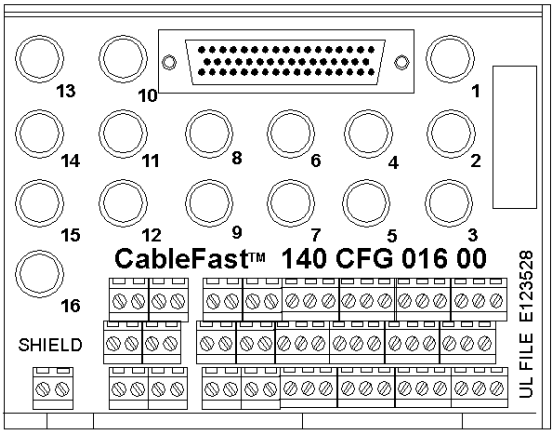
# 140 CFG 016 00 Quantum CableFast Cabling Block

## Overview

The G CableFast cabling block is a high-power output block used on both AC and DC circuits requiring up to 2 A. Individual fusing is provided and may be used in both 1- and 2-wire installations. It is also used for isolated AC modules.

## Terminal Block

The following figure shows the 140 CFG 016 00 terminal block.



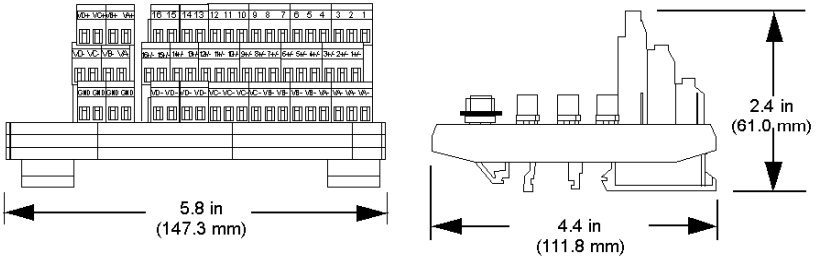

## Application Notes

The following are the application notes for the 140 CFG 016 00 module.

1. **Configuration** – Arranged in 16 isolated I/O points.
2. **Compatibility** – This terminal block provides individual 16-point 4 A fused connection points for the following modules: 140 DAI 340 00, 140 DAI 440 00, 140 DAI 540 00, 140 DAO 840 00, 140 DAO 840 10, 140 DAO 842 10, 140 DAO 842 20, and 140 DDO 843 00.

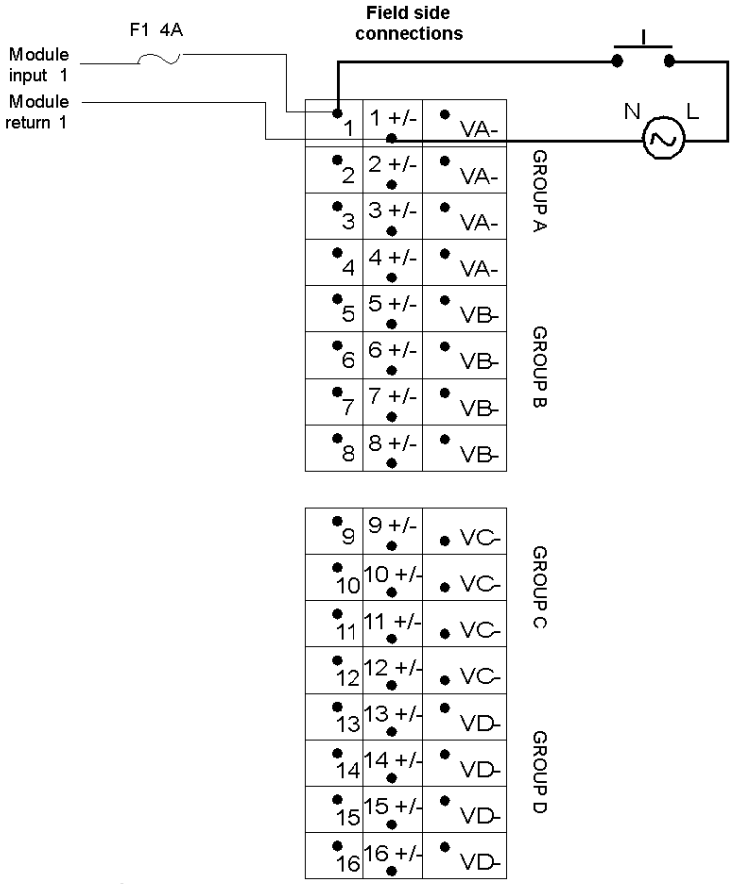
## Dimensions

The following figures show the dimensions for the 140 CFG 016 00 module.



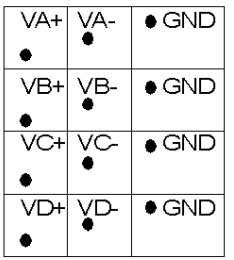
## Wiring for Isolated AC Input Mode

The following figure shows the 140 CFG 016 00 wiring for the input (isolated AC input mode) modules: 140 DAI 340 00, 140 DAI 440 00, and 140 DAI 540 00.




**Fuse part numbers**

Modicon # - 140CFU40000 (Qty. 10)
Wickman # - 3701400041 (UL VDE, SEMK IEC 127-3)

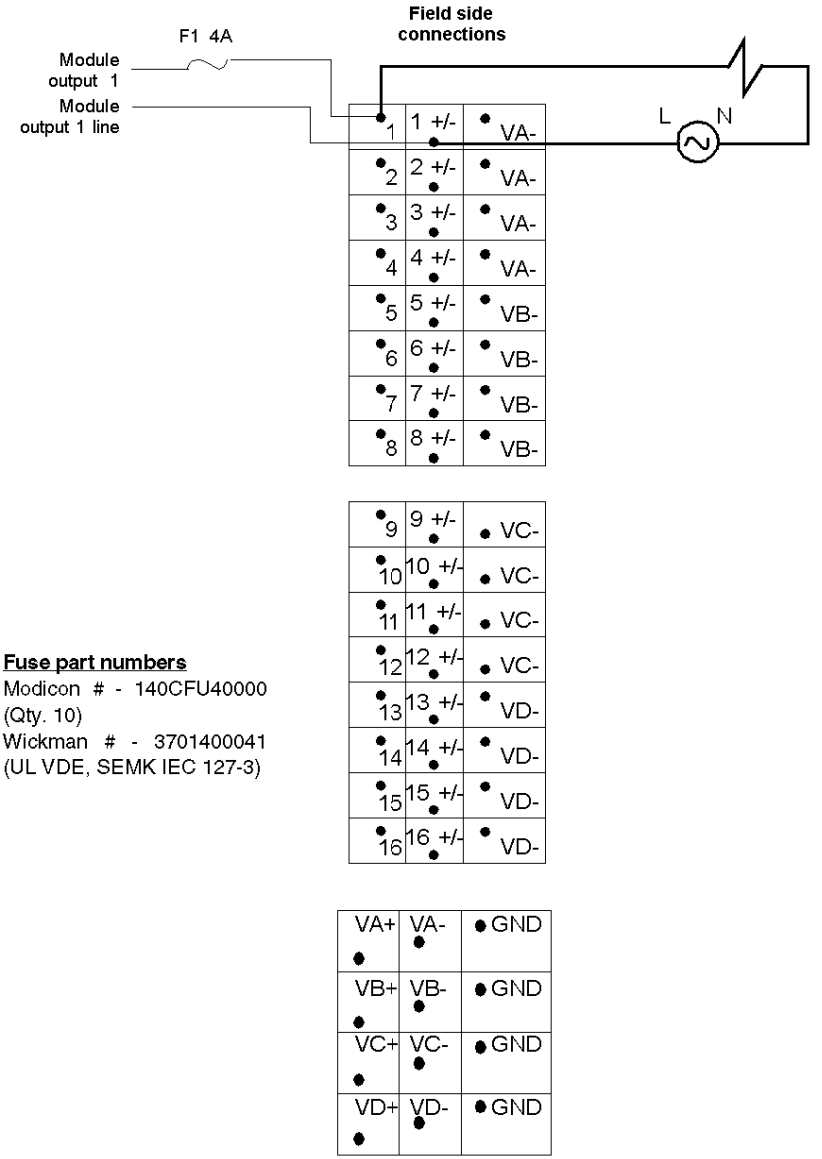


NOTE:

1. The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.
2. The GND (ground) terminal points are not connected.

## Wiring for Isolated Output Mode

The following shows the 140 CFG 016 00 wiring for the 140 DAO 840 00 and 140 DAO 840 10 output modules (isolated output mode).




**Fuse part numbers**

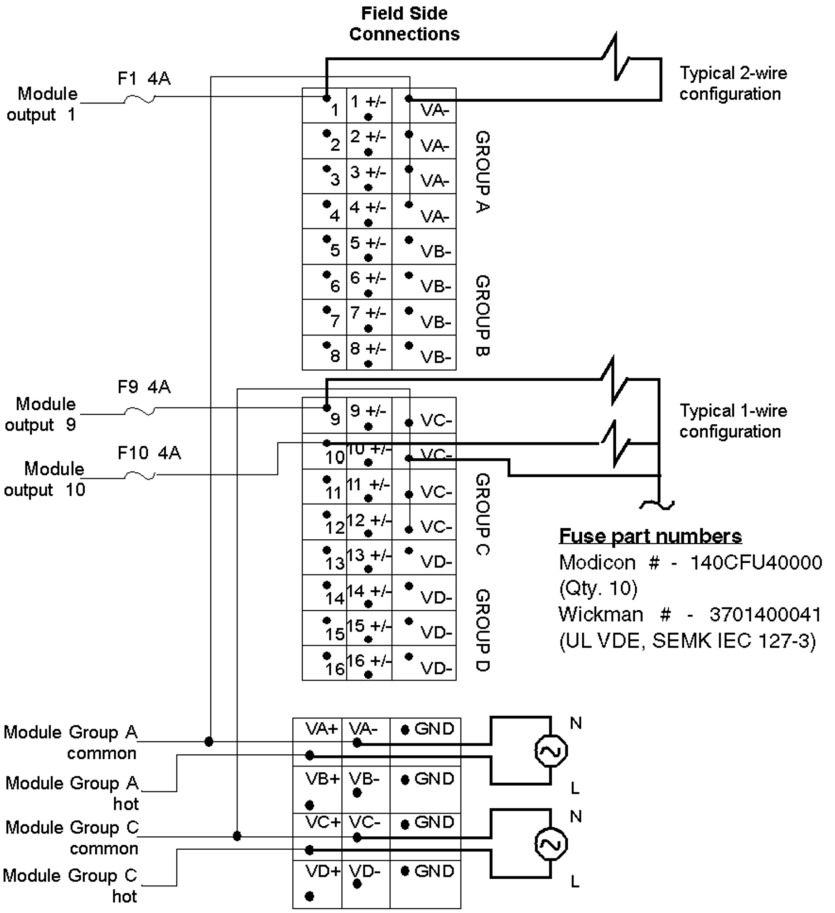
Modicon # - 140CFU40000
(Qty. 10)
Wickman # - 3701400041
(UL VDE, SEMK IEC 127-3)

NOTE:

1. The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.
2. The GND (ground) terminal points are not connected

## Wiring for Grouped AC Output Mode

The following figure shows the 140 CFG 016 00 wiring for the 140 DAO 842 10 and 140 DAO 842 20 output modules (grouped AC output mode).

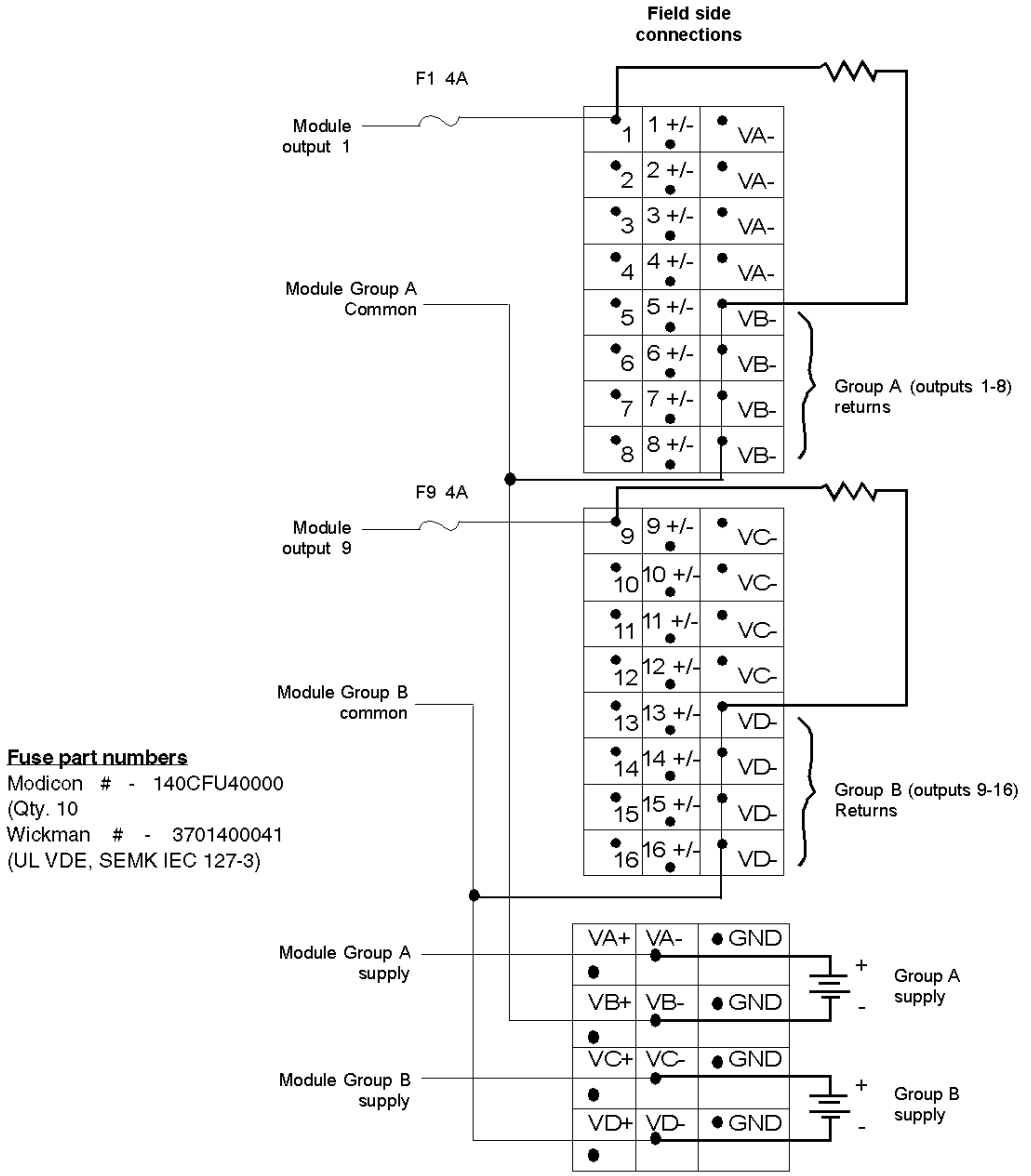



NOTE:

1. The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.
2. The GND (ground) terminal points are not connected.

## Wiring for Grouped DC Output Mode

The following figure shows the 140 CFG 016 00 wiring for the 140 DDO 843 00 (grouped DC output mode) module.




**Fuse part numbers**
Modicon # - 140CFU40000
(Qty. 10
Wickman # - 3701400041
(UL VDE, SEMK IEC 127-3)

NOTE:

1. The terminal block commoning strip, Modicon # 140 CFX 001 10 (Qty. 10), can be used to jumper the power between groups.
2. The GND (ground) terminal points are not connected.

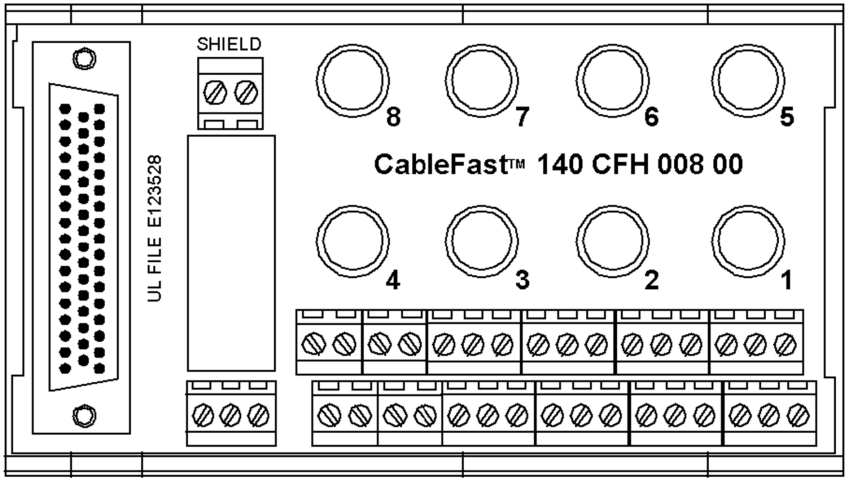
# 140 CFH 008 00 Quantum CableFast Cabling Block

## Overview

The H CableFast cabling block is used for analog inputs, with individual fusing provided per channel. This interface provides plus, minus, shield, and power supply interface for both field and loop power configurations.

## Terminal Block

The following figure shows the 140 CFH 008 00 terminal block.



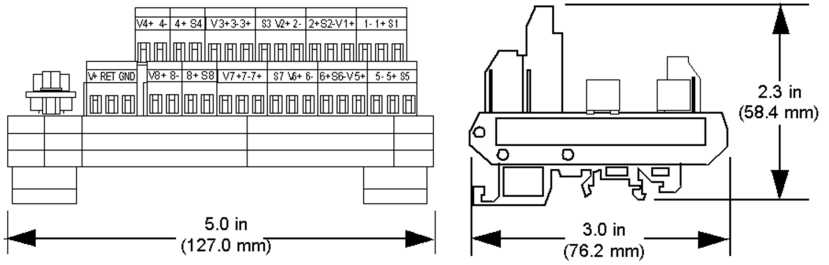

## Application Notes

The following are the application notes for the 140 CFH 008 00 module.

1. **Configuration** – eight analog inputs with a common loop supply. Each point is allocated four terminals.
2. **Compatibility** – this terminal block provides individually 0.063 A fused connection point sets for the 140 ACI 030 00 and 140 AVI 030 00 analog input modules.

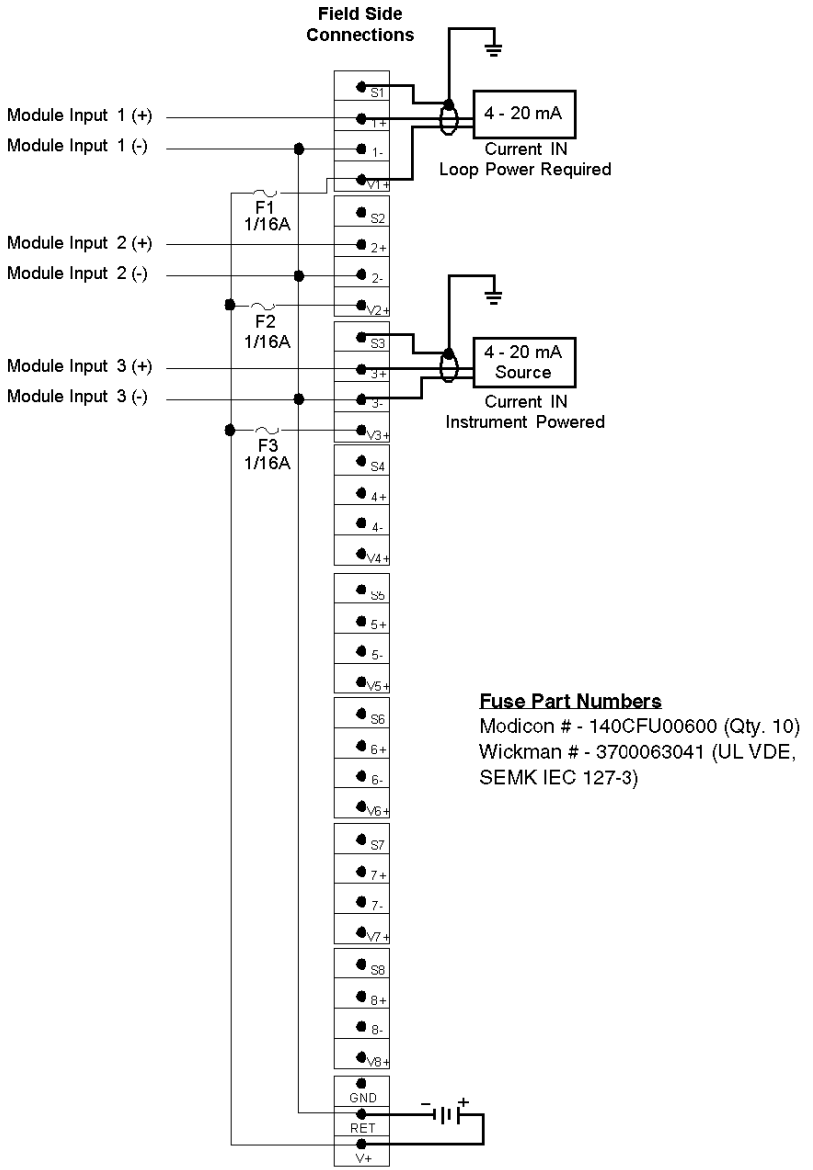
## Dimensions

The following figures show the dimensions for the 140 CFH 008 00 module.



## Wiring Diagram (Source Grounding)

The following figure shows the wiring for the 140 CFH 008 00 (source grounding) module.

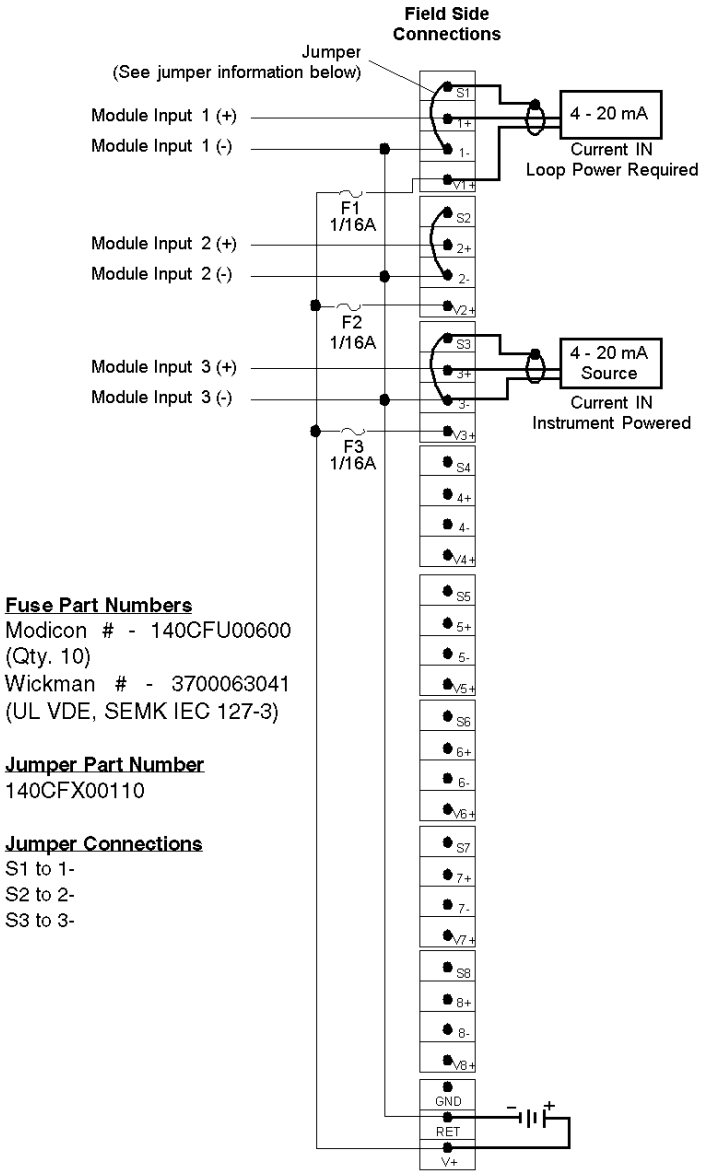


NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140 ACI 030 00 and the 140 AVI 030 00, see the wiring diagrams for said modules.
3. The GND (ground) terminal point is not connected.

## Wiring Diagram (Instrument Grounding)

The following figure shows the wiring (instrument grounding) for the 140 CFH 008 00 module.




**Fuse Part Numbers**
Modicon # - 140CFU00600
(Qty. 10)
Wickman # - 3700063041
(UL VDE, SEMK IEC 127-3)

**Jumper Part Number**
140CFX00110

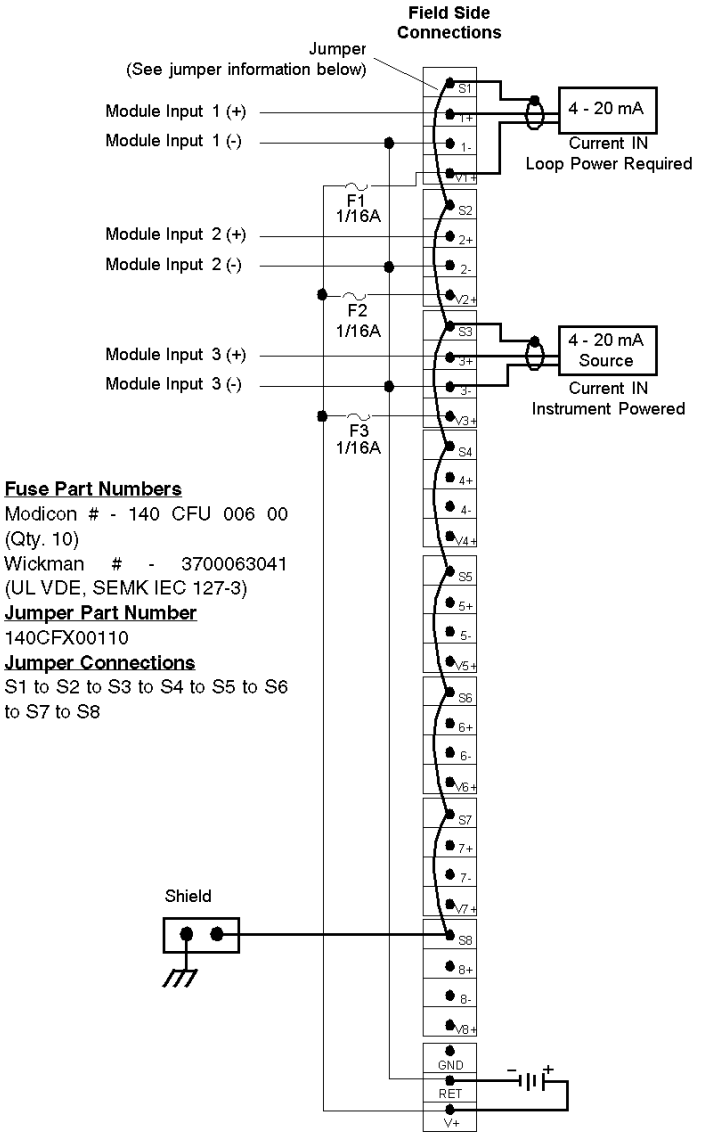
**Jumper Connections**
S1 to 1-
S2 to 2-
S3 to 3-

NOTE:

- If you use a single power supply, there will be no channel-to-channel isolation of the input points.
- For the required jumper options for the 140 ACI 030 00 and the 140 AVI 030 00, see the wiring diagrams for said modules.
- The GND (ground) terminal point is not connected.

## Wiring Diagram (Chassis Grounding)

The following figure shows the wiring (chassis grounding) for the 140 CFH 008 00 module.




**Fuse Part Numbers**
Modicon # - 140 CFU 006 00 (Qty. 10)
Wickman # - 3700063041 (UL VDE, SEMK IEC 127-3)

**Jumper Part Number**
140CFX00110

**Jumper Connections**
S1 to S2 to S3 to S4 to S5 to S6 to S7 to S8

NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140 ACI 030 00 and the 140 AVI 030 00, see the wiring diagrams for said modules.
3. The GND (ground) terminal point is not connected.

# 140CFI00800 Quantum CableFast Cabling Block

## Overview

The I block is used for analog inputs. This interface provides plus, minus, shield, and power supply interfaces for both field and loop power configurations.

See Common Features of the CableFast Cabling System *(see page 396)* for information on common specifications and features of CableFast cabling blocks.

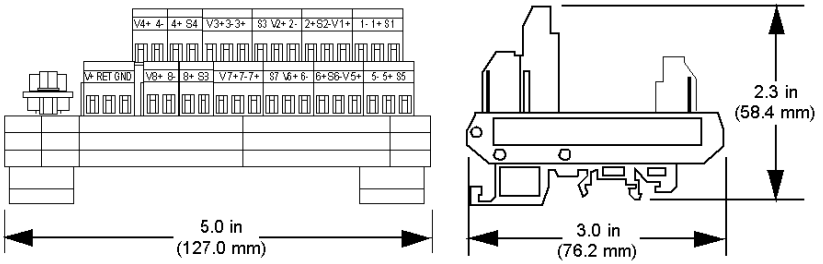
## Application Notes

The following are the application notes for the 140CFI00800 module.

1. **Configuration** – Eight analog inputs with a common loop supply. Each point is allocated four terminals.
2. **Compatibility** – This terminal block provides eight connection point sets for the 140ACI03000 and 140AVI03000 analog input modules.

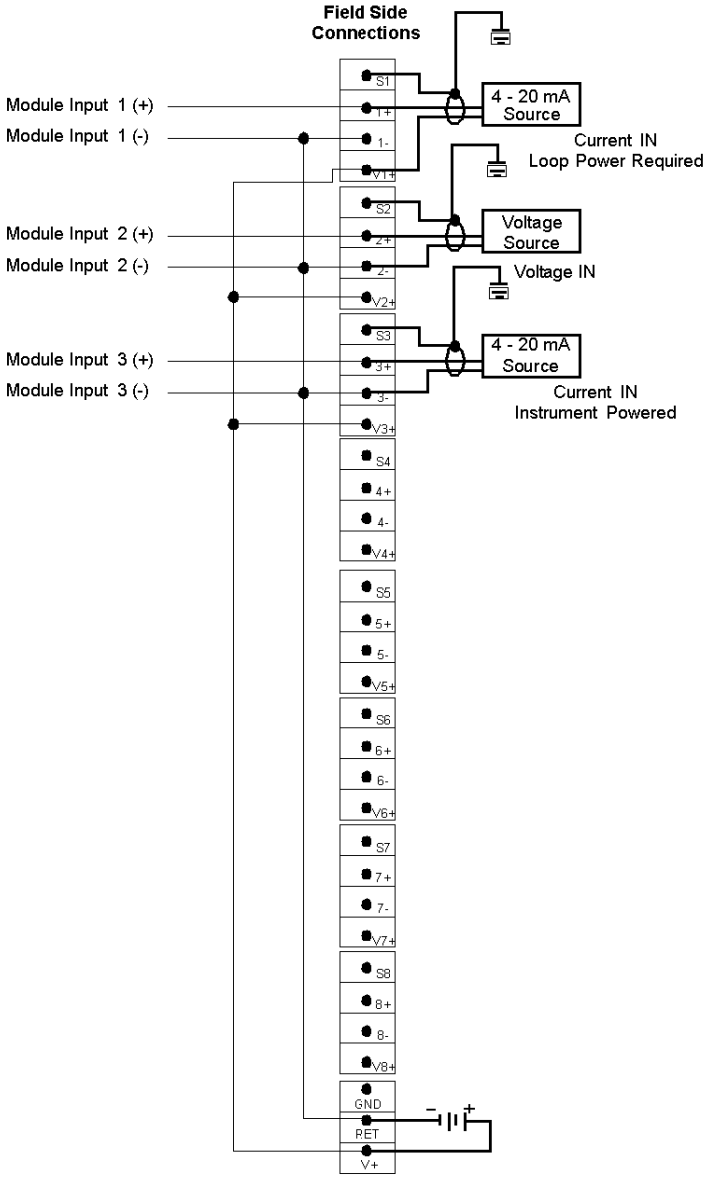
## Dimensions

The following figures show the dimensions for the 140CFI00800 module.



## Wiring Diagram (Source Grounding)

The following figure shows the wiring for the140CFI00800 (source grounding) module.

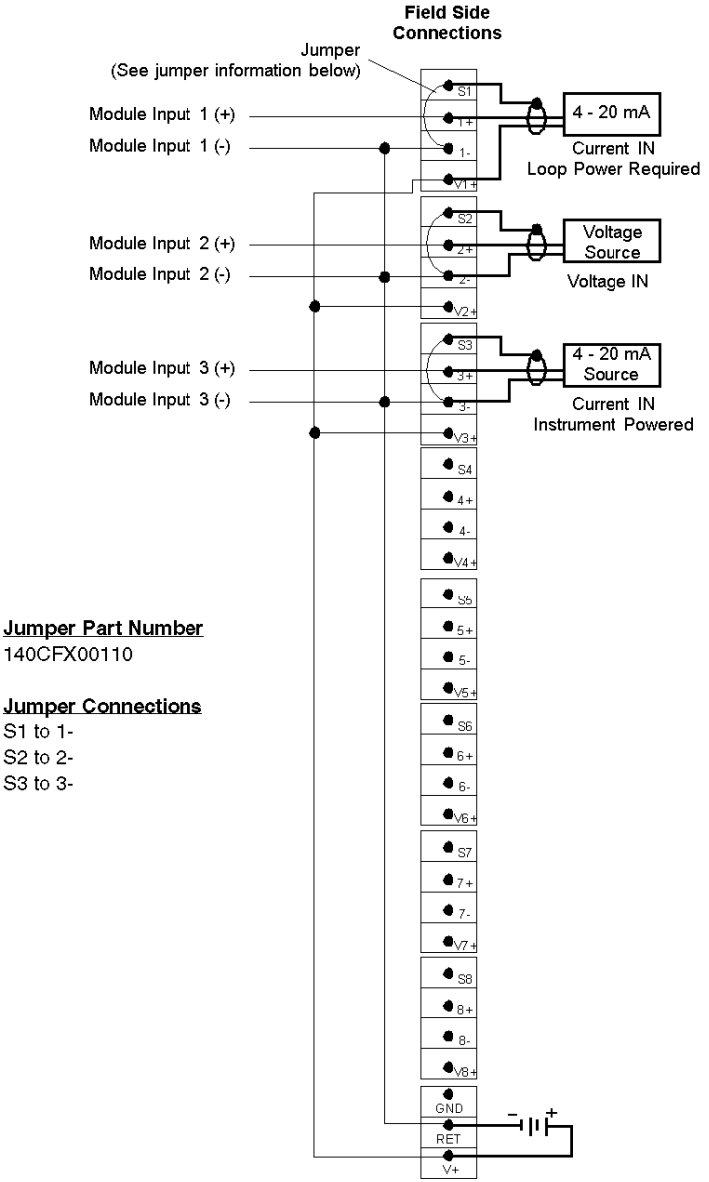


NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140ACI03000 and the 140AVI03000, see the wiring diagrams in ACI03000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* and AVI03000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* .
3. The GND (ground) terminal point is not connected.

## Wiring Diagram (Instrument Grounding)

The following figure shows the wiring for the140CFI00800 (instrument grounding) module.

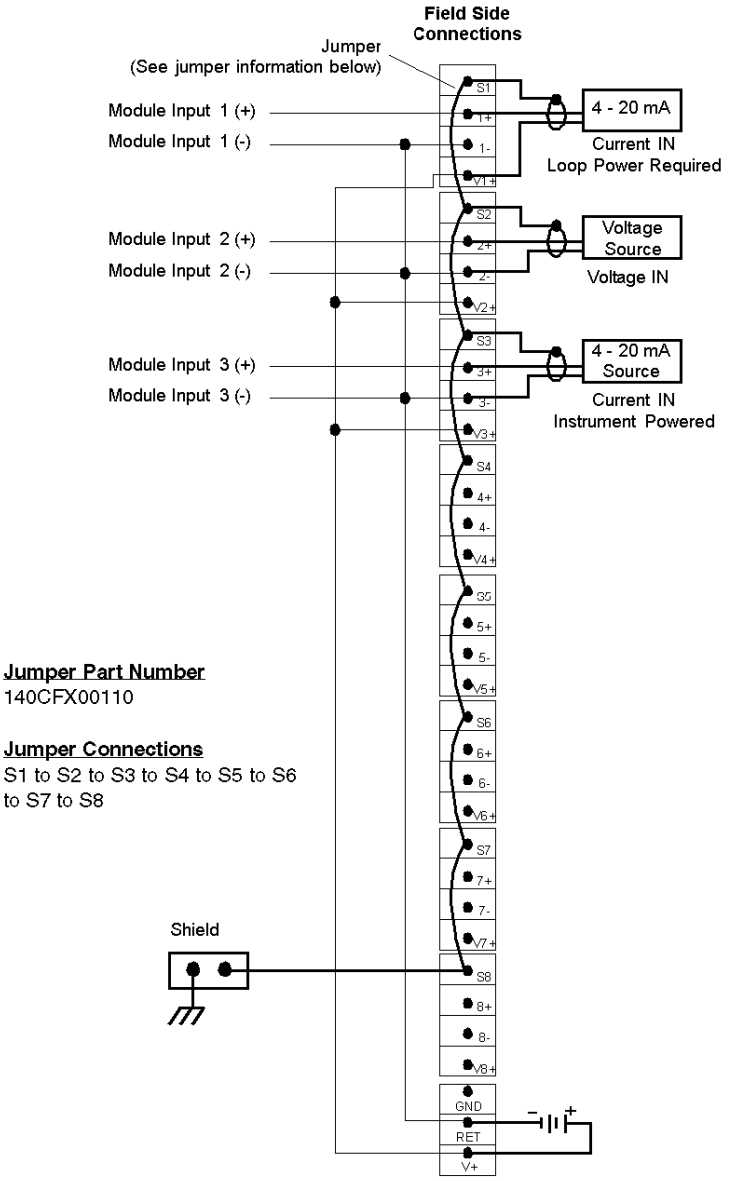


NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140ACI03000 and the 140AVI03000, see the wiring diagrams in ACI03000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* and AVI03000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*..
3. The GND (ground) terminal point is not connected.

## Wiring Diagram (Chassis Grounding)

The following figure shows the wiring for the 140CFI00800 (chassis grounding) module.



NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140ACI03000 and the 140AVI03000, see the wiring diagrams in ACI03000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* and AVI03000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* .
3. The GND (ground) terminal point is not connected.

# 140CFJ00400 Quantum CableFast Cabling Block

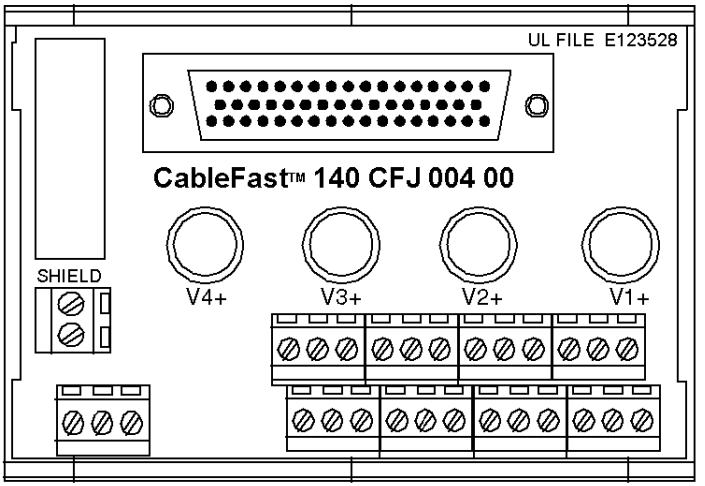
## Overview

The J block is used for analog outputs, with individual fusing provided per channel. This interface provides plus, minus, shield, and power supply interfaces for both field and loop power configurations.

See Common Features of the CableFast Cabling System *(see page 396)* for information on common specifications and features of CableFast cabling blocks.

## Terminal Block

The following figure shows the 140CFJ00400 terminal block.



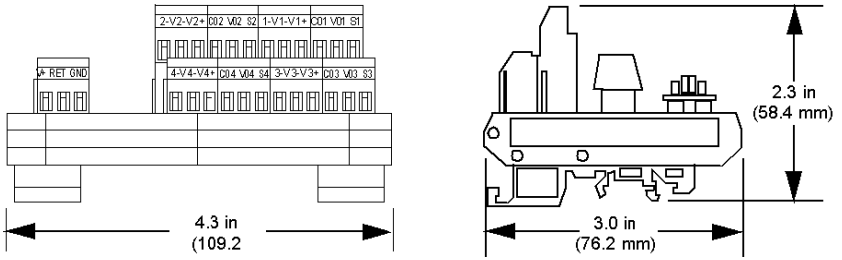

## Application Notes

The following are the application notes for the 140CFJ00400 module.

1. **Configuration** – Four analog outputs with a common loop supply. Each point is allocated six terminals.
2. **Compatibility** – This terminal block provides four individually 0.063 A fused connection point sets for the 140ACO02000 analog output module.

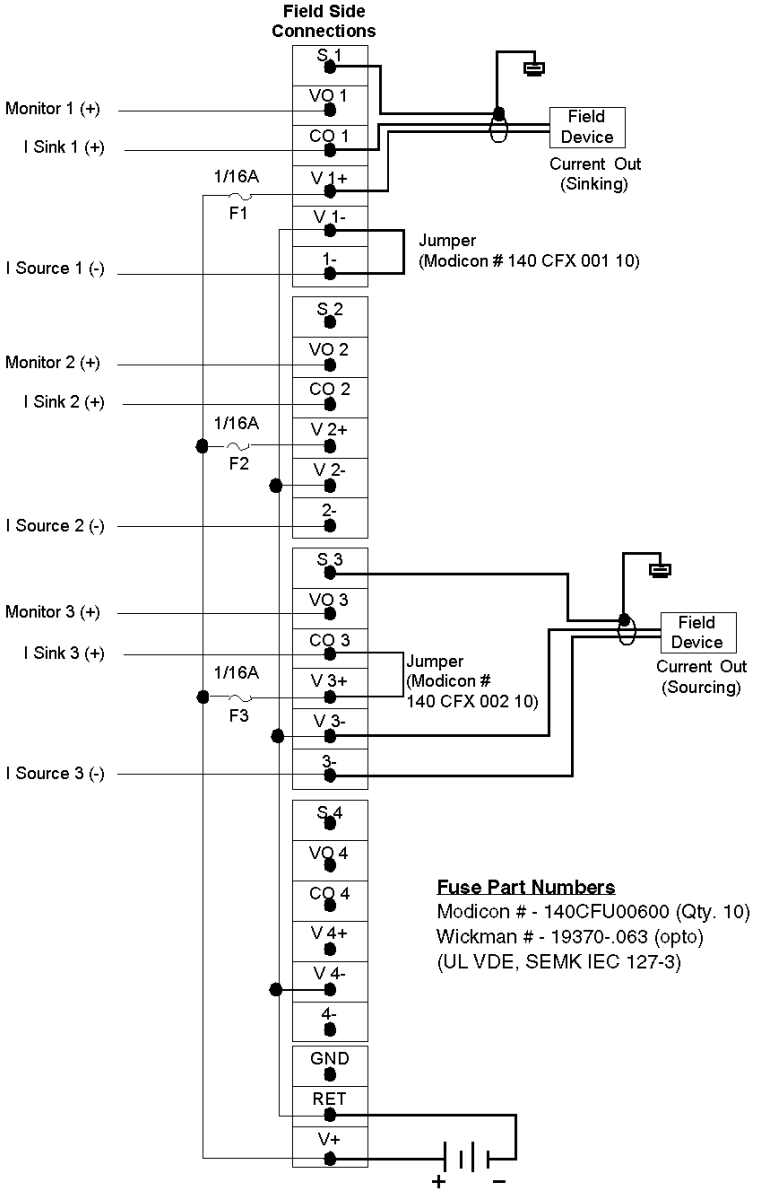
## Dimensions

The following figures show the dimensions for the 140CFJ00400 module.



## Wiring Diagram (Source Grounding)

The following figure shows the wiring for the 140CFJ00400 (source grounding) module.

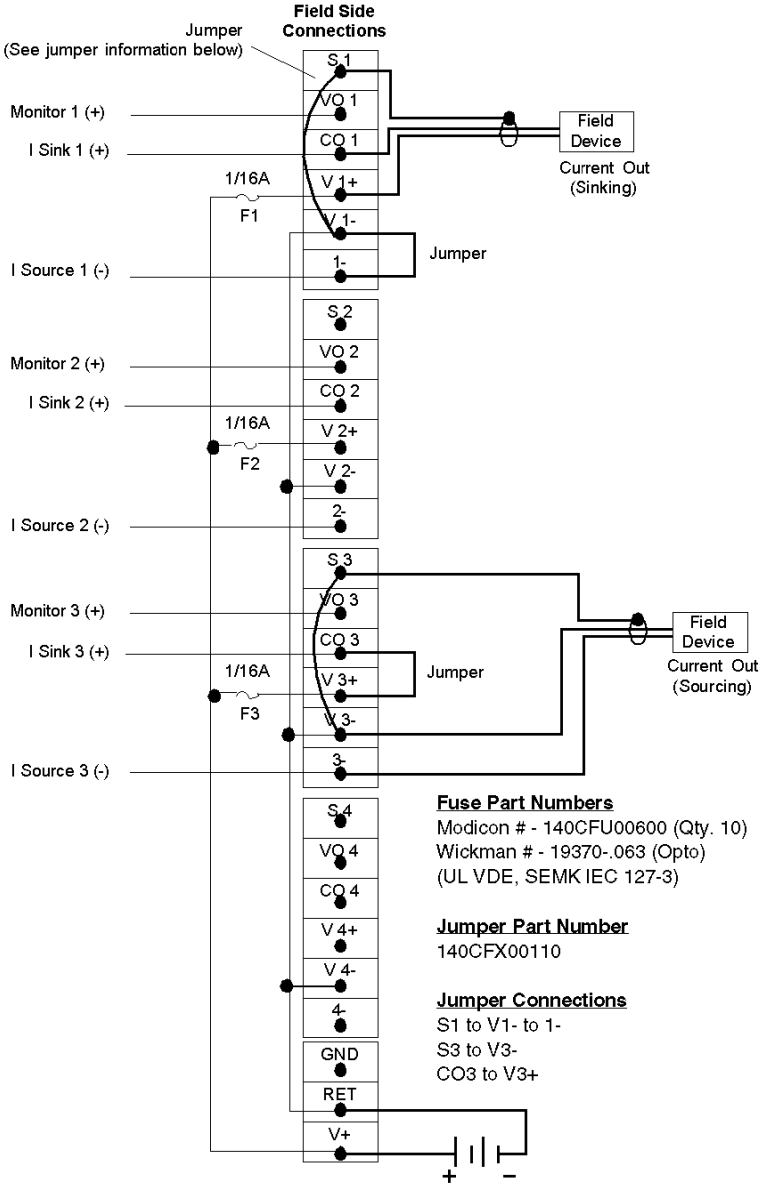


NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140ACO02000, see the wiring diagrams in ACO02000 map, wiring diagram *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*.
3. The GND (ground) terminal point is not connected.

## Wiring Diagram (Instrument Grounding)

The following figure shows the wiring for the140CFJ00400 (instrument grounding) module.

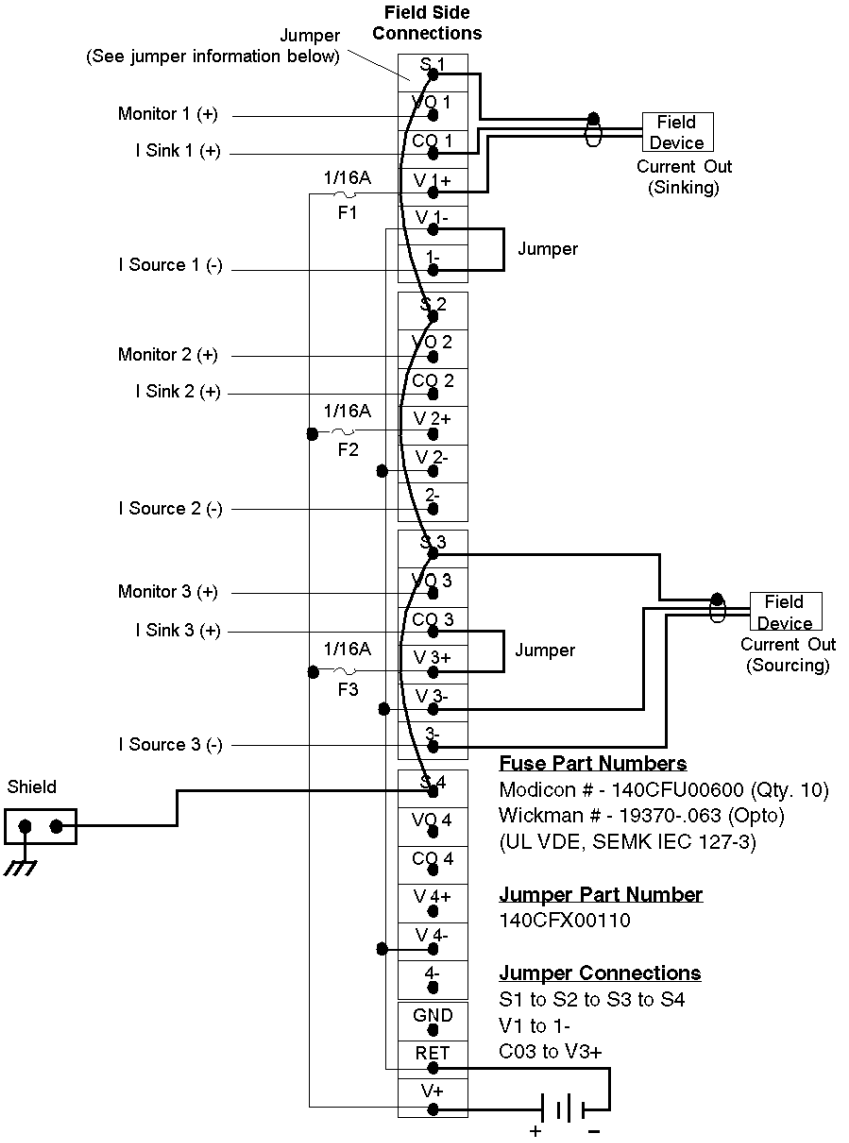


NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140ACO02000, see the wiring diagrams in ACO02000 map, wiring diagram *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*.
3. The GND (ground) terminal point is not connected.

## Wiring Diagram (Chassis Grounding)

The following figure shows the wiring for the 140CFJ00400 (chassis grounding) module.



NOTE:

1. When using a single power supply, there will be no channel-to-channel isolation of input points.
2. For the required jumper options for the 140ACO02000, see the wiring diagrams in ACO02000 map, wiring diagram *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*.
3. The GND (ground) terminal point is not connected.

# 140CFK00400 Quantum CableFast Cabling Block

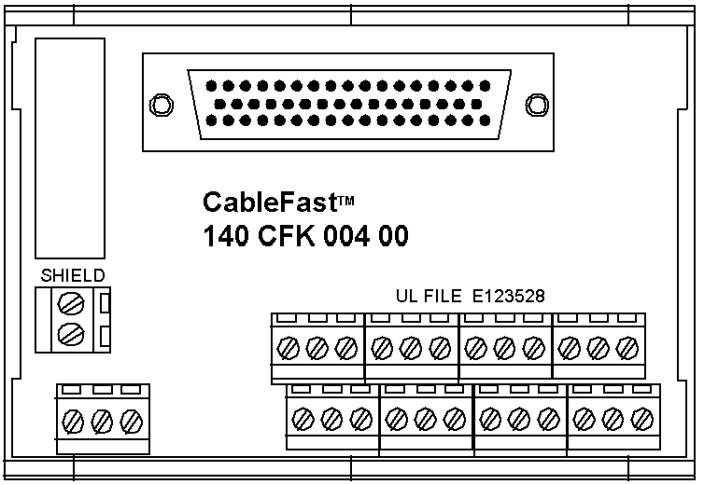
## Overview

The K block is used for analog outputs. This interface provides plus, minus, shield, and power supply interface for both field and loop power configurations.

See Common Features of the CableFast Cabling System *(see page 396)* for information on common specifications and features of CableFast cabling blocks.

## Terminal Block

The following figure shows the 140CFK00400 terminal block.



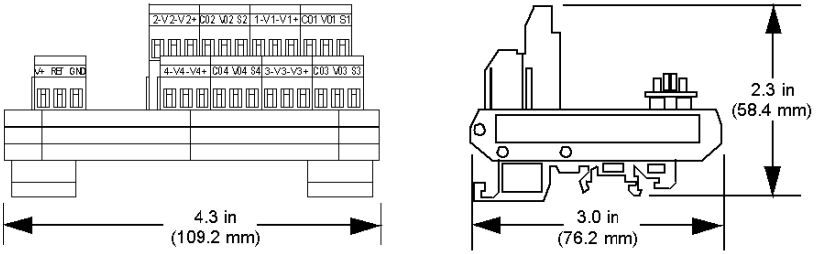

## Application Notes

The following are the application notes for the 140CFK00400 module.

1. **Configuration** – Four analog outputs with a common loop supply. Each point is allocated four terminals.
2. **Compatibility** – This terminal block provides four individually unfused connection point sets for the 140ACO02000 and 140AVO02000 analog output modules.

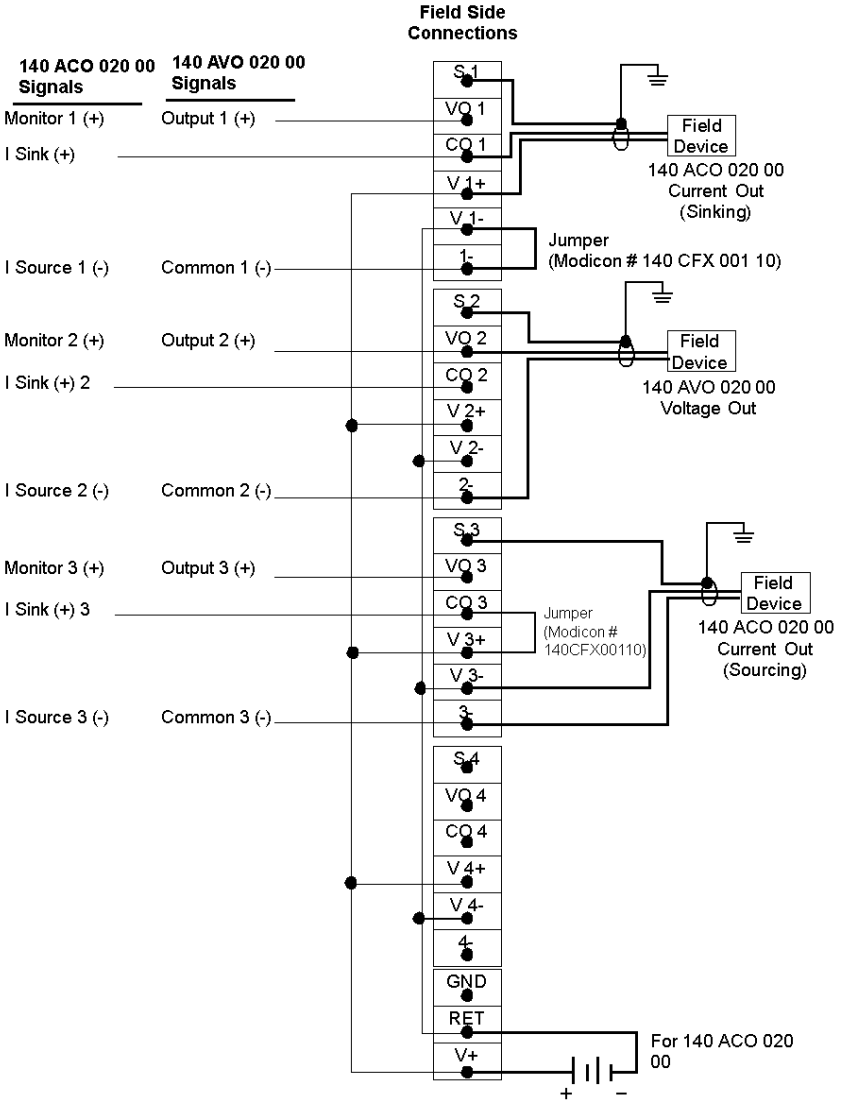
## Dimensions

The following figures show the dimensions for the 140CFK00400 module.



## Wiring Diagram (Source Grounding)

The following shows the wiring for the 140CFK00400 (source grounding) module.

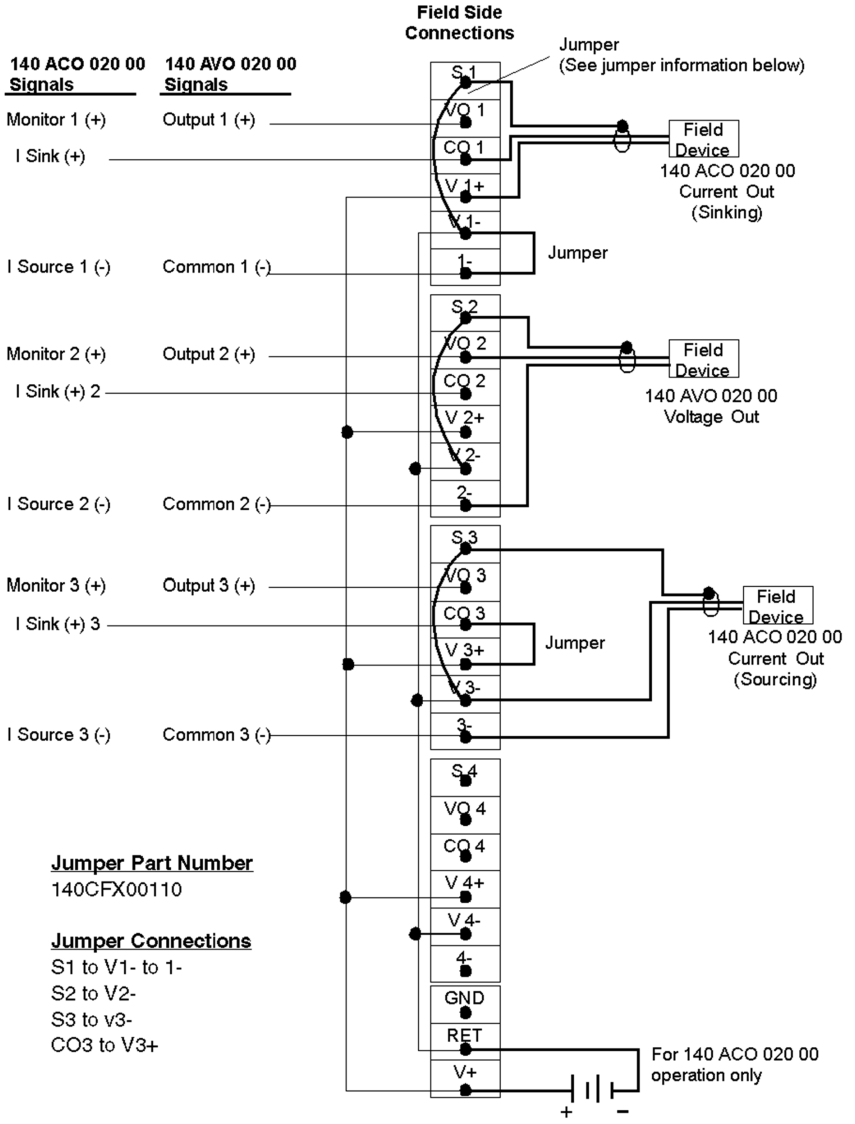


NOTE:

1. When used with the 140AVO02000 analog voltage out module, the master override connections and range select must be made on the Quantum I/O connector.
2. When using a single power supply, there will be no channel-to-channel isolation of input points.
3. For the required jumper options for the 140ACO02000, see wiring diagram in ACO02000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*.
4. The GND (ground) terminal point is not connected.

## Wiring Diagram (Instrument Grounding)

The following figure shows the wiring for the 140CFK00400 (instrument grounding) module.

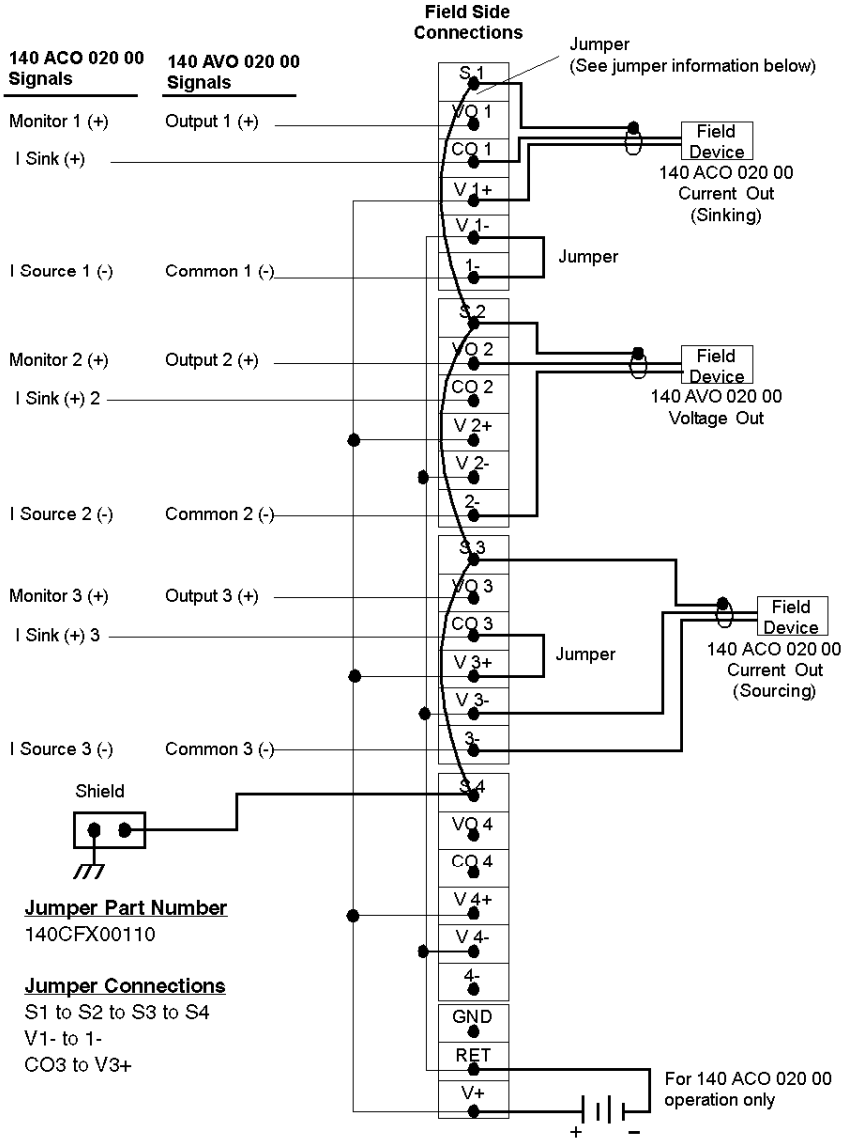


NOTE:

1. When used with the 140AVO02000 analog voltage out module, the master override connections and range select must be made on the Quantum I/O connector.
2. When using a single power supply, there will be no channel-to-channel isolation of input points.
3. For the required jumper options for the 140ACO02000 and the AVO02000, see wiring diagrams in ACO02000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* and the AVO02000 module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*.
4. The GND (ground) terminal point is not connected.

## Wiring Diagram (Chassis Grounding)

The following shows the wiring for the 140CFK00400 (chassis grounding) module.



NOTE:

1. When used with the 140AVO02000 analog voltage out module, the master override connections and range select must be made on the Quantum I/O connector.
2. When using a single power supply, there will be no channel-to-channel isolation of input points.
3. For wiring the 140ACO02000 and the 140AVO02000, see the wiring diagrams in ACO02000 I/O Module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)* and AVO02000 module *(see Quantum with Unity Pro, Discrete and Analog I/O, Reference Manual)*.
4. The GND (ground) terminal point is not connected.

# CableFast Cables

## Cable Specifications

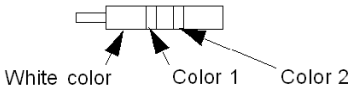
| | |
|---|---|
| Standard Power | |
| Cable Diameter | 0.43 in. nominal (10.9 mm) |
| Number of Conductors | 8-#20 AWG (0.8 mm), 7/28 tinned annealed copper; semi rigid PVC32-#26 AWG (0.4 mm), 7/34 tinned annealed copper; semi rigid PVC |
| Bend Radius (I.D.) | 0.75 in. min. (19.0 mm) |
| High Power | |
| Cable Diameter | 0.55 in. nominal (14.0 mm) |
| Number of Conductors | 8-#18 AWG (1.0 mm), 16/30 tinned annealed copper; semi rigid PVC32-#20 AWG (0.8 mm), 10/30 tinned annealed copper; semi rigid PVC |
| Bend Radius (I.D.) | 38.1 mm (1.50 in) min. |
| Common Specifications | |
| Cable Jacket | Jacket color: black, 0.040 in wall min, flexible PVC |
| Wire Strip Length | 8 mm (0.32 in) |
| Wire Marking | See the wire color coding table |
| Wire Rating | 300 V, 105 °C UL rated 2517, CSA Type AWM 1/2 FT1 |
| Cable Rating | 300 V, 105 °C rated |
| Shielding | Aluminum/polyester tape (aluminum side out) attached at connector body (360 °).#22 AWG, 7/30 drain wire.Shield resistance 16.55 Ω/M ft nominal |
| Agency Approval | UL-758; AWM style 2517 VW-1 and CSA C22:210.2; AWM I/II A/B FT1 |

## Cable Lengths

| Cable Lengths | Terminated | | Pigtail |
|---|---|---|---|
| | Standard Power | High Power | High Power |
| 0.91 m (3 ft) | X | X | |
| 1.82 m (6 ft) | X | X | X |
| 2.73 m (9 ft) | X | X | |
| 3.64 m (12 ft) | X | X | |
| 4.6 m (15 ft) | | | X |

## Inner Wire Color Codes

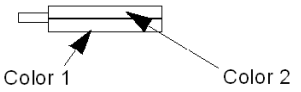
The following diagram represents the physical mapping of colors on the standard cablefast wires:




The following table describes the colors mapping of the standard cablefast wires:

| Wire/ Pin # | AWG for Standard Power Cable | AWG for High Power Cable | Color | Wire/ Pin # | AWG for Standard Power Cable | AWG for High Power Cable | Color |
|---|---|---|---|---|---|---|---|
| 1 | 26 | 20 | Black | 21 | 26 | 20 | White/Blue |
| 2 | 26 | 20 | Brown | 22 | 26 | 20 | White/Violet |
| 3 | 26 | 20 | Red | 23 | 26 | 20 | White/Gray |
| 4 | 26 | 20 | Orange | 24 | 26 | 20 | White/ Black/ Brown |
| 5 | 26 | 20 | Yellow | 25 | 26 | 20 | White/ Black/ Red |
| 6 | 26 | 20 | Green | 26 | 26 | 20 | White/ Black/ Orange |
| 7 | 26 | 20 | Blue | 27 | 26 | 20 | White/ Black/ Yellow |
| 8 | 26 | 20 | Violet | 28 | 26 | 20 | White/ Black/ Green |
| 9 | 20 | 18 | Black | 29 | 20 | 20 | Yellow |
| 10 | 20 | 18 | Brown | 30 | 20 | 18 | Green |
| 11 | 26 | 20 | Gray | 31 | 26 | 18 | White/ Black/ Blue |
| 12 | 26 | 20 | White | 32 | 26 | 20 | White/ Black/ Violet |
| 13 | 26 | 20 | White/ Black | 33 | 26 | 20 | White/ Black/ Gray |
| 14 | 26 | 20 | White/ Brown | 34 | 26 | 20 | White/ Brown/ Red |
| 15 | 26 | 20 | White/ Red | 35 | 26 | 20 | White/ Brown/ Orange |
| 16 | 26 | 20 | White/ Orange | 36 | 26 | 20 | White/ Brown/ Yellow |
| 17 | 26 | 20 | White/Yellow | 37 | 26 | 20 | White/ Brown/ Green |
| 18 | 26 | 20 | White/ Green | 38 | 26 | 20 | White/ Brown/ Blue |
| 19 | 20 | 18 | Red | 39 | 20 | 18 | Blue |
| 20 | 20 | 18 | Orange | 40 | 20 | 18 | Violet |

The following diagram represents the physical mapping of colors on the substitution cablefast wires:



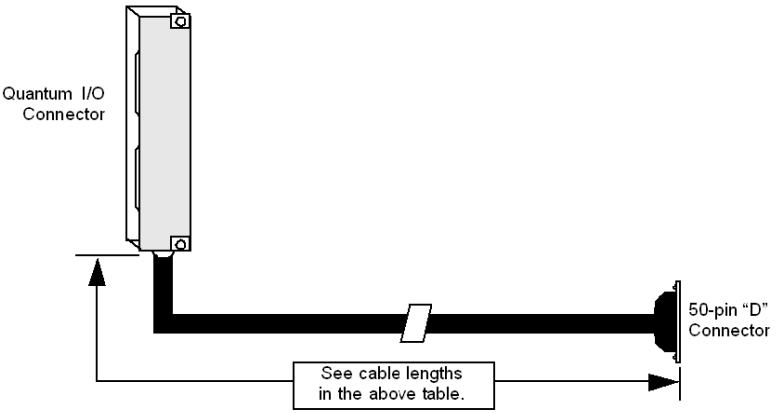

The following table describes the colors mapping of the substitution cablefast wires:

| Wire/ Pin # | AWG for Standard Power Cable | AWG for High Power Cable | Color | Wire/ Pin # | AWG for Standard Power Cable | AWG for High Power Cable | Color |
|---|---|---|---|---|---|---|---|
| 1 | 26 | 20 | Black | 21 | 26 | 20 | White/Blue |
| 2 | 26 | 20 | Brown | 22 | 26 | 20 | White/Violet |
| 3 | 26 | 20 | Red | 23 | 26 | 20 | White/Gray |
| 4 | 26 | 20 | Orange | 24 | 26 | 20 | Black/ Brown |
| 5 | 26 | 20 | Yellow | 25 | 26 | 20 | Black/ Red |
| 6 | 26 | 20 | Green | 26 | 26 | 20 | Black/ Orange |
| 7 | 26 | 20 | Blue | 27 | 26 | 20 | Black/ Yellow |
| 8 | 26 | 20 | Violet | 28 | 26 | 20 | Black/ Green |
| 9 | 20 | 18 | Black | 29 | 20 | 20 | Yellow |
| 10 | 20 | 18 | Brown | 30 | 20 | 18 | Green |
| 11 | 26 | 20 | Gray | 31 | 26 | 18 | Black/ Blue |
| 12 | 26 | 20 | White | 32 | 26 | 20 | Black/ Violet |
| 13 | 26 | 20 | White/ Black | 33 | 26 | 20 | Black/ Gray |
| 14 | 26 | 20 | White/ Brown | 34 | 26 | 20 | Brown/ Red |
| 15 | 26 | 20 | White/ Red | 35 | 26 | 20 | Brown/ Orange |
| 16 | 26 | 20 | White/ Orange | 36 | 26 | 20 | Brown/ Yellow |
| 17 | 26 | 20 | White/Yellow | 37 | 26 | 20 | Brown/ Green |
| 18 | 26 | 20 | White/ Green | 38 | 26 | 20 | Brown/ Blue |
| 19 | 20 | 18 | Red | 39 | 20 | 18 | Blue |
| 20 | 20 | 18 | Orange | 40 | 20 | 18 | Violet |

## 140 XTS Terminated Cables

| Part Number | Cable Type | | Cable Description |
|---|---|---|---|
| | Standard Power | High Power | |
| 140 XTS 002 03 | X | | CableFast system cable with Quantum I/O connector, 0.9 m (3 ft) and D-sub connector |
| 140 XTS 012 03 | | X | |
| 140 XTS 002 06 | X | | CableFast system cable with Quantum I/O connector, 1.8 m (6 ft) and D-sub connector |
| 140 XTS 012 06 | | X | |
| 140 XTS 002 09 | X | | CableFast system cable with Quantum I/O connector, 2.7 m (9 ft) and D-sub connector |
| 140 XTS 012 09 | | X | |
| 140 XTS 002 12 | X | | CableFast system cable with Quantum I/O connector, 3.7 m (12 ft) and D-sub connector |
| 140 XTS 012 12 | | X | |

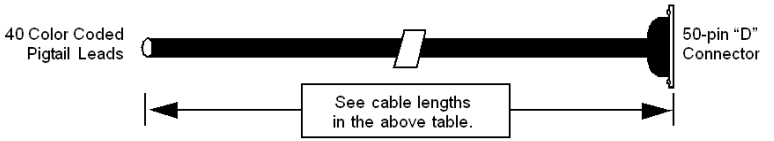
## I/O Connector




## 140 XCA 102 Pigtails

| Part Number | Cable Type | | Cable Description |
|---|---|---|---|
| | Standard Power | High Power | |
| 140 XCA 102 06 | | X | CableFast system cable, 6 ft (1.8 m), with D-sub connector and pigtails |
| 140 XCA 102 15 | | X | CableFast system cable, 4.6 m (15 ft), with D-sub connector and pigtails |

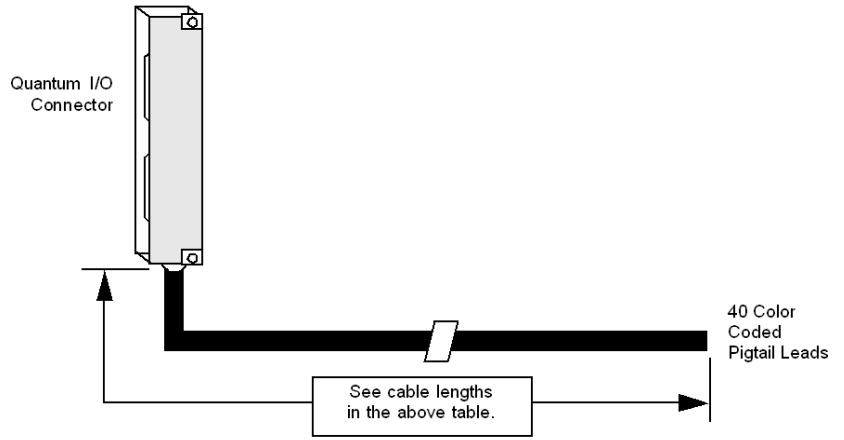
## Pigtail Leads




## 140 XTS 102 Pigtails

| Part Number | Cable Type | | Cable Description |
|---|---|---|---|
| | Standard Power | High Power | |
| 140 XTS 102 06 | | X | CableFast system cable with Quantum I/O connector, 1.8 m (6 ft), and pigtail cable |
| 140 XTS 102 15 | | X | CableFast system cable with Quantum I/O connector, 4.6 m (15 ft), and pigtail cable |

## I/O Connector for Pigtail Leads



# CableFast Accessories

## Overview

The following information pertains to the CableFast accessories.

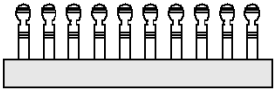
## Accessories

The following table shows the part numbers and descriptions for CableFast Accessories.

| Part Number | Description | Quantity |
|---|---|---|
| 140CFU40000 | Fuse Kit, Wickmann 4 A | 10 |
| 140CFU08000 | Fuse Kit, Wickmann 0.8 A | 10 |
| 140CFU00600 | Fuse Kit, Wickmann 0.063 A | 10 |
| 140CFX00110 | Terminal Block Common Strip, 10 Position (see below) | 10 |

## Terminal Block Common Strip

The following figure shows the terminal block common strip.

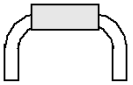


## Jumper, Fuse Replacement

Fuse replacement information is given in the following table.

| Part Number | Description | Quantity |
|---|---|---|
| 140 CFX 002 10 | Jumper, Fuse Replacement (see below) | 10 |

The following figure shows a jumper.



**NOTE:** The jumper is used instead of fuses as a disconnect device.

# Appendix E
# ConneXium Ethernet Cabling System

## Purpose

This appendix provides an overview on the ConneXium Ethernet Cabling System.

**NOTE:** For further details see *ConneXium Ethernet Cabling System Quick Reference Guide*

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Introduction | 470 |
| Configuration | 472 |

# Introduction

## The ConneXium NxS Switch

The internal switch control intelligence of a ConneXium NxS permits the redundant coupling of several network segments to a higher network segment.

The switch learns up to 2000 addresses, allowing the connection of several independent subnetworks. When reset, the switch deletes the learned addresses.

Ports' link integrity is provided through link testing in accordance with IEEE 802.3. Additionally, if the receive cable pair signals are incorrectly connected (if RD+ and RD- are switched), the NxS automatically detects it and reverses the connection polarity. The ConneXium NxS switch uses shielded RJ-45 connectors that are internally wired in the accordance with the MDI-X specification.

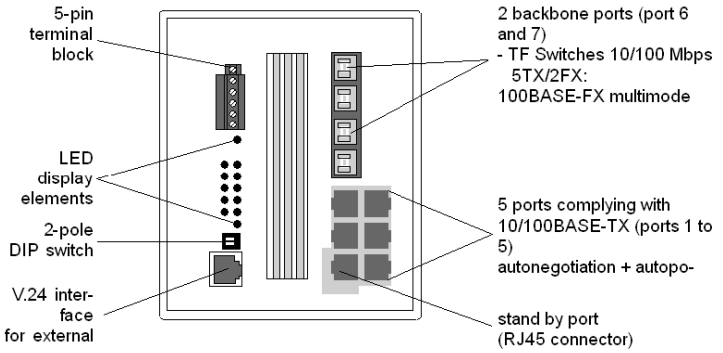
## Installation and Security Instructions

| *NOTICE* |
| --- |
| **IMPROPER POWER SUPPLY**<br>As electricity is used to operate this equipment, comply in every detail with the safety requirements specified in the operating instructions regarding the voltages to apply.<br>**Failure to follow these instructions can result in equipment damage.** |

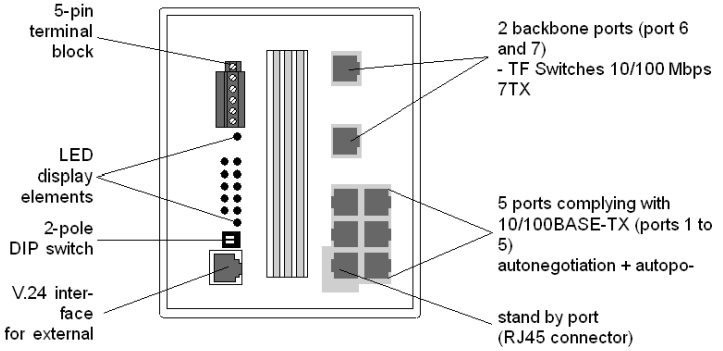
| *NOTICE* |
| --- |
| **NON-SELV SUPPLY VOLTAGES**<br>The ConneXium Switches 10/100 Mbps 5TX/2FL or 7TX units are designed for operation with safety extra-low voltage. Accordingly, connect only safety extra-low voltages (SELV) conforming to IEC950/EN60950/VDE0805 to the supply voltage connections.<br>**Failure to follow these instructions can result in equipment damage.** |

## Hardware

The following figure describes the interfaces, display elements, and controls of the ConneXium Switch 10/100 Mbps 5TX/2FX.




The following figure describes the interfaces, display elements, and controls of the ConneXium Switch 10/100 Mbps 7TX.



# Configuration

## Line Structure

The NxS switches enable backbones in line structures to be built up. Cascading is carried out using the backbone ports.

## Redundant Ring Structure

With the redundancy manager function of the NxS modules you can close the two ends of a line structured backbone to a redundant ring. The NxS switches are integrated into the ring via the backbone ports (ports 6 and 7). If one section becomes inoperable the reaction time comes to less than 0.5 s at up to 50 NxS modules being cascaded.

## Redundant Coupling of Network Segments

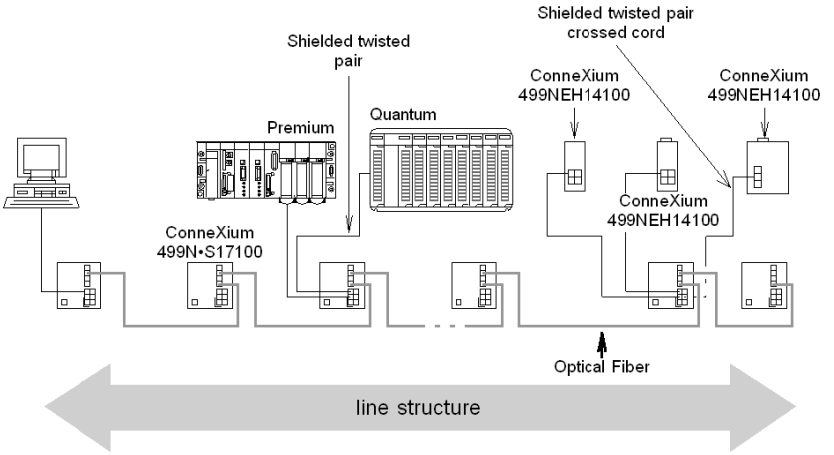
The built-in control intelligence of the NxS allows the redundant coupling of network segments.

The connection of two network segments is realized via two separate paths. The NxS switches in the redundant line get the redundancy function assigned by the DIP switch setting standby.

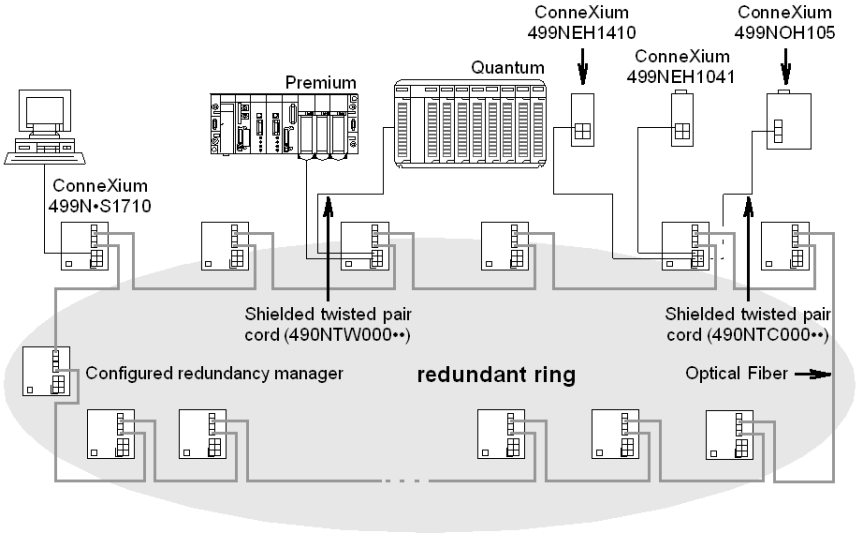
The NxS modules in the redundant line and the NxS switches in the main line share their operating states via the control line (crossover Ethernet cable).

If the communication can not be established on the main line the redundant NxS modules enable the redundant line within 0.5 s. If the main line is okay again, the NxS switches in the main line inform the redundant Nxs modules about this. The main line will be enabled and the redundant line will be disabled within 0.5 s.

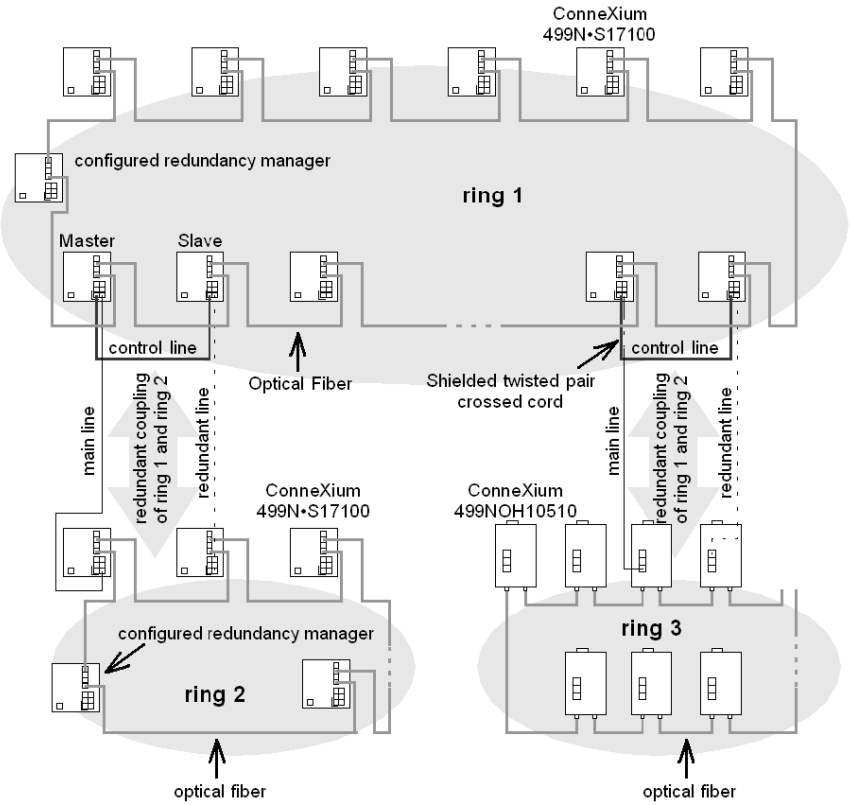
The following figure shows an optical line structure.



The following figure shows a redundant ring structure.



The following figure shows a redundant coupling of optical rings.



# Appendix F
# Agency Approvals and Conformal Coating

## Introduction

This appendix provides information on agency approvals and includes conformal coating availability of the indicated Quantum products.

The table below describes the headings used in the tables in this chapter.

| Heading | Description |
|---|---|
| UL 508 | certified by Underwriters Laboratories according to UL 508 |
| c-UL (UL for Canada) | certified by Underwriters Laboratories according to relevant Canadian standards (CSA) |
| CSA 22.2-142 | certified by Canadian Standards Association according to CSA 22.2 n$^{o}$ 142 |
| FM (Class I, Division 2) | certified by Factory Mutual for Hazardous Locations Class 1 Division 2 |
| CE | compliant to the EMC and Low Voltage European Directives |

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Agency Approvals: Power Supply | 476 |
| Agency Approvals: CPU | 477 |
| Agency Approvals: I/O | 478 |
| Agency Approvals: DIO Drops | 480 |
| Agency Approvals: RIO Heads and Drops | 481 |
| Agency Approvals: Ethernet Modules | 482 |
| Agency Approvals: NOM | 483 |
| Agency Approvals: Motion Modules | 484 |
| Agency Approvals: Battery and Simulator Modules | 485 |

# Agency Approvals: Power Supply

## Power Supplies Table

The following table provides the agency approvals and conformal coating availability for the power supplies of the indicated Quantum products.

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 CPS 111 00 | y | y | y | y | y | y |
| 140 CPS 114 00 | y | y | y | y | y | y |
| 140 CPS 114 10 | y | y | y | y | y | y |
| 140 CPS 114 20 | y | y | y | y | y | y |
| 140 CPS 124 00 | y | y | y | y | y | y |
| 140 CPS 124 20 | y | y | y | y | y | y |
| 140 CPS 211 00 | y | y | y | y | y | y |
| 140 CPS 214 00 | y | y | y | y | y | y |
| 140 CPS 224 00 | y | y | y | y | y | y |
| 140 CPS 414 00 | y | y | y | y | y | y |
| 140 CPS 424 00 | y | y | y | y | y | y |
| 140 CPS 511 00 | y | y | y | y | y | y |
| 140 CPS 524 00 | y | y | y | y | y | y |

# Agency Approvals: CPU

## CPUs Table

The following table provides the agency approvals and conformal coating availability for the CPUs of the indicated Quantum products.

| Quantum Part Numbers | Availability of a Conformal Coated Version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 CPU 311 10 | y | y | y | y | y | y |
| 140 CPU 434 12U | y | y | y | y | n | y |
| 140 CPU 534 14U | y | y | y | y | n | y |
| 140 CPU 651 50 | y | y | y | y | *CSA | y |
| 140 CPU 651 60 | y | y | y | y | *CSA | y |
| 140 CPU 651 60S | y | Scheduled | Scheduled | Scheduled | Scheduled | Scheduled |
| 140 CPU 652 60 | y | y | y | y | *CSA | y |
| 140 CPU 658 60 | y | y | y | y | y | y |
| 140 CPU 670 60 | y | y | y | y | *CSA | y |
| 140 CPU 671 60 | y | y | y | y | *CSA | y |
| 140 CPU 671 60S | y | Scheduled | Scheduled | Scheduled | Scheduled | Scheduled |
| 140 CPU 672 60 | y | Scheduled | Scheduled | Scheduled | Scheduled | Scheduled |
| 140 CPU 672 61 | y | Scheduled | Scheduled | Scheduled | Scheduled | Scheduled |
| 140 CPU 678 61 | y | y | y | y | y | y |

***CSA** Certified by Canadian Standards Association for Hazardous Locations Class 1 Division 2 according to CSA 22.2 n°213.

# Agency Approvals: I/O

## I/O Table

The following table provides the agency approvals and conformal coating availability for the I/O of the indicated Quantum products

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 ACI 030 00 | y | y | y | y | y | y |
| 140 ACI 040 00 | y | y | y | y | y | y |
| 140 ACO 020 00 | y | y | y | y | y | y |
| 140 ACO 130 00 | y | y | y | y | y | y |
| 140 AII 330 00 | y | y | y | y | y | y |
| 140 AII 330 10 | y | y | y | y | y | y |
| 140 AIO 330 00 | y | y | y | y | y | y |
| 140 AMM 090 00 | y | y | y | y | y | y |
| 140 ARI 030 10 | y | y | y | y | y | y |
| 140 ATI 030 00 | y | y | y | y | y | y |
| 140 AVI 030 00 | y | y | y | y | y | y |
| 140 AVO 020 00 | y | y | y | y | y | y |
| 140 DAI 340 00 | y | y | y | y | y | y |
| 140 DAI 353 00 | y | y | y | y | y | y |
| 140 DAI 440 00 | y | y | y | y | y | y |
| 140 DAI 453 00 | y | y | y | y | y | y |
| 140 DAI 540 00 | y | y | y | y | y | y |
| 140 DAI 543 00 | y | y | y | y | y | y |
| 140 DAI 553 00 | y | y | y | y | y | y |
| 140 DAI 740 00 | y | y | y | y | y | y |
| 140 DAI 753 00 | y | y | y | y | **n** | y |
| 140 DAM 590 00 | y | y | y | y | y | y |
| 140 DAO 840 00 | y | y | y | y | y | y |
| 140 DAO 840 10 | y | y | y | y | y | y |
| 140 DAO 842 10 | y | y | y | y | y | y |
| 140 DAO 842 20 | y | y | y | y | y | y |
| 140 DAO 853 00 | y | y | y | y | **n** | y |
| 140 DDI 153 10 | y | y | y | y | y | y |

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 DDI 353 00 | y | y | y | y | n | y |
| 140 DDI 353 10 | y | y | y | y | y | y |
| 140 DDI 673 00 | y | y | y | y | y | y |
| 140 DDI 841 00 | y | y | y | y | n | y |
| 140 DDI 853 00 | y | y | y | y | n | y |
| 140 DDM 390 00 | y | y | y | y | y | y |
| 140 DDM 690 00 | y | y | y | y | y | y |
| 140 DDO 153 10 | y | y | y | y | y | y |
| 140 DDO 353 00 | y | y | y | y | y | y |
| 140 DDO 353 01 | y | y | y | y | y | y |
| 140 DDO 353 10 | y | y | y | y | y | y |
| 140 DDO 364 00 | y | y | y | y | y | y |
| 140 DDO 843 00 | y | y | y | y | y | y |
| 140 DDO 885 00 | y | y | y | y | y | y |
| 140 DII 330 00 | y | y | y | y | y | y |
| 140 DIO 330 00 | y | y | y | y | y | y |
| 140 DRA 840 00 | y | y | y | y | y | y |
| 140 DRC 830 00 | y | y | y | y | y | y |
| 140 DSI 353 00 | y | y | y | y | y | y |
| 140 DVO 853 00 | y | y | y | y | y | y |
| 140 SAI 940 00S | y | Scheduled | Scheduled | Scheduled | Scheduled | Scheduled |
| 140 SDI 953 00S | y | Scheduled | Scheduled | Scheduled | Scheduled | Scheduled |
| 140 SDO 953 00S | y | Scheduled | Scheduled | Scheduled | Scheduled | Scheduled |

# Agency Approvals: DIO Drops

## DIO Drops Table

The following table provides the agency approvals and conformal coating availability for the DIO drops of the indicated Quantum products.

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 CRA 211 10 | y | y | y | y | y | y |
| 140 CRA 211 20 | y | y | y | y | y | y |
| 140 CRA 212 10 | y | y | y | y | y | y |
| 140 CRA 212 20 | y | y | y | y | y | y |

# Agency Approvals: RIO Heads and Drops

## RIO Heads and Drops Table

The following table provides the agency approvals and conformal coating availability for the RIO Heads and Drops of the indicated Quantum products.

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 CRA 931 00 | y | y | y | y | y | y |
| 140 CRA 931 01 | y | y | y | y | y | y |
| 140 CRA 932 00 | y | y | y | y | y | y |
| 140 CRP 931 00 | y | y | y | y | y | y |
| 140 CRP 932 00 | y | y | y | y | y | y |

# Agency Approvals: Ethernet Modules

## Ethernet Modules Table

The following table provides the agency approvals and conformal coating availability for the Ethernet modules of the indicated Quantum products.

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 NOE 211 00 | y | y | y | y | y | y |
| 140 NOE 211 10 | y | y | y | y | y | y |
| 140 NOE 251 00 | y | y | y | y | y | y |
| 140 NOE 251 10 | y | y | y | y | y | y |
| 140 NOE 311 00 | y | y | y | y | y | y |
| 140 NOE 351 00 | y | y | y | y | y | y |
| 140 NOE 511 00 | y | y | y | y | y | y |
| 140 NOE 551 00 | y | y | y | y | y | y |
| 140 NOE 771 00 | y | y | y | y | y | y |
| 140 NOE 771 01 | y | y | y | y | n | y |
| 140 NOE 771 10 | y | y | y | y | y | y |
| 140 NOE 771 11 | y | y | y | y | y | y |
| 140 NWM 100 00 | y | y | y | y | y | y |

# Agency Approvals: NOM

## NOM modules Table

The following table provides the agency approvals and conformal coating availability for the NOM modules of the indicated Quantum products.

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 NOM 211 00 | y | y | y | y | y | y |
| 140 NOM 212 00 | y | y | y | y | y | y |
| 140 NOM 252 00 | y | y | y | y | y | y |

# Agency Approvals: Motion Modules

## Motion Modules Table

The following table provides the agency approvals and conformal coating availability for the Motion Modules of the indicated Quantum products.

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 MMS 425 01 | y | y | y | y | y | y |
| 140 MMS 535 02 | y | y | y | y | y | y |
| 140 MSB 101 00 | y | y | y | y | y | y |

# Agency Approvals: Battery and Simulator Modules

## Battery and Simulator Table

The following table provides the agency approvals and conformal coating availability for the battery and simulators modules of the indicated Quantum products.

| Quantum Part Numbers | Availability of a conformal coated version | Agency Approval Status | | | | |
|---|---|---|---|---|---|---|
| | | UL 508 | c-UL (UL for Canada) | CSA 22.2-142 | FM Class I, Div 2 | CE |
| 140 XCP 900 00 | y | y | y | y | y | y |
| 140 XSM 002 00 | n | y | y | y | n | y |
| 140 XSM 010 00 | n | y | y | y | n | y |

# Appendix G
# System Specifications

## Purpose

This appendix provides an overview on the Quantum System Specifications.

## What Is in This Chapter?

This chapter contains the following topics:


| Topic | Page |
|---|---|
| Mechanical and Electrical Specifications | 488 |
| Power Supply Specifications | 489 |
| I/O Module Specifications | 490 |
| Operating and Storage Conditions | 491 |

# Mechanical and Electrical Specifications

## Mechanical Specifications

| | |
|---|---|
| Weight | 1 kg (2 lb) max |
| Dimensions (H x D x W) | 250 x 103.85 x 40.34 mm (9.84 x 4.09 x 1.59 in) |
| Wire Size | 1-14 AWG or 2-16 AWG max. 20 AWG min. |
| Material (Enclosures and Bezels) | Polycarbonates |
| Space Requirements | 1 rack slot (only HE-CPUs need 2 rack slots) |

## Electrical Specifications

| | |
|---|---|
| RFI Immunity (IEC 1000-4-3) | 80 ...1000 MHz, 10 V/m |
| Ground Continuity (IEC 1000-4-5) | 2 kV shield to ground |
| Electrostatic Discharge (IEC 1000-4-2) | 8 kV air / 4 kV contact |
| Flammability | Wiring connector: 94V-0 module enclosure: 94V-1 |

## Agency Approvals

| |
|---|
| UL 508 |
| CSA 22.2-142 |
| Factory Mutual Class 1, Div 2 |
| European Directive on EMC 89/336/EEC (CE) |

**NOTE:** Quantum system modules contain static-sensitive components. Each module is labeled with the following static-sensitive symbol.



# Power Supply Specifications

## AC/DC Power Supplies

| | |
|---|---|
| Fast transients (IEC 1000-4-4) | 2 kV common mode |
| Damped ocillatory transients | 2 kV common mode. |
| | 1 kV differential mode |
| Surge withstand capability (transients) (IEC 1000-4-5) | 2 kV common mode |
| | 1 kV differential mode |
| Nonperiodic peak input voltage | 2.3 times nominal for 1.3 ms<br>(Nominal = DC average or AC peak) |

# I/O Module Specifications

## I/O Modules with Operating Voltages Less than 24 V

| | |
|---|---|
| Fast transients (IEC 1000-4-4) | 0.5 kV common mode |
| Damped oscillatory transients | 1 kV common mode |
| | 0.5 kV differential mode |
| Surge withstand capability (transients) (IEC 1000-4-5) | 1 kV common mode |
| | 0.5 kV differential mode |

## I/O Modules with 24 to 48 V Operating Voltages

| | |
|---|---|
| Fast transients (IEC 1000-4-4) | 1 kV |
| Damped oscillatory transients | 2 kV common mode |
| | 1 kV differential mode |
| Surge withstand capability (transients) (IEC 1000-4-5) | 1 kV common mode |
| | 0.5 kV differential mode |

## I/O Modules with Operating Voltages Greater than 48 V

| | |
|---|---|
| Fast transients (IEC 1000-4-4) | 2 kV |
| Damped oscillatory transients | 2 kV common mode |
| | 1 kV differential mode |
| Surge withstand capability (transients) (IEC 1000-4-5) | 2 kV common mode |
| | 1 kV differential mode |

# Operating and Storage Conditions

## Operating Conditions

| | |
|---|---|
| Temperature | 0 ... 60 °C (32 ... 140 °F) |
| Humidity | 90 ... 95% RH non-condensing at 6 °C |
| Chemical interactions | Enclosures and terminal strips are made of polycarbonates. This material can be damaged by strong alkalis and various hydrocarbons, esters, halogens and ketones in combination with heat. Common products containing these include detergents, PVC products, petroleum products, pesticides, disinfectants, paint removers, and spray paints. |
| Altitude | 2,000 m. When the altitude exceeds this, reduce the 60 °C maximum operating temperature by 6 °C/1000 m of additional elevation. |
| Vibration | 10 ... 57 Hz at 0.075 mm constant displacement amplitude 57 ... 150 Hz at 1 g |
| Shock | +/-15 g peak, 11 ms, half-sine wave |

## Storage Conditions

| | |
|---|---|
| Temperature | -40 ... 85 °C. C -40 ... 185 °F |
| Humidity | 0 ... 95% RH noncondensing at 60 °C |
| Free Fall | 1 m (3 ft) |

## Gas Resistance in Conformally Coated Modules

This table shows the results of the mixed flowing gas test, 22 days exposure

| Standard | Gas | Test Requirement | Actual Exposure |
|---|---|---|---|
| EIA364-65 Level lll | CL2 (chlorine) | 20 PPB, +/- 5 PPB | 20 PPB |
| | NO2 (nitric oxide) | 200 PPB, +/- 50 PPB | 1250 PPB |
| | H2S (hydrogen sulfide) | 100 PPB, +/- 20 PPB | 100 PPB |
| | SO2 (sulfur oxide) | N/A | 300 PPB |
| ISA-S71.04 (GX Severe) | CL2 (chlorine) | 10 PPB | 20 PPB |
| | NO2 ((nitric oxide) | 1250 PPB | 1250 PPB |
| | H2S (hydrogen sulfide) | 50 PPB | 100 PPB |
| | O2 (sulfur oxide) | 300 PPB | 300 PPB |

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